

EXHALE: Update Log (stage 2, from 2026-09-05)

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This log starts fresh on 2026-09-05. The two earlier logs are kept unchanged under their new names:

- docs/Update_EXHALE_stage0.{tex,pdf}: the early phase (the ATES fork up to 2026-08-15).
- docs/Update_EXHALE_stage1.{md,tex,pdf}: sections 1-171, 2026-06 to 2026-09-05. Every "Update_EXHALE section N" reference in the source comments and in the older memos points into that file; the references were rewritten to the new name on 2026-09-05.

Each entry below is one dated section: what changed, why, which files, the gate that was run and its result, and the movement it produced against the scratch baseline or the goldens. Section numbers restart at 1. The plan the entries execute is docs/development_plan_20260905_rev3.md (sections 1-10) with the order of docs/development_plan_20260905_execution.md; item labels (A1, A8, 10.1-1, D0, C1 ...) are the plan's.

1. Plan, audit and decisions that open stage 2 (2026-09-05)

State of the tree at the start: HEAD 35d9dd5 (clean apart from docs/), binary rebuilt from it with gfortran 16.2 (md5 16cde610347c10c0a48db5c2a7fa6cbb). All twelve regression goldens differ from the case outputs (they are the pre-section-170 snapshots); by decision they are not refreshed until the end of Phase 1 (below).

Documents written today, all under docs/:

- physics_numerics_audit_20260905.md (external audit of 35d9dd5), its diagnostics in audit_20260905/.
- development_plan_20260905.md, _rev1.md, _rev2.md, _rev3.md and the three reviews _review.md, _review2.md, _review3.md: the plan and its revisions after each review. Rev 3 is the plan of record; its section 10 holds the independent five-group source audit (three new P0, twelve new P1, about fifty P2), with the reports and the verification probe in audit_20260905/source_audit/.
- development_plan_20260905_execution.md: Steps 0-7, who acts, gates.

Decisions taken by the user today (recorded in rev 3 sections 8 and 10.5 and in the execution document, Step 0):

1. Goldens are refreshed once, at the end of Phase 1; until then a scratch baseline (backup/regression/baseline_post170_<date>/, read through a REGRESSION_GOLDEN_DIR variable to be added to run_check.sh) is the control for byte-identity. golden/ is not touched before that.
2. ROE interface: speed_estimate_ROE returns a status and, in the vacuum case, the two gas-edge speeds; Num_flux gains an HLLE vacuum branch. Signature shown before coding.
3. One spectrum type builds every band. Power-law input: power law on the whole grid, XUV and below 13.6 eV alike (the present construction is the stated meaning of the key). A new Spectrum type: Planck gives the Planck photosphere from Stellar Teff and Stellar radius on the whole grid. Load: the table on the whole grid. The Balmer continuum follows the run's type. The setup report states the sub-13.6 eV source and the integrated grid flux.
4. $m_{\text{He}}/m_{\text{H}} = 3.9715$ in the species table; $R_{\text{J}} = 7.1492e9$ cm (IAU nominal equatorial) in the code and the Wind-AE bridge alike (R_0 of every run +2.26%); MJ and MSUN to the IAU nominal values.
5. The du stop uses the signed spread of $\rho v r^2$ over the mean, as flux_spread does.
6. Every planet folder, benchmark and planet example runs on the SED its reference literature used or assumed, one file per planet under inputdata/sed/ with provenance; the input switch is a configuration change, the reruns stay instruction-gated.

7. A loaded SED that ends above the band an active absorber needs (He 2³S at 2583 Å, a low-IP metal at its threshold) stops the run; the message offers the two remedies (turn the absorber off in the input, or supply a covering SED). No continue key.

Open: decision 4 (A3 retry design: carrier substep retry or outer step controller); 2, 5, 6 inside D0; 7 before B2; 10 in Phase 5.

Documentation restructuring done today (this section): Update_EXHALE.* renamed to Update_EXHALE_stage1.*, Update_EXHALE_early_phase.* to Update_EXHALE_stage0.*; every reference in src/, docs/, python/, README*.md, the notebooks and the workspace CLAUDE.md rewritten to the new names. Check: `git diff --numstat` shows additions equal to deletions in every one of the 112 touched files (no line moved; the three exceptions are the renamed logs themselves, which git sees as deleted paths). The rebuilt binary's md5 moved from 16cde610347c10c0a48db5c2a7fa6cbb to e779298d0b482bfb40ae2f9aa80d60a because four of the rewritten references are inside printed message strings (EXHALE_main.f90 2602, ionization_equilibrium.f90 2948, input_read.f90 1004, diffusion_tests.f90 1292), which is text the binary carries; no executable statement changed. The pointers in README.md, README_HOWTO.md and the workspace CLAUDE.md name this file as the current log and the stage 1 file as the history.

Scratch baseline created: backup/regression/baseline_post170_20260905/, the Hydro_ioniz.txt, Ion_species.txt, *_adv.txt and run.log of the twelve cases as they sat in the case directories (48 files, 26 MB; provenance in the headers: git=6d07d48afd41 tree=dirty, the working tree that became 35d9dd5, runs of 2026-09-05). It is the byte-identity control for Phase 1 and is deleted after the golden refresh at the end of Phase 1.

2. Phase 0: the tests exist and are red on the known defects (2026-09-05)

Plan reference: development_plan_20260905_execution.md Step 1. Four workers, disjoint files; every result below was re-run by the advisor after the worker's report (times are the advisor's runs).

2.1 src/tests/physics_probe/ (W1): 41 assertions, 15 PASS, 26 FAIL, 3.9 s

Derived from the frozen diagnostics in docs/audit_20260905/ (which stay); links the production modules through a computed use closure (source_closure.py), so no module can be left out of the link by hand. Drivers named for the physics: riemann_wave_speeds.f90 (LLF signal speed 1 against the required 3, mass flux -3.5 against -4.5, mirror symmetry; ROE normalization invariance broken by 0.661 and 3.162 in the two branches; NaN star speeds on the equal-pressure expansion; the production ROE flux leaves thermal energy -2.93 after one Euler update on the vacuum-producing expansion), h2_channel_detector_ratio.f90 ((S+D)/M 0.2545 against the measured 0.2827, D/(S+D) 1/9 against 0.20; channel sum and stoichiometry GREEN), h3p_cooling_limits.f90 (cooling independent of n(H₂) at 0, 1, 100 cm⁻³; joins +0.430, -0.00113, -0.0235 at 300, 800, 1800 K; factor jump 1.5e-3 at 1e14), h2_rovibrational_identity.f90 (chemistry T² dlnZ/dT against the EOS u_{rv}: -0.63% to -8.29% over 300-8000 K), photoevent_energy_ledger.f90 (H event at 20 eV closes exactly through the production photoelectron_share; the H₂ double event at 80 eV accounts for 60.28 of 80 eV, 19.72 eV unassigned), radiation_exchange_ledger.py (two-cell ownership rule, GREEN by construction).

2.2 src/tests/grid_and_gates/ (W2): 6 programs, all RED, 13.4 s

grid_width_identity.f90 (production define_grid): $dr_j(j)$ against $r_{\text{edg}}(j) - r_{\text{edg}}(j-1)$ off by 1.0145 on the Mixed grid at $r_{\text{max}} = 10$, 1.0100 at $r_{\text{max}} = 2$, 1.0046 on Stretched, exact on Uniform (where every cell has the same width, which is why the defect was invisible). photon_grid_quadrature.f90 (production set_energy_vectors, J_inc, cross sections; wasp_full spectrum): code/exact P_HI 1.0951 (floor 4.8 eV), 1.0890 (5.139 eV), 1.0003 (13.6 eV); P_HeI 1.0157 and P_HeII 1.0299 in every configuration; heating 0.99984. New from this test: P_HeI carries the same threshold-bin defect (24.6 eV is a grid point whatever the floor), and the X-ray block ends at 1184.19 eV, not 1240 (share of P_HI 1.9×10^{-10}). mass_flux_spread_functional.py: the coded du and the signed flux_spread on the baseline states (lower_profile 823.4 vs 3.61 with two sign changes in the window; mol_base_handoff 1.29 vs 0.696 with none; wasp_full 9.85×10^{-4} vs 9.85×10^{-4}), and $du = 0$ on a synthetic +F/-F window. output_state_consistency.sh (200-step mol_base_handoff): header sec_ion=T, heat_total/heat median 0.2752, largest difference 0.822. base_level_single_statement.sh (lower_profile, 2 steps): setup 2.0035×10^{-5} bar against the profile's 1.00×10^{-6} bar, factor 20.035. restart_grid_guard.sh (roundtrip on Mixed, restarted on Uniform with the same N): exit 0, the restart's r equals the file's Mixed radii exactly and sits 0.60 relative from the run's own grid.

2.3 inputdata/sed/ (W4): the SED inventory of decision 16

Four files built from data already in the workspace, each with a provenance header, a build script under inputdata/sed/build/, and the coverage test sed_coverage_and_order_check.py (31 assertions, all PASS): WASP-52 b (eps Eri, MUSCLES v22, reproduces the existing planet-folder file to round-off), hot Uranus (the benchmark's Ribas band table continued to 2995 Å from the Gueymard composite it came from), HD 209458 b (WHI 2008 solar minimum at 0.04707 au, the p-winds file reproduced exactly), LHS 1140 b (GJ 1132 proxy). Not built, data missing: WASP-121 b (Huang 2023's LLmodels F5V spectrum above 1700 Å) and HD 189733 b (Salz 2016's composite; its solar table is not here). inputdata/scaled_solar_hd189.sed judged an orphan (no provenance, hardness inconsistent with its own normalization, ends at 1899.5 Å). Verbatim paper quotes and the open items are in inputdata/sed/README.md; rev 3 decision 16 records the status. Also found: WASP-52b/input.inp points at a directory that no longer exists (EXHALE/), the top-level Koskinen benchmark runs at 0.048 au against the paper's 0.05, HD209458b/input.inp at 0.048 au against the literature 0.047.

2.4 Regression harness and new cases (W3)

run_check.sh: REGRESSION_GOLDEN_DIR (default golden/; golden mode still writes only to golden/); the rebuild is fatal with its log kept in backup/regression/.build.log; the summary counts only the cases of the run; the binary copy is status-tested; FILES gains the two _adv.txt profiles (the post-processing pass was outside the regression); a file without a reference is MISS, not FAIL, and a run that compared nothing prints NO VERDICT; a restart case keeps its IC pair in <case>/IC/ and the script restores it after clearing output/; phase0-cases prints the new case list. Makefile: make check CASES=...; make test runs element_census_tests, diffusion_tests, residual_determinism and every executable src/tests/*/run.sh found at run time, every suite even after a failure, nonzero exit if any failed. New cases, not in DEFAULT_CASES (PHASE0_CASES): hydrostatic_column (hot-Uranus gravity, irradiation 18 orders

below normal, no key exists to switch cooling off so the gas is held where it is negligible; 300 steps, isothermal to 2%, but draining through the outflow boundary at 0.8 Mach, so it is a configuration for the Phase 4 D2 measurement, not a hydrostatic state), oxygen_chemistry (examples/18_oxygen_chemistry verbatim, 1000 steps, 3 min 52 s single-threaded under load), hp_front, hp_zero_seed, hp_trace_seed (restarts of the mol_base_handoff state with Ionization transport: True; the H II column set to 0 and to $1e-12$ of the H nuclei by moving the protons into H I, since load_IC refuses a changed He/H; the three 100-step states are mutually distinct, so the seed reaches the solver). No dump hook asserts the seed at the entry of solve_carriers; the README names the two lines where one belongs. Measured (advisor re-run of two cases: 6.2 s, byte-identical on all four files): check wasp_full mol_base_handoff against the scratch baseline is byte-identical on all four files per case (W3, 13 min); the five new cases reproduce their own baseline copies; make check CASES=mol_base_handoff against the stale golden/ reports the known $8.3e-4$ and $3.6e-4$ movements and MISS on the two _adv files; make test (31 s) runs five suites: diffusion_tests and residual_determinism PASS, grid_and_gates and physics_probe RED on the known defects as intended, and **element_census_tests does not compile at HEAD** (pre-existing: carrier_state gained nrho and wfac in section 158 and the test's two calls still pass ten arguments; carrier_write_back likewise takes nrho where the test passes ntot). Fixed in section 3 below.

Also found by W3: with the flux made negligible, the He recombination coupling warning (ionization_equilibrium.f90 947, 952) fires in every cell with a ratio of $1e80$ because both sides sit at their guards; diagnostic only.

3. element_census_tests compiles again (2026-09-05)

src/tests/element_census_tests.f90: the two carrier_state calls pass the nrho and wfac outputs the routine has had since stage 1 section 158, and the two carrier_write_back calls pass nrho (the density the carrier fraction is a fraction of) in place of ntot. Test only; no production source. Result: builds, 20 assertions PASS (E1 stoichiometry and charge, E2 elemental closure through the write-back including the CO collapse, E3 the H2 photoevent nuclei ledger, E4 the base H2 fraction). Until now the suite had been unbuildable, so its assertions had not been checked since section 158.

4. Phase 1 batch 2a: bounded corrections with no golden movement (2026-09-05)

Seven items, seven workers on disjoint files in one tree (private OBJDIR/EXE builds, no shared regression runs); one rebuild and one gate by the advisor afterwards. Plan references: rev 3 sections 4.1 (A2), 10.2 items 7, 8, 10, 12, 10.3, 10.5 decisions 3, 13, 17.

- **A2, ROE estimator and caller** (speed_estimate_ROE.f90 rewritten; Num_Fluxes.f90 ROE branch; input_read.f90; EXHALE_main.f90): docs/a2_roe_interface.md implemented as approved: roe_star_state, status OK/VACUUM/INADMISSIBLE, the vacuum test first, PVRS kept only when admissible, two-rarefaction exponent $1/z$, two-shock densities times the upstream densities, HLLC flux on faces without an admissible star state with a printed counter, refusal of Numerical flux: ROE with molecular chemistry and the ladder EOS. Four deviations accepted: a positivity guard on the input states at the top of the

- estimator; the refusal placed in the consistency block after the optional keys (where `thereis_mol` is known); the old `p_min` floor removed (the guard supersedes it); the counter printed for ROE runs only. Test: 30 ROE assertions GREEN (scale invariance, equal-pressure expansion $c_* 0.9576611$, stationary contact, asymmetric shock and rarefaction against exact Riemann solutions at 5%, vacuum edges $-6.127017/+6.127017$, positive thermal energy after one Euler update).
- **2a-SED, decision 17** (`sed_read.f90`): a table that stops above the band an active absorber needs stops the run with the file, the missing band in eV and A, the absorbers (He 2^3S ; each active neutral metal below the file minimum, K I at 4.341 eV included) and the two remedies; no continue key. Test `sed_coverage_stop.sh` (six assertions) GREEN; no configuration of record trips it (the Koskinen runs have the metastable off).
 - **2a-OPAC** (`util_ion_eq.f90` `PH_heat_H/PH_heat_HHe`, `opacity_models.f90`): model P's factor multiplies the shared flux weight `int_f` of every absorption integrand (rates, heating, secondary ionization, `acc_q`), so the beam's loss equals the local absorption; test `opacity_model_p_ledger.f90`: a cell with $f = 10$ closes energy and H I ionizations to $3e-8$ (was a factor 10 off).
 - **2a-LYA** (`lya_rt.f90`, `util_ion_eq.f90` `fuv_lw_photon_field`, `water_photolysis.f90` header, plus the two call sites in `ionization_equilibrium.f90` and `diffusive_photochemistry.f90` that the new `nHI` argument required): band B2 is attenuated by $\text{lya_stellar_beam_transmission}(T, \tau) = \text{erfc}(x1/(\text{sqrt}(2) X_s))$, the expression `jlya_escape_prob` carried inline and now calls (bitwise equal at 4941 test points). Test `lya_band_transmission.f90` (ten assertions) GREEN. Movement, oxygen_chemistry 1000-step snapshot: du 903 -> 596, log10 Mdot 9.62 -> 9.61, B2 transmission $1e-291$ at the base, $1e-3$ at $1.07 R_p$, 0.98 at $1.5 R_p$; H₂O moves up to 6.2x and OH 7.3x in the layer. Scratch baseline of that case re-snapshotted (previous copy kept).
 - **2a-WAE** (`wae_exhale_input.f90`): mass fractions from the same $m_{\text{He}}/m_{\text{H}}$ the solver is handed; test: reconstructed He/H equals the input to $1e-12$ (was 1.0071696x). RJ/MJ/MSUN untouched (decision 14 is 2c).
 - **2a-HPFIX** (`constrained_chemical_equilibrium.f90`): `hp_is_fixed` pins `species_fixed(is_HII)` like H₂ and the oxygen carriers; new suite `src/tests/constrained_network_layout/` (eight assertions) GREEN. The three `hp_*` cases never enter the continuation (every cell is a class-1 root), so no run exercises the fix yet.
 - **2a-PLANCK** (`J_inc.f90`, `set_energy_vectors.f90`, `input_read.f90`, `write_setup_report.f90`, `excited_hydrogen.f90`, `docs/input_schema.md`): Spectrum type: Planck (requires Stellar `Teff` and Stellar `radius`) builds the whole grid from $\pi B_{\nu}(T_{\text{eff}}) (R_{\text{star}}/a)^2$; the Balmer routines call the same shared function (unchanged result); the setup report states the spectrum type, the source of the band below 13.6 eV and the integrated grid flux against the nominal `J_XUV`. New suite `src/tests/spectrum_type/` (12 assertions) GREEN. Measured on the wasp_full star: Planck/power-law 1642 at 4.8 eV, 358 at 6.2 eV, 2.9 at 10 eV, 0.014 at 13.6 eV; He 2^3S rate 45.2 s^{-1} (Planck) vs 0.100 (power law); integrated grid flux / nominal 1.42 (power law, floor 4.8 eV), 0.995 (floor 13.6), 112.8 (Planck).
 - **2a-GUARDS** (`define_grid.f90`, `set_IC.f90`, `EXHALE_main.f90`, `viscous_conduction.f90`, `steady_newton.f90`, `input_read.f90`, `diffusive_photochemistry.f90`, `water_photolysis.f90`, `oxygen_rates.f90`, `lyman_werner.f90`, `parameters.f90`, `molecular_infrared_data.f90`): `cell_nearest_radius` (the bare `minloc` read `r(503) = 0` out of bounds on the default grid), `j_min >=`

1, dead report_unmet_steady_gates removed, Mdot guarded for small N and non-outflow, conduction floor counted and reported, okres read in the ray test, correct sweep count, get_word initialized, dead close(unit=1) removed, adv_corr zeroed on allocation, dead Tcode argument removed from four carrier routines and their callers, carrier_mass_amu through the named species indices, wp_ready used as the guard it names, dead constants and stale headers fixed, molecular_infrared_data's i private. Test grid_window_indices (seven assertions) GREEN.

Gate, advisor's single rebuild (md5 b52c79ebb15ee9014858425bfea4cb1c): make test runs seven suites; every assertion added in this batch is GREEN and only the known defects of later items remain RED (two test drivers had linked the program object element_census_tests.o from build/; their link lists now exclude program objects). Regression, sixteen cases against the scratch baseline, single-threaded: fifteen byte-identical on all four files (the eleven of DEFAULT_CASES, hydrostatic_column, the three hp_*); oxygen_chemistry moved as stated under 2a-LYA.

Follow-ups recorded for batch 2b and the docs pass: carrier_mass_amu returns 0 for a species missing from the table (needs a stop); write_output.f90 806-816's FUV ledger closed form carries no line transmission for B2 (its rel_diff reads 1.0 now; the LW row was already 0.84); inputdata/sed/README.md describes the old warning in six places; docs/input_schema.md row 11a should state the coverage requirement; docs/a2_oxygen_option_design.md 402 describes the pre-2a B2 field; the physics_probe/README.md summary counts are stale; J_XUV (used by the Wind-AE bridge), the FUV band fluxes and the monochromatic flux do not yet follow the spectrum type (decision 13 is enforced for the photon grid only); backup/regression/parse_golden/ was already stale at 35d9dd5; the three audit probe scripts under docs/audit_20260905/ no longer compile against the tree and are marked as frozen records.

5. Phase 1 batch 2b: bounded corrections that move some outputs (2026-09-05)

Six workers (five code items and one documentation pass), disjoint files, private builds; one rebuild (md5 d2173f0210abdc84f41cb6fb5076e18) and one gate by the advisor. Plan references: rev 3 sections 4.1 (A1, A3 increment (a), A8) and 10.3 (k_lw cell mean, He I singlet, He II recombination cooling).

- **A1, LLF signal speed** (Num_Fluxes.f90, RK_rhs.f90, EXHALE_main.f90): $a1 = \max(\text{abs}(vL)+aL, \text{abs}(vR)+aR)$, the spectral radius of the Euler flux Jacobian, replacing $\max(\text{abs}(vL+aL), \text{abs}(vR+aR))$ (1 instead of 3 on the inflow test pair); the header's positivity claim now describes the coefficient the code carries. Reporting: calls of the positivity repair, repaired interfaces, repaired interfaces in accepted steps, and LLF faces, all printed even when zero. Test: the three LLF assertions GREEN. Movement: none (no case selects LLF and the repair path is never entered in the matrix: flux correction: 0 positivity repair call(s) in every run).
- **A3 increment (a), carrier reference scales and status** (diffusive_photochemistry.f90, write_setup_report.f90, EXHALE_main.f90): one carrier_element_reference_density(ic, nH, nO, nC) (H2 and H+ hydrogen, OH and H2O oxygen, CO min(O, C)) read by both the Jacobian step (carrier_source_derivative_step) and the residual floor; the reported residual is that of the state after limit_to_element_budget; solve_carriers returns converged / iteration cap / line search failed and the caller stops on a failed status in a carrier the limiter did not move (rejection and retry remain increment

- (b), decision 4 open); a cumulative CO-ceiling record (cells, applications, molecules removed) in the run summary and EXHALE_resolved.out; carrier_mass_amu stops on a species missing from the table. New suite src/tests/carrier_reference_scales/ (14 assertions; before the change the proton row's diagonal derivative was exactly zero at $H^+ = 0$ and $1e-12$ n_H because the step had collapsed to $1e-300$). Movement of this item alone: hp_zero_seed up to $7.8e-4$ at the ionization front, hp_trace_seed $2.8e-5$, hp_front round-off, mol_carrier and mol_base_handoff byte-identical. Cost of the extra residual assembly +1.5% wall time on hp_zero_seed.
- **A8, Chung detector inversion** (h2_photo_channels.f90): Table II is the detector event ratio $(N_S+N_D)/N_M$ (Chung et al. 1993 p. 886, one pulse per event, quoted at the code site), $q_D = N_D/(N_S+N_D) = 0.20$ at 80 eV; $\sigma_M = (1-f_{di}) \sigma_i$, $\sigma_S = (1-q_D) f_{di} \sigma_i$, $\sigma_D = q_D f_{di} \sigma_i$; yan_rho converted with $q_D = \rho_D/(1+\rho_D)/f_{di}$; an inadmissible q_D stops the run instead of being clipped; frac_H2_double_of_protons renamed frac_H2_double_of_proton_events. At 80 eV: D 0.02254 -> 0.04408, M 0.7972 -> 0.7796, S 0.1803 -> 0.1763. Test: 18 assertions GREEN including the stored-channel and yield integration checks; E1 test 4 prints the same $6.951e-3$. Movement of this item alone: mol_base_handoff n_e up to $1.8e-2$ at $r = 1.018$, x(H⁺) $1.9e-2$ at the base, T $4.6e-4$; mol_sec_ion at the $1e-3$ to $5e-3$ level, with a 0.078 cm/s velocity shift moving the $v = 0$ crossing by one cell at $r = 1.024$ - 1.026 and flipping the advection-correction branch of the _adv files there. Mdot unchanged.
 - **k_lw cell mean** (lyman_werner.f90, util_ion_eq.f90 fuv_lw_photon_field): new lyman_werner_dissociation_rate_cell_mean(F, N_out, N_in, T, n_H, tau_out, tau_in), a composite three-point Gauss rule on segments of at most 0.05 dex of column and 0.5 of continuum depth (the level-resolved cross section is a piecewise power law with slope up to 3.3, so the water bands' exponential mean does not apply; measured accuracy $2.5e-4$ worst over 1500 random cells); the star-ward face carries NH2col(j+1). Test: seven assertions GREEN (the old inner-face rate was 3.18x too small on a cell spanning 1.12 dex). Movement: mol_lyman_werner k_LW +1.4% to +28.1% (median +2.7%), n(H2) -20% and n(H3+) -20% at the front $r = 1.071$, T up to $8e-3$, n_e $1.7e-2$; mol_base_handoff and wasp_full identical (no LW flux). Cost under 0.5%.
 - **He I singlet and He II recombination cooling** (util_ion_eq.f90 eval_cool/eval_cool_cells, Cool_coeff.f90, ionization_equilibrium.f90 comment): the 24.6 eV collisional ionization and the Cen (1992) excitation of He I are charged to the ground singlet he_ground_singlet_density(nhei, nheiTR) (the metastable had paid both levels); $\lambda_{rec_HeII} = kT \alpha_{rec_HeII_total}(T)$ with $\alpha_{rec_HeII_total} = \alpha_{11S} + \alpha_{23S}$ when the metastable is tracked (the coefficient the balance removes He+ at) and alpha_B otherwise; the coefficient is built inside eval_cool from T and the run's flags so that energy_semi_implicit's finite-difference dC/dT sees one function (a coefficient-passing form would have left an offset the derivative multiplies by $1e5$). At $1e4$ K the He II coefficient rises $2.807e-13 \rightarrow 3.640e-13$ (+30%); the He recombination coupling's own weight (y_net alpha_1) is still not charged (about 10% of this channel, stated at the code site). Test: 15 assertions GREEN. Movement of this item alone: wasp_full median $2e-5$ / max $1.7e-4$ in rho, T max $5.9e-4$, Ca I and Na I up to $4.7e-3$; mol_base_handoff cool column median 1% / max 2.6% (the recombination channel is 64% of the cooling there), state $1e-4$.

- **Documentation pass:** inputdata/sed/README.md (seven places, the stop), docs/input_schema.md (row 11a coverage rule; rows 11, 11b, 11d, K3, K4 line references corrected), docs/a2_oxygen_option_design.md (B2 row), src/tests/physics_probe/README.md (two summaries: 41/15/26 at 35d9dd5; 83 assertions, 69 PASS, 14 FAIL after 2a with the LLF fix already in the tree) and grid_and_gates/README.md, rev 3 section 10.2 and 10.5 "landed" notes, the execution document Steps 1 and 2a marked done, README.md and README_HOWTO.md (Spectrum type: Planck, the coverage rule, make test, the regression section's case list corrected from eight to eleven).

Gate (advisor): make test runs eight suites; every 2b assertion GREEN, RED only on the 2c and Phase 3 targets. Regression, sixteen cases against the post-2a scratch baseline, single-threaded: wasp_he23off and hydrostatic_column byte-identical (metastable off, no carriers); the other fourteen moved, Mdot unchanged to two decimals in all sixteen. Largest relative differences over all cells and columns (single cells; the large values are v near a zero crossing or trace species in one cell, see the items):

case	Hydro_ioniz	Ion_species	Hydro_ioniz_adv	Ion_species_adv
wasp_full	1.06e-3	4.63e-3	1.37e-3	4.63e-3
wasp_full_newton	1.48e-3	4.69e-3	1.91e-3	4.69e-3
mol_base_handoff	2.54e-2	1.99e-2	5.58e-2	4.21e-2
mol_metals	2.16e-2	2.20e-2	2.70e-2	2.20e-2
mol_lyman_werner	1.25e-1	2.06e-1	5.63e-2	2.06e-1
mol_diffusion	2.54e-2	1.99e-2	5.86e-1	1.95e-2
mol_ir_bands	2.16e-2	2.20e-2	9.65e-2	2.20e-2
mol_sec_ion	1.70 (v at a zero crossing)	6.31e-3	1.70	1.0 (trace ion, one cell)
mol_carrier	2.52e-2	1.98e-2	3.87e-2	1.95e-2
lower_profile	5.38e-1	5.19e-2	5.38e-1	4.96e-2
oxygen_chemistry	1.94	1.0	1.94	1.0
hp_zero_seed	8.13e-2	9.54e-2	4.20e-2	9.54e-2
hp_trace_seed	4.25e-1	1.0	4.24e-2	1.0
hp_front	2.59e-2	6.30e-3	4.42e-2	6.29e-3

The scratch baseline was re-snapshotted from these outputs; the post-2a state of every case is kept in backup/regression/baseline_post2a_20260905/.

Follow-ups recorded: T_equation.f90 (the temperature root of the advection post-process) carries no He 2³S cooling channel and charges the He I ground-state coefficients to the summed neutral column, so the _adv temperature and the hydro temperature are not solved with one cooling function (Phase 3 A5/A6 or the _adv consistency item of Phase 6); post_process_adv.f90 392 calls eval_cool without nheitr; cross_sec.f90 519-528, parameters.f90 475-501 and input_read.f90 856-858 still describe the H⁺ column as a proton count and quote a "factor 1.6" between the two double-ionization models that is now reversed and much smaller; write_output.f90 200-203 says the f_shield column is the factor the rate carries (the rate is now the cell mean); pct_limited counts only overshoots above 1e-

10 while the clamp acts at any size (same class as the CO counter, now recorded cumulatively); the give-up path of `retry_step` leaves the loop through the same exit as an accepted step (decision 4); `set_oxygen_coeffs` is called with strided sections that create array temporaries in the carrier Jacobian; `docs/EXHALE_user_manual.tex` is stale on the spectrum types and the SED stop.

6. Phase 1 batch 2c: corrections that move every output (2026-09-05)

Fourteen worker items on disjoint files plus advisor edits, every one measured alone against a frozen snapshot of the tree as it stood when the item started (private builds; the items therefore chain, and the "before" of one item is the "after" of those that landed earlier). One rebuild by the advisor with the conda gfortran 16.2 and its OpenBLAS (md5 5845df587aa7948153f1bed3becb28ac), `make test`, and one regression run against the post-2b scratch baseline. Plan references: rev 3 sections 10.1 (items 1-3), 10.2 (items 1-6, 9, 11), 10.5 (decisions 13-15, 17, 2c-CONST). Movement is quoted as the largest relative change over the 500 physical cells of `wasp_full` (WASP-121b, He 2^3 S + metals, du stop, no Newton finish, so its `Mdot` carries the path-dependent spread of that stop) and `mol_base_handoff` (hot Uranus, 12000-step relaxation snapshot, not steady, so its outer-tail and base-cell numbers compare two points of one transient).

- **2c-DRJ, the cell width is the cell's own width** (`define_grid.f90`, `set_IC.f90`): `dr_j(j) = r_edg(j) - r_edg(j-1)`, an exact identity of the finite-volume grid; the stored width had been the width of cell `j+1`, 0.7 to 1.25 percent too large over the stretched region. Every consumer confirmed to use `dr_j(j)` for cell `j`; none changed. `grid_width_identity` GREEN. `wasp_full`: T $2.0\text{e-}3$, v $5.4\text{e-}3$ (base cell), n_e $6.3\text{e-}3$, $x(\text{H}^+)$ $1.1\text{e-}2$, mass flux +0.65 percent, printed `log10 Mdot` 13.24 both. `mol_base_handoff`: bulk profiles $1\text{e-}2$ class with the transient tail at $1\text{e-}1$, mass flux -0.96 percent, printed `Mdot` 10.52 -> 10.51.
- **2c-QUAD, thresholds on bin edges** (`set_energy_vectors.f90`): the photon grid is built from bin edges, `e_v` the geometric center of each bin and `de_v` its exact width; H I, He I, He II, the metastable edge and `e_mid`, `e_top` are edges, and the X-ray block reaches `e_top`. `photon_grid_quadrature` GREEN except P_HeI at $1.2\text{e-}3$ (closed by 2c-THR). `wasp_full`: T $7.5\text{e-}2$ at 1.12 Rp, n_e $7.8\text{e-}2$, $x(\text{H}^+)$ 0.23, base v 0.47, mass flux +0.82 percent, `log10 Mdot` 13.24 -> 13.25. `mol_base_handoff`: T $4.7\text{e-}2$ (tail), n_e $2.5\text{e-}2$, mass flux +0.08 percent, `Mdot` 10.51 both. This is the single largest movement of the batch after the Jupiter radius, and it is the removal of the 9.5 percent overcount of P_HI found by the audit.
- **2c-BALMER, the Balmer continuum follows the run's spectrum type** (`excited_hydrogen.f90`, `J_inc.f90`; decision 13): the $n = 2$ photoionization rate and heating integrate the field the photon grid uses, the power law for a power-law run, the Planck field for Planck, the loaded table for Load (`stellar_flux_eV`, `spectrum_covers_eV`); a table that does not reach 3647 Å stops the run with the decision-17 message naming H($n=2$). `balmer_continuum_field` GREEN (11 assertions). `wasp_full` ($n = 2$ coupling armed): `gamma_2` falls by a factor 2939, n_e at 1.02 Rp -12.5 percent, $x(\text{H}^+)$ at the base -27 percent, T $9.6\text{e-}3$, wind above 1.2 Rp and `Mdot` unchanged (13.25). `mol_base_handoff`: byte-identical (no stellar lines).
- **2c-KICK, the momentum kick removed** (`EXHALE_main.f90`): the `u(2,:) + 1e-16` diagnostic aid is gone; the relative step change skips cells whose old momentum is exactly zero. `momentum_row_from_fluxes_only`

- GREEN. Movement round-off on `wasp_full` ($1e-9$), $4e-6$ on `mol_base_handoff`, $4.4e-3$ in `v` on `lower_profile` (cold-start snapshot with a sign-changing base flux amplifies it); `Mdot` unchanged everywhere.
- **2c-STATE, one state per output file** (`EXHALE_main.f90`): when the secondary-ionization coupling never switched on during the march, the final write no longer forces it on; the header says `sec_ion=F`, and the `_adv` files and the heating breakdown are built from the relaxed state. `Hydro_ioniz.txt` / `Ion_species.txt` byte-identical in every case; the `_adv` files and `Heating_breakdown.txt` of the fixed-step snapshots whose flip never fired (`mol_base_handoff`, `lower_profile`, and the other molecular gates) change by up to 0.97 in `n(H II)_adv` at the base. `output_state_consistency` stays RED at $1.2e-5$: the heat column is assembled from the pre-sweep rates and the breakdown re-evaluates on the post-sweep composition (Phase 3, together with the `excited_H_update` call between the loop and the final write).
 - **2c-DU, the signed mass-flux spread** (`EXHALE_main.f90`; decision 15): one functional `mass_flux_spread`, $(\max - \min) / \max(|\text{mean}|, 1e-30)$ of the signed $\rho \ v \ r^2$ over `j_min..N`, read by the stop, the Newton arming and the stage flip, and written to `run.log` at the final step (`mass_flux_spread_functional.py` GREEN). `wasp_full` byte-identical (every threshold crossing on the same step); `mol_base_handoff` byte-identical (fixed steps; the reported `du` falls from 1.29 to 0.70); `wasp_full_newton` flips and hands off one step earlier, solution moves by $7e-8$.
 - **2c-RESTART, a restart keeps the run's grid** (`load_IC.f90`): the file's cell centers are compared with the run's over the physical cells and a relative difference above $1e-10$ refuses the restart, naming both grids. `restart_grid_guard.sh` GREEN; `roundtrip` and the three `hp_*` cases byte-identical on the same grid. Consequence: after the grid-moving items of this batch (RJ +2.26 percent of the domain in `R_p` units, `dr_j`) the shipped `hp_*` IC pairs are refused, and are regenerated with `backup/regression/make_hp_initial_conditions.py` from a fresh `mol_base_handoff` output (below).
 - **2c-BASE, one base level with a lower-atmosphere profile** (`input_read.f90`, `write_setup_report.f90`): with a profile, `n0` follows from `p_match_bar` exactly as from `p_base` of `base.inp`; a density key beside a profile is refused unless it agrees within 1 percent (`base_level_single_statement.sh` GREEN). `lower_profile` starts at the profile's level: base density $1.0e14 \rightarrow 5.0e12 \text{ cm}^{-3}$, its 12000-step snapshot changes entirely (printed `log10 Mdot` 8.44 $\rightarrow$ 7.92, ionization front inward, base `n_e` $\times 8$). The case README and `examples/17_lower_profile/input.inp` drop the density key. Nothing else moves.
 - **2c-CI, the C I Voronov row** (`Cool_coeff.f90`): `P = 0`, `X = 0.193`, `K = 0.25`, `A = 6.85e-8`, `dE = 11.26` (Voronov 1997 Table 1; the transcription had X and K in the wrong slots, 2.56x at 2000 K, 1.69x at 1e4 K). `voronov_carbon_ionization` GREEN. `wasp_full` $1e-5$ class, `x(C I)` $6e-4$, `Mdot` $-1e-5$ dex; `mol_metals` bitwise identical (collisional ionization of C I is negligible at its temperatures).
 - **2c-MASS and 2c-MASS2, helium mass and Jupiter radius** (decision 14; `species_table.f90`, `parameters.f90`, `wae_exhale_input.f90`, `wae_ic_writer.f90`, `binary_element_diffusion.f90`, `diffusive_photochemistry.f90`, `lower_column.f90`, `utilities.f90`, `examples/exhale_io.py` and the Python tools): He I / He II / He III / He 2^3S carry 3.9715 `m_H` (HeH+ 4.9715), `RJ = 7.1492e9` cm, `MJ = 1.8982e30` g, `Msun = 1.98842e33` g (IAU 2015 nominal), one definition each, the Python tools importing `RJ_CM`, `MJ_G`, `MSUN_G` from `exhale_io`. Tests `atomic_mass_and_radius_constants`,

jupiter_radius_python_tools.py, suite species_masses GREEN. Measured as two sub-steps: the helium mass alone moves wasp_full by $2.3\text{e-}3$ in T, $1.1\text{e-}2$ in $x(\text{H}^+)$, $\text{Mdot} +0.095$ percent, and mol_base_handoff by $1.2\text{e-}2$ (tail), $\text{Mdot} -0.004$ percent; the Jupiter radius then rescales every physical length (R_0 , the domain top r_{max} in R_p units, the gravity) and moves wasp_full by $7.3\text{e-}2$ in T, 0.16 in n_e , 0.34 in $x(\text{H}^+)$, raw $\text{Mdot} +11.0$ percent (printed $13.24 \rightarrow 13.29$), and mol_base_handoff raw $\text{Mdot} +15.8$ percent (printed $10.52 \rightarrow 10.58$). MASS2 alone (the remaining copies): mol_diffusion $2.8\text{e-}5$, mol_carrier $6.5\text{e-}8$, wasp_full byte-identical.

- **2c-THR, one definition of each threshold** (parameters.f90, cross_sec.f90 and the files that restated a threshold): $e_{\text{th_HI}} = 13.598434599$, $e_{\text{th_H2}} = 15.425927$, $e_{\text{th_HeI}} = 24.587389$, $e_{\text{th_HeII}} = 54.417765$, $e_{\text{th_HeTR}} = 4.767775$ eV (NIST ASD), every cross section turning on exactly there; $\text{sigma}(E, Z)$ takes the ion's measured potential as an optional argument so He II is not restated as $13.6 Z^2$. ionization_threshold_turn_on and threshold_literal_uniqueness.py GREEN; the P_HeI residual of QUAD closes ($0.99878 \rightarrow 0.99995$). wasp_full $7\text{e-}4$ at most, mass flux $+0.044$ percent; mol_base_handoff $1.3\text{e-}3$ (tail); printed Mdot unchanged.
- **2c-N2FLOOR, H(n=2) is an absorber of the grid-floor rule** (sed_read.f90 photon_grid_floor_ev, J_inc.f90, set_energy_vectors.f90, excited_hydrogen.f90): with the excited-hydrogen coupling armed the grid floor is $e_{\text{th_HI}}/4 = 3647$ Å, a loaded SED is read down to it and the decision-17 stop is about the file, not the type (WASP-52b with the eps Eri table to 3700 Å now runs). Tests sed_coverage_stop.sh and photon_grid_n2_floor GREEN: the 20 bins added below the lowest bulk-gas absorber change no rate the grid computes, exactly. wasp_full $2\text{e-}4$ at most (the edge written as $e_{\text{th_HI}}/4$ instead of 3.40 eV), Mdot 0.0000 dex; wall clock $+2.3$ percent for the 20 bins.
- **2c-CONST, the particle masses and the astronomical constants in one place** (parameters.f90, species_table.f90, lower_column.f90, Cool_coeff.f90, binary_element_diffusion.f90, diffusive_photochemistry.f90, lyman_werner.f90, oxygen_rates.f90): $m_{\text{He_atom}} = 6.6464790722\text{e-}24$ g (4.002603254 u $\times$ CODATA 2018 u; the brief's $6.6464731\text{e-}24$ was $9.2\text{e-}7$ low and is not the helium-4 atom), amu , m_e , m_p in parameters.f90, $m_{\text{He_over_m_H}}$ and amu_over_m_H derived, bsp_mass built from them, $\text{melem_A} = \text{melem_A_u} * \text{amu}/\mu$. This closed a UNIT ERROR: the standard atomic weights in u had been added to mass sums carried in hydrogen atoms, every metal nucleus 0.78 percent too heavy in ρ , the mean molecular weight, the base composition, the census closure and the metal element diffusion. The advisor closed the remaining local copies (Cool_coeff line-center opacity now takes the atomic weight in u and uses the global amu; $\text{amu_C/N/O} = \text{melem_A_u}(\text{iel_}*)$; the local $m_{\text{amu_g}}$ and kb_erg/m_h2 copies of the diffusion, Lyman-Werner and oxygen modules replaced by the globals). wasp_full $1.0\text{e-}3$ at most (base v), mass flux $+3.2\text{e-}5$; mol_base_handoff $2.8\text{e-}5$; printed Mdot unchanged.
- **2c-METALEDGE, every active threshold a bin edge** (set_energy_vectors.f90): each of the six bands keeps its bin budget and is cut at every ionization threshold strictly inside it (H2 when the molecular chemistry is on, every photo-ionizable stage of an element with nonzero abundance), the budget shared over the sub-intervals in proportion to their logarithmic width with at least one bin each; N_L is derived from the partition and equals the previous count in every configuration (240 wasp_full, 220 mol_metals, 220 mol_base_handoff). A band with

no interior threshold keeps exactly the partition it had. photon_grid_threshold_edges GREEN: seventeen thresholds at relative distance 0 from an edge (eighteen with H2), P_m of C I 1.045 $\rightarrow$ 1.000, Ca I 0.933 $\rightarrow$ 0.997, Na I 1.030 $\rightarrow$ 0.998, Fe I 0.965 $\rightarrow$ 0.994 of a 400001-point reference, total H2 photoabsorption 0.985 $\rightarrow$ 1.000. wasp_full: T 9.6×10^{-3} , n_e 5.2×10^{-3} , $x(\text{C I})$ 0.14 at the base, mass flux -0.17 percent, $\log_{10} \dot{M}$ 13.2860 $\rightarrow$ 13.2852. mol_metals: bulk 2×10^{-3} , n_e 4.5×10^{-2} in the neutral layer at 1.1 R_p (ionization degree 1×10^{-6}), $\dot{M}$ unchanged. mol_base_handoff: 1×10^{-4} class, by the H2 edge alone. Open (decision needed, below): Mg I, Ca I, Na I, Fe I still miss 1×10^{-3} by 0.25 to 0.58 percent, the rectangle-rule error of the 20-bin sub-Lyman band.

- **Advisor edits after the worker:** the H2 channel thresholds $e_{\text{th_H2_di}}$ (18.076 eV) and $e_{\text{th_H2_dd}}$ (51.4 eV) join the edge set when the molecular chemistry is on, because the channel shares of one cross section switch on there (the 32-41.5 eV neutral-dissociation window is continuous and is not cut; the comment at the code site says so); the setup report prints the grid floor as the lower edge of the first bin (photon_grid_floor_eV, or the table floor for a loaded SED) instead of the first bin center, which the new partition had moved to 4.95 eV against a floor of 4.768 eV. The documents that restated the changed quantities were brought in line: docs/input_schema.md (profile level, R_J), docs/a2_oxygen_option_design.md (molecular masses in m_H units), rev 3 and the execution document (landed marks).

Decision handed to the user: the sub-Lyman band budget

The band [4.768, 13.598] eV has 20 bins (edge ratio 1.054), 4.4 times coarser in $d \ln E$ than the pure-H I band (50 bins, 1.012), and it is where every low-IP metal is photoionized; Fe I's cross section falls by a factor 2.5 across the single bin above its threshold. Measured ladder (code/exact):

num_TR	20	30	40	50	80
Mg I	0.99753	0.99871	0.99915	0.99955	0.99982
Ca I	0.99737	0.99877	0.99937	0.99961	0.99984
Na I	0.99752	0.99912	0.99958	0.99967	0.99987
Fe I	0.99417	0.99685	0.99805	0.99883	0.99953
Nl (wasp_full)	240	250	260	270	300

Decided by the user 2026-09-05: num_TR = 89, the budget that gives the band the bin ratio of the pure-H I band (0.0118 in $\ln E$). Measured with the same driver: Mg I 0.999853, Ca I 0.999870, Na I 0.999902, Fe I 0.999615, so the four rates are asserted at the 1×10^{-3} of every other ion. Nl becomes 309 for wasp_full (was 240) and 289 for the hot-Uranus cases (was 220); the H($n=2$) band keeps its 20 bins. Every He 2^3S and metals-on case moves again by this, and the gate below was re-run on the 89-bin binary (md5 b0a4ada9f84ee8410c84fda6780ba097) before the baseline re-snapshot. The user also decided the LHS1140b/examples/photochem_profile_base/input.inp density key: deleted (the profile fixes the level).

Gate (advisor)

make test on the single rebuild: every assertion of this batch GREEN (threshold_edges, n2_floor, balmer_continuum_field, atomic_mass_and_radius_constants, species_masses, ionization_threshold_turn_on, threshold_literal_uniqueness, voronov_carbon_ionization, grid_width_identity, photon_grid_quadrature, restart_grid_guard, base_level_single_statement, mass_flux_spread_functional, momentum_row_from_fluxes_only, sed_coverage_stop); diffusion_tests had 13 of 35 assertions fail because its own references carried the helium mass as 4 m_H ($7.2e-4 = 1 - 3.9715/4$), and now derives every helium mass from m_He_over_m_H (35 of 35, no tolerance changed, the T9 ambipolar targets written as functions of the mass ratio). RED, as intended, only on the Phase 3 targets: output_state_consistency ($1.2e-5$), h3p_cooling_limits, h2_rovibrational_identity, h2_double_photoevent_energy_sum, helium_collisional_ionization_by_level.

Regression, sixteen cases against the post-2b scratch baseline, single-threaded: the thirteen cases that ran all moved beyond the $1e-3$ tolerance, as every item of this batch predicts; the three hp_* restart cases were refused by the new grid guard (their IC pairs sat on the old grid: domain top 4.7253 -> 4.6215 R_p from R_J, base spacing from dr_j) and were regenerated with make_hp_initial_conditions.py from the fresh mol_base_handoff/output (nuclei conserved row by row; H II as written / 0 / $1e-12$ of the nuclei) and re-run. Movement of the thirteen (the cell-wise maxima compare the same cell index, whose radius moved by up to 2.2 percent with the domain, so they overstate the change at fixed radius; the Mdot column is the run's printed value):

case	log10 Mdot before -> after	mass flux at cell N-20	max rel T	max rel v	max rel rho	domain top [R_p]
wasp_full	13.24 -> 13.29	+6.2 %	6.7e-02	7.6e-01	4.1e-01	1.6014 -> 1.5666
wasp_he23off	13.24 -> 13.29	+6.2 %	6.7e-02	7.7e-01	4.1e-01	1.6014 -> 1.5666
wasp_full_newton	13.21 -> 13.26	+6.7 %	1.3e-01	1.9e-01	3.4e-01	1.6014 -> 1.5666
mol_base_handoff	10.52 -> 10.57	+8.3 %	2.5e-01	1.1e-01	4.5e-01	4.7253 -> 4.6215
mol_metals	10.52 -> 10.58	+8.3 %	2.4e-01	1.1e-01	4.4e-01	4.7253 -> 4.6215
mol_lyman_werner	10.50 -> 10.55	+8.2 %	1.0e-01	1.2e-01	2.5e-01	4.7253 -> 4.6215
mol_diffusion	10.52 -> 10.57	+8.3 %	2.5e-01	1.1e-01	4.5e-01	4.7253 -> 4.6215
mol_ir_bands	10.52 -> 10.58	+8.3 %	2.4e-01	1.1e-01	4.4e-01	4.7253 -> 4.6215

case	log10 Mdot before -> after	mass flux at cell N-20	max rel T	max rel v	max rel rho	domain top [R_p]
mol_sec_ion	10.47 -> 10.52	+8.2 %	2.1e-01	3.7e+02 (sign change of a stagnant base cell)	3.8e-01	4.7253 -> 4.6215
mol_carrier	10.52 -> 10.58	+8.4 %	2.5e-01	1.1e-01	4.6e-01	4.7253 -> 4.6215
lower_profile	8.44 -> 7.95	-68.9 %	2.0e+00	1.5e+03 (sign change)	9.7e-01	4.1487 -> 4.0577
hydrostatic_column	8.25 -> 8.27	-0.1 %	5.7e-02	4.3e-01	4.9e-01	2.9600 -> 2.9595
oxygen_chemistry	9.61 -> 9.60	-6.9 %	7.3e-01	3.1e+01 (sign change)	8.9e-01	4.1487 -> 4.0577

The +0.05 dex of every WASP-121b and hot-Uranus case is the Jupiter radius (item MASS, +11 and +16 percent in the raw flux at fixed cell, partly compensated by the shrunk domain at cell N-20); lower_profile is item BASE (the run now starts at the profile's 1 microbar level); the oxygen_chemistry and hydrostatic_column moves are the grid and the threshold edges. The baseline was re-snapshotted after the hp_* re-run (the post-2b copy kept as baseline_post2b_20260905/).

Golden refresh: prepared, awaiting the user's command

golden/ was archived to backup/regression/golden_pre170_20260905/ with a README (provenance from the headers, unknowns marked), roundtrip was re-run on the post-2c binary, and DEFAULT_CASES of run_check.sh now lists the five Phase 0 cases as well. The refresh itself, run_check.sh golden over the seventeen cases followed by make check and one run_fcheck.sh, is the user's command to give; until it is given the reference of record is the pre-2c state and make check reports every case as moved.

Gate on the 89-bin band (2026-09-06)

Fourteen cases re-run against the post-2c (20-bin) baseline with md5 b0a4ada9f84ee8410c84fda6780ba097: hydrostatic_column byte-identical (no sub-Lyman absorber), every other case within 1.3e-3 in any cell (mol_metals rho 1.3e-3, wasp_full v 1.3e-3 at the base) and printed Mdot unchanged, the size the 20 -> 89 ladder predicts; the hp_* IC pairs were regenerated on the 89-bin mol_base_handoff output and the three cases run (Mdot 10.56).

oxygen_chemistry stopped on the 89-bin grid, diagnosed. At step 13, cell 7, the carrier Newton reached its iteration cap (30) with a relative OH row residual of 0.63 and the status stop of item 2b-A3a ended the run. Diagnosis (worker, measured on frozen snapshots): the Newton is correct (a root exists; the same solver with a larger budget reaches it in 141 iterations, the last three 5.4e-2, 3.1e-9, 3.9e-12) and its residual scale and the element limiter are not involved. The stiffness comes from upstream: at step 12 the ionization sweep accepted a non-root at cell 7 under the relaxation

amnesty (info 4, normalized residual 5.2e4), which put $n_e = 3.5e10$ and $n(H+) = 2.6e8$ into the frozen background where the previous step had 1.1e3 and 0, so the $H+ + H_2$ channel asked for two thirds of the cell's H_2 in one 2.85 s step. On the 20-bin grid cell 7 never took that path. The "7.8e-2 carrier row" of the 20-bin summary is the STEADY residual of a relaxation snapshot (dt $\rightarrow$ infinity), not the Newton's; the backward-Euler Newton converged to below 6e-12 on all 999 steps there. Two things landed: the iteration budget is 200 (the comment at the code site states the measurement), and the attribution test of the status stop, which asked "did an element constraint act" by carrier over the whole grid, now asks it by cell (the CO ceiling, acting in 160 cells of this case, had made any CO failure anywhere excusable, and had left OH rows it moved by 49 percent through the oxygen closure reported as the solver's); new suite `src/tests/carrier_constraint_attribution/`, 7 assertions, RED before. Open for the user: decision 4 (reject and retry the step on a shorter dt, the structure that also covers a step without a root) and the ionization sweep's relaxation amnesty, which handed the carrier system a non-root (acceptance policy, `ionization_equilibrium.f90` class 4).

Golden refresh executed (2026-09-06, on the user's instruction)

`run_check.sh` golden over sixteen cases: the eleven of the previous DEFAULT_CASES, hydrostatic_column, the three `hp_*` restart cases and roundtrip, all from the 89-bin binary `b0a4ada9f84ee8410c84fda6780ba097` (conda gfortran 16.2, OpenBLAS); the previous golden is `golden_pre170_20260905/`. `golden/` now carries the `*_adv.txt` files as well. `oxygen_chemistry` has no golden entry yet (it stops, above); it stays in DEFAULT_CASES, so `make check` reports it as the one failing case until the carrier solver question is settled. `make check` and one `run_fcheck.sh` follow in this section.

After the refresh, same day (2026-09-06): the loaded-SED grid, the carrier budget, hygiene, the SED files

- **2c-SEDGRID, the photon grid of a loaded SED** (`sed_read.f90`, `J_inc.f90`, `parameters.f90`, `set_energy_vectors.f90`; item C5 of the D0 draft): the Load path kept the table rows as bin points with central-difference widths, so no threshold was a bin edge there, on the path every planet calculation of record uses (decision 16). Now the bins are built from edges (geometric means of neighboring rows, the table span as the two end edges), every active ionization threshold inside the span is inserted as an edge with the split bin's two halves carrying the row's flux (exact for the integrated flux), `e_v` the geometric center, `de_v` the exact width; `read_sed` receives the threshold list from `ionization_thresholds_active`, one definition. The table rows themselves are kept in `e_sed_node/F_sed_node` (undiluted), which `stellar_flux_eV`, `loaded_table_floor_eV` and `spectrum_covers_eV` now read, so the loaded field no longer depends on the binning. Test `sed_edges` (`src/tests/spectrum_type/`, 15 assertions) on the eps Eri table of WASP-52 b: thresholds at distance 0, flux identity to 2e-16, rate deviations from the exact table integral P_{HI} 1.8e-3 $\rightarrow$ 4.6e-7, P_{HeI} 1.1e-3 $\rightarrow$ 1.1e-6, P_{HeII} 8.4e-3 $\rightarrow$ 8.5e-6, worst metal 3.2e-3 $\rightarrow$ 5.5e-4 (Mg II, the table's own 1 Å resolution). Movement: benchmarks/wasp52 2000-step marching stop, mass flux at cell N-20 2.3e-6, $\log_{10} \dot{M}$ 9.68 both, profiles up to 2e-2 at a relaxation point; `wasp_full` (power law) byte-identical on seven output files. No regression case loads a table, so no golden moves. Noticed, not fixed: `e_mid` is not an edge of the loaded grid (the LX/LEUV split it feeds moved 0.02 percent); `stop_if_balmer_band_unstated` prints bin centers as the band.

- **Carrier Newton budget** (diffusive_photochemistry.f90): newton_maxit 30 -> 200 after the diagnosis above; oxygen_chemistry runs its 1000 steps on the 89-bin grid (log10 Mdot 9.60) and is added to golden/ below. The constraint attribution of the status stop, by cell, and its test suite carrier_constraint_attribution (7 assertions) landed with it.
- **Hygiene, D0 items C16 and C4:** the dummy alpha, threaded through reconstruction_continuation_rhs, RK_rhs and Num_flux and never read, is removed from the three signatures and ten call sites (hydrostatic_column byte-identical against a same-compiler control); output_state_consistency.sh and its README describe the rate lag (measured RED 1.19e-5 with the header now reading sec_ion=F) instead of the removed post-loop block. docs/audit_20260905/roe_flux_vacuum_probe.f90 is an archived probe that no longer compiles against speed_estimate_ROE; left as the frozen record it is.
- **Document sync** (docs/EXHALE_user_manual.tex + PDF rebuilt, docs/p44_base_sawtooth.md, write_output.f90 and util_ion_eq.f90 comments, write_setup_report.f90): the manual carries the Planck type, the decision-17 stop, the six-band edge grid, the IAU constants and NIST thresholds, the signed du, the profile base level and the restart grid guard; NLTR is now counted on the Load path (the "no grid point below 13.6 eV" sentence had been printed for every loaded SED); src/utis/glob.py, which shadowed the standard library module, is src/utis/EXHALE_interface_state.py with its two importers updated. Reported, not changed: the GUI offers no Planck type; global_parameters exports an integer named count that shadows the intrinsic.
- **Stellar SED files** (inputdata/sed/, decision 16, source rule of 2026-09-06: MUSCLES data normalized as the planet's reference paper did): wasp121b_solar_huang2023.txt (recommended; WHI 2008 solar shape below 1700 Å, $F(1-912 \text{ Å}) = 1.6e6$ and $L_{\text{ya}} = 1.0e5$ as Huang et al. 2023 section 2.2 prescribe, 6459 K photosphere above; the 1.6e6 confirmed as the $\lambda < 912 \text{ Å}$ band by Salz et al. 2019 section 7), wasp121b_wasp17_huang2023.txt (MUSCLES WASP-17 alternative; its 1700-2583 Å photosphere is 1.8x fainter than the blackbody, the size of the He 2^3S photoionization bias of the solar candidate), hd189733b_epseri_salz2016.txt (eps Eri, MUSCLES v22, normalized to Salz et al. 2016 Table 3; the proxy pairing is the one Salz et al. 2018 section 4.1.1 make), and hd189733b_bourrier2020.txt (the released MOVES III tables, mean of Visits B-E, measured to be the flux at 1 au, scaled to 0.031 AU, eps Eri photosphere above 1600 Å identical to the Salz file's; 30 percent fainter in the Lyman-Werner band, 1.4x brighter in the EUV). Each file was read by EXHALE.x for two steps on the final binary without error. User decision: HD 189733 b runs at $a = 0.031 \text{ AU}$ (Salz 2016 Table 2, the modelling paper of the comparison); the planet folder, benchmarks/hd189, examples 01-11 and 13, and params_table.txt carry it (the regression inputs jfnk_hd189*, tpm_hd189 and the archived copies keep 0.0330). No input.inp has been switched to a loaded SED yet (2d).

Phase 1 closing gate (2026-09-06)

make test on the final binary b2c15e53c888d09fa90ffdad80443c3a: ten suites, eight PASS; RED only on the Phase 3 targets (output_state_consistency 1.19e-5, h3p_cooling_limits, h2_rovibrational_identity, h2_double_photoevent_energy_sum, helium_collisional_ionization_by_level). make check: sixteen

DEFAULT_CASES against the refreshed golden/, all four files byte-identical in every case (REGRESSION PASS); the sixteen goldens taken from the earlier 89-bin binary reproduce on the final one, which is the measured statement that the loaded-SED grid, the carrier budget, the alpha removal and the report edits leave the analytic-spectrum matrix untouched. `run_fcheck.sh` result recorded below. `run_fcheck.sh`: the `-fcheck=bounds,do,mem` build ran 1500 bounded HD 209458 b steps with no runtime trap (fcheck regression: CLEAN, log10 Mdot 9.59), and the production build was restored to b2c15e53c888d09fa90ffdad80443c3a. **Phase 1 is closed.** Open for the user before Phase 3: decision 4 (carrier step rejection/retry), the relaxation amnesty of the ionization sweep that handed the carrier system a non-root, the D0 draft (docs/d0_governing_system_20260906.md, 33 consistency items, 15 questions) for acceptance, the switch of the planet inputs to their loaded SEDs (2d), the WASP-121 b SED choice (solar shape, WASP-17 shape, or the two joined at 1700 Å), and the sub-Lyman band budget of the Load grid (the table's own resolution is kept; Mg II stays at 5.5e-4).

7. PLAN_20260906_rev2, Step A: acceptance contracts and run modes (2026-09-06)

Plan of record for this stage: docs/PLAN_20260906_rev2.md (rev 0 and rev 1 and their two reviews are kept beside it; the execution document of 2026-09-05 points at rev 2). The decisions of `To_be_determined_by_user_20260906.md` were judged against `To_be_determined_by_user_recommend_20260906.md` and adopted as rev 2 section 2 states.

Documents written (advisor): docs/a0_run_mode_contract_20260906.md (the three run states; Time stepping: Local and PTC are initialization only; physical time advances only by accepted global steps; headers gain `mode=` and `t_phys=`; the proposed Run mode: `init | phys` key awaits the user's confirmation before A0-impl); docs/a2_certification_contract_20260906.md (one evaluator, four contexts: probe, stationary trial, physical step, certification; statuses `not_applicable / evaluated / unavailable`; one condition per active balance; what replaces what). Documents drafted by workers and verified for structure and wording: docs/b1a_active_equation_inventory_20260906.md (flags, equations per flag, constraints, five validity states with producers, the prerequisite table for eight configurations; finding: only the hydro rows have a state-consistent evaluator; the composition projection, the rho rewrite and thirteen closures with stated domains are recorded nowhere) and docs/b1_target_system_20260906_draft.md (energy ledger with the shared H2 zero at $v=0$, $J=0$; chemistry mass-preserving; Lyman-alpha loss and return from one 2p balance, an escape-probability multiplier on the existing cooling explicitly not the target; the zero-net-current closure derived from Koskinen 2022 B9/B10 and shown to give exactly the ambipolar field `binary_element_diffusion` already computes; the trace-metal counter-flux; the rollback table of fourteen state-changing operations showing `u_old` is not a checkpoint; CO alternative A blocked on rate literature, alternative B specified and recommended; ten open questions).

- **A1 (carrier half)** (`diffusive_photochemistry.f90`; suite `carrier_returned_state_acceptance`, 36 assertions, plus one in `carrier_constraint_attribution`): the carrier update is accepted on the RETURNED state, every unconstrained row of every cell, $|res|/\max(\text{row_terms}, 1e-300) \leq 1e-8$ with the absolute floor inside the scale, finiteness of every row, independent of the termination branch; the relative-drop exit is `carrier_solve_stagnated`, a diagnostic; the decision is `carrier_returned_state_verdict`, a side-effect-free function returning verdict and reason for A3 to consume. Defect found and fixed:

`limit_to_element_budget` marked a cell constrained only when the clamp exceeded `limit_report = 1e-10`, so smaller clamps left cells presented as Newton-measured. Impact: `mol_carrier` byte-identical (no changed branch reachable: all 499 transport steps of the first 500 ended at the floor); **oxygen_chemistry refuses at step 4**: the CO thermal ceiling, a clamp without a rate, leaves a neighboring cell's CO row at $1.0e-7$ of its terms (accepted steps before it: $1.1e-9$, $4.2e-9$), and the code no longer certifies it. That is the plan's position (B1 alternative B: a ceiling-active atmosphere is outside the validated set); the case can run only in A0's initialization mode once A0-impl exists, so it fails make check until then.

- **A1 (class half)** (`steady_residual.f90`, `steady_newton.f90`, `ionization_equilibrium.f90` comment block; suite `acceptance_classes`, 12 assertions): `n_cells_without_chemical_root(acc_n) = acc_n(4)`, one definition for the trial handling (a local-closure validity check, no stationary condition) and for the three points that declare a steady state solved; a class-4 cell in the state handed back is written but reported "NOT certified". Impact: `wasp_full_newton` and `mol_base_handoff` identical (no class-4 or class-5 acceptance in any ledger). Open: the marching stop cannot pass the count because the loop never receives the sweep ledger; A2 closes it.
- **C1** (in progress) and **A2** (steps 1-3 of the contract, in progress).
- **C1** (`inputdata/sed/README.md`, `build/sed_coverage_and_order_check.py`, suite `spectrum_type` driver `loaded_sed_table_semantics`, 78 assertions): a table value is a bin average on every shipped file (MUSCLES const-res, WHI, Gueymard: 1 Å or 1 nm bins; Bourrier 0.016 Å samples with no stated convention) except the blackbody part of the solar WASP-121 b candidate (point sample); the reader treats a row as a histogram in energy, exact for the band energy of a bin-averaged table, second order in the bin width otherwise (MEASURED order 2.0 to 2.9 through the production consumers). Coverage is now checked per configuration from the active threshold inventory (2600.5 Å metastable, 3647 Å when both stellar keys arm H(n=2), the lowest active metal), 12 configurations. Findings closed the same day (advisor): `sed_read.f90` single-precision $1e-8$ at two sites ($6e-9$ relative shift of every loaded energy); the HD 209458 b WHI file rebuilt to 5000 Å (its 3000 Å cut stopped the run in `read_sed` naming H(n=2)); WASP-52b/`input.inp` SED path from before the 2026-07-14 move; the two solar build scripts' headers (2583 Å / 4.80 eV). Remaining: the WASP-17 file begins at 10.0 Å because the MUSCLES v24 table does (the grid top becomes 1239.84 eV; recorded).
- **BALMERQ** (`excited_hydrogen.f90`; suite `spectrum_type` driver `balmer_band_quadrature`, 21 assertions; `balmer_continuum_field` Planck references refined): the Balmer-continuum rate and heating integrals follow the field's own nodes in closed form (power-law segments of a loaded table and of the power-law type; 5-point Gauss on a geometric subdivision for Planck), replacing a fixed 400-point uniform rule that stepped over emission lines. On the eps Eri table of benchmarks/wasp52 the old rule was 2.9 percent low in the rate and 2.7 percent high in the heating (the Lyman-alpha line at 10.20 eV lies in the band); on the power law it carried $9.5e-5 / 6.0e-5$ of truncation. Movement: benchmarks/wasp52 `gamma_2` +2.99 percent, `n_e` $5.3e-4$ at 1.09 Rp; `wasp_full` $1e-8$ class (sub-0.1 percent, no golden refresh). One worker reported two scratch runs killed by a `kill` matching a shared basename: the common brief now says kill by PID after checking `/proc/<pid>/cwd`, never by name.

- **A2, steps 1-3** (src/modules/time_step/certification.f90 new, steady_residual.f90, steady_newton.f90, EXHALE_main.f90 call sites, utilities.f90 header, diffusive_photochemistry.f90 save/restore pair; suite certification, 18 assertions): one evaluator builds a 19-entry inventory from the configuration flags and returns per entry a status (not_applicable / evaluated / unavailable), the row measure (max over physical cells of $|res|/\max(scale, floor)$), the volume average, the worst cell, finiteness and the scale text; hydro rows and carrier balances are evaluated (carriers inside an isolated workspace that saves and restores every module array carrier_steady_residual writes, asserted after every call), elemental transport, level equations and the seven System_HeH_* closures are unavailable until step 4. The certification context runs at the marching stop (which now receives the sweep ledger, closing the A1 open item), the JFNK final and the PTC declaration; probe and trial decisions read one record, the merit rule unchanged. Exit status (user decision): a run that DECLARED a stationary state and failed certification exits 2 (stop 2); a run ending on a cap makes no claim, writes certified=F cert_reason=no_stationary_claim and exits 0; run_check.sh never tests the status. Five shell gates of grid_and_gates that asserted exit 0 from a full run now accept 2 with the reason stated at the code site (advisor). Impact: numeric content of wasp_full, wasp_full_newton, mol_carrier identical; headers gain certified=. **No configuration certifies today**, by construction (the closures are unavailable), and two measured findings stand on their own: the JFNK info = 0 is set at the loop top and never revised when the returned state is re-measured at its own composition (wasp_full_newton energy row $4.1e-4$ against the $1e-5$ target, 40x), and the du stop accepts hydro rows of order one on their own terms (wasp_full at $du = 1.0e-3$: mass $1.3e-2$, momentum 1.8, energy 0.72; at $du = 2e-2$ the same rows read 1.7, 2.0, 0.96, so the spread and the rows measure different things). Not implemented in this item (it would change the solve): the probe rule that a class-4 chemistry makes a Newton direction unusable. Tolerances inherited; the convergence study is owed.
- **B1 accepted (2026-09-06).** The user decided the fourteen open questions of docs/b1_target_system_20260906_draft.md as recommended (metal reservoir zero; electron mass convention; ioniz_eq density intent(in); the residual H I excitation cooling under the Lyman-alpha ledger; trace metals in the diffusion matrix; H+ in the B9 set; CO exclusion adopted with the destruction model held for rate literature; H3+ low-collider policy; one H2 partition function; rollback of diagnostic extrema; identity row placement; the CO ceiling's validity category; O(1D) reservoir; the refused _adv product rule). The accepted specification is docs/b1_target_system_20260906.md (the draft kept and marked superseded). This is the Phase 2 gate of PLAN rev 2; B2 (the residual-controlled temperature solve) started the same day.
- **A3**(diffusive_photochemistry.f90; suite carrier_retry, 32 assertions): the carrier subsystem's checkpoint (carrier_checkpoint_take / restore / matches: carrier fractions, the module arrays of the A2 save/restore enumeration, every constraint and solve diagnostic, the verdict, the cumulative CO record), and the controller: a refused returned state restores the substep's entry state, halves the carrier substep with the background explicitly frozen (listed at the routine), covers the interval with accepted substeps, and on exhaustion (nine refusals, $floor(dt/2^8)$) restores the ENTRY state and stops; f_sp is written only for a covered interval. Two CO records (accepted, restored on refusal; attempted, never restored) and controller attempt statistics with the A0 ledger-

family hook. Retrieved result equals the explicitly subdivided integration bitwise; step doubling gives first order (ratio 2.001). Impact: mol_carrier (0 retried intervals in 11999), mol_base_handoff, oxygen_chemistry (3 steps) byte-identical. **Finding:** the retry recovered the oxygen_chemistry refusal ($1.04\text{e-}7 \rightarrow 7.9\text{e-}9$ after ten halvings) because the row scale carries the $1/\text{dt}$ time term, so any fixed clamp imbalance passes a fixed relative floor at a short enough substep; that is a defect of the acceptance measure shared by A1 and A2, fixed next (A1scale: the returned-state and stationary measures are judged against the physical terms of the row, transport and reaction, with the absolute floor; the time term stays only in the Newton's own iteration test).

- **B2** (energy_semi_implicit.f90; suite energy_update, 20 assertions): the implicit temperature step of a cell, $R(T) = u(T) - u(T_{\text{old}}) - a [\text{heat} - \text{cool}(T)] = 0$ with $u(T)$ from the run's equation of state (caloric H2 included), is solved by a safeguarded Newton-bisection hybrid on a physical bracket (0.01 T_0 to $5\text{e}4 \text{ K}$ with H2, $1\text{e}7 \text{ K}$ otherwise) to $|R|/\text{scale} \leq 1\text{e-}9$, the cooling written back is $\text{cool}(T_{\text{returned}})$, and every non-converged outcome is a named failure (ITER_CAP / NO_BRACKET / FLOOR / NONFINITE) with the cell, the residual and the missing volumetric source; no clamp remains, a floor is a failure, and the interim action is a stop that B3a replaces by rejecting the attempted step. The caller is untouched (optional status argument, energy_update_last_status). RED kept in the suite: the two-iteration update on a falling cooling branch left $|R|/\text{scale} = 2.9\text{e-}2$ and a 6.9 percent temperature error. Impact: wasp_full transient $5.4\text{e-}5$, converged (same stop step 16257, $\log_{10} \text{Mdot}$ 13.29) $1.6\text{e-}6$; mol_base_handoff $4.9\text{e-}5$; hydrostatic_column identical; no floor or failure fired anywhere; cost +9 percent end to end on wasp_full. Follow-ups assigned: the identical clamp in the conduction stage (viscous_conduction.f90 572-589; B2b), the certification's counting of energy-floor hits as an accepted correction (A2 step 4), the stale end-of-run floor report (A0-impl).
- **A1scale** (diffusive_photochemistry.f90; carrier_retry 42 assertions): the returned-state acceptance and the stationary carrier measure are judged against the row's PHYSICAL terms ($| \text{transport} | + | \text{reaction} | + \text{floor}$, the floor in rate units without the $1/\text{dt}$ form); the time term $|n - n_{\text{old}}|/\text{dt}$ stays only in the Newton's own iteration scale (row_terms, bitwise unchanged), beside which row_terms_phys is accumulated. RED: a residual of $1\text{e-}7$ of a row's physical terms read $3.3\text{e-}8$ at dt and $5.6\text{e-}9$ at $\text{dt}/16$ on the old scale (crossing the $1\text{e-}8$ floor with the defect unchanged); now $1.0\text{e-}7$ at every dt . Impact: mol_carrier, mol_base_handoff identical; oxygen_chemistry is refused at step 2 (CO row $5.5\text{e-}8$ beside a clamped cell; 37 retry attempts to the floor, all refused, entry state restored), and the old measure only postponed that refusal to step 88 on today's tree. Finding carried forward (A1scale2, needs its own impact measurement): with a dt -independent acceptance there is a shortest certifiable substep, since the Newton's stopping test is still relative to the full terms and the round-off it leaves grows as $1/\text{dt}$ against the physical terms; the same measure in both places closes it. Also: cterms in steady_newton.f90 (534, 645) is written and never read. **Consequence for the matrix:** oxygen_chemistry cannot run under the physical acceptance while the rateless CO ceiling is active (B1 decision 7: exclusion); once A0-impl lands, the exhaustion action reads the run mode (initialization: record the refusal in the attempted ledger, write the state flagged, continue; physical: stop), which is the item A3b, and the case runs in init mode as the diagnostic it is.

- **B3a and B4 designs** (docs/b3a_attempted_step_controller_design_20260906.md,

docs/b4_spatial_operator_design_20260906.md, read-only workers; advisor decisions in their last sections). B3a: eight of the fourteen step operations have no returned-state verdict today (RK/BC, element diffusion, excited H, ioniz_eq, the projection, Apply_BC, conduction, Shapiro); checkpoint about 2.1 kB per cell. B4: species mass fluxes are not consistent with the mass flux today (equilibrium species are not transported and rewrite rho; carrier advection is a cell-centered product; element diffusion is in advective form; no species enthalpy transport); increments B4-1 to B4-6 with their expected movement. **Advisor correction to the B4 draft:** the formation-energy flux $F_{\text{form}} = \sum_s \epsilon_s F_s$ is the RESERVOIR's transport and enters no thermal equation (photoheating already carries $h_{\text{nu}} - I$; recombination emits I); the draft had written it as a thermal source, which would heat a cell by the ionization energy leaving it. B4-2 becomes the diagnostic reservoir balance that the ledger identity T1.5 tests, plus the face-composition enthalpy (T-B4.5). The B4-2 pre-measurement (src/utls/formation_energy_flux_diagnostic.py, section 7 of the design) stands as the size of that reservoir transport: on wasp_full_newton 28 percent of the absorbed heating leaves the column as advected ionization energy (median 14 percent of the local heating, 52 percent at the ionization front at 1.23 Rp); the present energy face flux is $v(E + p)$ with the caloric H2 internal energy already inside E, addressed by the owning cell (T-B4.5). Findings: lower_profile's golden mass flux is not flat ($1.1e-2$), so continuity-based diagnostics on it discriminate nothing until it is Newton-finished; benchmarks/hd209/output integrates to cooling 24 percent above heating on its saved state.

- **B2b** (viscous_conduction.f90; energy_update suite second driver conduction_floor_tests, 24 assertions): the Crank-Nicolson conduction stage no longer clamps; a solved temperature at or below the shared EOS floor (imported from energy_semi_implicit, one declaration) is a FAILURE with status, cell, the solved temperature and the energy the floor would have injected; the tridiagonal solve refuses a zero or non-finite pivot; no partial state is written. The operator is in flux form and conserves thermal energy exactly on a closed column (MEASURED 0.0); as boundary-conditioned in a run it exchanges heat through the base Dirichlet anchor by design ($3.4e-15$ up to that flux). Impact: byte-identical on wasp_full, mol_base_handoff, hydrostatic_column and on two conduction-on variants in which the stage runs; zero floor activations anywhere. B1 section 7.4 rows 9 and 11 updated (no clamp state left). Noticed: benchmarks/koskinen2022_model_a/* cannot restart (stored IC on the pre-2c grid, $4.5e-4$ off; the same class as the hp_* regeneration).
- **B4-5 measurement** (src/tests/grid_and_gates/hydrostatic_residual.f90, design section 8): on the analytic isothermal column, through the production Reconstruct, Num_flux, RK_rhs and Source, the pressure-gravity pair is at design order in the interior ($dr^{1.96}$ PLM, $dr^{1.98}$ WENO3 at $r = 2$; base cell $1e-4$ of its weight, first order in the base spacing), and the $O(1)$ hydrostatic imbalance of the mechanical column sits at the OUTER free-outflow cell: 0.75 of its own weight at every N under PLM (the zero-gradient ghost zeroes the MC forward difference, so cell N carries no pressure gradient), $6.4e-2$ falling at order 1.4 under WENO3. A well-balanced pair (B4-5b) would not remove the drain; the outer boundary fill is the item (B4-4a, started). Two latent defects found: define_grid's `tol = 1.0` initializer implied SAVE, so a second call in one process skipped the stretch-factor Newton (production calls it once; fixed by the advisor: `tol` reset before the loop); the case README's single Grid cells ladder does not refine the base cell (README updated).

- **A2 step 4** (certification.f90, binary_element_diffusion.f90, excited_hydrogen.f90, ionization_equilibrium.f90; certification suite +7 driver and +15 shell assertions): the five entries that were unavailable are measured on the state without solving or writing anything (He/H partition and trace-element transport balances, He 2^3S and H(n=2) level balances, the eliminated-species closure as the sweep's own normalized reaction residual at the given composition), each restoring every module array it installs, asserted on every call. Output content identical on seven cases. **What refuses certification now:** the three hydrodynamic rows on every marching case; the energy row alone on wasp_full_newton ($4.1e-4$ at cell 376, the JFNK info = 0 finding); the H2 carrier row on mol_carrier; and, new, the elemental transport on the diffusive runs (mol_diffusion partition $6.4e-5$ at cell 203; lower_profile partition $4.8e-3$, trace metals $1.2e-2$, the first measurement of what the Picard alternation reaches). The closures and level balances are within tolerance everywhere ($1e-16$ to $1e-12$ of their turnover), so the equations that lacked an evaluator are the solved ones. Both floors are reported as attempt diagnostics (0 everywhere). cert_tol_element is inherited, not measured, and is now decisive for the diffusive runs: the convergence study of contract section 4 is owed.
- **A1scale2** (diffusive_photochemistry.f90; carrier_retry 51 assertions): the carrier Newton stops when its residual is within its own tolerance of the full terms AND within the acceptance tolerance of the physical terms, or when the arithmetic forbids the second (residual at 64ϵ of the full terms), a named outcome; the interim reason unreachable_at_substep for a refusal whose rows are all at round-off was superseded the same day by A1scale3 (below); the stop message reports the halvings made. Option (a) (stopping on the physical scale alone) was built, measured (one extra iteration every step, no rescue of the short substep) and declined. Correction of the A1scale report: the Newton reached one epsilon in two iterations on that column; the $7.7e-5$ was the ninth attempt's state. Impact: mol_carrier, mol_base_handoff identical; oxygen_chemistry still refused after step 2 (now the unreachable reason, OH at cell 90, 15 attempts). Decision taken from its finding (A1scale3, started): a row at the arithmetic round-off of its own terms is not evidence of a defect and is accepted as "round-off limited" with a flag and a count, so a trace carrier at the element floor does not refuse an interval; the physical-terms acceptance stays for every other row.
- **A0-impl** (input_read.f90, parameters.f90, EXHALE_main.f90, utilities.f90, load_IC.f90, write_setup_report.f90, ionization_equilibrium.f90 ledgers; suite run_mode, 22 assertions): the run mode is in. Run mode: init | phys, default init in every configuration (advisor decision replacing the draft rule: the ordinary use is a stationary solution reached by marching and certified, which is initialization in the contract's terms; a physical integration is asked for); phys with Time stepping: Local refused at startup; in phys the handoff state is tested at step 0, the clock advances only after a complete accepted step (measured equal to the sum of accepted dt to the last bit, with and without forced rejections), two ledger families (ledger_family in global_parameters), headers gain mode= and, in phys, t_phys=; a restart of an init snapshot in phys mode is the handoff at $t = 0$ (contract section 3 clarified), and only a mode=phys header without a finite t_phys is refused; init never exits nonzero for certification. Impact: no input changed; wasp_full, hp_front, hydrostatic_column, mol_base_handoff bitwise identical. Both floor reports of EXHALE_main.f90 now state the B2/B2b behavior.

- **COLit** (docs/co_destruction_rates_literature_20260906.md; publisher PDFs of Westley 1980 NSRDS-NBS 67, Tsai 2017, Wakelam 2012 added to references/): B1's CO choice is settled on physics: every neutral CO destruction channel carries a barrier of 6.7 eV or more, so the one-sided destruction model's $\tau_{\text{dest}} \ll \tau_{\text{res}}$ fails at the ceiling's onset at every density (factor 9 to $1.5e8$) and the exclusion (decision 7) is the physics. Findings: molecular_reaction_heat.f90 (H2, H+, He+, H2+, H3+, HeH+) and the oxygen Shomate table of oxygen_rates.f90 carry different reference zeros (T1.9 one-table item, assigned as B3b-ENTH); CO2 is not in the species table (removes the one evaluated CO channel below 2500 K and an OH sink); the workspace VULCAN network's CO reactions all produce species EXHALE does not carry, VULCAN's stated validity is 500 to 2500 K, KIDA's 10 to 300 K. Requests for the user: the A&A papers Visser et al. 2009 (503, 323; the CO shielding functions), McElroy et al. 2013 (550, A36
 - RATE12 file), Venot et al. 2012 (546, A43) and 2020 (634, A78), plus Moses et al. 2011 (ApJ 737, 15) and Tsai et al. 2021 (ApJ 923, 264), which this host cannot download.
- **A1scale3** (diffusive_photochemistry.f90; carrier_retry 61 assertions): a row at the arithmetic round-off of its own full terms (64 epsilon) is counted "round-off limited", flagged per row, and left out of the acceptance decision; every other row is judged on the physical terms at $1e-8$ as before; the controller halves only for genuine refusals; the stop prints both scales for every refused row. Impact: mol_carrier, mol_base_handoff bitwise identical (0 round-off limited rows); oxygen_chemistry still refused at the same step and row (CO at cell 281, $|res|/phys = 3.2e-8$, $|res|/full = 2.2e-12$, four orders above the round-off bound); the element limiter's imbalance one cell beyond a clamped cell, correctly refused. The count's accessor exists; its line in the run summary and the certification carrier entry are B3a's (owner of both files).
- **B3b-H2Q** (caloric_eos.f90, mol_rates.f90; h2_rovibrational_identity 19 assertions): one H2 partition function over the observed Roueff et al. 2019 ladder (302 levels, zero at $v=0, J=0$) gives $u_{rv}(T)$ as its first moment and the chemistry's K_{eq} as its free energy; the Huber-Herzberg model constants are deleted. The RED (0.6 to 8 percent in u_{rv}) was the model ladder's level placement (no centrifugal distortion: J^4 signature, 100 percent of the gap to 2000 K), not a defect of the sum; independent check: K_{eq} against NIST-JANAF (Shomate) now 0.9997 to 1.0008 over 600 to 5000 K (was 1.010 to 1.101). Impact: wasp_full identical; mol_base_handoff T $1.3e-5$, n(H2) $3.1e-4$; mol_diffusion T $3.2e-5$; mol_metals at 400 steps n(H2) $5.6e-3$ at 1.17 Rp; Mdot unchanged. Findings: a third copy of the ladder sum in molecular_infrared_cooling.f90 212-215 (to fold into h2_partition_function); element_census_tests does not compile against the live tree (no_free argument, another worker's signature change; to fix at the next single build); the load average reached 38 with the parallel long runs, so future briefs pace the 12000-step impact runs.
- **A3b** (diffusive_photochemistry.f90; carrier_retry 89 assertions): the carrier controller's exhaustion reads the run mode and returns a status (carrier_interval_covered / exhausted) the caller reads. phys: unchanged (restore, print, stop until B3a takes the status). init: the entry state is restored, the interval is recorded in the attempts ledger (count, first/last step, worst row and both scales), the carrier history is marked not certifiable (carrier_history_certifiable()), and the march continues from the entry carriers. The accepted CO

record is indexed by A0-impl's ledger_family. Impact: mol_carrier, mol_base_handoff bitwise identical; oxygen_chemistry (init, its default) now runs its 1000 steps, exit 0, with 998 exhausted intervals (first at step 2, worst CO at cell 313), log10 Mdot 9.61 against the 20-bin 9.60, and a final certification refusing on four carrier rows at 1.0: its carriers are frozen at the step-2 values for the whole run, which is exactly what the mark and the ledger say. The refused row is the real imbalance beside a clamped cell ($|res|/full$ 4e-12 against a 1.4e-14 round-off bound), open physics of the excluded CO model, not a controller defect. Its golden is re-taken at the end of this series with the mark noted in the case README.

- **B5pre** (steady_newton.f90; suite steady_completion_flag): the JFNK and PTC completion flag is decided AFTER the certification of the returned state, from the same record the certification prints; a solve that met the gate at the loop top and not at the returned state ends info = 2 naming the row. Measured on wasp_full_newton: loop-top $|R| = 5.5e-6$, returned state 4.2e-4; the existing own-composition consistency pass opens the gap (0, 1, 2, 3, 8 sweeps: 5.5e-6, 1.9e-4, 3.2e-4, 3.9e-4, 4.2e-4), so the hydro-only Newton converges to a state that is not a root of the coupled hydro + composition system: the loop-top gate measures a lagged residual. Consequence today: EXHALE_main treats info = 2 as a solver failure and falls back to marching, which ends on a du plateau with a state that certifies worse than the JFNK state; assigned to B3a (info = 2 = a failed stationarity claim, written uncertified, exit 2, no fallback) and to B5a (steady_newton.f90: the residual evaluated on the self-consistent composition of the current iterate, so gate, merit and returned state measure one thing). Also found: the JFNK stop message prints an unset resid_th (B3a); certification_row_measure and relnorm_over_cells are one expression written twice.
- **B3b-PE** (h2_photo_channels.f90; photoevent_energy_ledger 105 assertions): the 19.725 eV missing from the H2 double photoionization event is the kinetic energy release of the two protons: 51.4 eV is a vertical threshold (Yan et al. 1998 section 4), the asymptotic product energy is $D_0(H_2) + 2 I(H) = 31.675$ eV, and energy conservation at the threshold fixes the fragment kinetic energy with no free parameter (Franck-Condon check $e^2/R_e = 19.42$ eV, 1.5 percent). Second defect found: the 33 to 41 eV neutral window dissociates into $H(2p) + H(2s)$ (Chung et al. 1993 p. 887), so 20.398 eV per event is internal excitation leaving as prompt Lyman-alpha and two-photon continuum, which the integrand charged to heat. h2_channel_energy_recipients (reservoir, electron, fragment kinetic, radiated; sums to $h \nu$) with the per-channel table and sources. Channels M and S already closed; every atomic absorber and all 17 metal ions correct (wasp_full byte-identical). Impact: mol_sec_ion T 4.2e-4, heat_H2 4.5e-2; mol_base_handoff T 2.1e-5; Mdot unchanged. The four-line wiring of the two new terms into the heating integrand of util_ion_eq.f90 is handed to B4-6 (owner of that file). Open: suprathermal H2 dissociation credits heat but destroys no H2; the double channel has no absorber row of its own; channel S has no source for a fragment kinetic split.
- **B3b-H3** (h3p_cooling.f90; h3p_cooling_limits 17 assertions): the Miller et al. 2013 fits as the paper defines them: Table 5 is the LTE emission of one molecule per steradian ($\Lambda = n(H_3^+) 4 \pi E(T) s$), Table 6's $s(T, n_{H_2})$ is scaled LTE with H2 the only collider (Oka and Epp coefficient; the paper's upper-limit caveat at the code site); below the table's $1e6 \text{ cm}^{-3}$ the non-LTE factor follows the exact collisional limit (linear in n_{H_2} , zero at zero), replacing the clamp at the lowest column; above $1e14$ the table edge is held; the four Table

5 segments do not join (measured with the paper's coefficients: +43 percent at 300 K, -0.11 percent at 800 K, -2.4 percent at 1800 K) and the code joins them by a stated linear blend in $\log_e E$ over the 5 percent below each join (Table 6's 300 K value supports the upper polynomial); B1 T6.3 reworded to say so. Published values reproduced per segment to the paper's precision. Impact: the H3+ cooling peak unmoved; T moves $1e-5$ to $2.7e-4$ in the three molecular cases; the clamp removal changes cells beyond 1.22 Rp where $n_{H2} < 1e6$ by more than 50 percent at $1e-10$ of the peak (emission from gas with no collision partners, gone); wasp_full identical. Four domain counters exposed (B6 wiring is the certification owner's).

- **COLit, completed with the user's PDFs** (Visser 2009, McElroy 2013, Venot 2012, 2020; docs/co_destruction_rates_literature_20260906.md, 1419 lines): CO photodissociation as Visser et al. define it (37 bands inside EXHALE's 912 to 1110 Å beam, unshielded $2.6e-10 s^{-1}$ in the Draine field, shielding tables in $N_{CO} \times N_{H2}$ for four parameter sets, T_{ex} raised only to 512 K because no data exist for dissociation out of $v'' = 1$); evaluated on the oxygen_chemistry run's own base columns ($\log N_{CO}$ 18.96, $\log N_{H2}$ 21.80) it gives $\tau_{dest} = 2.8e8 s$, twelve decades faster than the barrier channels and still two to six decades short of $\tau_{dest} \ll \tau_{res}$ at the base; zero in mol_base_handoff, whose spectrum carries no flux longward of 911.8 Å. RATE12's paper prints no entries (the data file is still needed for the ion channels); Venot 2012 is valid 300 to 2500 K and 0.01 to 100 bar (EXHALE's base at $1e-6$ bar is four decades below); none of the six sources covers the 3000 to 4000 K band where the ceiling fires. The CO exclusion (B1 decision 7) stands, now quantitatively. If a CO photodissociation term is ever added it is a fourth absorber on the existing Lyman-Werner beam with a second column (N_{CO}) and the cell-mean form of lyman_werner_dissociation_rate.
- **TESTINFRA** (grid_and_gates/run.sh, physics_probe/run.sh, their READMEs): both suites take EXHALE_OBJDIR, EXHALE_EXE, EXHALE_TEST_OUT with the semantics of spectrum_type/run.sh; default behavior byte for byte as before; against a private build both run without touching build/ objects (the six grid_and_gates shell tests still place their run copies under build/tests/). A dead fallback that wrote a build stamp the closure never compiled is fixed. Also (advisor): the two Python source scanners now skip .ipynb_checkpoints/ directories, which hold editor auto-save copies of four modules (git-ignored) and had made threshold_literals_written_once red on a stale copy of cross_sec.f90; the Makefile source list is explicit and never reached them.
- **B3b-ENTH** (molecular_reaction_heat.f90, oxygen_rates.f90; 78 Fortran
 - 37 Python assertions): one table species_formation_energy(isp) for every species column (H/He stages, 27 metal stages cumulative from the neutral, H2/H2+/H3+/HeH+, OH/H2O/CO, plus free electron, H(n=2), O(1D)) from the single B1 reference; the 17 reactions of the H2/He network are a stoichiometry table and each heat a difference of table entries; the Shomate table stays with its reference stated, every use of it being a balanced-reaction difference (the zero cancels), and the oxygen and carbon entries of the one table are differences of it; one D0(H2) (mol_rates:h2_dissociation_energy_eV, reproduced by the fit to 1.5 meV). Physics defect fixed: the oxygen photolysis thresholds were 298.15 K reaction enthalpies;

they are now 0 K dissociation energies (the lowest Shomate interval's F coefficient is the 0 K enthalpy), 0.9 to 1.0 percent lower, deposited heat +0.8 percent at the oxygen base cell. Impact: mol_base_handoff, mol_sec_ion, mol_metals byte-identical; oxygen_chemistry T +0.4 percent at 1.02 Rp, Mdot 9.61. Open for B3c: the undeposited collisional oxygen heats and the He 2³S associative branch (D0 C25; the table exists); the B4-2 Python diagnostic lacks four oxygen entries; formation_energy_report has no caller; a stale threshold comment in water_photolysis.f90:281.

- **C3** (six input.inp and READMEs; inputdata/sed/README.md): the planet folders and benchmarks of HD 209458 b, HD 189733 b and WASP-121 b load their literature SED files (Spectrum type: Load from file.; Power-law index removed, LX/LEUV kept because mandatory and overwritten by the grid integral, which the setup report shows); every README records the configuration of record with the existing power-law products listed by mtime and the reruns left to a separate instruction; read_sed accepts every file at its configuration's floor (the H(n=2) edge in all six). log F_XUV at the orbit moves 3.193 -> 3.127 (HD 209458 b), 4.438 -> 4.311 (HD 189733 b; the recomputed LX/LEUV reproduce Salz Table 3), 6.205 -> 6.204 (WASP-121 b). Advisor follow-ups: HD 209458 b's orbital distance set to 0.04707 AU (Oklopčić and Hirata 2018, the reference the run is compared with; the input said 0.0480 while the file was scaled to 0.04707); the setup report's flux ratio format (F8.3 overflowed when the sub-Lyman band dominates); the three benchmark directories added to the coverage script's inventory (all pass). Known: Load IC? True in the two planet folders is broken by the grid guard (pre-rebaseline IC), the same class as the hp_* regeneration.
- **COLit, revised with RATE22, Moses 2011, Tsai 2021** (document now 1956 lines): RATE22 (Millar et al. 2024, A&A 682, A109) carries 'He+ + CO -> C+
 - O + Heat 1.6e-9 cm³ s⁻¹, measured, accuracy A, all products EXHALE species; it has noH+ + CO, CO + MorCO + H2destruction entry, and itsTu = 41000on 6562 entries is an undocumented "no upper bound" marker. On the oxygen_chemistryprofile the He+ channel with the Visser shielding givestau_dest/tau_resfrom 6.7e2 at the base to 9.6e-5 at 1.14 Rp: **the one-sided destruction model is constructible above the helium ionization front and not below it**; the boundary is compositional, not the ceiling's thermal switch. Decision 7 (exclusion) stays until T5.1 is implemented; B1 section 5.2 revised, B3b-CO scheduled after B3c. Moses 2011: no stated envelope, one fit interval 500 to 2500 K, no shielding. Tsai 2021: 0.1 nm continuum opacities, no CO self-shielding; EXHALE'slyman_werner.f90' is the working precedent. The literature requests are closed except Baulch 1976 vol. 3, which blocks nothing.
- **B4-6** (utilities.f90 cell_mean_attenuation, util_ion_eq.f90; xuv_cell_mean_attenuation 56 assertions): the photoionization and photoheating sums of a cell use the exact cell mean of the attenuation, exp(-tau_out) (1 - exp(-dtau))/dtau (composite Gauss when the 2D correction is on), instead of the inner-face value the top-down columns gave; one definition for H, He, He 2³S, H2's four channels, the metals, the single-absorber rates, q_abs and the secondary-ionization integrands. Inner-face error: 0.5 percent at dtau =

0.01, 42 percent at

1. Photon conservation on the written states: cell mean 1.000000000 of the incident flux, inner face 0.9725 (wasp_full), 0.9843, 0.9849 (the loss sat at the ionization front). Impact: wasp_full (24000 steps) x(H+) 8.3e-2 at 1.32 Rp, Mdot +0.74 percent; mol_base_handoff Mdot +0.93; mol_sec_ion +1.01; hydrostatic_column unchanged. B3b-PE's two heating terms wired into the integrand (its own numbers reproduced cell for cell). Advisor: the single-precision 1.0e-18 cross-section unit literal (33 sites, 4e-8 relative on every rate) made double. Finding to resolve at the gate: wasp_full on the mid-series tree plateaus at $du = 4.2e-3$ from step 22000 and never reaches $1e-3$ (the golden stopped at 16230), identical in the item's before binary, so it belongs to an in-flight change (B3a edits EXHALE_main.f90) or to an earlier landed one; measured after B3a.
- **A2tol** (measurement; a2_certification_contract_20260906.md section 9): the du stop and the hydrodynamic row measures are not measures of one thing (over a decade of du on wasp_full the mass row falls at order 0.2, the momentum row rises, the energy row is not monotone; mol_carrier's carrier row falls at order 0.08 over 12000 to 48000 steps); the JFNK returns an energy row 41 to 47 times its target at every rung of a two-decade Resid tol ladder (structural; B5a); the elemental transport rows cannot be measured on the marching route (the operator advances $1e-6$ of the way per CFL step) and the diffusive cases never reach the steady route (du stalls at 5 to 6e-2 against the $1e-2$ hand-off). Round-off floors on the converged JFNK state: mass $1.0e-13$, momentum $5.1e-12$, energy $1.6e-11$ (residual bitwise reproducible). Recommended and adopted (applied by B3a in certification.f90, provisional until B5a): cert_tol_mass $3e-12$, cert_tol_momentum $1e-8$, cert_tol_energy $5e-5$, cert_tol_element $1e-8$ (with the homopause regime caveat), carrier, closure and level tolerances kept. At these values wasp_full_newton certifies with Resid tol: $1.0e-7$ (marginally at $1e-6$) and nothing as shipped. Noted for the user: the shipped default Resid tol $1e-5$ is one to two decades looser than the solve reaches at no cost.
 - **B5a** (steady_newton.f90; suite steady_selfconsistent_residual): eval_residual eliminates the composition AT the state it evaluates (the equilibrium sweep iterated at fixed conserved variables until the composition stops moving), so the loop-top gate, the line-search merit, the returned state and the certification measure one object; the gate equals the certification to $1e-12$ by construction. Finding: the eliminated residual's Jacobian carries the summed Picard response $(I - M)^{-1}$, and a finite-difference probe with a convergence test stops after one pass (the seed already is the state's composition), so the Newton direction ascended; the probes now run a FIXED sweep count taken from an elimination seeded at a different state (about 15 terms), and the "frozen-composition Jacobian" the brief offered as a fallback is exactly the arrangement that fails. wasp_full_newton: info = 0 at $||R|| = 4.4e-7$ in 9 iterations and CERTIFIED (was info = 2 at $4.8e-5$ after 26), state moved $1.0e-4$ at most, log10 Mdot 13.27, cost 1.49x. mol_base_handoff identical. Open: Resid tol values were calibrated on the lagged measure; the other Solver: Newton cases and the PTC route are measured at the series gate; n_selfconsistent_max comment stale (B3a); the duplicated row-measure expression (B3a).
 - **HYG-absorbed** (water_photolysis.f90, utilities.f90 comment; driver absorbed_fraction_switch, 5

assertions): the FUV band rates take the XUV beam's absorbed_fraction_per_unit_depth by use association; the local copy was bitwise the same over 1501 points and is deleted; the series switch and its error bound (about $1e-8$ on the closed-form side of $d = 1e-8$) stated once; mol_base_handoff, oxygen_chemistry byte-identical. Advisor: the B4-6 driver's reference carried the same single-precision $1.0e-18$ the production code had; made double with the production change, the driver is green again (51 assertions had turned red on the $4e-8$ mismatch).

- B3a** (EXHALE_main.f90, new attempted_step.f90, certification.f90, parameters.f90; suite attempted_step, 87 assertions): a marching step is one trial of the fourteen operations from a checkpoint covering everything rows 1 to 12 write, evaluated at one adoption boundary before update_map_end_step, retaken at half dt from the checkpoint when refused (cap 8, floor $dt/2^8$, a reduced dt bounding the next step); the clock and the accepted-step count move only there; exhaustion stops in phys and continues flagged in init. Impact: wasp_full (300 and its du stop at 14496 steps, log10 Mdot 13.33), hydrostatic_column, mol_base_handoff, mol_carrier bitwise identical against a control built from the same snapshot; cost within run-to-run spread. Measured and reported, not gating: the time-discrete hydro row (1.92 at a cold-start base where the momentum scale vanishes) and the energy identity T1.5 (-0.20 to -0.74 on wasp_full, the size of the row-8 projection, the RED of AT-1a); the step-doubling estimate gives order 0.89 for the whole step (the operator split is first order), cost 12 percent at n_err_every = 20. Two defects of the item caught by its own impact runs and fixed. The mid-item requests landed (A2tol tolerances, info = 2 written uncertified with no fallback, the ledgers in the summary, the H3+ domain counters, one row-measure definition, the n_selfconsistent_max comment). The B4-6 "du plateau" was the mid-edit tree; the final tree reaches its du stop. Advisor decision on the exit status: a run that DECLARES a stationary state and fails certification exits 2 in every mode (A0's init exemption covers physical-step claims only); the mode gate is removed (A2exit). Small items of its report assigned (HYG-B3a: operator status on every path, conduction accessor, step-level carrier verdict, optional volume measure) and the -00 floating-point exception on hydrostatic_column under diagnosis (O0FPE).
- B4-6b** (lya_rt.f90, excited_hydrogen.f90 field evaluation; driver lya_beam_cell_mean, 37 assertions): the stellar Lyman-alpha depth was a trapezoid between cell centers (neither the cell's face depth nor its own, every depth short by half the outermost cell); it is now the rectangle rule of calc_column_dens on the cell's own opacity, with the cell's own depth and its star-ward face as outputs. The beam transmission is $\text{erfc}(c \sqrt{\tau})$, not exponential, so its cell mean is a 3-point Gauss rule in $u = \sqrt{\tau}$ (mean and mean of the square, since the trapping makes the field quadratic in the transmission); the photosphere factor's mean is closed form with log1p. jlya_escape_prob keeps the cell center by Neufeld 1990 section II (the escape fraction is a point property of where the photon is created). Impact: hydrostatic_column, mol_base_handoff byte-identical; wasp_full reaches its du stop at step 14442 on both binaries, wind unmoved ($1e-7$), the field $1.6e-3$ and the Balmer proton source $6.5e-4$; the face rule over-counted the column's scatterings by 2.4 to 2.7 percent. Carried forward: the FUV band B2 line transmission in util_ion_eq.f90 still reads a face value (B4-6c).
- HYG-B3a** (diffusive_photochemistry.f90, certification.f90, steady_newton.f90, viscous_conduction.f90; suites carrier_retry 97, energy_update 48, certification 29): the car-

rier operator sets its interval status on every return path (already true since A3b; now stated and tested); carrier_last_verdict_of_run renamed for what it is, plus a step-level carrier_step_verdict() reset at every transport step; conduction_floor_cell_hits exported like the energy update's; the volume-weighted companion of certification_row_measure optional and no longer computed where discarded. All impact cases bitwise identical (wasp_full_newton finish: info = 2 at $||R|| = 7.3e-6$ against the new cert_tol_momentum = $1e-8$, re-anchored by A2tol2). Left for the gate (advisor): the now-unneeded pre-set in EXHALE_main.f90 and a stale field comment in attempted_step.f90.

- **A2exit** (EXHALE_main.f90 exit block; run_mode suite 30 assertions, certification shell gate): the certification exit status follows the CLAIM, not the mode: a run that declared a stationary state (du stop, residual gate, JFNK finish) and was refused exits 2 in every mode with its outputs written; a capped or plateaued run exits 0. wasp_full_newton now exits 2 with info = 2; run_check.sh never reads the status. Advisor: run_newton.sh and run_baseline.sh (set -e loops) now report a status 2 instead of aborting. Open: the EXHALE_PTC=1 route stops before the certification and so claims a stationary state with no verdict (B5).
- **B4-4a** (Apply_BC.f90; hydrostatic_residual +1 and new free_outflow_boundary 18 assertions): the outer free-outflow ghost is the isothermal hydrostatic continuation of the last cell ($\rho_g = \rho_N \exp[-(\phi_g - \phi_N)/(p_N/\rho_N)]$, p_g by the same factor, $v_g = v_N$), one rule for PLM and WENO3, replacing the zero-gradient copy (PLM) and the linear extrapolation (WENO3). The outer cell's momentum residual on the analytic column falls from 0.750 of its weight at every N to $4.4e-3$ to $1.0e-4$ over N = 250 to 2000 (order 1.8 PLM, 1.6 WENO3); interior and base unchanged; four ghost rules measured, this one 30 to 40x better. Velocity: the copy (a continuity-exact rule stopped fastest but divides by a ghost density the stratification makes small and was the one rule under which wasp_full_newton segfaulted at -O3). Two premises of the brief corrected: wasp_full's edge is subsonic (Mach 0.18 to 0.96; mol_base_handoff is the supersonic case), and a supersonic edge blocks the ghost only through the Riemann solution, not the stencil, so byte identity is not available. Impact: mol_base_handoff one cell above $1e-4$; hydrostatic_column $3.5e-3$ at cell N; wasp_full du stop 16208 -> 17135 steps with the base-breathing phase moved (11 percent in rho at 1.27 Rp, mean mass flux above r_flux 0.078 percent); wasp_full_newton Mdot 13.27 both. Removing the outer residual does not by itself stop the marched hydrostatic column's drain (still 0.79 c_s at 300 steps). Reported: a reproducible -O3-only segfault reachable from the rejected velocity rule (latent; O3SEGV to diagnose); EXHALE_main.f90 2538-2540 ghost-counter comment stale (B3c); B4 design section 8.3 (ii) historical.
- **O0FPE** (util_ion_eq.f90; shell test fpe_traps in grid_and_gates; run_fcheck.sh header): the -O0 failure was two defects. (1) An integer division by zero in EXHALE_main.f90 about 1234: mod(count, n_err_every) with n_err_every = 0 in init mode, reached because .and. does not short-circuit (-O1 and above happen to skip it); fix handed to B3c (owner of the file). (2) A real floating-point overflow in he_rec_coupling (util_ion_eq.f90:2813): $1100 \exp(1.2/T4) \sqrt{T4}$ at 17 K is $3.9e309$, an Inf consumed as ne/Inf = 0 into the correct low-density limit; the product is now formed only above $T4 = 1.71e-3$ and the limit value taken below, byte-identical on hydrostatic_column, wasp_full, mol_base_handoff. A third site of the same class (the stall detector's huge sentinel) also handed to B3c. run_fcheck.sh states why it builds at -O1.

- B4-6c** (`util_ion_eq.f90` `fuv_lw_photon_field`; driver `lya_band_transmission` rewritten, 16 assertions): band B2's line factor is the cell mean of the stellar Lyman-alpha transmission over the cell's own line-center depth (product of the two means with the continuum mean formed inside `water_photolysis_rate`; the covariance shortfall $6.0\text{e-}6$ against a $1.4\text{e-}4$ bound, the face rule $1.2\text{e-}2$ high on the budget); the top cell carries its own mean instead of a neutral 1. The face rule was 1.0018x on the `oxygen_chemistry` state, 1.095x on an exponential H I layer and 18.2x in the base cell of a molecular base whose line-center column reaches $1\text{e}7$ in one scale height. Consumers: the H₂O and OH photolysis in B2 (both `ionization_equilibrium` call sites and the carrier photolysis), the output columns, the FUV heating column; the H₂ channel reads LW only. Impact: `wasp_full`, `mol_base_handoff`, `mol_sec_ion` identical; `oxygen_chemistry` moves by factors at 45 steps, shown by a control (a $4\text{e-}5$ perturbation of the face value diverges the same way) to be the case's transient amplification, not the size of the change; the case is not usable as a controlled point until it converges. Finding: `write_output.f90`'s FUV band ledger closed form $N_{\text{ph}} (1 - \exp(-\tau))$ carries no line factor for B2 and its other rows are 0.79 to 0.98 off (LEDGER-FUV assigned).
- O3SEGV** (diagnosis, no change): the -O3-only segfault B4-4a reported was a null F0 argument of `jv_product` in a `pgmres` clone the compiler had specialized for a null pointer, i.e. a defect of the source text of one intermediate build (13:08) between two edits of `steady_newton.f90`; every later build carries a plain `pgmres`, HEAD plus the rejected ghost rule completes `wasp_full_newton` three times, `AddressSanitizer` finds nothing over a full run. Correction to B4-4a's premise: `-g` changes no generated code (`.text` bitwise identical at -O3 with and without it). Noted: at the time of the diagnosis the working tree stopped `wasp_full_newton` at step 0 on the energy floor (B3c in progress).
- LEDGER-FUV** (`write_output.f90` `write_oxygen_chemistry`; suite `fuv_band_ledger`, 15 assertions): the FUV band ledger is three labeled statements: (a) closure of the rated absorptions against the beam's loss between the cell faces (holds for any rate form, $1\text{e-}16$); (b) the beam budget with each band's own line factor and the un-rated share named per band (none for B1/B3/B4, which is therefore the cell-mean test; H I resonance scattering for B2, a share; for LW a sign-changing disagreement); (c) energy; (d) state drift. Two defects of the ledger fixed: it multiplied post-sweep densities by a pre-sweep field, and summed the two inner ghost cells. Data rows of every output identical; only the `FUV_bands.txt` trailer changes. **Finding (LW-NORM assigned):** the LW absorbers take 45 percent MORE photons than the beam loses ($1.27\text{e}13$ against $8.73\text{e}12$) and $249 \text{ erg cm}^{-2} \text{ s}^{-1}$ against an incident 171.5: two independent normalizations of one absorption, the level-resolved dissociation table over `p_lw_absorbed` against the DB96 eq. 39 band share (`lyman_werner.f90`, `h2_self_shielding_table.f90`). Also: `docs/a2_oxygen_option_design.md` 882 documents columns that no longer exist; the `a2_m2` kinetics check asserts the beam identity for every band.
- A2tol2** (measurement and decision; `a2_certification_contract_20260906.md` section 10): the section-9 anchors re-measured on the self-consistent JFNK residual of B5a. The returned row is now 0.36 to 0.93 of the solve's target (section 9 measured 41 to 47 above it). `wasp_full_newton` at `Resid tol: 1.0e-8` certifies (`info = 0`, 14 iterations, 658 s, `log10 Mdot` 13.27; rows mass $2.6\text{e-}13$, momentum $1.3\text{e-}9$, energy $6.275\text{e-}10$); the PTC route stalls from a cold start; `wasp_he23off_newton`, `newton_rsw01`, `newton_rsw05`

at the default $1e-5$ refuse on the mass and momentum rows at cell 499; the four Load IC Newton cases (jfnk_hd189, jfnk_hd189_tight, ptc_warm, armD_D2_newton) are stopped by the grid guard and need their IC regenerated on the current grid before they can be measured. **Applied:** cert_tol_energy $5e-5 \rightarrow 1e-6$ in certification.f90 ($5e-5$ had been anchored on the lagged-gate energy row $4.700e-06$ and could refuse nothing; $1e-6$ is a decade above $1.212e-07$, the largest energy row of any state whose solve met its own target, $5.7e4x$ above the floor $1.765e-11$); mass $3e-12$ and momentum $1e-8$ kept; the report header now names sections 9 and 10. backup/regression/wasp_full_newton/input.inp gains Resid tol: $1.0e-8$ as its configuration of record, the only setting at which that case certifies (Mdot unchanged at this precision, cost +11 percent); its golden moves at the series gate with everything else. Verified: private build, certification suite green (29 assertions). Not changed: the other three Newton cases stay at the default so that the matrix keeps a refused claim (exit 2) under test.

- LW-NORM** (diagnosis; no executable statement changed): the 45 percent excess of the FUV band ledger is two WAVELENGTH BANDS, not two fits. The self-shielding table's line list runs to 1200 Å while its sigma_pump is normalized per photon of 912-1110 Å; the beam loss is the DB96 eq. (39) equivalent width of 912-1110 Å. MEASURED: the column integral of sigma_diss/p_eff reaches 1.51538 at the top of the column axis against 1.519288 for the photon content of 912-1200 Å over 912-1110 Å of a flat spectrum (0.26 percent), is exactly density-independent, and reproduces the shipped oxygen_chemistry ratio 1.4544 at its base column where eq. (39) clamps to 1. p_eff already separates dissociations from absorptions, so the heating takes fragment energy only (not cause 2). Landed: lyman_werner.f90 header section 3a with the invariant and the measurement, its section 3 "2 percent" statement corrected to the live table's thin-limit cross section $2.6396e-17 \text{ cm}^2$; lyman_werner_cell_mean probe gains two assertions and a photon-conservation diagnostic (9/9); fuv_band_ledger/run.sh comment. Advisor follow-up: the table's generator inputs (nine abs_T####.npz, 27 CLOUDY rate files, the line-by-line scripts, the Meudon level and line data, 29 MB) moved from the 2026-09-03 scratch directory into src/utls/h2_shielding_lbl/ with a README, and h2_shielding_table_line_by_line.py's template brought in line with the hand-renamed axis dummy of h2_shield_locate: the generator now reproduces the shipped module BYTE FOR BYTE (MEASURED). The repair, table on 912-1110 Å or beam widened to 912-1200 Å with the B1 flux joined, is item 6 of To_be_determined_by_user_20260906.md (recommendation: widen), because the second redefines the Stellar LW flux key. Still to correct after B3c releases write_output.f90: the ledger's LW label says "two fits" and should say "two bands" (with the header_names_lw_fits assertion of fuv_band_ledger).
- HYG-HeCI** (test hygiene, advisor): physics_probe was red on three helium_collisional_ionization_by_level assertions ($1e-13$ at $1e4 \text{ K}$ to $1.0e-4$ at $5e4 \text{ K}$). Not a code defect: batch 2c replaced the single-precision literal $3.940e-11 \text{ erg}$ (24.6 eV) of eval_cool with the global e_th_HeI_erg (24.587389 eV), and the test's reference still carried the old literal with a comment calling it "a separate open item". The reference now uses e_th_HeI_erg and e_th_HeTR_erg; physics_probe PASSED on the current tree (private build). The LW-NORM worker's attribution of this to a concurrent Cool_coeff.f90 edit was wrong: that file's uncommitted diff is the landed 2c-CI / MASS2 work.

- **LW-NORM-B** (user decision 2026-09-06: recommendation (B) adopted; landed): ONE Lyman-Werner band, 912-1201 Å, with one normalization.
 - **The band.** `oxygen_rates.f90` carries four FUV bands, not five: the Lyman-Werner interval is 912-1201 Å and band B1 (1110-1201 Å) is gone into it. 1110 Å was an edge because Draine & Bertoldi end the Solomon process there for interstellar $v = 0$ gas; at the 700-3200 K of a planetary base the vibrationally excited levels pump longward of it, and the self-shielding table's line list always ran to 1200 Å. The two halves carried the same H₂O yield triplet, so no quantum yield moves; the band-averaged cross sections are re-formed over the whole interval and MEASURED again from the same Photochem files: $\sigma(\text{H}_2\text{O})$ $8.51337\text{e-}18$, $\sigma(\text{OH})$ $4.74371\text{e-}18$ cm², (H₂O) 12.0526 eV, (OH) 11.3829 eV, $= 2hc/(912+1201 \text{ Å}) = 11.7354$ eV. They agree to seven digits with the merge of the two shipped halves, and they reproduce the values the band carried before it was split in 2026-08.
 - **The key.** Stellar LW flux now means the 912-1201 Å band flux at the orbit; Stellar FUV B1 flux is retired and a file that states it stops the run with a message naming the merge and asking for the two numbers added. Spectrum type: Load integrates 912-1201 Å. The resolved dump loses `fuv_band_B1_flux`. The inputs that carried both keys were restated to their sum: $343.0 + 137.9 = 480.9$ for the HD 209458 b spectrum (Update_EXHALE_stage1.md records the 912-1201 Å integral of that spectrum as 481.0 and its halves as those two numbers), and the LW-only cases (`mol_lyman_werner`, `armA_LW`, `armD_D2_LW`, `armD_D2_LW_newton`) from 343.0 to 480.9 so that they keep the same star. The README_HOWTO HD 189733 b recipe goes from LW 600.1 + B1 648.4 to LW 1248.5.
 - **The table.** `src/utls/h2_shielding_lbl/lbl_table.py` normalizes per photon of 912-1201 Å (`wl_HI`, `wl_max`, and `E_LW_PHOTON_ERG` the mean photon energy of that band); `F_BAND_ERG` is restated as $343.0 * 289/198 = 500.6$ with the derivation in the file, and the file now records that it cancels out of `sigma_pump` exactly. The nine `abs_T####.npz` were recomputed (`run_abs.py`, 23 to 251 s each) and `h2_self_shielding_table.f90` regenerated. MEASURED: the generator reproduced the shipped module byte for byte before the change, so the only difference is the band. `sigma_pump` at the bottom of the column axis is $1.701\text{e-}17$ (700 K) to $1.887\text{e-}17$ cm², which is DB96's own $2.557\text{e-}17$ per 912-1110 Å photon restated per 912-1201 Å photon ($1.676\text{e-}17$) to 1-13 percent, the same agreement as before.
 - **The beam loses what the table rates.** The band share A is no longer the DB96 eq. (39) equivalent width; it is the column integral of the pump cross section `sigma_diss/p_eff` of the same table, in closed form (`h2_lw_band_photon_fraction_absorbed`, exact for the module's own interpolant knot interval by knot interval). MEASURED: A at the top of the column axis is 0.6903 to 0.9974 over the whole (T, n_H) grid, where it used to reach 1.5154; the closed form agrees with an independent 20000-step quadrature of the two public accessors to 2e-10. The band ledger's photon count is now the cell mean of the same pump cross section (`lyman_werner_band_absorption_rate_cell_mean`), on the quadrature the dissociation rate already used, and not the cell mean of `sigma_diss` divided by `p_eff` at one column: on cells spanning 2.5 H₂ scale heights the second stood 8.0 percent above the beam loss and the first closes on it to 1e-6 (`physics_probe/lyman_werner_cell_mean` sec. 6). `h2_band_equivalent_width`

- and eq. (39) are removed; the two published self-shielding fits stay, called by nothing, for the comparison of docs/h2_self_shielding_cloudy.md.
- **The ledger.** The LW row of output/FUV_bands.txt is a closure like B3 and B4, and its label says so. MEASURED on the oxygen_chemistry state, 5 steps: unrated_frac -3.62e-1 before, -1.18e-2 after, and the LW absorbed energy goes from 233.5 against an incident 171.5 (36 per cent over the band) to 202.3 against 200.0. The residual left is the part of the column above the top of the table's column axis: this case reaches $6.4e21 \text{ cm}^{-2}$ against an axis top of $5e21$, where the clamped edge cross section drives A past 1 and the beam is capped while the rate is not. The run already warns about that column.
 - **Impact.** mol_lyman_werner at its own 12000 steps, before at 343.0 against after at 480.9: log10 Mdot 10.53 -> 10.54, T and rho under $5e-3$, cooling $6.5e-2$, the molecular densities and the shielding $1.2e-1$ at the H2 front ($r = 1.075$), the base H2 column $3.3875e20$ -> $3.4067e20 \text{ cm}^{-2}$ and the base x_{H2} 0.96288 -> 0.96365. The optically thin rate at the top of the column moves by $480.9/500.6 = 0.9606$ exactly: the band's photon content grew 46 percent and the cross section fell by the same factor, so what is left is that the real spectrum gives 480.9 over 912-1201 Å where the flat continuation of the 912-1110 Å deck would give 500.6. Deeper cells move -4.7 to -6.0 percent, the wider band's extra lines shielding differently; slightly less destruction leaves slightly more H2.
 - **Not repaired here, and it blocks the fuv_band_ledger suite.** With its restated 480.9 the oxygen_chemistry case stops at step 0 on the energy floor. MEASURED, it is not the band merge: the same case stops with the OLD binary at the same restated flux, and both binaries stop or run at chance along the flux axis (old: runs at 343.0 and 440.0, stops at 360.0, 400.0 and 480.9; new: runs at 343.0 and 400.0, stops at 360.0, 440.0 and 480.9). That is the step-0 acceptance knife edge of items B3a and 2 of To_be_determined_by_user_20260906.md. Every ledger assertion is GREEN on the same case at a flux where it runs.
 - Also: p_lw_absorbed is now written by the sweep and checkpointed but read by nothing, since the ledger takes the pump rate directly. Removing it needs attempted_step.f90.
- **B3b-IR** (molecular_infrared_cooling.f90; h2_infrared_line_populations and h2_partition_sum_uniqueness probes): the third copy of the H2 ladder sum (212-215) is gone; the H2 infrared line populations are normalized by caloric_eos's h2_partition_function, the one Boltzmann sum over the Roueff et al. 2019 ladder whose energy the equation of state carries and whose free energy sets the $\text{H} + \text{H} \leftrightarrow \text{H}_2$ equilibrium. The probe that greps the tree for a second sum went RED (1) -> GREEN (0). Impact (advisor measured, 12000-step snapshots, OMP_NUM_THREADS=1): mol_ir_bands (9 output files) and mol_base_handoff (7 files) BYTE-IDENTICAL before and after, Mdot 7.99 both; the retired copy had an $\exp(-\min(E/T, 700))$ clamp whose only difference from the shared sum's $\exp(-E/T)$ is below double rounding. No golden moves. The worker's draft report stopped at "impact runs in progress"; this entry closes it.
 - **LW-NORM-B, advisor verification** (2026-09-06): private build of the landed tree, physics_probe PASSED (554 assertions incl. the rewritten section 6: column integral of sigma_pump at the top of the axis 0.69 to 0.997 on the whole grid, was 1.515; photons spent / photons lost 0.999999), spectrum_type PASSED (154),

fuv_band_ledger NOT RUNNABLE: its case is oxygen_chemistry, whose Stellar LW flux restated to 480.9 (the same spectrum over 912-1201 Å) stops at step 0 with energy update FAILURE at cells 85-88 ($T = 2.1 \times 10^4$ K, ionization sweep NON-ROOTs accepted by the relaxation amnesty with residual 1 to 4, then the balance asks for T below the 14.5 K floor). MEASURED by the worker along the flux axis with both binaries: 343 runs, 360 stops, 400 runs (after) / stops (before), 440 stops (after), 480.9 stops: a step-0 acceptance knife edge of the OLD code at cells with no H₂, items 1 and 2 of To_be_determined_by_user_20260906.md (decision 4: reject-and-retry; amnesty cap), not of the band. The flux is NOT tuned to make the case run; the case and fuv_band_ledger stay red until those two decisions land. Code read: $a_lines = \min(A, 1)$ with transmission $1 - A$ (a band share, not an optical depth) from the closed-form power-law integral of the tabulated σ_{diss}/p_{eff} ; both the rate and the share are read at the local T and n_H of the cell for the whole column above it (homogeneous-column table), a comment in util_ion_eq.f90 now says so. Accepted. Goldens untouched; every LW case moves at the series gate (mol_lyman_werner 500-step pair: $T = 3.3 \times 10^{-3}$, H₂ 0.125 at the front, $k_{LW} \text{ thin-limit} \times 0.9606 = 480.9/500.6$).

- **ACCEPT-1/2** (user decisions 2026-09-06, items 1 and 2 of To_be_determined_by_user_20260906.md, both as recommended; brief plan_rev2_brief_ACCEPT12.md, to start when B3c releases EXHALE_main.f90, energy_semi_implicit.f90, ionization_equilibrium.f90): (1b) a failed energy update refuses the attempted step in both modes and the controller retakes it at half dt, exhaustion exits 2; (2a) the relaxation amnesty accepts a non-root only below a residual cap measured on the matrix, above it the cell keeps the composition it entered the sweep with (new acceptance class 6). Motivation: oxygen_chemistry at its restated LW flux dies at step 0 on exactly this pair (LW-NORM-B verification entry above).
- **B3b-IR impact runs, advisor correction** (2026-09-06): the worker's before/after pairs were run with binaries built from the tree as it stood at 12:30, when the B3a controller still refused on the time-discrete hydrodynamic row in init mode: MEASURED from their run logs, 12000 steps accepted of 104389 attempted (mol_ir_bands) and 104594 (mol_base_handoff), 103697 refusals all "the Runge-Kutta stages and the fluxes: the time-discrete hydrodynamic row was out of tolerance", dt collapsed and T stayed within 1 K of the 1140 K initial condition over the whole grid (golden mol_ir_bands reaches 2553 K at 2.18 R_p). The byte-identity of the pair therefore says little about the layer; the statement that the IR refactor changed no arithmetic rests on the emission assertions at five temperatures with identical digits and on retired/ $Q - 1 = 0$ at every probe temperature, which stand. On the current tree the same case accepts 12000 of 12000 attempts with no refusal (MEASURED, private build). The H₂ IR tabulation grid extension (advisor, this entry's neighbor above): n_tab 401 → 496 at the same log spacing to $t_tab_hi = 30146$ K, old nodes reproduced by the same expression, emission at 10000 K: $clamped = direct \text{ sum} = 1.57875 \times 10^{-18}$ (was 1.181×10^{-18}); molecular_infrared_bands.py template carries the private :: i of the generated data module and regenerates it byte for byte (MEASURED).
- **B3c** (EXHALE_main.f90, ionization_equilibrium.f90, energy_semi_implicit.f90, molecular_reaction_heat.f90; new suite coupled_source_step, 30 assertions in the advisor's run): rows 7 to 9 of the enumerated step are one local source step per cell at fixed volume, the temperature and the composition its unknowns, iterated to a fixed point ($|dT|/T \leq 1 \times 10^{-8}$, $\max |df_{sp}| \leq 1 \times 10^{-8}$, cap 20 passes). The composition projection (comp_p_from_T + W_to_U, which rebuilt the energy row at the new particle count)

is gone; the energy row is anchored on `u_th_old`, the thermal energy the cell had at its old composition; `rho` is intent(in) to the chemistry and the mass sum is a checked diagnostic (worst 6.4e-12, T2.2). The two collisional deposits D0 C25 named (oxygen O1/O1r/O2/O2r/O6, associative He 2³S) are in as differences of the one formation-energy table. RED reference kept in the suite: the removed update changed `u(3)` by -0.600 with `heat = cool = 0`. **Advisor decision on the worker's deviation (accepted):** Delta `u_form` is NOT added to the energy row. In this code `heat` and `cool` are the net THERMAL sources (photoheating deposits $h\nu - E_{th}$, collisional ionization cooling removes E_{th} , every molecular, Penning, Lyman-Werner and oxygen reaction heat is a difference of the same table), so the formation reservoir is already accounted for and the term would double count; MEASURED with it added, `wasp_full` dies at step 0 with 368 of 504 cells at the floor. Consequence: T1.5/T1.6 as an identity on `u_th + u_form` against `dt(heat - cool)` cannot close; the B3a instrument (`attempted_step.f90`) is to be restated as `du_th = dt(heat - cool) +` the transport contribution of the step, or evaluated on the source substep alone (item T15-INSTRUMENT, pending). Impact (control = same snapshot with the edits reversed): `wasp_full` to its `du` stop, `log10 Mdot` 13.34 -> 13.34, `T` 2.5e-3, `rho` 2.4e-3, metals 1.5e-2, one sign change of `v` at `r = 1.062` (2.5e-4 of the wind); `wasp_full` 300 steps front one cell apart; molecular 12000-step snapshots up to 40 percent in `rho` beyond 4 `Rp`, `Mdot` 10.58->10.57, 10.53->10.51, 10.58->10.58; `hydrostatic_column` <= 6e-9. **Cost: 5 to 12 passes per step, 8.1x on wasp_full 300, 10.3x to the du stop (18:59 -> 3:15:07)**. Assigned as B3c-COST (re-sweep only the cells not yet at the fixed point; tolerance anchored on a measured convergence study, not chosen to pass). Also in and byte-identical: the carrier status pre-set removal (HYG-B3a) and the two O0FPE sites; `fpe_traps` passes at -O0 with traps. Left red on purpose: `heat_column_matches_breakdown_total` 6.886e-3, because `write_heat_breakdown_eq` (`util_ion_eq.f90` 2280) is a second hand-maintained copy of the heating assembly that lacks the new deposit, and `post_process_adv.f90` ~847 a third: assigned as HEATBRK (one assembly, three callers), not the ten-line patch. Advisor verification: private build of the live tree, `coupled_source_step` 30/0, `energy_update` 48/0, `attempted_step` 31/0, `certification` 29/0, `run_mode` 30/0, `acceptance_classes` 12/0, `grid_and_gates` 92/1 (the HEATBRK red).

- **HEATBRK, first pass BLOCKED** (2026-09-06): the heating assembly the sweep uses is 210 inline lines of `ioniz_eq` (`ionization_equilibrium.f90` 2119-2331), not a routine, so the one-assembly fix has to extract it there, a file ACCEPT-1/2 owned at the time; the worker stopped with no edit, as briefed. Re-measured RED: worst 6.886e-3 (cell 376, the associative He 2³S and collisional oxygen deposits missing from the breakdown) over a 1.24e-5 median floor (the breakdown RE-SOLVES the field and rates on the written state while the heat column carries the rates the sweep produced it with). Interface proposed: `heating_of_composition(...)` in `utils_ion_eq` returning `heat` and a non-optional `heat_chan(:,17)` (channel list and header owned by the module; LW dissociation and fluorescence as two columns), called by the sweep, by `write_heat_breakdown_eq` (which then forms no rate) and by `post_process_adv` (which today omits the Balmer and associative terms from its theat). **Advisor decision:** `Heating_breakdown.txt` writes the channel array the sweep computed for the written state, not a recomputation; this is the "one state per output file" rule (rev3 section 10.2 item 2, landed) applied to this file, and the only form in which the breakdown total can equal the heat column to round-off. Resumes when ACCEPT-1/2 releases `ionization_equilibrium.f90`.

- ACCEPT-1/2, advisor verification** (2026-09-06): private build of the live tree, energy_update 53/0 (the driver was killed by error stop 1 before), acceptance_classes 22/0 (did not compile before: class 6 did not exist), attempted_step green (10 RED before: exit status, history lines, reason breakdown), coupled_source_step 30/0, certification 29/0, run_mode 30/0, carrier_retry 97/0. Code read: the kept state of class 6 is x_entry, the composition the cell entered the sweep with (built once, also the solve's starting point), heat and cool of the returned composition as section 170 requires. Accepted, including the two deviations: the two-line steady_residual.f90 change that makes n_cells_without_chemical_root count class 6 (without it the controller would not see a refused non-root), and as_op_projection -> as_op_composition (operation number 8 unchanged; B3c had removed the projection the name described). Findings taken over: cap 1e2 MEASURED on the whole default matrix at 400 steps (the amnesty population is a cold-start population: 111 events 1e-6 to 8.853 at step 0 of oxygen_chemistry, nothing until 8.6e4, then the four cells 85-88 whose energy update failed); with the cap oxygen_chemistry runs its 1000 steps and exits 0 (log10 Mdot 9.61), no step refused, fifteen of sixteen cases byte-identical; the comment's "res ~28 in the metals-on molecular gates" is no longer reproduced (they accept no non-root today) and was corrected in place. The energy-refusal path is exercised by the default-off hook EXHALE_ENERGY_FAIL_AT_STEP. Outside scope, assigned: "class 4" wording in certification.f90/steady_newton.f90 and steady_newton.f90:815 acc_resmax(4) missing class 6 (HYG-CLASS6, folded into HEATBRK's resume); an H-only run has no acceptance classes at all (pre-existing, B6 list); oxygen_chemistry cannot run in phys mode (56 handoff non-roots refused by A0), a statement about that configuration, recorded.
- HEATBRK landed, advisor verification** (2026-09-06; measured on mol_base_handoff 200 steps, wasp_full 300, oxygen_chemistry 1000, the mol_metals impact pair still running when this was written): heating_of_composition (util_ion_eq.f90 1503-1780) is the one heating assembly, 17 named channels, the total the running sum of the columns in the sweep's original order (sweep heat bit-identical); the sweep calls it once (181 inline lines of ioniz_eq gone), write_heat_breakdown_eq writes the module array heat_channel_state the sweep filled and forms no rate, post_process_adv calls the same routine. Private build of the live tree: grid_and_gates 93/0 with heat_column_matches_breakdown_total 1.678e-16 (was 6.886e-3), physics_probe 557/0 incl. heating_channel_closure and the scanner heating_sum_uniqueness (one exception recorded: the band ledger of write_output.f90 restates the Lyman-Werner and FUV photolysis event energies, a fourth copy, assigned HYG-LEDGER-EPS). Heating_breakdown.txt 18 -> 21 columns; _adv profiles move by up to 1.8e-2 (the Balmer and associative deposits the third copy lacked). heat_channel_state is class-1 state and enters the attempted-step checkpoint through B3c-COST (owner of attempted_step.f90), assigned. HYG-CLASS6 (class 4/6 wording, acc_resmax over both classes) in the same landing.
- B3b-CO landed: the CO row carries two published destruction rates** (2026-09-06). src(ic_CO) is no longer 0. It is $-(k_{D1} n(\text{He}^+) + k_{\text{CO}}) n_{\text{CO}}$ with $k_{D1} = 1.60\text{e-}9 \text{ cm}^3 \text{ s}^{-1}$ (UMIST RATE22 entry 4068, measured, accuracy better than 25 percent, temperature independent, READ) and k_{CO} the cell mean of $(F_{\text{LW}}/\langle h\nu \rangle) \sigma_{\text{CO}} \Theta(\text{N}_{\text{CO}}, \text{N}_{\text{H2}}) \exp(-\tau_{\text{cont}})$ on the 912-1201 Å beam. New: co_self_shielding_table.f90 (the 12CO block of Visser, van Dishoeck & Black 2009 Table 6, $T_{\text{ex}}(\text{CO})$

= 50 K, with their Table 5 beside it for the spread) and `co_photodissociation.f90` (the point rate, the two-column cell mean, the fragment heat). `fuv_lw_photon_field` gains the CO column and returns `k_CO` and `Theta`; `carrier_photolysis` rebuilds `k_CO` from the current carrier columns, so the CO layer shields itself inside the relaxation. Constants, all MEASURED: `sigma_CO_band = 1.0160e-17 cm^2 per beam photon` (Heays et al. 2017 cross section over the 912-1118 Å window CO absorbs in), `<hv> = 12.8674 eV` (oscillator-strength weighted over Visser's 37 bands), `D0(CO) = 11.1157 eV` and `q(D1) = +2.2117 eV`, both differences of the one formation-energy table; the deposit is 1.7517 eV per dissociation. Both deposits are channels 18 and 19 of `heating_of_composition` (17 -> 19 channels) and are formed nowhere else; `heat_per_co_dissociation` is registered in the uniqueness scanner with the band ledger of `write_output.f90` as its one recorded exception. `output/FUV_bands.txt` gains `N_CO`, `k_CO` and `Theta_CO` and a CO row in the beam ledger (CO is printed apart from `rated_ph` because `beam_loss_ph` carries no CO term: MEASURED, folding it in moved the LW budget residual from -1.8e-2 to -8.2e-2). `output/Oxygen_chemistry.txt` gains the domain record `tau_dest <= 0.1 tau_res` with the worst ratio and its radius, `tau_res/tau_form` from RATE22 8597, the count of cells above the shielding table's 512 K excitation-temperature limit and the count in which the He+ channel outruns He+ recombination. Tests: `physics_probe co_destruction` (27 assertions, RED on a mutated table entry and a mutated photon energy), `fuv_band_ledger` extended. **Impact, MEASURED on oxygen_chemistry at its maxsteps of 1000, the only regression case that carries CO** (a control binary of the same tree with the CO source and the two deposits reverted): both run 1000 steps, exit 0 and give `log10 Mdot = 9.61`; the after binary costs 628 s against 11 percent less for the control at a matched 150 steps. The He+ + CO charge-transfer deposit is **65 percent of the total heating at r = 1.089**, the helium ionization front where CO still survives, and under 1 percent outside 1.05-1.17; the layer's temperature moves by up to 46 percent at `r = 1.023` and its CO by up to 52 percent. `mol_carrier` at 300 steps is byte-identical in every output file, which is the gate that the new terms are on the same `thereis_oxychem` as the rest of the oxygen path; no other regression case sets Oxygen chemistry. The domain record of that run: 401374 out-of-domain cell visits, worst `tau_dest/tau_res` 1.34e5 at `r = 1.0002`, worst `tau_res/tau_form` 0.219, 305125 cell visits above the 512 K excitation-temperature limit and 281325 in which the He+ channel outruns He+ recombination. Goldens not refreshed. **P0 recorded:** the He+ ledger is not closed. This reaction consumes a helium ion and neither the carrier row (ion stages frozen) nor the sweep's He+ balance returns it, so where `k_D1 n_CO` exceeds the He+ loss rate both the destruction and that heating peak are over-stated. The fix is a sink in the He+ row of `mol_heh_rows` (`System_HeH_mol.f90`), not this item's file; `dom_cells_HeP` is the observable. **Not done, carried over:** the thermodynamic CO ceiling is still in `limit_to_element_budget`; deleting it changes `EXHALE_main.f90`, `certification.f90` and `write_setup_report.f90`, none of them this item's files. The exact lines are in the item's report.

- **HYG-LEDGER-EPS closed in the same landing** (2026-09-06): the FUV band ledger of `output/FUV_bands.txt` is formed by `utils_ion_eq::fuv_band_absorption_ledger`, beside the heating assembly and reading the same imports, and `write_output.f90` writes what it returns; the `heat_LW` column of `output/Lyman_Werner.txt` is now the assembly's own channel (`heat_channel_state(:, ih_H2_LW_dissoc)`) instead of a second evaluation of `k_LW n_H2 e_lw_fragment_erg`. The `heating_sum_uniqueness` scanner's exception list is EMPTY: every deposited energy is named only in the module that defines it and in the one

assembly. RED before (the pre-move import line, two deposits reported outside the assembly), GREEN after. FUV_bands.txt is byte-identical across the move (MEASURED, oxygen_chemistry at 5 steps); the heat_LW column moves by at most $5.57e-4$ relative, which is the one-sweep lag between the state the file wrote and the state the energy equation deposited, and the new number is the deposited one.

- B3c-COST, the cost of the coupled step measured and the skip specified, not implemented** (2026-09-06; EXHALE_main.f90, attempted_step.f90, the coupled_source_step and attempted_step suites). WHERE THE COST IS, MEASURED on wasp_full 300 steps, one thread: the run timers give ioniz_eq 83.6 percent and the energy solve 16.3 percent of the marching wall time, and a gprof build puts **78 percent of the whole marching time inside the MINPACK solve of one cell's chemistry** (one Jacobian and one QR per cell per pass; the warm start already reaches each root in one Newton step), against about 13 percent in the whole-grid rate and field work a pass repeats. So the cost is the number of CELL solves and not the field, and reusing the field between passes is worth at most that 13 percent. THE CONVERGENCE STUDY CONTRADICTS THE EXPECTATION THE ITEM WAS WRITTEN FROM: after pass 2, 308 of the 500 cells are still above the tolerances, after pass 5 still 130, and the cell that sets the pass count walks inward one cell per pass with the ionization front. Contraction 0.191 median over 2729 passes. A cell-local skip is therefore worth 2.41x on the cell solves with a one-pass rule -- but that rule is WRONG: a cell at rest moves again on 177 of the 3029 passes, 690 of the 809 wake-ups at pass 8, because the column keeps moving a cell after the cell has stopped. **Two consecutive passes of rest: zero wake-ups, 1.96x on the cell solves, 1.55x on the step.** The skip has to be a mask in the cell loop of ioniz_eq, which this item does not own (HEATBRK was writing to that file at 20:30 and 20:35), so it is SPECIFIED and not implemented: the optional cell_at_rest argument, the class-6 path (sys_x = x_entry straight to the extraction) as the body of the skip, and the one consequence that is not a pure skip -- the acceptance ledger describes the cells the sweep SOLVED, so a mask needs an acceptance class stored for each cell and inherited by masked cells, or n_cells_without_chemical_root reads a different population from the one it believes. TOLERANCE ANCHORED AND KEPT AT $1e-8$: the anchor the comment gave (the $1e-6$ heat-column gate) stopped depending on it when HEATBRK made the two assemblies one; the anchor that binds is cert_tol_carrier = cert_tol_element = $1e-8$ on the composition this step returns. Measured $1e-10$ to $1e-5$ on wasp_full 300 and mol_base_handoff 1500, the adopted state moves by about the tolerance and a decade costs 1.3 passes and 15 percent of the wall time, **so the tolerance is not the cost lever**; the step's own truncation error (step-doubling, phys, $5.3e-3$ relative) is five decades above the loosest value and does not bind. Newton on T with the composition's response (item 4) stays live, and the CHEAP version of it is refuted by measurement: the contraction ratio spreads by 122 percent globally and by 200 percent cell by cell over the tail of a step, so the sequence is geometric on average over a run and not within one step at one cell -- which is why B3c's Aitken failed, and why the acceleration has to come from $dc/dT = -(dF/dc)^{-1} (dF/dT)$ against the factorization hybrid already forms, not from the iterates. ALSO FIXED, a defect of my own file found outside the task: the run's cost line divided passes accumulated over every ENTRY of the coupled loop by the number of adopted STEPS, two different populations, so it could report a mean above the worst single entry (MEASURED on the control, wasp_full 300 in phys mode: "13.98 passes per step on average, worst 11"). It now divides by the entries and names them. ALSO: heat_channel_state is in the attempted-step

checkpoint (save/restore/compare/perturb), the four lines HEATBRK's report asked for, with two assertions RED on the control (the moved array is not seen, and the restore leaves it 3.0 off). Tests: coupled_source_step 36/0 (33 before), attempted_step 97/0 (95 before). Impact, control = the same tree snapshot with this item's two production files reverted: wasp_full 300, mol_base_handoff 1500 and mol_metals 1500 all **byte-identical** in Hydro_ioniz.txt and Ion_species.txt.

- B3c-COST** (measurement; EXHALE_main.f90 comments and debug counters, attempted_step.f90 checkpoint; numerically inert, wasp_full 300, mol_base_handoff 1500, mol_metals 1500 byte-identical): the coupled source step's cost is not where the brief assumed. MEASURED on wasp_full 300 (3029 passes): after pass 2 still 308 of 500 cells above tolerance, after pass 5 still 130; the pass count is set by the ionization front walking inward one cell per pass, not by a fixed handful of cells, so a cell-at-rest skip is worth 1.96x on the cell solves and 1.55-1.62x on the step, and the naive rule (rest on the last pass) is wrong: 177 of 3029 passes wake a resting cell, two consecutive passes of rest wake none. gprof: 78 percent of marching time is the MINPACK solve of one cell's chemistry (Jacobian + QR per cell per pass), the whole-grid field and rate work 13 percent, the energy solve 16 percent (already deactivates converged cells). Tolerance anchored and KEPT at 1e-8 (the binding anchor is cert_tol_carrier = cert_tol_element = 1e-8 on the returned composition): a decade costs 1.3 passes and 15 percent, not the lever. Aitken and any acceleration built from the iterates fail because the contraction ratio spreads 200 percent cell by cell within a step. **The lever is a Newton on (T, composition) taken cell by cell with $dc/dT = -(dF/dc)^{-1} dF/dT$ from the factorization hybrid already forms, inside the sweep (ionization_equilibrium.f90), plus the two-pass rest mask with an acceptance class inherited by masked cells:** assigned as COST2, after B3b-CO releases that file. Also fixed: the cost line divided passes over loop entries by adopted steps (could report a mean above the worst); heat_channel_state is in the checkpoint with two assertions (attempted_step 97/0, coupled_source_step 36/0; advisor rebuild and rerun green). Outside scope recorded: EXHALE_ERR EVERY read in both modes but the probe runs only in phys; mol_base_handoff in phys mode refuses 7 of its first 20 steps at the fixed point (pass cap at step 0), recovers.
- B3b-CO, advisor verification and follow-up** (2026-09-06): private build of the live tree, physics_probe 584/0 (incl. co_destruction 27 and the scanner heating_sum_uniqueness with an EMPTY exception list after HYG-LEDGER-EPS moved the band ledger into utils_ion_eq::fuv_band_absorption_ledger), fuv_band_ledger 24/0, grid_and_gates 93/0. Accepted with the four stated departures from the design (cross section as the mean over the 912-1201 Å beam of one that ends at 1118 Å; the same flat-band cross section for a loaded spectrum, the shape-aware integral belonging to sed_read.f90; q_D1 as a table row because mion_fsp is a parameter; CO on its own ledger row, co_frac 6.39e-2 the size of decision 2's approximation). **P0 opened by the item itself and assigned at once as B3b-CO2:** the He+ + CO channel deposits its heat (65 percent of the total at the He+ front of oxygen_chemistry, r = 1.089) while no He+ is consumed and no C+ or O delivered anywhere; brief plan_rev2_brief_B3b-CO2.md (He+ sink in both molecular systems with Jacobian, products to the free O and carbon pools, heat rate identical to the species rate from one place, element census). Also recorded: the thermodynamic CO ceiling is still present because its deletion touches EXHALE_main.f90 34, 3641-3648, certification.f90 84, 1072-1076 (n_active_unvalidated_physics loses its only producer),

write_setup_report.f90 18, 925-934 (CEILING-DEL, after B4-1 releases main); heat is deposited for carrier reactions whose products a never-covered carrier interval does not deliver (oxygen_chemistry in init mode never covers one), an operator-split defect that B4-1c and B5 address and that leaves the oxygen case without an end-to-end gate for its CO source; heat_channel_state now has 19 columns (checkpointed by B3c-COST); python/paper_data.py HEAT_FIXED stale.

- B3b-CO2 landed: the He+ + CO channel now consumes the helium ion** (2026-09-06). Row (2) of mol_heh_rows (System_HeH_mol.f90) loses k_D1 n_CO n(He+) with $k_{D1} = rk_{D1_Hep_CO}() = 1.60e-9 \text{ cm}^3 \text{ s}^{-1}$, and its turnover scale (set_mol_turnover_rates) gains k_D1 n_CO n_He; the helium leaves neutral, so the free neutral He that closes the helium budget receives it and no other row moves. One row serves every path: System_HeH_mol_metals and the constrained-equilibrium residual both call it, and neither molecular system has an analytic Jacobian to extend (hybrd1, finite differences), so the test asserts the difference derivative the solver actually builds. CO is a background density of that row and not an unknown, because the one-sided model gives it no formation and therefore no local equilibrium; the reaction is split across the two operators, each reading the other's frozen value from the one rate, which makes the heat of channel 18 the heat of exactly the events the species ledger performs. The products needed nothing added: the free-atomic-oxygen closure of carrier_source and nC_free - n_CO in carrier_write_back already carry the two nuclei and element_census already counts them in CO. Tests: new physics_probe driver co_helium_ion_sink (8 assertions, RED 4 FAIL against the pre-change module relinked into the same suite, GREEN 8 PASS; the RED reads hep_row_co_sink 0.0 against -3.84e6, hep_row_derivative_co_term 0.0 against -38.4, hep_turnover_scale_co_term 0.0 against 3.84e13); physics_probe suite PASSED after; element census with EXHALE_ELEMENT_ASSERT=1 on oxygen_chemistry 150 steps exits 0 with no budget violation (a guard, not a RED/GREEN pair: the change moves no C or O nucleus). Impact, MEASURED at 1000 steps of oxygen_chemistry: the largest share of the total heating taken by channel 18 falls from 0.672 at $r = 1.081$ to 0.035 at $r = 1.117$, channel 18 at $r = 1.0894$ from $6.90e-7$ to $9.75e-9 \text{ erg cm}^{-3} \text{ s}^{-1}$, $n(\text{He}^+)$ there to 0.015 of its former value and to 0.0073 at $r = 1.011$, T at $r = 1.0233$ from 1156.4 K to 1088.2 K, and $\log_{10} \text{Mdot}$ stays 9.61 with exit 0. dom_cells_HeP does NOT fall (281325 -> 281382): the counter compares k_D1 n_CO with recombination and still says this reaction is the leading He+ loss of the layer, so what it now reports is the operator split's lag rather than a missing sink. mol_carrier 300 steps and mol_metals 400 steps byte-identical in every output file. Goldens not refreshed. Reported for owners elsewhere: the dom_cells_HeP header text in write_output.f90 558-562 still says the frozen He+ "over-states that channel"; the mirror decomposition hydrogen_helium_row_terms in constrained_chemical_equilibrium.f90 needs the same term after line 1859 or it stops summing to row 2; the n_co comment in ion_cell_state.f90 still calls CO an equilibrium closure.
- B3b-CO2 landed, advisor verification and follow-ups** (2026-09-06): the He+ row of mol_heh_rows (System_HeH_mol.f90, the one row every caller uses) loses k_D1 n_CO n(He+), its turnover scale gains the same bound; CO stays a background density of the solve because the one-sided model gives it no formation channel (not a balance row). Products needed no code: the O nucleus goes to the free-oxygen closure and the C nucleus

to $n_{\text{C_free}} - n_{\text{CO}}$ in the carrier write-back, `element_census` already counts both; the carbon CHARGE is set by the sweep's stage equilibrium, the design of record ("the step moves nuclei, the sweep moves charge"), so the reaction's C+ is not imposed (recorded as a design choice, not a defect). MEASURED, `oxygen_chemistry` 1000 steps: channel-18 share of the total heating 0.672 at $r = 1.081 \rightarrow 0.035$ at $r = 1.117$; $n(\text{He}^+)$ at the front 0.015 of before (0.0073 at $r = 1.011$); T at $r = 1.0233$ 1156 $\rightarrow$ 1088 K; $\log_{10} \dot{M}$ 9.61 both; constrained-continuation fallbacks 932 $\rightarrow$ 1880, all accepted; `mol_carrier` 300 and `mol_metals` 400 byte-identical. `dom_cells_HeP` did not fall (281325 $\rightarrow$ 281382): it measures dominance of the channel over recombination, not over-statement; its header text now says so. Advisor follow-ups applied directly: the $\text{D1 He}^+ + \text{CO}$ term added to the He^+ decomposition of `constrained_chemical_equilibrium.f90` (hydrogen_helium_row_terms, whose header requires every `mol_heh_rows` reaction to appear), the `FUV_bands.txt/oxygen_chemistry.txt` `dom_cells_HeP` header rewritten, the stale `n_co` comment of `ion_cell_state.f90` replaced. Private build: `physics_probe` 592/0 (incl. `co_helium_ion_sink` 8), `fuv_band_ledger` 24/0. Remaining: the operator split is now a LAG (the sweep destroys He^+ at the entry CO while the CO row destroys CO at the previous sweep's He^+) and `oxygen_chemistry` never covers a carrier interval in init mode, so this row has no end-to-end gate: B4-1c/B5 territory, recorded.

- **HEATBRK closed** (2026-09-06): the `mol_metals` pair finished at its 12000 steps, `Hydro_ioniz.txt` and `Ion_species.txt` byte-identical, gate $1.683\text{e-}16$ with 21 columns, `_adv` profiles moved by at most $8.45\text{e-}3$ (the deposits the post-process copy lacked). All three impact cases of the item therefore confirm: the sweep's heat is unchanged, only the two consumers of the assembly moved.
- **B4-1** (`species_face_flux.f90` new; `Reconstruction.f90`, `PLM_rec.f90`, `RK_rhs.f90`, `EXHALE_main.f90`, `binary_element_diffusion.f90`, `Makefile`, new suite `src/tests/species_face_flux/`): the advective half of the ELEMENT transport now rides on the hydrodynamic face mass flux, inside the Runge-Kutta stages, on the same faces, areas, volumes and time step as the mass row. The face composition is the reconstruction of the mass fraction by the same limiter as the primitive variables, taken from the side the face mass flux selects, and the closing member of the set is one minus the others, so $\sum_s F_s(j) = F_{\text{rho}}(j)$ holds by construction; the operator asserts it at every face of every stage (largest residual MEASURED in the unit suite: $2.1\text{e-}16$ relative, against a tolerance of 64 ulp). The cell-velocity upwind advection of `binary_element_diffusion.f90` is gone from the marching path: the operator advects only when it is given an advecting mass flux, which the fixed-wind relaxation supplies and the marching path does not, and the trace-metal solve lost its advective term the same way. The carriers of `photochemical_transport_step` are NOT in this increment (that file was held by another item): they keep their present advective term and are B4-1c. Tests, all MEASURED with a private build: a uniform composition is preserved to 2 ulp on a compressing flow, an expanding flow and one whose face mass fluxes alternate in sign, for PLM and for WENO3; the mass fraction stays in $[0,1]$ on a step profile carried across the domain in 300 steps; the closing member of the set differs from an independently reconstructed $1 - Y$ by $1.1\text{e-}16$; the face reconstruction converges at order 1.97 (PLM) and 1.92 (WENO3); the helium nucleus total of the domain changes only by its boundary fluxes to $4.9\text{e-}17$, where the cell-velocity form it replaces leaves $1.6\text{e-}9$ on the same column and the same step (RED and GREEN in one run); the injected violation of the face identity stops the operator with the face named; the new step is refus-

able at its injection point and the run goes on. The element-diffusion suite `diffusion_tests` is 35 of 35 after T11 was rewritten onto the new operator (its premise, that a face-averaged upwind decouples a cell whose face velocities straddle zero, is what the face-flux form removes). Impact, MEASURED on scratch copies, `OMP_NUM_THREADS=1`, `mol_*/hp_*` at 200 steps and `mol_diffusion/lower_profile` at 1000 rather than the 12000 their `maxsteps` pins, `wasp_*` at 300: 14 cases byte-identical, every one of them without element transport (`wasp_full`, `wasp_he23off`, `wasp_full_newton`, `mol_base_handoff`, `mol_metals`, `mol_lyman_werner`, `mol_ir_bands`, `mol_sec_ion`, `mol_carrier`, `oxygen_chemistry`, `hp_zero_seed`, `hp_trace_seed`, `hp_front`, `hydrostatic_column`); `mol_diffusion` moves by at most $7.1\text{e-}5$ in `Ion_species.txt` and $3.6\text{e-}3$ in `Cooling_breakdown.txt`, `lower_profile` by $1.2\text{e-}3$ and $1.1\text{e-}2$. The transported quantity, $n_{\text{He}}/n_{\text{H}}$, moves by $1.9\text{e-}10$ at the base and $3.5\text{e-}6$ at the molecular front of `mol_diffusion`; the base-layer mass flux $\rho \ v \ r^2$ moves by at most $9.2\text{e-}7$ over $1.00 \leq r \leq 1.16$. Those are smaller than the design's estimate because at 1000 steps neither case has separated its elements far from the reservoir ratio, and a uniform composition is what both forms carry exactly; a converged pair is what would measure the difference at the separation those cases reach, and is named as open in `docs/b4_spatial_operator_design_20260906.md`. No golden was refreshed.

- B4-1 landed (element rows), advisor verification (2026-09-06):** new module `species_advective_transport` (`species_face_flux.f90`): the face mass fraction by the primitives' own limiter (PLM_rec_scalar, Reconstruct_scalar, the MC limiter and the WENO3 geometry coefficients factored, not duplicated), upwind side = sign of F_{ρ} (coincides with the HLLC contact selection wherever both are defined, ROE/LLF offer only the sign), the closing member one minus the others so $\sum_s F_s = F_{\rho}$ by construction and asserted at every face and stage, advanced inside the three SSP-RK3 stages with the mass row's dt , dV and areas; `positivity_limited_fluxes` writes a replaced first-order flux back into `face_flux` so the species flux built on it is the repaired one. Transported set: helium and the hydrogen component (normalized), the trace metals riding on the same faces inside `m_1` at the reservoir abundance (trace limit stated). `binary_element_diffusion.f90` keeps the diffusive half only (`element_transport_residual` keeps its stationary advective term, equal where $r^2 \rho \ v$ is constant). Carriers NOT migrated (`diffusive_photochemistry.f90` was B3b-CO's): B4-1c. Suite `species_face_flux` 13 + 2 whole-binary rows GREEN on the advisor's private build: uniform composition preserved to 2 ulp (PLM, WENO3, three flows incl. a sign-alternating face flux), face identity $2.1\text{e-}16$, bounds exact, face reconstruction order 1.97/1.92, helium nucleus total changes only by the boundary flux to $4.9\text{e-}17$ where the replaced form gave $1.58\text{e-}9$ (B1 AT-4d RED/GREEN in one run), identity assertion fires, element step refusable; `diffusion_tests` 35/35 (T11 rewritten: its premise was the defect removed); `attempted_step`, `grid_and_gates` 93/0, `carrier_retry` 97/0 green. Impact (worker, control = same snapshot with the stage calls removed): 14 cases without element transport BYTE-IDENTICAL (`wasp_*` at 300, `mol_*` at 200 steps); `mol_diffusion` and `lower_profile` at 1000 steps move by at most $3.6\text{e-}3 / 1.1\text{e-}2$ (cooling breakdown), $n_{\text{He}}/n_{\text{H}}$ by $\leq 6\text{e-}5$, base $\rho \ v \ r^2$ by $\leq 1\text{e-}6$: far below the design's estimate because at 1000 steps the elements have not separated from the reservoir ratio; the converged 12000-step pair is open in the design document. Outside scope recorded: ESWENO3 measures second order on a smooth profile (1.92 against PLM 1.97, face error 8x PLM), a property of the primitive reconstruction to look at (WENO3-ORDER); `hp_*` cases need output/`Hydro_ioniz_IC.txt` in place; `element_flux_profile.txt` has a trailing text column.

- CEILING-DEL** (diffusive_photochemistry.f90, write_output.f90, write_setup_report.f90, EXHALE_main.f90, certification.f90; tests carrier_retry (+ scanner no_CO_ceiling_identifier_in_src), carrier_constraint_attribution rewritten on the carbon clamp): the thermodynamic CO ceiling (clamp to co_equilibrium_density, four counters, attempted twins, three accessors, Oxygen_chemistry.txt header line, three co_ceiling_* resolved keys, the end-of-run line, validity state 4.1's only producer) is deleted per design decision 10; the domain record of B3b-CO stands in every place (seven co_domain_* resolved keys under oxygen chemistry + carrier transport, the CO items under class 4.2 beside H3+ in the certification, n_active_unvalidated_physics = -1 "not produced", kept because the five validity states are a fixed set). MEASURED before deletion on oxygen_chemistry 1000 steps: co_ceiling_cells 0, applications 0, CO removed 0.0 (with the published destruction rates in the row and the He+ sink in the sweep the carried CO never exceeds its equilibrium), so every numerical column is byte-identical before/after; mol_carrier 300, mol_metals 400, wasp_full 300 byte-identical. The design's "of order 160 cells" predates the CO rates and is retired with the clamp. Advisor verification: private build, carrier_retry, certification 29/0, carrier_constraint_attribution 11/11, fuv_band_ledger green, no co_ceiling identifier left in src/. Recorded for the docs sweep (DOCS-CEIL): the checkpoint enumerations and validity category 12 in a0_run_mode_contract, a2_certification_contract, b1a_active_equation_inventory, b3a_attempted_step_controller_design, b1_target_system (1246-1554), development_plan_20260905_rev2/rev3/review3, physics_numerics_audit_20260905, and b3b_co_destruction_design sections 2.3/4/7 still describe the ceiling as present. The domain record is not checkpointed (a statement about the state, valid whether or not the step is accepted), unlike the ceiling record it replaces.
- DOCS-CEIL** (documentation only, 2026-09-07): sixteen documents brought level with CEILING-DEL, the current state READ from the code (destruction row $\text{src}(\text{ic_CO}) = -(\text{k_D1 } n(\text{He}+) + \text{k_CO}) n_{\text{CO}}$, conservation-only limit_to_element_budget, carrier_co_domain_record, certification class 4.2 CO block, n_active_unvalidated_physics = -1, seven co_domain_* keys); no measurement deleted, the superseded ones ("of order 160 cells"; "still cuts CO above 3000 K") kept with the measurement that replaced them beside them. Files: a0/a2 contracts, b1a inventory (4.1 row retired, 4.2 row the domain record, open question 3 closed), b3a design (checkpoint enumerations), b1 target system (5.1 split into the 2026-09-06 READ and the current one, AT-5 superseded, decision 12 moot), b3b_co design (sections 2.3, 3(iv), 4, 6, 7, 7.4 as done), development plan rev2/rev3/ review3/execution, physics_numerics_audit (P6 resolved), code_status, d0 (C26 closed, question 5 answered), co_destruction_rates_literature (banner), To_be_determined_recommend, PLAN rev2. Left as dated records: the superseded plan revisions and external reviews, and the b1 draft. Recorded: line-number citations in the 2026-09-06 design documents have drifted by up to a thousand lines (a re-anchoring sweep on routine names is a separate item, DOCS-LINES); the stale checkpoint header comment in diffusive_photochemistry.f90 goes to B4-1c.
- WENO3-ORDER, measurement and fix** (2026-09-07): the ESWENO3 reconstruction was SECOND order on every grid the code runs on. Probe weno3_reconstruction_order (physics_probe; exact volume averages by

8-point Gauss quadrature, face error against the point value, $N = 50$ to 800): production rate 1.997 (uniform in r), 1.988 (the define_grid Mixed production grid), 2.940 on a grid uniform in the volume coordinate, where the defect is invisible. Cause, isolated by variants: not the ESWENO nonlinear weights ($S = 1$ gives the same 1.997), not the floor $\text{eps} = \text{dr}_j^2$ (rate 3.000 at kappa 0.1 to 10 once fixed), not the ideal ratios $D1, D2$ ($1/2 + O(h)$, harmless); TWO of the four stencil coefficients applied the NEIGHBOUR's volume share where cell j 's own belongs: $C1(j) = c/(b+c)$ on the jump to $j+1$ at the right face and $C2(j-1) = a/(a+b)$ on the jump to $j-1$ at the left face, each $1/2 + O(h)$ where $1/2 - O(h)$ belongs, an $O(h)$ coefficient error on an $O(h)$ jump. Inherited character for character from ATES v2.0 (ATES/ATES-Code-main). Advisor check of the derivation: a linear interpolant of two cell averages in the volume coordinate, evaluated at a face $dV(j)/2$ from cell j 's centre, weights the jump by $dV(j)/(dV(j) + dV(nb))$; with the corrected shares a profile quadratic in the volume coordinate is reproduced to $5.9e-16$. **Fixed (advisor, two symbols in each of Reconstruct and Reconstruct_scalar, Reconstruction.f90):** $C1(j) \rightarrow C2(j)$ on dWp in WL , $C2(j-1) \rightarrow C1(j-1)$ on dWm in WR ; the probe's production mapping and header updated, the two waiting assertions enabled: weno3_production_third_order rate 3.000, _mixed 2.991 (RED before: shortfalls 0.803, 0.812); physics_probe 603/0. **Every golden whose case reaches the WENO3 stage moves, and not by rounding** (face error down two decades at $N = 800$); PLM-only cases are untouched. Consequence for the method: until now the two-stage PLM $\rightarrow$ WENO3 recipe was a continuation between two second-order schemes with the second having the larger error constant (1.3x uniform, 3.8x on the production grid). Mdot values are to be quoted from Newton-finished runs after the gate. Attribution of the weight form and the floor to Yamaleev and Carpenter (2009, JCP 228, 3025 and 4248) stays unverified: both papers are paywalled here; requested from the user. Also recorded: stale .ipynb_checkpoints copies of Apply_BC.f90 and UW_conversions.f90 under src/modules/ define the same module names as the live files (not built; to remove, HYG-CKPT).

- **HYG-CKPT** (advisor, 2026-09-07): ten untracked, gitignored Jupyter checkpoint copies of production modules under src/modules/*/ .ipynb_checkpoints/ (Apply_BC, Reconstruction, Source, base_boundary, caloric_eos, low_mach_dissipation, cross_sec, UW_conversions, h2_photo_channels, element_census) removed: not built, but each defined the same module name as a live file and two scanners had to skip that directory (batch 2c) to stay green.
- **T15-INSTRUMENT** (attempted_step.f90 +289/-165, suite, b3a design section 9, b1 T1.5 wording; advisor verification: private build, attempted_step 43/0 with the two whole-binary rows skipping until the main-loop hook lands, coupled_source_step, certification green): the B3a instrument now states the THERMAL identity of the step, $\sum_j dV_j [u_{th}^{n+1} - u_{th}^n] = \sum_j dV_j dt (\text{heat} - \text{cool})_j + (\text{transport contribution})$, the transport contribution MEASURED between two marks on either side of the coupled source step, the residual the closure of the source step alone, Δu_{form} reported as the reservoir's change and never added (the B3c reason stated at the routine). GATE in phys mode at the energy row's own energy_res_tol = $1e-9$ (reason as_reject_source_energy = 14 at operation 9): MEASURED on a hooked scratch tree, worst lresidual/scale 7.44e-1 $\rightarrow$ 5.40e-10 (wasp_full 300 phys), 8.63e-1 $\rightarrow$ 1.88e-10 (mol_base_handoff), zero refusals, states byte-identical in both modes. The two hook lines in EXHALE_main.f90 (before/after the

coupled_source: loop) are assigned to COST2, the file's current owner, with the rewording of the print block. **Defect found and fixed in passing:** attempted_step_checkpoint_take saved `u` and `f_sp` through a saver that reallocates from index 1, so a reader indexing the saved grid arrays by cell compared cell `j` with cell `j - Ng`; energy_identity was that reader, so every `d_u_th/d_u_form` printed so far (incl. the 28 percent reservoir share of the B3c report: with bounds fixed, same step, `d_u_th` 2.006e-5, `d_u_form` 1.298e-5) was formed against a state shifted by two cells; the state restore itself was unaffected (whole-array assignment). The second private copy of the `eps_s` table in `attempted_step.f90` (no OH/H₂O/CO rows) is deleted in favor of `species_formation_energy`. Recorded: `n_projection_applied` can no longer be nonzero since B3c but is still printed by the report and refused by the certification (HYG-PROJ, spans `global_parameters`, `certification.f90`, main); design section 9 decision 8 still names the projection as a live correction (dated record).

- **COST2** (measurement; the only production change is the T15 hook and print in `EXHALE_main.f90`, byte-identical on `wasp_full` 300 in both modes and `mol_base_handoff` 60 phys; attempted_step whole-binary rows now run, identity residual -1.8e-10, zero refusals): the brief's premise is refuted twice. (1) MEASURED with T frozen through the coupled loop on `wasp_full`: the composition sweep alone still takes 10-13 passes with the same contraction 0.15-0.20 and the same front walking one cell per pass; freezing the lagged coefficients as well collapses it to 2 passes; freezing only the radiation field leaves 3-4, only the rate coefficients 9-13. The pass count is the self-consistency of the photoionization field with the composition that attenuates it; the temperature is worth at most one pass. (2) A timer inside `ioniz_eq` (60 steps, 29.95 s): `PH_heat` 11.40 s (38 percent of the run), the whole cell loop 7.37 s (25 percent), pre-sweep `eval_cool` 2.44 s, post-sweep assembly 2.40 s; skipping the entire cell body for resting cells saves 4.7 percent. B3c-COST's `gprof` attribution (78 percent in MINPACK) was inverted; the `cs_max_pass` comment still carries it (to be rewritten by COST3). (3) The two-pass rest mask was implemented, measured (6.0 percent on `wasp_full`, 0 on `mol_base_handoff`, state moved 5.5e-7, step 0 of `mol_base_handoff` in phys turned into an exhausted attempt) and removed. **The lever: the field dependence is one-directional (a cell's P_HI is set by the attenuation of the cells outside it) and the sweep already runs outside in, so updating the column with the composition just produced turns the Jacobi iteration into a forward substitution: same fixed point, about one pass instead of ten, and PH_heat no longer repeated per pass.** Assigned as COST3 (`util_ion_eq.f90` `PH_heat_HHe/PH_heat_H`, `ionization_equilibrium.f90` sweep order; the OpenMP parallel cell loop becomes sequential in the field direction, so the design must weigh 10 parallel passes against 1-2 sequential ones and measure both at 1 and 16 threads). Also recorded: the pre-sweep `eval_cool` is run for its coefficients and its cooling discarded, then a second traversal forms the cooling (16 percent together); `mol_base_handoff` in phys refuses 7 of its first 60 steps at the fixed point on the unmodified tree.
- **B4-1c landed: the carriers ride on the species face flux** (2026-09-07): the carriers of `photochemical_transport_step` (H₂ always, OH/H₂O/CO under Oxygen chemistry, H⁺ under Ionization transport) join the transported set of B4-1 and are advected as `F_rho Y_c/m_c` inside the Runge-Kutta stages. `carrier_set_init` declares them once the keys are parsed (`advected_carrier_register`

in `binary_element_diffusion.f90`, which also holds the columns and hands them back through `advected_carrier_fractions`); `EXHALE_main.f90` is untouched, the existing three calls carry both sets. The carriers ride OUTSIDE the normalized set ($n_{\text{norm}} = 0$): a carrier is one species inside an element, the element is already the closing member, and normalizing it beside the element would count that mass twice; what keeps carrier and element consistent is the operator's own write-back, which restores the element totals cell by cell. Their rows are advanced from cell 1 (the elements' start at 2), with the inflow composition taken from the inner ghosts: the handoff's `q_H2_base` where one is stated, the base cell's own partition where none is, one rule for both branches of decision D6 (B4-4a). `diffusive_photochemistry.f90` loses the cell-velocity advective term, `carrier_advection_correction` and the base Dirichlet branch from its MARCHING rows and their Jacobian; two paths keep the advective term, `carrier_steady_residual` (the stationary balance is the whole equation) and the fixed-wind relaxation `relax_photochemical_composition` (it takes many transport steps with no hydrodynamic stage between them, so nothing else is taking that transport), through the switch `carrier_rows_advect`. Tests: two rows added to `species_face_flux`, `uniform_carrier_partition_preserved` 0.83 ulp (bound 8) on a sign-alternating face mass flux and `h2_nucleus_total_changes_only_by_boundary_flux` 1.36e-15 (bound 1e-12), RED before in the strongest sense (the driver does not compile against the pre-increment module); suite 15 + 2 GREEN; `carrier_retry` 101, `carrier_returned_state_acceptance` 37, `carrier_constraint_attribution` 11, `carrier_reference_scales` 15, `acceptance_classes` 22 GREEN with no reference re-derived. Impact MEASURED against a binary built from the same tree snapshot with the two files at their pre-increment text: 12 cases BYTE-IDENTICAL, every one without Molecular carrier transport (`wasp_full`, `wasp_he23off`, `wasp_full_newton`, `mol_base_handoff`, `mol_metals`, `mol_lyman_werner`, `mol_ir_bands`, `mol_sec_ion`, `mol_diffusion`, `lower_profile`, `oxygen_chemistry`, `hydrostatic_column`); the four that do transport carriers move. `mol_carrier` at 1000 steps: `Ion_species` 5.0e-1 (H3^+ at $r = 1.100$), `Cooling_breakdown` 6.2e-1, `Hydro_ioniz` 2.6e-2 (v at $r = 1.186$); `hp_zero_seed/hp_trace_seed/hp_front` at 100 steps: 8e-3 to 1.1e-1, `Hydro_ioniz` 1.3e-3 to 1.8e-3. **The front does not move; its tail does.** The $x_{\text{H2}} = 0.5$ crossing of `mol_carrier` is $r = 1.0698$ at 1000 steps before and after and $x_{\text{H2}}(1.05)$ is 0.7308 against 0.7310, while $x_{\text{H2}}(1.10)$ falls from 3.19e-3 to 2.02e-3 (-37%): the molecular tail the cell-velocity form was carrying by its own numerical diffusivity, which is where the design said the difference lives, and every large entry above is a consequence of that column. Every carrier interval covered and the element limiter moved no cell, in both runs. Recorded and not fixed: `oxygen_chemistry` has `carrier_transport` on by default with the oxygen chemistry, but at 200 steps in initialization mode 201 of 201 intervals are uncovered before AND after, so the write-back never runs and the case is byte-identical for that reason, not for an absence of effect (its rows did change: worst CO row of step 1 $|res|$ 6.03e-5 against 5.68e-5). No golden was refreshed. Also in this increment: the stale clause saying the carrier checkpoint restores "the cumulative CO record" removed from two comments in `diffusive_photochemistry.f90` (the `carrier_checkpoint` type carries no such field since CEILING-DEL).

- **B4-1c landed, advisor verification** (2026-09-07; the worker's own entry carries the numbers): every carrier of the photochemical transport operator (H2 ; $\text{OH}/\text{H2O}/\text{CO}$ under oxygen chemistry; H^+ under ionization transport) rides on $F_{\text{rho}} \ Y_{\text{c}} / m_{\text{c}}$ inside the RK stages through the registry of

binary_element_diffusion.f90 (carriers outside the normalized set, from cell 1, base inflow composition = the inner ghosts'), species_face_flux.f90 unchanged (species-agnostic), EXHALE_main.f90 untouched; the marching rows of diffusive_photochemistry.f90 keep diffusion, drift and chemistry only; the advective term stays in carrier_steady_residual (the whole stationary equation) and in the fixed-wind relaxation relax_photochemical_composition (no hydro stage takes it there), switch carrier_rows_advect. Advisor: private build, species_face_flux 15 + 2 rows (carrier partition 0.83 ulp, H2 nucleus budget 1.4e-15), carrier_retry, carrier_returned_state_acceptance, acceptance_classes, grid_and_gates green. Impact (worker, same-snapshot control): 12 cases without carrier transport BYTE-IDENTICAL; mol_carrier 1000 steps: the H2 front does not move ($x_{H2} = 0.5$ at $r = 1.0698$ both), the tail the cell-velocity form carried by its numerical diffusivity does ($x_{H2}(1.10) 3.19e-3 \rightarrow 2.02e-3$, -37 percent), hence H3+ 0.50, cooling 0.62, heating 0.37 relative there; $hp_* 100$ steps $\leq 1.7e-2$; base $\rho v r^2 \leq 9.7e-4$. Recorded: element_advection_* names are now too narrow (HYG-NAME, six call sites in main, after COST3); oxygen_chemistry in init mode covers 0 of 201 carrier intervals so it cannot judge carrier transport; a refused carrier interval discards a transport the mass row already took (B3a/B5 question); adv_corr still checkpointed though only the residual writes it; 52 documents under docs/ carry em-dashes (DOCS-DASH, launched).

- **DOCS-DASH** (documentation only, 2026-09-07): 1354 em-dashes and 237 en-dashes removed from 60 files (docs/*.md, READMEs, test and data READMEs), each replaced by the punctuation the sentence needs (colon, comma, parentheses, or a table placeholder hyphen), en-dash ranges and joined names to the hyphen the repository already writes; 108 mechanical substitutions corrected by hand on read-back; grep over the 187-file set clean. Left: Update_EXHALE_stage1.md (append-only log, 319/59), and 1220 LaTeX --- in 27 .tex sources that render as the same dash (DOCS-DASH2, launched: the .md/.tex twins realign there; the built PDFs are regenerated at the gate's docs pass), plus docs/lower_atmosphere_figs/README.md (11) and three READMEs under backup/ (5) outside the first sweep's paths.
- **DOCS-DASH2** (documentation only, 2026-09-07): 1283 LaTeX --- in 29 .tex sources (docs, paper, posters) and 30 spaced -- replaced by the punctuation each sentence needs; four exceptions kept and named (a direct quotation of Frelikh and Murray-Clay 2026 compared letter for letter, the AASTeX \keywords separator, a literal metals.inp listing, hyphen rules); the four READMEs of the first sweep's remainder cleared. The 24 built docs/*.pdf, literature_survey.pdf, paper/ms.pdf and the poster PDFs are now dated and are regenerated in the gate's docs pass. Left: the append-only stage logs (Update_EXHALE_stage1.tex 1614, stage0.tex 37).
- **COST3** (measurement; util_ion_eq.f90, ionization_equilibrium.f90; default output BYTE-IDENTICAL on wasp_full 300, mol_base_handoff 1500, mol_metals 1500, hydrostatic_column 300 at one thread; advisor rebuild: coupled_source_step 36, physics_probe 862/0 incl. the new photoionization_field_substitution driver (259 assertions), grid_and_gates 93/0): the star-ward columns can now be carried forward cell block by cell block (advance_starward_columns, reproducing calc_column_dens* bitwise; photoionization_field_at_cell_*, the PH_heat_* cell bodies extracted verbatim; xuv_field_block_cells, default 0 = whole grid). MEASURED on wasp_full 300: the substitution stops the front's walk (worst-moving cell 206 $\rightarrow$ 177 over the passes on the lagged scheme, 202 $\rightarrow$ 196 with

the columns carried) and cuts the pass count 10.10 -> 7.13, but a pass costs the same field work either way, and the field of a whole grid is one loop sixteen threads solve at once: block width 8 buys 1.43x at one thread (136 -> 103 s) and LOSES 2.8x at sixteen (17.0 -> 42.5 s); width 64 keeps the parallel front and saves nothing. The default therefore stays the whole grid, and the width is not tied to the thread count (the widths reach states differing by $1.5e-7$ at the $1e-8$ tolerance, the thread count by $7e-14$). **The remaining lag is the cell's OWN optical depth**, which enters its own field only on the next pass (contraction 0.17): closing it means the cell residual carries its own field, `System_HeH*`, i.e. B5. Folded: the pre-sweep `eval_cool` run for its coefficients only is now `chemical_rate_coefficients` (-3.6 percent `wasp_full`, -5.6 percent `mol_metals`, +3.5 percent `mol_base_handoff`, the last from the 27 metal slots looped with metals off: guard on `thereis_metals`, HYG-COL). Recorded: `docs/openmp_parallelization.md` (2.07x at 16 threads, bit-identical across threads) is stale: measured 8.2x and round-off $6.6e-14$ (DOCS-OMP). A `cp -irp` of another item blocked on an interactive prompt since 2026-09-06 was killed by PID (23818, cwd checked).

- **WENO3-REF** (comments only, `Reconstruction.f90` and the probe's References block; the user supplied the publisher PDFs, read with `pdftotext -layout`): the nonlinear weights are EXACTLY Yamaleev and Carpenter 2009, JCP 228, 3025, Eqs. (18), (20), (21), (22) (published τ already squared; $d_0 = 2/3$, $d_1 = 1/3$ entering as the ratio 1/2 that D1, D2 carry); the scheme is NOT their ESWENO: the energy-stability dissipation of Eq. (48) is absent, their analysis is a uniform-grid finite-difference flux reconstruction, and for systems only in characteristic variables (Section 5 quoted), so the code inherits the accuracy result (Eq. 24) and not the energy estimate; the geometry (C1, C2, D1, D2 on the stretched finite-volume grid) is this code's own. $\epsilon = dr_j^2$: a defensible stand-in, not a defect: the published h^2 scaling of Eqs. (62), (64) without the solution scale of Eq. (65); MEASURED on the production grid the face error is flat to every digit for $\kappa \geq 0.1$ over four decades (rate 2.991) and deteriorates below $1e-2$ per Eq. (57), so in smooth regions the departure is not measurable; where it acts is the ENO biasing (b_r/ϵ spans $2.2e-6$ to $8.4e3$ over the `wasp_full` grid; the published floor would make the base LESS nonlinear), a candidate experiment for the base sound-wave work, not a change here. Case label and comments now say "third-order WENO"; identifiers unchanged. Advisor: `paper/ms.tex` 144 "the energy-stable ESWENO3" reworded to the verified attribution and the ESWENO3 name replaced by WENO3 there and in `docs/code_comparison.tex`. **Recorded for the user:** every `src/tests/*/` suite of this series (19 directories) and the 2026-09-05/06 design and plan documents are untracked in git; a fresh clone of the remote carries none of them.
- **References received** (2026-09-07, from the user, publisher PDFs in `references/`): Zhang and Shu 2010 (JCP 229, 8918), Poinot and Lele 1992 (JCP 101, 104), Thompson 1987 (JCP 68, 1) and 1990 (JCP 89, 439), Neufeld 1990 (ApJ 350, 216), Mao and Kaastra 2016 (A&A 587, A84), Verner et al. 1996 (ApJ 465, 487), Yelle 2004 (Icarus 170, 167), Larsson et al. 2008 (CPL 462, 145), Tully 1975 (JCP 62, 1893), Slinger and Black 1982 (JCP 77, 2432). Five verification items launched against them: REF-ZHANGSHU (the positivity limiter), REF-LODI (the characteristic boundary design document), REF-NEUFELD (the Ly-alpha escape-probability closure), REF-METALS (every metal photoionization and recombination fit parameter against the published tables, a transcription error being a physics fix), REF-CHEM (the H3+ network transcriptions, the H3+ dissociative re-

combination coefficient, O6 and the Ly-alpha H2O yields). The series gate waits for them, since REF-METALS and REF-CHEM may move coefficients.

- Reference data fetched** (2026-09-07, at the user's direction, each with a provenance README, third-party material unchanged): references/verner_photo/ (D. A. Verner's distribution: phfit2.f reference routine with its fit parameters, table1.dat of Verner et al. 1996, the 1995 inner-shell fits), references/mao2016_cds/ (CDS J/A+A/587/A84 rr_spex3.dat, 39696 rows), references/badnell_rr/ and badnell_dr/ (Strathclyde TAMOC total RR and DR fit coefficients clist_K/clist_eV, tabulated cfout_*, reference routines). REF-METALS now compares every metal stage's photoionization, RR and DR coefficients against these by direct evaluation of the reference routines, and reports which source each stage uses today and any missing DR contribution.
- HYG-MAIN** (2026-09-07; numerically inert, wasp_full 300, mol_carrier 200, mol_diffusion 200, mol_base_handoff 300 byte-identical; advisor rebuild: certification, attempted_step, species_face_flux green): HYG-NAME element_advection_* -> species_advection_* in binary_element_diffusion.f90, the main program (use list, six calls) and the two test drivers, the two design documents that still named them (advisor); HYG-PROJ n_projection_applied deleted from global_parameters, the attempted_step_report block and both certification.f90 references (MEASURED: every reference in src/ was a read, no assignment anywhere, so the counter was provably zero since B3c), rep%n_unbudgeted_accepted_correction is now n_shapiro_applied alone, b3a design decision 8 dated closed for row 8; HYG-COL the 27-slot metal column loop of advance_starward_columns guarded on thereis_metals (inert; COST3's +3.5 percent NOT reproduced: guarded 53.7/52.8 s against unguarded 55.7/52.5 s on mol_base_handoff 300, within the run-to-run spread, kept as dead-work removal); DOCS-OMP docs/openmp_parallelization.md dated and brought to COST3's measured state (8.2x at 16 threads, 6.6e-14 between thread counts). Recorded: no reduction on a physical quantity exists in src/, so the 6.6e-14 thread-count difference comes from block boundaries set by omp_get_max_threads() in chemical_rate_coefficients/eval_cool; which routine turns a moved boundary into a moved last bit is not localized (THREAD-DET: until it is, the thread count is a silent input to every result at round-off); photoionization_field_at_cell_HHe makes an array temporary for nm_j on every call (runtime warning in physics_probe); EXHALE_main.f90:452 is 141 characters.
- REF-ZHANGSHU** (comments, docs/p49_front_energy_mode.md 6.1, probe positivity_limiter_scaling 15 assertions; user-supplied PDF read): the citation was wrong as stated: positivity_limited_faces is not the limiter of Zhang and Shu 2010 (JCP 229, 8918) but one built on the same idea, departing on three axes (primitive, not conserved, variables; one theta per face state, not per cell, Eqs. (2.4)-(2.5), (2.10)-(2.12); a relative one-ulp floor, not their absolute 1e-13). What carries over: Lemma 2.5 (convex combination with an admissible anchor); what does not: Theorem 2.1 (the next cell average admissible under their Gauss-Lobatto set and CFL, N = 2 covers k = 1 only; Remark 2.6 names finite-volume WENO as open) and conservativity of the reconstruction (a theta taken face by face moves the cell's end-point mean by 8.3e-2 where a theta taken cell by cell gives 0). Along their conserved path p(t) is concave, so the code's theta is

never larger than their t_{eps} (MEASURED 0.856 against 0.949 at Mach 60): the anticipated negative-internal-energy failure cannot occur. **Defect found and fixed (advisor applied the worker's proposal):** with eps one ulp of the average the bare update $q_{\text{avg}} + \theta (q_{\text{rec}} - q_{\text{avg}})$ is a cancellation of the size of the floor and lands on EXACTLY ZERO for 1013 of 200000 reconstructed values crossing zero (never negative), the value the floor exists to prevent (division in $v = m/\rho$ and in the sound speed); `positivity_limited_faces` now clamps the scaled density and pressure to eps after the update (same number in exact arithmetic). Moves only runs in which the limiter fired, by one ulp except where the zero would have divided; goldens at the gate. Probe: `bare_update_reaches_zero` (the reason), `production_limiter_never_delivers_zero` (the production routine over 4000 grids of zero-crossing face states) GREEN. A θ taken cell by cell (Eq. 2.11), which would restore the reconstruction average, is recorded as a change of scheme, not applied.

- **REF-LODI** (documentation, docs/phaseC_characteristic_base_bc.md +282/-50; the three user-supplied journal PDFs read as page images since their text layers lose the displayed equations): the wave-amplitude normalization (Poinot and Lele Eqs. (15)-(23); Thompson 1990 (44)-(47) with $l_1 = (0, 1, -\rho c, 0, 0)$; Thompson 1987 (19)), the LODI relations ((24)-(36), equal to Thompson 1990 (48) term for term), the subsonic inflow two-condition closure (P&L section 3.1, Tables III/IV; Thompson 1990 Eq. (68)) and all four branch-table rows CONFIRMED; no constant wrong, the constraint count unchanged. Two statements of the note corrected: pair (D) is NOT singular at stagnation (Thompson 1990 Eqs. (69)-(70) have denominator $u_1 + c$; the note's singularity was its own closure $\rho_b = F_c/(r_b^2 v_b)$; D1 reopened with a pointer), and D6's claim that no transparent condition is expressible as a fixed point is too strong (Thompson 1990 Eq. (64) $L_5 = \rho c g_1$ and P&L Eq. (40) $L_1 = K (p - p_{\text{inf}})$ are algebraic in the face state; D6 stays the user's decision with a third option). One gap stated: the source projection $l_1 C$ (gravity and the spherical divergence, Thompson 1990 footnote 1 and Eq. (52); the steady form is $L_1 = -\rho c g_1$, not 0) survives only because both states of (C-) sit at the same radius; any reformulation relating cell 1's centre to the face must restore it. Carlson's "but not U and p" marked unverified against the primaries.
- **REF-CHEM** (comments and documents only; `mol_rates.f90` and `oxygen_rates.f90` function bodies byte-identical; probes `h3p_recombination_branching` 12 and `water_photolysis_lyman_alpha_yields` 6, RED with the competing published values): every `mol_rates` constant sourced to Yelle 2004 matches his Table 1 digit for digit; Yelle's R7 ($\text{H}_3^+ + \text{H}$) reference column reads "Estimated", so Frelih's barrier-free $2.0\text{e-}9$ transcribes an estimate faithfully; the code's H_3^+ dissociative recombination R_6/R_7 ($2.16\text{e-}8 / 5.04\text{e-}8 \times (300/T)^{0.65}$) is already Larsson et al. 2008's 300 K total $7.2\text{e-}8$ split by their branching 0.70, with the 0.65 index inherited from the Sundstrom/Datz fits Yelle prints (Larsson gives no temperature dependence; he states the CRYRING/ASTRID values the older $2.9\text{e-}8$ rests on were too high from rotational excitation), so the code stays; the Slanger and Black 1982 Ly-alpha H_2O yields 0.78/0.10/0.12 are exact (p. 2435) with one label corrected, $\Phi_2 + \Phi_4 \rightarrow \Phi_1 + \Phi_2$, and the paper's coupling of the three yields and the 8 percent OH(A) bound recorded; Tully 1975's $2.87\text{e-}10$ is a 300 K statistical-model value (Table II) that VULCAN flattens over 100-2100 K, and its agreeing "experiment" is a 1973 relative-rate evaluation, strengthening the audit's verdict for O6. Documentation fixed in passing: `methodology_comparison.md` 6.4 listed three-body H_2 formation as identical in value while the code's

Cohen and Westberg 1983 $2.8e-31 T^{-0.6}$ differs from Ham et al. 1970 by 1.142 at every temperature; Baulch 1992 is in references/ (its $H_2 + H_2$ entry valid 2500-8000 K, above the whole layer); section 10 item on H_2 dissociative photoionization closed (the code carries it, `h2_photo_channels.f90` channel S).

- **REF-NEUFELD** (comments and documents; `lya_rt.f90` machine code identical before/after; `physics_probe` 1118/0): the Ly-alpha escape-probability closure does NOT use Neufeld 1990 or Harrington 1973. $\beta_{esc} = \pi^{(-1/4)} \sqrt{a/\tau}$ is the one-flight wing escape chance under complete redistribution; Ly-alpha scatters with partial redistribution and the published static-slab damping-wing solution is Harrington 1973 eq. (40), $\langle N \rangle = 0.9093161$ B scatterings for a mid-plane source, so $\beta = 1/\langle N \rangle$, proportional to $1/\tau$ and independent of a ; MEASURED on the stored `wasp_full` column the code exceeds it by 234x at the base and 14x at 1.20 R_p , crossing at 1.42 R_p . Neufeld contains no expression of the code's form (his escaping fractions are escape against destruction, eq. 2.25 partitions the two faces; the code's two-face PRODUCT is unpublished); the escape frequency scales as $(a \tau)^{(1/3)}$ (Harrington eqs. 37, 47; Neufeld eq. 4.32), not the code's $(a \tau)^{(1/2)}$; Neufeld's validity $(a \tau)^{(1/3)} > 10$ holds in 320 of 504 cells (inside 1.156 R_p); Neufeld is static, the Sobolev factor is Sobolev 1960 radial-ray. The header's `J_int` formula lacked the $(1 - \beta)$ factor the code applies (fixed). **Advisor decision: adopt the published solution (LYA-BETA, launched):** $\beta = 1/$ of eq. (40) joined to the thin limit, the face partition of eq. 2.25 in place of the product, the $(a \tau)^{(1/3)}$ escape frequency, a validity record, `A_2p1s` $6.3e8 \rightarrow 6.2648e8$ (spectroscopic), every case with `Jlya` escape-prob: True moves (expected at frozen state: `J_int/n_2p` up to 90x higher inside 1.02 R_p , 14x at 1.20 R_p , up to 7x lower beyond 1.47 R_p ; `Mdot` to be measured to the du stop). The same question for the metal lines' `line_escape_probability_one_face` (`Cool_coeff.f90`) is LYA-BETA-METALS, after REF-METALS.
- **REF-METALS** (`cross_sec.f90`, `Cool_coeff.f90`; probes `metal_photoionization_fits`, `recombination_coefficient_fits`, 255 assertions; advisor rebuild green apart from LYA-BETA's in-progress probe): all 132 fit parameters of the seventeen metal photoionization cross sections match Verner et al. 1996 Table 1 (`verner_photo/photo.dat`) digit for digit and `metal_photoion_sigma` reproduces Verner's own `phfit2` exactly wherever the fit is the one `phfit2` uses; 220 Badnell RR and DR parameters (fourteen daughters; `cfout_K` witness within 0.045 dex), the H/He rows and the Mao and Kaastra $n = 1$ rows match. **Two fixes:** the Mg II turn-on gated at Verner's rounded 15.04 eV while the grid edge is the NIST 15.035 (`e_th_MgII` now; byte-identical on `wasp_full`, the band geometric means miss the sliver); `alpha_rr_metal('CaI')` used Verner `rrfit.f`'s `rrec(:,20,19)`, the row that FORMS Ca II (second index = the recombined ion's electron count, fixed by `rnew(:,1,1) = H+ + e` and `rnew(:,2,2) = He+ + e`), 6.05x the Ca I row ($1.120e-13$, 0.900) at $1e4$ K; corrected. Impact (worker, scratch): `wasp_full` 300 Ca I 0.87 relative at its 1.196 R_p maximum, Ca II +2.8 percent at the base, T $4.3e-3$, v $9.9e-3$; `mol_metals` 400 Ca II/III x2.5 at 1.38 R_p , cool $2.9e-2$; `wasp_full`, `mol_metals`, `lower_profile` move at the gate through calcium. **Two physics items opened:** (a) the outer-shell fit is evaluated on the whole photon grid to 1240 eV while every row carries an `E_max` (34-558 eV) above which `phfit2` hands over to the 1995 subshell fit and adds the inner shells: the code carries 0.1-12 percent of the true metal photoabsorption there, and Fe I's row ($Q = 1.54$) RISES with energy to 49x the truth at 1240 eV; the handover of `phfit2` itself is adopted (METALS-OUTER, launched); (b) the inner-shell absorption

and its Auger cascade (an atom two or more stages up) cannot be represented by the three-stage metal ladder: To_be_determined_by_user_20260906.md item 7. Also: verner_photo/README.md corrected (photo.dat is the 1996 Table 1, table1.dat the 1995 fits); rev3 line 944 closed; e_th_MgI/e_th_MgII provenance lines in parameters.f90 to be annotated (METALS-OUTER, after LYA-BETA releases the file).

- METALS-OUTER** (cross_sec.f90; generator src/utls/metal_photoion_table.py from verner_photo/photo.dat + table1.dat; probes metal_photoionization_fits 188 and metal_photoion_table_transcription 15; docs/photoion_cross_sections.tex + PDF; advisor rebuild: physics_probe green): the seventeen sigma_XX routines are one parameter block, and metal_photoion_sigma makes phfit2's handover: the 1996 outer-shell fit below E_max, the 1995 fit of the same shell at and above it (MEASURED: einn of phfit2 equals photo.dat's E_max for all sixteen metal rows; the production routine reproduces phfit2's outer shell to 1e-10 at 17 ions x 12 energies where 110 of 204 rows disagreed before, worst Fe I at 1240 eV 24.9 Mb against 0.00172 Mb, a factor 14464). Inner shells still absent by decision item 7 (option (a) recommended; the measured sizes now in that item). Impact (scratch, metals-off mol_base_handoff byte-identical): wasp_full 300 v 0.15, T 4.7e-2, heat 7.8e-2, Fe III 0.99, C III 0.60 relative (hard photons now reach the base, the top stages fall at small radius; metal share of the heating at the base 17.4 -> 8.5 percent); mol_metals 400 v 0.56, heat 0.42; lower_profile 200 heat 8.3e-3; log10 Mdot unchanged to 0.01 dex in the bounded snapshots. Both Mg thresholds already agree with NIST to their digits (item 4 needed no move; provenance note pending parameters.f90).
- LYA-BETA-METALS** (measurement by the worker, fix applied by the advisor; Cool_coeff.f90, probe fine_structure_escape_probability, docs/resonance_line_trapping.md sections 6.1 and 10; publisher scans of de Jong, Boland and Dalgarno 1980 and Osterbrock 1962 fetched into references/): line_escape_probability_one_face is de Jong et al. 1980 eq. (B-7), both branches to round-off, applied to the eight ground-term fine-structure lines only; NO metal resonance line uses an escape probability (Mg II h&k, Ca II H&K, Na I D, Fe I/II at beta = 1, justified in that memo's sections 1-5 by $n_{crit, eff} = \beta A/q \sim 1e10-1e15$ against $n_e \leq 4.3e9$), and the transit post-process uses none. The fine-structure lines have a $\sim 1e-8$ and a $\tau \leq 1e-9$, so the Ly-alpha correction does not transfer. **Defect:** (B-7) is written in the frequency-integrated depth $\sqrt{\pi} \tau_0$ (normalized profile; equal to Hollenbach and McKee 1979 eq. (5.10) once the convention is matched, 3.4 percent at worst over $\tau = 3$ to $1e6$) while the caller passed the line-centre column: beta too large by 1 (thin) to 2.11 (branch point) to 1.77 (asymptotic). **Fixed:** each cell's depth is now $\sqrt{\pi}$ times the line-centre depth in fine_structure_line_transfer. MEASURED regime on the shipped columns: largest line-centre depth 0.14 ([O I] 63 um at the hot-Uranus base), so the factor there is 1.12 and the change of the total radiative losses is below 0.1 percent (mol_metals 0.096 percent), but a factor of two in any base thick in these lines and in the Base IR field incident term the same beta scales. physics_probe 1291/0 after the fix. Metals cases move at the gate at that level.
- Decision item 7, literature check** (2026-09-07; the user supplied Locci 2022 PSJ 3, 1; Locci 2024 PSJ 5, 58; Cecchi-Pestellini 2006 A&A 458, L13 and 2009 A&A 496, 863; Fox 2008 SSRv 139, 3; all in references/): the published exoplanet X-ray photochemistry models take the inner-shell absorption as opacity summed over

all ionization channels (Cecchi-Pestellini 2009 Eq. 1 with Verner 1993 cross sections), put the photoelectron and the Auger electrons into the secondary-electron cascade (Locci 2022 Eq. 3), and carry singly charged ions only, i.e. option (b) of the item with the charge under-count accepted; no escape hydrodynamics code includes inner shells. Recommendation revised from (a) to (b) in `To_be_determined_by_user_20260906.md`; the user decides.

- METALS-INNER** (decision 7 option (b), user 2026-09-07; `cross_sec.f90` subshell block from `phfit2.f`'s PH1 data (its thresholds differ from `table1.dat` in six rows, `phfit2` being the reference), `set_energy_vectors.f90` (every subshell turn-on of an active metal ion a bin edge; bin count unchanged, 17 -> 60 threshold cuts on `wasp_full`), `util_ion_eq.f90` comment only (the existing rate σ/e_v and heat `photoelectron_share(E_th)` sigma integrals already implement "one stage, one electron of $h\nu - E_{th}(\text{outer})$ to the SvS85/Dalgarno cascade" once the cross section is the total); generator `metal_photoion_table.py` --check; probes 392 metal assertions, `physics_probe` 1479/0; docs `photoion_cross_sections.tex` (+PDF), `code_status` new L row): the metal photoabsorption is the SHELL SUM `phfit2` returns, reproduced to 1e-10 at 17 ions x 12 energies (130 of 204 rows disagreed before). It also closed a cliff METALS-OUTER's handover had left at `E_max` where the 1996 fit stood for a valence complex the 1995 outer row does not cover (Fe I x0.012 and Fe II x0.0066 steps, now 1.23 and 1.13; genuine K edges C/N/O x19-21), and put `E_max`, a step of the cross section, on a bin edge (it was not). Approximation stated at the code site: ion advanced one stage, one electron of $h\nu - E_{th}(\text{outer})$ to the cascade (MEASURED with the degradation routine: 0.95-1.09x the heat and 1.10-1.37x the secondary H I ionizations of the true two-electron event; the un-spent `I_2` is 24-35 eV of a 300-550 eV photon), fluorescence neglected (with `e_top` = 1240 eV only C, N, O, Na have K holes; the high-Z ions contribute L shells only). Impact (worker, same-snapshot control; metals-off `mol_base_handoff` byte-identical): `wasp_full` 300 the wind 1-3 percent hotter and faster, base heating +26 percent, T at most +9.5 percent, Fe II/I/Mg III/N III at the base x74-160 from vanishing values, **Mg II falls 84 percent at $r = 1.178$** (the h&k transit line); `mol_metals` base heating +26 percent, T 1.4e-3; `lower_profile` heating +21 percent at the base, T 2e-5; metals' share of the 100-1240 eV absorption 2.2 -> 3.5 percent (`wasp_full`), the band 0.38 percent of the whole; `log10 Mdot` unchanged to 0.01 dex in the snapshots. Krause 1979 (JPCRD 8, 307) K/L fluorescence yields could not be obtained (NIST 503, AIP 403): requested from the user; no yield is quoted anywhere. Advisor verification: `physics_probe` 1479/0, `grid_and_gates` 93/0; `spectrum_type` segfaulted in `loaded_sed_threshold_edges` (its threshold array was sized `n_mion + 16 = 43` while the active list now reaches 126), fixed by sizing it with the module's `n_thr_max`; suite green after. (`n_thr_max` was a local parameter of `set_energy_vectors`; hoisted to module scope, public, so every caller of `ionization_thresholds_active` sizes its array by the one bound; `spectrum_type` 154/0, `physics_probe` 1479/0, `grid_and_gates` 93/0 after.)
- METALS-INNER addendum, Krause 1979 yields** (comments and documents only; user-supplied `Krause_1979_JPCRD_8_307.pdf`, read from page images where the OCR is noisy, alignment checked on $\omega_K(\text{Fe}) = 0.340$): Table 3 ω_K for C/N/O/Na 2.8e-3/5.2e-3/8.3e-3/0.023, Mg/Si/S/K/Ca/Fe 0.030/0.050/0.078/0.140/0.163/0.340; Table 5 effective L1/L2 yields (Coster-Kronig included per the text, $\nu_3 = \omega_3$) for Mg to Fe all below 6.5e-3; Table 2 uncertainties (ω_K 40-10 percent at Z 5-10, footnote

(a) on molecules and solids). With $e_{\text{top}} = 1240$ eV only C, N, O and Na I can hold a K hole, so the neglected radiative branch is at most 2.3 percent of a K-hole event (under 1 percent for the three coolants) and under 0.7 percent of any L-shell event; raising e_{top} past 1.3 keV (Mg, Si K) or 4 keV (Ca, Fe K) is the condition under which the approximation must be revisited, stated at the code site, in `photoion_cross_sections.tex` (new tab:kedge), `code_status` and decision item 7.

- LYA-BETA landed, advisor verification** (`lya_rt.f90`, `hydrogen_n2_rates.f90` ($A_{2p1s} 6.3e8 \rightarrow 6.2649e8$, the nonrelativistic dipole value times the reduced-mass factor, = NIST ASD's Wiese and Fuhr 2009 value), `parameters.f90` comments; probe `lya_escape_probability` 24 assertions; advisor rebuild `physics_probe` 1479/0, `certification` 29/0): the escape probability is the published static-slab solution, $\langle N \rangle =$ Neufeld 1990 eq. (3.27) at zero destruction = Harrington 1973 eq. (40) ($4 \sqrt{6}/\pi^2$) $u_2 B = 0.909316 B$ for a mid-plane source (the 0.9093161 of the REF-NEUFELD entry was a transcription slip), $\beta = 1/(1 + \langle N \rangle)$ (exact at both limits: a generated photon is emitted $1 +$ times, and the budget $n_{2p} A \beta = P$ wants the escape chance per emission), the two faces sharing one escaping population by Neufeld eq. (2.25) instead of the former product, $\Phi(\xi)$ summed in closed form; the Voigt parameter has left the escape probability. a still enters only `lya_wing_penetration_coefficient` (where an incident stellar photon crosses in one flight, a different question, judged at the code site). Defect found in passing: $\beta_{\text{tot}} = 1 - (1 - \beta_{\text{esc}})(1 - \beta_{\text{sob}})$ lost digits at the base ($4.8e-9$ relative), now $\beta_{\text{esc}} + \beta_{\text{sob}} (1 - \beta_{\text{esc}})$ (closure $2.2e-16$). RED before $4.79e2$ / GREEN $2.19e-16$ on the end-to-end identity. **Impact, wasp_full to its du stop (2 h 59 m): log10 Mdot 13.34 $\rightarrow$ 13.34 (+1.6 percent, 0.0067 dex); Jint/n_2p x84 at the base, x73 at 1.05, x37 at 1.10, x10 at 1.20, x0.75 at 1.50 R_p ; T within 2 percent; the old `min(1, .)` cap had zeroed the internal field from 1.15 R_p outward. $H(n=2)$ chord column x31.5 at $b = 1.00$, x2.8 at 1.20, x1.005 at 1.45: every H-alpha/H-beta transit number in this repository from a Jlya escape-prob: True run is superseded (He 10830 not affected). Validity $(a \tau)^{1/3} > 10$ holds in 349 of 504 cells, out to 1.193 R_p , exactly where the change is an order of magnitude. Advisor follow-ups applied: the second A_{2p1s} in `exhale_transit_lib.py` and `docs/transmission_spectrum.tex` set to 6.2649e8 (one value with the wind model), the stale $E = \min(\text{boost}, 1/\beta)$ comment in `parameters.f90` replaced by the implemented $1 + (\text{boost} - 1)(1 - \beta) T_{\text{star}}$. Pending: the B6-style print of the Ly-alpha domain record in `certification.f90` (the record and accessor exist; LYA-DOM, with the gate's follow-ups).**
- LYA-DOM** (advisor; `certification.f90`): the Ly-alpha damping-wing domain record (`lya_wing_domain_record`: cell visits below the $(a \tau)^{1/3}$ limit, visits seen, the smallest value, the limit) joins the out-of-domain closure count and is printed beside the H3+ and CO records, informational like them. `certification`, `run_mode`, `attempted_step` green. **Every worker item of the series has landed; the series gate begins (single build, make test, the 16-case matrix, movement table, one golden refresh, fcheck, docs pass).**
- Series gate, 2026-09-07** (PLAN rev2 steps A to C and the items above, one series): single build of the shared tree (make -q up to date, EXHALE.x md5 ba3d34e6eace); make test green apart from the two steady suites, which then passed on the wasp_full_newton run log (EXHALE_STEADY_RUNLOGS); the 16 default cases run against

that binary (the harness holds a single-instance lock, so `wasp_full` ran through `run_check.sh` check and the other fifteen through a runner replicating its steps for each case, all `OMP_NUM_THREADS=1`); every case moves against the previous goldens, as every item of the series said it would. Movement of the physical columns over the cells where the golden value is at least $1e-3$ of the column maximum (max relative; T median; the change of $\log_{10} \dot{M}$ from the mean of $r^2 \rho v$ over 1.20-1.45 R_p):

case	ρ max	v max	p max	T max	T median	$d\log_{10} \dot{M}$	$\log_{10} \dot{M}$ new
<code>wasp_full</code>	2.97e-01	1.28e+00	1.65e-01	9.18e-02	3.8e-02	+0.0095	13.35
<code>wasp_he23off</code>	2.95e-01	1.30e+00	1.62e-01	9.18e-02	3.9e-02	+0.0084	13.35
<code>wasp_full_newton</code>	6.96e-01	3.51e-01	4.30e-01	2.35e-01	8.7e-02	+0.0414	13.30
<code>mol_base_handoff</code>	1.60e-01	2.50e-01	1.52e-01	2.33e-01	2.2e-02	-0.0440	10.57
<code>mol_metals</code>	1.51e-01	2.25e-01	1.53e-01	2.07e-01	2.3e-02	-0.0476	10.58
<code>mol_lyman_werner</code>	2.12e-01	4.87e-01	1.99e-01	2.55e-01	2.6e-02	-0.0011	10.54
<code>mol_diffusion</code>	1.59e-01	2.50e-01	1.52e-01	2.33e-01	2.2e-02	-0.0440	10.57
<code>mol_ir_bands</code>	1.51e-01	2.26e-01	1.53e-01	2.07e-01	2.3e-02	-0.0475	10.58
<code>mol_sec_ion</code>	2.16e-01	4.03e-01	1.96e-01	2.53e-01	3.0e-02	+0.0149	10.51
<code>mol_carrier</code>	1.54e-01	6.37e-01	1.44e-01	2.24e-01	1.9e-02	+0.0106	10.56
<code>lower_profile</code>	9.40e-03	3.22e+01	7.50e-03	9.77e-01	8.8e-03	-0.0335	7.29
<code>hydrostatic_column</code>	2.09e-12	2.81e-03	3.54e-12	1.40e-03	2.5e-14	-0.0000	8.27
<code>oxygen_chemistry</code>	8.62e-01	1.29e+02	9.76e-01	7.61e-01	3.7e-01	+nan	9.61
<code>hp_zero_seed</code>	2.25e-05	1.81e-04	1.61e-04	3.20e-03	4.4e-05	+0.0000	10.56
<code>hp_trace_seed</code>	2.24e-05	1.81e-04	1.61e-04	3.20e-03	4.4e-05	+0.0000	10.56
<code>hp_front</code>	2.07e-05	5.77e-04	3.66e-04	1.00e-02	6.7e-05	+0.0000	10.56

The maxima sit at the fronts, which move by a cell; the T medians (2-9 percent on the wind cases) and the $\dot{M}$ changes (within ± 0.05 dex everywhere it is defined; `oxygen_chemistry` has a reversed flow in the window) are the state of the physics after B3c, B4-1/1c, LYA-BETA, the WENO3 coefficient fix, the metals (REF-METALS, METALS-OUTER/INNER), the LW band, the CO channels, ACCEPT-1/2 and the rest. `wasp_he23off` exits 2 (its stationary claim refused by four inventory entries, as A2tol2 measured at the default tolerance; outputs written in full); `wasp_full_newton` exits 0, certified, $\log_{10} \dot{M}$ 13.30. PDFs of the 25 dated documents, the paper and both posters rebuilt. **Golden refresh pending the user's hand:** the archive copy golden $\rightarrow$ `golden_pre_series_20260907` and `run_check.sh` golden were refused by the session's permission classifier; the commands are in the report to the user. The final confirmation (make check against the refreshed goldens, serial) follows the refresh. **Golden refresh done by the user (2026-09-07 13:19):** golden/archived as `golden_pre_series_20260907/` (17 cases) and re-snapshotted from the case outputs of this gate (checked: golden equals output on `wasp_full`, `mol_metals`, `oxygen_chemistry`, `lower_profile`). The serial make check against the refreshed goldens runs next as the bitwise confirmation, then `run_fcheck.sh`.

- **User manual, environment-variable table** (user report, 2026-09-07): the fixed table float of the environment hooks (67 rows) overflowed page 11 into the footer; it is now a longtable with the caption at its head, breaking across pages 10-11 with a "continued" foot. PDF rebuilt. Pages 20-21 (user report): the keyword table's long file paths were unbreakable `\texttt` tokens and ran past the right margin; 18 such paths are now `\path{}` (url package, breaks at / and _), one inside a caption kept as `\texttt` (a moving argument). Rebuilt with latexmk, so the longtable column widths and cross-references are resolved (a single pdf_{latex} pass leaves both unresolved, which is what produced the "Table ?" and the first overflow the user saw).

- **Update log TeX twin** (user request, 2026-09-07): docs/Update_EXHALE.tex and its PDF now exist, GENERATED from this Markdown by src/utils/update_log_to_tex.py (pandoc body, the memo-class preamble of the stage logs); regenerate after appending here. 56 pages at this entry.
- **Decision item 4 closed** (user, 2026-09-07): candidate C, wasp121b_composite_huang2023.txt (solar XUV shape below 1700 Å rescaled to Huang et al. 2023's F_XUV and Ly-alpha, the MUSCLES WASP-17 photosphere above), is the WASP-121 b spectrum of record, as both input.inp files already state; A and B remain sensitivity inputs. No decision item of To_be_determined_by_user_20260906.md is open. **make check against the refreshed goldens: REGRESSION PASS, all 16 cases byte-identical (64 files), 13:21-16:25 KST 2026-09-07** (the harness runs the cases eight at a time, one thread each). run_fcheck.sh follows. **run_fcheck.sh: CLEAN** (1500 bounded steps of the HD 209458 b folder case under -fcheck=bounds,do,mem,log10 Mdot 9.56) after one harness fix: the scratch copy of the planet input named its spectrum by the folder's relative path, which the script now rewrites to the repository's absolute path for the Spectrum file and Lower atmosphere profile keys (the first run failed on "Cannot open file ../inputdata/sed/...", not on a runtime check). Production EXHALE.x restored, make -q up to date, md5 ba3d34e6eae. **The series is closed:** goldens refreshed once, make check byte-identical, fcheck clean, docs and PDFs rebuilt. What follows is in docs/session_handoff_20260907.md.
- **HYG-PY** (advisor, 2026-09-07; python/paper_data.py, python/make_struct_figures.py): HEAT_FIXED (the headerless-file fallback) and HEAT_CH_LABEL/HEAT_CH_ORDER (the heating-budget panel) now carry all 19 deposit channels of heat_channel_name with the descriptive brackets stripped the way _clean_name strips them (heat_He23S_assoc, the two Lyman-Werner halves, heat_mol_chem, heat_FUV_photolysis, heat_oxygen_collisional, the two CO channels); the panel loop already skips a channel a run's file lacks. Both scripts parse; the loader reads the file header, so the fallback list was never what a real file was read by.
- **THREAD-DET** (advisor, 2026-09-07; util_ion_eq.f90): the thread count is no longer an input to any result. Mechanism, MEASURED: Cool_coeff.o imports glibc libmvec's two-lane _ZGVbN2v_exp, _ZGVbN2v_log, _ZGVbN2vv_pow (gfortran -O3 vectorizes the rate-coefficient loops), and the vector variants agree with the scalar libm to one ulp, not bitwise; inside a block of cells the loop takes lane pairs from j_lo and an odd remainder by the scalar call, so where a block STARTS decides which cell meets which variant. chemical_rate_coefficients and eval_cool cut the grid into 2*omp_get_max_threads() blocks, so the block boundaries, and with them the last bit of every rate, moved with the thread count (the 6.6e-14 of COST3/HYG-MAIN). Fixed: both sweeps use blocks of the fixed even length xuv_rate_block = 32 (16 blocks on the 504-cell grids); the threads decide only who takes which block. MEASURED on wasp_full 300: old binary at 1 thread, new at 1 thread and new at 16 threads are byte-identical in Hydro_ioniz, Ion_species, Heating_breakdown, Cooling_breakdown (the old single-thread partition happened to start both its blocks at even offsets, so the refreshed goldens are untouched). Shared EXHALE.x rebuilt from the tree (make -q up to date).
- **DOCS-LINES** (documentation only, 2026-09-07): every file.f90:NNN citation in the nine design and status documents (d0 922 tokens on 309 lines, b1a 127, b1 64, b3b 24, a2 7, a0 5, b3a 5, b4 2) re-anchored to file

+ routine, declaration or labelled comment block, read against the live tree; the step tables of d0 6.1 and b1 7.1 index by the `as_op_*` operation tag of `attempted_step.f90`; the convention is stated in each document. Citations of removed code (the composition projection, the two-iteration energy update, the CO ceiling, the cell-velocity carrier advection, the element advective term, DB96 eq. (39), the H3+ clamp, the second H2 partition function) kept as dated records naming the removing item. Stale values corrected in passing: `e_lw_photon_erg` 1.88021e-11 erg (11.7354 eV) over 912-1201 Å, `binary_element_diffusion.f90` 3037 lines, a2 section 1's chemical count now passed by the marching caller. All 190 names introduced grepped present in `src/`. Recorded: pre-existing markdown table pipe misalignments (unescaped | in math) in a2, b4, `code_status`, d0; two markdown line citations of `grid_and_gates/README.md` left.

- B5, carrier arm** (`steady_newton.f90`, `diffusive_photochemistry.f90`, `input_read.f90`; new suite `steady_species_rows` 41 + 8 rows; advisor rebuild: it and certification, `run_mode`, `carrier_retry`, `coupled_source_step`, `species_face_flux` green): the Newton unknown vector carries one row and one unknown per solved carrier (H2, OH, H2O, CO, H+) through a row registry in place of the H2-only `nvar = 4` special case; band geometry derived, `kl = ku = 3 nvar - 1`, `ncolor = 6 nvar - 1` (reproduces the 8/17 and 11/23 the code carried); the completion flag reads the species rows of the system it solved; EXHALE_SPECIES_JAC_TEST asserts `Jr` inside the solve (the run-level test builds its Jacobian before the registry exists). Ionization transport + Solver: Newton supported by derivation when the proton is a Newton unknown (`x_hp_fix` handed to the sweep), refused with the reason otherwise; MEASURED `nvar = 5`, Jacobian action 7.2e-7 (all rows), 4.6e-5 (species rows), 0 unresolved colors. Two pre-existing defects fixed: `frozen_residual` left the species slots of an `intent(out)` array unwritten (directional check relative error 1.0 from stale memory) and `n_headroom_iterate` was never assigned (the trial-admissibility threshold stood at 0; with the four oxygen balances as rows all 41 colors came back unresolved). `wasp_full_newton` stays certified (13.30, byte-identical); H2-only control byte-identical; marching path byte-identical (`wasp_full`, `mol_base_handoff` 300). **Findings:** no regression case enters the steady solver, so every golden is a marching snapshot and `cert_tol_carrier/cert_tol_element` still have no anchor (contract section 11 records two rows from solves that did not meet their target, 7.46e-3 and 5.82e-3, not anchors). **Open, and a physical-consistency defect:** the stationary carrier row still uses the cell-velocity advective form while the marching rows ride on $F_{\rho} Y_s^{\text{face}} / m_s$ since B4-1c (two operators, two fixed points): B5b. Also open: the element rows (`element_transport_residual` must return a residual per element, not the worst at each cell), the cell's own field in the cell residual (interface written: FIELD-SELF), reload consistency, three EXHALE_main.f90 sites (the registry name, EXHALE_JAC_TEST coverage, the PTC refusal text).
- B5b** (`species_face_flux.f90`, `steady_residual.f90`, `steady_newton.f90`, `certification.f90`, `binary_element_diffusion.f90`, `diffusive_photochemistry.f90`, three sites of EXHALE_main.f90; suite `steady_species_rows` 67 rows; advisor rebuild: it and `species_face_flux`, certification, `carrier_retry`, `coupled_source_step`, `run_mode`, `acceptance_classes` green): ONE discretization of the material advective term exists in the code: the RK stages, the stationary carrier rows, the fixed-wind carrier relaxation and the stationary element rows all form it as the divergence of the same face species fluxes (`species_face_fraction`,

species_face_flux, new species_flux_divergence) on the mass row's faces, areas and volumes; the cell-velocity upwind form, its deferred van Leer correction (carrier_advection_correction, adv_corr, carrier_slope), the base Dirichlet record it needed and the advective coefficient ntv are gone from diffusive_photochemistry.f90 (RED 76.7 relative against a stage-1 update on a sign-changing face flux, GREEN 7.5e-14). element_transport_residual returns one residual per element (the certification prints one entry per element: lower_profile C 0.977, O 0.921, N 0.968, seven elements "not applicable"); rows srow_element_he and one srow_element_trace per metal element registered before the carriers, project_element_mass_fractions the write-back. mol_diffusion reloaded with Coupled carrier solve: nvar = 4, one element row, Jacobian action $4.9\text{e-}3 / 1.3\text{e-}3$, 0 unresolved colors, the solve refused at the He/H row (0.9987) with no descent direction: the element system is expressible and its Jacobian right, not yet solved from that start (the element unknown moves the whole column's opacity, which the banded model does not carry; the "all rows" bound of the suite widened $1\text{e-}4 \rightarrow 1\text{e-}2$ with the measurement at the site). Reload consistency asserted on the binary. EXHALE_main.f90: set_transported_species_rows (not carrier-specific), the two EXHALE_JAC_TEST inserts, the PTC text; at that site two pre-existing defects fixed: the species slots of Yvec were read uninitialized ($2.4\text{e}2$ against 1.85) and the probe direction was unscaled ($3.3\text{e}1$ at nvar = 4); now $3.2\text{e-}6 / 4.5\text{e-}6$. **By construction the stationary species rows now inherit the mass row's imbalance** ($\text{div}(\mathbf{F}_{\text{rho}} \mathbf{Y}) = \mathbf{Y} \text{ div } \mathbf{F}_{\text{rho}} + \mathbf{F}_{\text{rho}} \text{ grad } \mathbf{Y}$; the old form was the second term alone), so on marching snapshots whose mass rows read 1.1-1.9 of their terms the rows read $\mathcal{O}(1)$ where they read $4\text{e-}4$ to 0.4 before. Marching path byte-identical on wasp_full, mol_base_handoff, mol_carrier, mol_diffusion, lower_profile at 300 steps; wasp_full_newton reloaded certifies (13.30, $2.6\text{e-}9$ of the stored state). Anchors still UNCONFIRMED (contract 11.3). Recorded: element_diffusion_step / relax_element_composition still advect on the smoothed steady mass flux, so the elemental Picard alternation and the elemental row are two operators (B5c); carrier_rows_advect kept as the on/off switch of one operator; Coupled carrier solve now also governs the element rows (key not renamed); ionization_equilibrium.f90 cites carrier_base_state, which no longer exists (to FIELD-SELF). THREAD-DET confirmed by the official check: make check on the rebuilt binary, REGRESSION PASS, every case byte-identical (16:58-20:04 KST 2026-09-07; the binary of that check also carried whatever of B5 had landed by 16:58, and B5's marching path was measured byte-identical anyway). A further make check follows FIELD-SELF and B5c.

- **B5c** (binary_element_diffusion.f90; suite steady_species_rows 60/0; advisor rebuild: it and species_face_flux, certification, carrier_retry green): the elemental relaxation and the stationary elemental row are one operator. element_diffusion_step takes the FACE MASS FLUX and forms its advective term as the divergence of the face element fluxes through the three shared routines (element_advective_divergence, one place); relax_element_composition reads that flux from the mass row of the state (face_mass_flux_of_state) and no longer smooths a steady $\text{mdot}/(4 \pi r^2)$; element_transport_residual takes both halves of the helium row from composition_residual and the trace rows from the new trace_composition_residual, the routines the step itself solves, so the fixed point is the row's zero by construction; solve_trace_element_in_hydrogen is a deferred-correction loop that takes one pass, term for term the old direct solve, when the advection is off (the marching path). MEASURED on a

synthetic column relaxed to its own convergence: the row reads 8.65e-13 (He alone), 8.45e-13 / 5.38e-13 (He + ten trace elements) of its terms, the advection carrying 0.993 of them; the replaced form leaves 6.4e-3 on the same state. The smoothed-flux rationale does not survive: it repaired a form (the element equation minus X times the mass equation) that is gone, and its consequence is now a measured property of the equation: on a face mass flux that does not conserve mass the conservative balance has no bounded steady composition (X driven onto 0 and 1). Marching path byte-identical (mol_diffusion, lower_profile 300 steps); the direct-steady reload of mol_diffusion byte-identical, and its outer pass never reaches the relaxation (the solve ends info = 2, B5b 6.3), so the run-level RED/GREEN does not exist. **Found: diffusion_tests.x has not linked since B5b** (binary_element_diffusion now uses steady_residual_mod, whose chain the test's DIFT_SRC does not carry): a functions/ module depending on time_step/steady_residual is a layering inversion; the flux goes by argument (DIFT-LINK, launched), and make test's diffusion_tests row is red until it lands. Also recorded: the relaxation's convergence measure is helium-only (DIFT-LINK folds the trace drift in); the vanished-element write-back branch leaves a trace row at 1e-4 to 2e-2 where an element is driven to exactly zero (deliberate guard, measured, documented at the test site); element_transport_residual's dummy v is unread.

- **FIELD-SELF** (ionization_equilibrium.f90 self-field loop, default one pass; util_ion_eq.f90 and System_HeH* comments; probe photoionization_field_self_consistency 12; advisor rebuild: coupled_source_step 36/0, physics_probe 1491/0, certification, acceptance_classes green): the cell's own optical depth CAN now be closed inside the cell solve (one pass leaves a 4.06e-2 self-consistency defect in the ionized fraction of a cell one optical depth thick, the loop closes it to 2.8e-15 in 13 passes, contraction 0.080; the field routines already formed the B4-6 cell mean from the cell's own densities, so no field code changed), **and doing so buys nothing: COST3's attribution is refuted.** MEASURED on wasp_full 300: coupled passes 10.09 -> 10.01 for 1.65x the wall time (2 self-field passes) and 1.87x (8); mol_base_handoff 1500: 7.19 -> 7.24 for 2.0x; the decisive binary with BOTH levers (carried columns at block 8 AND four self-field passes): 7.20 passes, 190 s, against COST3's 7.13 / 103 s for the columns alone. (**Superseded by COST4 below:** with both stellar lags closed at the settings that close them the count is 8.18, not seven, and the two couplings named here are worth 0.08 of it; the count is the slowest rung of a ladder of modes.) Same fixed point: state movement $\leq 1.4\text{e-}7$, at the $1\text{e-}8$ tolerance's level. Default output byte-identical on wasp_full 300, mol_base_handoff 1500, mol_metals 1500, hydrostatic_column; wasp_full_newton uncapped CERTIFIED, both files byte-identical to its golden (that binary carried B5b too). Kept in the tree at its inert default on COST3's precedent (the switch is the record of a measured negative result). Comment fixes in passing: System_HeH_metals.f90 said the sweep is serial (untrue since the parallelization; its arrays are threadprivate); the carrier_base_state citation of ionization_equilibrium.f90 rewritten to carrier_base_composition_imposed. Recorded: he_ground_singlet_density's clamp counter needs a lock when the self-field loop runs (a slot per cell would be free, composition.f90); f_vib_quench/e_vib_bound stay at the entry composition inside the loop; the COST3 attribution in that report, its log entry above and the csm_max_pass comment of EXHALE_main.f90 are superseded by this measurement (comment to be corrected when DIFT-LINK releases main).

- DIFT-LINK** (binary_element_diffusion.f90 and its three call sites in EXHALE_main.f90, steady_newton.f90, certification.f90; suite steady_species_rows 62 rows; advisor rebuild: it and species_face_flux, certification, coupled_source_step green): the element module no longer uses steady_residual_mod; the face mass flux is a required argument of relax_element_composition and element_transport_residual, obtained by each caller with face_mass_flux_of_state. The layering inversion is gone and **make diffusion_tests links again with the Makefile and DIFT_SRC untouched: 35 passed, 0 failed** (MEASURED by the advisor on the live tree; it failed at the link since B5b). The relaxation's convergence measure now runs over every element it relaxed, each against its own reservoir (element_composition_distance): RED 5.99 / GREEN 0.0 on a column where Fe moves 1.78 while helium moves 0.25, so the helium-only measure returned a number 7x too small to bound the element that moved. Byte-identical on mol_diffusion, lower_profile 300 steps and the mol_diffusion direct-steady reload. Dead code removed (helium_mass_fraction, an unread element_mass_fractions call). **Advisor fixes on its findings:** the limiter of species_face_fraction divided by $Y_{av} - Y_f(j)$, which vanishes when the donor cell's average is itself negative (a trace species handed back at a small negative density; MEASURED as a SIGFPE under -ffpe-trap=zero): the anchor is now the average clamped into [0,1], which makes both denominators strictly positive and is what the comment already claimed; mol_diffusion 300 steps byte-identical, since at production flags the infinity was clamped to the same 0. The csm_max_pass comment of EXHALE_main.f90 and docs/steady_solver_design.md section 15 corrected to the FIELD-SELF and B5c measurements. Recorded: diffusion_tests T9 reads 1.098e-12 against its 1e-12 bound at -O1 -fcheck (an optimization-level rounding sensitivity of relative_settling_mass, 35/0 at -O3); the outer loop's under-relaxation damps helium only while the drift it tests can now be set by a trace element.
- Batch gate after B5/B5b/B5c/FIELD-SELF/DIFT-LINK/THREAD-DET** (2026-09-07/08): shared tree rebuilt (EXHALE.x md5 eba2d7f2260f, make -q up to date); make test green, the two steady suites passing on the wasp_full_newton run log and the diffusion_tests row restored; make check **REGRESSION PASS, all 16 cases byte-identical** (20:54 to 00:05 KST). Every item of this batch was designed to leave the marching path alone and the check confirms it: the goldens refreshed at the 2026-09-07 series gate still stand, and no golden was refreshed here.
- HYG-T9** (advisor, 2026-09-08; src/tests/diffusion_tests.f90): T9's ambipolar-limit bound sat exactly on the arithmetic's noise floor, so the suite passed at -O3 and failed at -O1 with checks (DIFT-LINK's finding). MEASURED at both levels: the neutral limit is exact (no electron term), the H+ plasma reads 8.207e-13 (-O3) and 7.740e-13 (-O1), the He++ plasma 7.083e-13 and 1.098e-12. Mechanism, written at the site: the limits are exact algebra in the masses, so the deviation is arithmetic, and the ambipolar term of the plasma limits comes from a centred difference of $\ln \rho$ over dr whose logarithm is $O(b_0/r)$, so each ulp reaches dm_{eff} amplified by $1/dr$. The bound is now one decade above the largest measured (1e-11), the anchoring rule the certification tolerances use, and the test still catches a wrong ambipolar limit by ten decades (a wrong limit misses its target by 0.1 to 1) with the neutral limit still asserted exact. 35/0 at -O3 and at -O1 -fcheck.
- B5d** (steady_newton.f90 only; suite steady_species_rows 77 rows; advisor rebuild: it and certification,

species_face_flux, coupled_source_step, carrier_retry green): the item's own brief expected the element rows to fail because an element unknown moves the whole column's opacity, too non-local for the banded model. **MEASURED and REFUTED:** the scaled Jacobian column of an element unknown carries at most 6.3 percent of its 2-norm outside the band, the same order as a hydrodynamic column (mass 0.0018, momentum and energy 0.0000), so the diagonal correction, the rank-one column-integral update and the identity block the brief offered would each have repaired a defect that is not there; none was written. What does break the model, located by the same measurements and fixed: (1) the base element row was structurally EMPTY, cell 1 being the Dirichlet reservoir of the element operator, so the banded model was singular by construction (its LU pivot grew to 8.413e19 against a largest band entry of 5.229e3; now 1.433e6) and it now carries the boundary condition the operator states; (2) the element ROW SCALE collapses on the abandoned iterate (helium rows spanning 1.277e-29 to 2.360, so an ordinary $dF/dY = 46.7$ became a scaled band entry of 4.890e22 and $||\text{grad merit}|| = 1.104e44$), and the row scale is now floored at the flow-time rate of the quantity the row transports, the hydrodynamic convention with no chosen constant. Result: the same case reaches $||R|| = 1.723$ against 1.974 at outer iteration 18 instead of 8 with $||\text{grad merit}|| = 4.813e3$, 40 decades lower and a number the model's own entries can produce. **The element solve still does not converge and the item does not claim it does;** cert_tol_element and cert_tol_carrier are unmoved and UNCONFIRMED. New diagnostics: banded_model_conditioning (empty rows and columns stated once per registry, pivots and scales under EXHALE_BAND_REPORT), jacobian_column_reach_beyond_the_band (EXHALE_JAC_COLUMN_REACH). mol_diffusion 300 steps byte-identical; wasp_full_newton byte-identical, info = 0, CERTIFIED, 13.30, same 6 outer iterations. **Named as what is left: the globalization** (the non-monotone search accepts steps that raise $||R||$ from 1.7 to 25; the trust-region route gives up at radius 3.8e-16 after 12 ray-test refusals while the merit is still falling): item B5e. Also recorded: carrier_row_term_scale has the same collapsing structure and was not measured (no carrier row in that case); the trace arm of the floor is covered by inspection only.

- **COST4** (measurement; NOTHING in the repository changed by the item, its four candidate files byte-identical to the snapshot it took; every experiment a private binary in a scratch tree whose no-switch build is byte-identical to the control): **the coupled loop's pass count is a ladder of nested modes and equals the SLOWEST of them, not their sum.** MEASURED on wasp_full 300, one thread, removing one composition coupling at a time: the production loop 10.09 passes; the stellar field's dependence on the composition removed, 7.16; the temperature removed as well (the energy solve replaced by its own converged answer), 4.94; the He recombination rates too, **2.00** (which reproduces, with a different probe, the 2 passes COST2 measured with every composition-dependent input frozen). Both hypotheses of the brief are refuted: the two diffuse couplings are worth at most 0.08 of the 10.09 (Balmer off 10.09, He rec off 10.01, both off 10.01, He rec frozen after pass 1 10.01, and on mol_base_handoff all three read the control's 7.19 to the last digit), and **the Balmer continuum is not a lag at all:** excited_H_update fills its arrays once BEFORE the loop, so they are constants of it, and refreshing it every pass costs 1.5 percent of wall for +0.01 passes. The stellar lag closed IN FULL for the first time (a one-cell block and eight self-field passes): 8.18 passes for 218 s against 10.09 for 139 s, the next rung, with freezing the field entirely giving 7.16. The temperature is not the residual either: the composition sweep

alone, at the temperature the step converged to, takes 10.94 passes (6.32 against 7.19 on mol_base_handoff).

Decision: change nothing (no loop-internal Balmer refresh, no self-consistent He recombination coupling, no default moved); to go below seven passes every rung must come out at once. Advisor: the three statements this refutes are corrected in place, the csm_max_pass comment of EXHALE_main.f90, the FIELD-SELF entry above (marked superseded) and open item 1 of the 2026-09-08 handoff. **The one remainder, now open item 1:** on mol_base_handoff the ladder is flat (freezing the field RAISES the count 7.19 -> 7.41) and everything the item could reach accounts for 0.6 of its 7.19 passes; the named candidate, READ and not measured, is n_tot, the third-body density of the three-body molecular reactions, built by calc_ntot from the ENTRY composition, and with it the H2 vibrational quench fraction and bound energy. Also recorded: xuv_field_block_cells and xuv_self_field_passes are compile-time parameters, so measuring the two together needs a rebuild; heating_of_composition calls he_rec_coupling a second time whole-grid every pass for its heating alone (both calls together bounded at 1.7 percent of wall); on mol_base_handoff the He recombination coupling is what keeps step 0 inside the pass cap.

- **COST5** (measurement; NOTHING in the repository changed by the item, its four candidate files byte-identical to its own snapshot; probe binaries byte-identical to the control at their default): the molecular remainder COST4 left is named, quantified, and the ladder reaches its bottom rung. **The count is arithmetic, and the law is measured and then tested:** passes = $2 + \log_{10}(\text{the step's own excursion} / \text{the stopping tolerance}) / (\text{decades per pass})$. On mol_base_handoff the composition contracts by 0.171 a pass and the temperature by 0.072 (0.77 decades a pass) from an excursion of 7.0×10^{-5} to a tolerance of 1×10^{-8} : 7.00 predicted, 6.93 MEASURED; on wasp_full 0.222 and 0.199 from an excursion seven times larger, 9.2 predicted, 10.09 measured (so the molecular cases are the cheap ones, not the pathological ones). Tested, not fitted: the law says the count depends on the stopping tolerance through one logarithm at 1.30 passes per decade, and moving both tolerances together gives 8.06 / 6.96 / 5.68 / 4.04 at 1×10^{-9} / 1×10^{-8} / 1×10^{-7} / 1×10^{-6} , i.e. 1.34 over three decades; and it is not the cell solve, six decades of hybrid1 tolerance leaving the count at 6.97 for up to 1.99x the wall. Of the four candidates only the named one is an iterate: n_tot, the third body of R12/R13/R15 (the ONLY composition dependence of the molecular network outside the field: of seventeen hoisted coefficients exactly two see it, both linearly). Holding it: mol_base_handoff 7.19 -> 6.03 at 1500 steps, mol_metals 6.99 -> 6.01; it limits the composition mode's RATE (0.171 -> 0.073) but the count falls only 6.93 -> 6.11 because the temperature mode then binds. The other three are entry state through the operator-split field (vibrational pair 6.96, electron density 6.94, Lyman-Werner 6.96, against 6.96), and the metal element totals are invariant across a sweep by construction (6.99 against 6.99, exact). With every input AND the temperature held the sweep needs **2.01 passes**, reproducing COST2's and COST4's atomic bottom rung on the molecular path. **Decision: nothing closed.** The mode that sets the count is the temperature/composition alternation (EXHALE_main.f90, energy_semi_implicit.f90, not that item's files); closing the n_tot lag exactly needs two sites in System_HeH_mol.f90 (also not its), and the in-file alternative costs a second cell solve per pass, which the cell-tolerance scan bounds at more than the 0.80 passes it returns. The production treatment is not wrong: at the fixed point the entry and returned compositions are one state, so the third body the network is solved with is the third body of the accepted state, to csm_comp_tol. Advisor: the law is now in the csm_max_pass comment; the misleading indentation of composition.f90's

calc_ntot branches fixed. Recorded: the loop's stopping test compares an ABSOLUTE mass-fraction change against csm_comp_tol, so it is a relative 1e-8 on H2 and no constraint on a 1e-12 species. **The one lever that keeps both the fixed point and the tolerance is sequence extrapolation** on the (T, c) pair after the third pass (the decay is geometric with a ratio stable from pass 2 and close across cases): ESTIMATED at three passes of seven on the molecular cases and four of ten on wasp_full, item COST6.

- **HYG-4** (advisor, direct; the four hygiene items of the 2026-09-08 handoff): (i) he_ground_singlet_density's run-wide clamp counter is incremented with !\$omp atomic update inside the function, so the named critical section around its one call inside a parallel region (ionization_equilibrium.f90, self-consistent field loop) is gone and the three comments that justified serial column formation by the counter are reworded (the columns are still formed before the blocks are solved, for the ordering, not for the counter); (ii) element_transport_residual no longer takes the cell velocity it never read (four call sites: certification.f90, steady_newton.f90, two test rows), the balance being the divergence of the FACE mass flux handed in; (iii) the outer Picard loop of steady_wind_with_element_diffusion halved the element under-relaxation omega whenever comp_drift failed to fall, but comp_drift is the worse of the element drift and the carrier drift while omega damps the element relaxation alone (the carrier pass is bounded by carrier_trust), so a carrier drift that failed to fall slowed the element alternation for nothing; the damping test now reads the element relaxation's own drift (elem_drift) and the exit test keeps the worst of the two. No regression case runs both He_diffusion and the carrier transport, so the matrix cannot move. (iv) f_vib_quench/e_vib_bound staying at the entry composition inside the self-field loop is CLOSED BY MEASUREMENT (COST5): they are entry state through the operator-split field, and holding them at the first-pass value is worth 0.00 passes (6.96 against 6.96). Suites: steady_species_rows 74/0, certification 29/0, diffusion_tests 35/0.
- **B5e** (steady solve globalization; steady_newton.f90, the steady_species_rows suite, design doc section 17, contract section 11.5): **the first stationary solve with a species row that meets its own target.** MEASURED on mol_diffusion reloaded with Coupled carrier solve: True (one element row, nvar = 4): the scaled trust region takes ||R|| from 1.986 to **4.345e-09** against a target of 1e-08 in 66 outer iterations (54 accepted), where the pseudo-transient line search aborts at iteration 18 on "no descent direction exists for the banded model" at 1.702. Two defects were named by instrumenting the eleven exits of trust_region_step and repaired where the measurement pointed: (i) the radius collapse was the CAUCHY LEG ASCENDING the model, because its gradient is the transpose of the BANDED A and has no sign (MEASURED $r_0 \cdot A g = -2.1e13, -2.2e19$ at the collapsing iterates, pred negative and exactly linear in the radius); where $r_0 \cdot A g \leq 0$ the leg is now dropped and the dogleg reduces to the Krylov leg cut to the ball, which always predicts a reduction (GMRES gives $r_0 \cdot A s_N < -||A s_N||^{2/2}$); and pred is formed as $-(r_0 \cdot A s) - ||A s||^{2/2}$ instead of the cancelling difference of two squared norms, whose sign on a short step was rounding; (ii) the best-iterate ledger ranked on ||R|| (three hydrodynamic rows) while the state is judged on those AND every species row, so it could hand back the iterate with the worse element balance; it now ranks on distance_from_certification, the largest of the judged row measures over their tolerances, formed once inside eval_residual under an optional argument; the three-unknown branch keeps its comparison character for character. **Refused by measurement:**

bounding the STEP by the judged rows (implemented in all three step controls) stalled the element solve at $||R|| = 1.92$ on 13 refusals; and the banded-gradient Cauchy length $(r_0 \cdot A_g) / ||A_g||^2$, more defensible on paper, left the same solve crawling at $1.8e-4$ after 81 iterations against $4.3e-9$ in 66 with the original length, so the original stays, recorded at the site with both numbers. **Default moved:** `use_tr = (nspec_row > 0)`, the trust region is the step control of the coupled route (EXHALE_TRUST_REGION=0 restores the line search for measurement); no regression case sets `Coupled carrier solve: True`, so no golden can move. **Anchor:** the elemental transport He/H row of the returned state reads **4.633e-11 against `cert_tol_element = 1e-8`** (215x inside), the first measurement under that tolerance; it is NOT moved, because contract sections 9 and 10 want a ladder (the same case at `Resid tol 1e-9` and `1e-10`, not run) to tell a target-limited row from one at its floor. The state is not certified: its cell-1 hydrodynamic mass row reads $6.2e-11$ against $3e-12$, a base-cell matter and not an element-row one. **The carrier arm is not rescued:** on `mol_carrier` the trust region stagnates at 1.873 (line search 1.855) and 18 of its 49 iterations end on "no Krylov direction could be sampled", so `cert_tol_carrier` stays UNCONFIRMED, and the measurement says it is not a step-length problem. Worker MEASURED `wasp_full_newton` byte-identical (it 6, $||R|| = 8.131e-09$, info 0, CERTIFIED, $\log_{10} \text{Mdot}$ 13.30) and the `mol_diffusion` trust-region run identical to the pre-item text under EXHALE_TRUST_REGION=1 (every landed change inert on a path that already worked). Recorded, not fixed: EXHALE_SPECIES_JAC_TEST=1 perturbs the run it measures (iterations 1-5 agree to every printed digit, iteration 6 differs, the two runs then diverge; some module state survives a probe `eval_residual`; the same root writes `n_eq_sweeps_model` and the frozen background); and the ray test can refuse a step that reduced the merit far MORE than the model promised (ratio = 7187), which Algorithm 4.1 of Nocedal and Wright never does, the obvious amendment (gate it on `ratio <= eta_accept`) left unmade for want of a control measurement. The worker was stopped by a rate limit after writing its report; advisor verified the tree: build clean; suites `steady_species_rows` 74/0 (six new driver rows), certification 29/0, `species_face_flux` 16/0, `carrier_retry` 101/0, `coupled_source_step` 30/0; one "per-iteration" phrase in `run.sh` reworded; `wasp_full_newton` re-run by the advisor against its golden with the B5e binary: all four files byte-identical, info 0, CERTIFIED, $\log_{10} \text{Mdot}$ 13.30.

- **Batch gate 2 (2026-09-08, 05:04 to 08:08):** make check on the tree as of B5e + HYG-4 (before COST6 landed; the harness built at 05:04, COST6's first edit is 05:17) **PASS, all 65 files of the matrix byte-identical** to the goldens refreshed by the user on 2026-09-07. make test in the same script failed 7 suites on their build-staleness precheck only (the shared build/ had not been remade after HYG-4 when the suites ran, the harness remade it afterwards); the suites themselves were run on the advisor's private build of the same tree: `steady_species_rows` 74/0, certification 29/0, `species_face_flux` 16/0, `carrier_retry` 101/0, `coupled_source_step` 30/0 then 39/0 with COST6, `diffusion_tests` 35/0, `physics_probe` 1491/0, and `energy_update`, `grid_and_gates`, `spectrum_type` 53/0, 93/0, 154/0. Every suite of make test is therefore green on this tree. No golden touched.
- **COST6** (extrapolation of the coupled loop's geometric decay; EXHALE_main.f90 +532 lines, `coupled_source_step` +3 whole-binary rows; **default OFF**, EXHALE_CSM_EXTRAP=1 turns it on): implemented, guarded, measured. The object is the GLOBAL pair (T, f_sp), which is exactly the loop's iterate (the

energy solve rewrites $u(3, :)$ and $W(3, :)$ in full from T and the entry anchor each pass, so an undo of the pair is exact, and the guard-forced-refusal row reproduces the off run byte for byte); one Rayleigh-quotient ratio per block over all cells and species, the projection onto the dominant mode COST5 measured stable, and one damped ray, so no cell can move in a direction the global mode lacks. That is how it differs from COST2's Aitken taken cell by cell (12.00 against 11.05 passes): a ratio taken cell by cell spreads 200 percent across the grid and cannot be tested for validity; the global one is tested ($||d_k - \theta d_{k-1}|| \leq 0.5 ||d_k||$) before use. Admissibility is by construction (fractions in $[0,1]$, He 2³S inside its He I column, T inside the energy solve's own bracket, all linear in the step, so the largest admissible multiple is taken); MEASURED, the worst violation is identically zero on every case at 300 and 1500 steps, 1 and 16 threads, and the step was never damped below 0.95. The stopping test and its tolerances are untouched: a candidate is kept only if the pass taken FROM it meets them. Three things the measurement taught: the mode ALTERNATES (cosine of consecutive increments -0.85 to -1.00 on `mol_base_handoff`, +0.71 to +0.98 on `wasp_full`; the first implementation demanded $\theta > 0$ and refused 1188 of 1188 candidates); the one-mode model does not reach back into early passes (the ratio drifts 0.77, 0.47, 0.31, 0.22, 0.17, 0.14, 0.11 over one `wasp_full` step), hence a REACH guard, take the limit only where the modelled remaining error is already inside the tolerance ball (reach scan 6.61 / 6.15 / 6.36 / 7.81 / 7.74 / 7.75 at 1/2/3/10/30/100, control 6.96); and undoing a mediocre candidate costs more than it saves (7.29 strict against 6.94 lenient, control 6.99 on `mol_metals`). MEASURED, sweeps control -> on: `mol_base_handoff` 1500 10780 -> 10102 (**-6.3 percent**, matched pair one binary 285.4 -> 268.0 s), `mol_carrier` 300 -2.3, `mol_metals` 400 -0.75, `wasp_full` 300 -0.63, `hydrostatic_column` 0; state movement $8.3e-8$ to $5.0e-7$ relative, $\log_{10}$ $\dot{M}$ unchanged. **The brief's estimate of three passes of seven is NOT reachable, and the reason is the item's real finding:** the stopping test measures the INCREMENT, and the error is $|\theta/(1-\theta)|$ times it, one to two decades smaller, so the loop is one to two decades stricter than its tolerances say and spends 1.0 to 1.6 passes on that; the extrapolation's ceiling is the last factor $1/|\theta|$. Redefining the test on the estimated error is a change of what the loop returns and is put to the user as decision item 8 of `docs/To_be_determined_by_user_20260906.md` (recommendation (b)). Tests: `coupled_source_step` RED 2 rows against the control, GREEN 39/0 (`extrapolation_lowers_the_pass_count` 6.09 against 6.96 at 300 steps; `refused_extrapolation_costs_nothing` identical at 6.80; `mean_passes_within_bound` re-measured 6.80, unchanged); `physics_probe` 1491/0. Worker-introduced defect fixed inside the item: a ledger write with six integer descriptors for seven items. Recorded, not fixed: `write_setup_report.f90` does not echo the switch; `n_csm_x_kept` can only count candidates the stopping test did not meet and reads 0 everywhere (the number that matters, met on the very next pass, is printed beside it); the `csm_max_pass` comment block is now some 60 lines of measured tables and belongs in a `docs/` memo. Advisor: the measured tables and the item history that had grown to some 230 comment lines around the coupled-step constants are moved to `docs/coupled_source_loop_cost.md` (sections 1-7, decision item 8 linked); the code site keeps each constant's conclusion and a section pointer. The setup report does not echo environment switches (none of them, by design), so the switch is reported by the run counter line alone. Advisor verification: `coupled_source_step` 39/0 and `physics_probe` 1491/0 on the advisor build; default-off byte-identity against the GOLDENS at matrix length (which checks B5e, HYG-4 and COST6-off together); `mol_base_handoff` 12000 steps and `wasp_full`

to its own stop, both Hydro_ioniz.txt and Ion_species.txt IDENTICAL to the goldens (both runs end with status 2, "not certified", exactly as the golden runs do).

- COST7** (decision item 8, option (b), user 2026-09-08; EXHALE_main.f90, coupled_source_step suite, docs/coupled_source_loop_cost.md sections 7-8): **the coupled source loop stops on the ESTIMATED REMAINING ERROR of the pair (T, composition), not on the last increment.** The estimate is $|\text{theta}/(1 - \text{theta})|$ times this pass's increment in each of the two norms the test uses, theta the global Rayleigh-quotient ratio, formed in ONE routine (coupled_pair_geometric_error_estimate) that the COST6 extrapolation now reads too (its reach guard is literally the stopping test at a loosened tolerance; shared objects renamed csm_x_* -> csm_geom_*, env hooks EXHALE_CSM_GEOM_*). Where the sequence gives no estimate (passes 1-2, a guard failing) the increment test stands, and that fallback (EXHALE_CSM_GEOM_REFUSE=1) is byte-identical to the pre-COST7 binary on all 10 written files of wasp_full 300 and all 7 of mol_base_handoff 1500. **The item's finding:** the estimate is an L2 quotient against two max-norm tolerances, and MEASURED with a probe that runs the loop on to 1e-12 and rolls back, the unfactored estimate accepts states OUTSIDE csm_comp_tol on 100 of 299 steps of mol_base_handoff (1.8x the tolerance at worst); a factor 2 leaves 1 of 299; **a measured safety factor csm_err_safety = 3 leaves none**, the accepted state's distance to its own fixed point reading 7.4e-9 and below (composition) and 2.1e-9 and below (temperature) on every one of the 2496 accepted states probed. The price is that almost none of the passes COST6 estimated survive: wasp_full 300 10.09 -> 10.01 passes (sweeps -0.9 percent), mol_base_handoff 1500 7.19 -> 6.96 (-3.1), mol_metals 400 6.99 -> 6.93 (-0.9), **mol_carrier 300 6.01 -> 5.01 (-16.6)**, hydrostatic_column 1.01 -> 1.01 (+1 sweep of 302: on its single three-pass step the estimate EXCEEDS the increment, so the new test is the stricter one there; it is not uniformly looser). log10 Mdot unchanged on every case; state movement 4.3e-10 to 1.1e-7 relative, at the tolerance level. What the change buys is the MEANING of the two tolerances, the decision's own first reason, not passes. The COST6 extrapolation under the new test is worth nothing (6.79 against 6.82 passes) and stays default off. Tests: coupled_source_step RED 41/3 against the control, GREEN 44/0; mean_passes_within_bound re-measured 6.23 (was 6.80), both at the site; extrapolation_lowers_the_pass_count rewritten as extrapolation_does_not_cost_passes; the fallback-is-one-path row replaces the brief's byte-identity row (a suite cannot hold a golden of the previous binary; the identity is measured in the report). physics_probe 1491/0. Recorded, not fixed: the composition increment of mol_base_handoff has a floor near 5e-9 (127 of 1500 probes cannot reach 1e-12 in 59 passes), so csm_comp_tol = 1e-8 is within a factor of two of what that case can reach at all; COST6's n_csm_x_kept still reads 0. **Every golden of the matrix moves at the 1e-8 level:** gate 3 below.
- Gate 3 (2026-09-08, 09:29 to 12:00)**, tree after COST7 (with COST6 default off, HYG-4, B5e): shared build remade, **make test all suites green**; make check verdict FAIL against the 1e-3 relative tolerance, as expected for a change of the stopping test, and every movement examined. log10 Mdot unchanged on all 14 cases that print one (13.35, 13.35, 13.30, 10.57, 10.58, 10.54, 10.57, 10.58, 10.51, 10.56, 7.29, 8.27, 9.61, 7.98). Movement table (largest relative change of any column, new against golden): wasp_full 6.6e-5 (v at the base, cell 136), wasp_he23off 3.8e-4 (v at cell 146), wasp_full_newton 1.1e-10, mol_base_handoff 2.8e-6, mol_metals 2.4e-6, mol_lyman_werner 4.1e-7, mol_diffusion 2.8e-6 (adv file 2.2e-3 at the outer boundary,

cell 504), mol_ir_bands 2.4e-6, mol_sec_ion 4.5e-3 ($v = 0.091$ cm/s at cell 177, a near-zero base velocity, adv 1.5e-3), mol_carrier 5.2e-7, lower_profile 9.3e-6, hp_zero_seed / hp_trace_seed / hp_front identical, roundtrip within. Two cases move by order one and both are pre-existing base artefacts, not COST7's: **oxygen_chemistry** (1000-step snapshot) reads 42x in $n(\text{H II})$ at cell 22 because the GOLDEN carries an odd-even sawtooth there (cells 21, 24, 25, 27 at $7\text{e}3\text{--}9\text{e}3$ against $2\text{e}5$ in their neighbors) and the new run is smooth ($1.6\text{e}5$ to $3.9\text{e}5$ across cells 18-26; ρ , p , T within $1\text{e}-6$); **hydrostatic_column** reads 0.999 in the adv file's T at cell 350 because the adv post-process gives $T_{\text{adv}} = 0.78$ K inside $1.40 R_p$ in this static column (the original T is 1124 K throughout) and the edge of that region moved by one cell; the non-adv files move by $4.6\text{e}-8$ (Ion_species by 0.16 at cell 2 on values of $1\text{e}-182$, denormal noise). The adv defect is recorded as an open item. **Goldens to be refreshed by the user** (archive first), then make check re-run.

- **ADV-STATIC** (post_process_adv.f90, new suite adv_static_limit, docs/postprocess_advection_validity.md): **the brief's premise is refuted and the 0.78 K of hydrostatic_column's _adv file is named.** MEASURED: the flow where it sits is Mach 0.23 to 0.82 (the case is a 300-step relaxation snapshot, not a static column: $\rho v r^2$ spans seven decades); the IONIZATION correction is held at equilibrium in 503 of 503 cells (Ion_species_adv.txt equals Ion_species.txt to the bit) and its Damkohler condition already gives the continuous $v \rightarrow 0$ limit the brief asked for, so nothing there was touched. The number is the exact solution of the equation the _adv ENERGY solve poses: the steady internal-energy equation with the enthalpy flux of the mass-flux divergence DROPPED, $\rho v \frac{de}{dr} - p v \frac{d \ln(\rho)}{dr} = \text{heat} - \text{cool}$, which equals the exact $\dots + h \text{div}(\rho v) = \text{heat} - \text{cool}$ only where $\rho v r^2$ is stationary; with the radiation off it reduces to the discrete adiabat $T_j/T_{j-1} = \rho_j/[\rho_j - (\gamma-1)(\rho_j - \rho_{j-1})]$, and integrating that on the run's density reproduces the reported temperature to $6\text{e}-11$ over the 347 cells before the first fallback ($1138 \text{ K} \rightarrow 0.78 \text{ K}$ at $1.396 R_p$; the dropped term exceeds the kept ones in 480 of 502 cells). With heat/cool negligible the equation is homogeneous in v , so no $v \rightarrow 0$ condition can change its answer (99.96 percent departure at $|v| \leq 1.4\text{e}-3 c_s$). Landed: a validity condition of the energy solve's own, the thermal Damkohler number $Da = t_{\text{cross}} |\text{heat} - \text{cool}| / u_{\text{th}}$ (new pure thermal_damkohler_number; above one the cell keeps the run's own T), the counterpart of ionization condition (ii) that the bare $v \leq 0$ sign test lacked; the stationary-mass-flux assumption stated at the site with its exact dropped term and MEASURED at run time (largest ratio, where, how many cells exceed one; nothing acts on it yet); the first eval_cool writes a named tcool_in instead of a discard slot; the stale "is not evaluated" sentence of the validity memo rewritten with both tables. Impact: byte-identical everywhere measured (hydrostatic_column full case all four files; wasp_full, mol_base_handoff, lower_profile post-processed from the same golden state by both binaries, all four files; max thermal Da over tested cells $0.951 / 0.534 / 0.055 / 3\text{e}-19$, 0 cells above one anywhere). Suite RED/GREEN (rows 1-7 cannot link before; row D measured=silent before), advisor 10/0. Also MEASURED: no matrix case has a stationary mass flux by the cell-to-cell measure ($\rho v r^2$ over outflowing cells spans 1.27x its median in wasp_full, 2.2x in mol_base_handoff, 83x in lower_profile); the du stop is taken on a restricted window and does not bound this. **Follow-up decided by the advisor under the physical-correctness rule: restore the dropped term (item ADV-ENERGY)**, since an energy equation that is exact only for a stationary mass flux is wrong on every

snapshot and at the $1e-3$ level on every converged wind; MEASURED worth on the static column 569 K at 1.40 R_p and 121 K at 2.96 R_p in place of 0.78 K and 6.0 K. The alternative, refusing the column where the assumption fails, would rewrite 162 of 502 `_adv` temperatures of `wasp_full` (median 5 percent) and is decision 14 (T8.1) of the target-system document, left to the user. Advisor fixed the stale provenance paragraph of the case README.

- ADV-ENERGY** (`T_equation.f90`, `ion_cell_state.f90`, `post_process_adv.f90`, `adv_static_limit +7` rows, `docs/postprocess_advection_validity.md`): the advection post-process's energy equation is now the EXACT steady internal-energy equation. The enthalpy flux of the mass-flux divergence, $h \operatorname{div}(\rho \mathbf{v})$ with $h = e + p/\rho$, enters `T_equation` through a new zero-default field of the cell record, `teq_cell%div_rho_v`, as a separate additive term in both branches (caloric $\mu_m \operatorname{div} \rho_v (E(x_{H2}, T) + T)$, monatomic $\mu_m \operatorname{div} \rho_v T$; μ_m , μ_p , `coeff` untouched; zero field an exact zero, tested to the bit). $\operatorname{div}(\rho \mathbf{v})$ is the mass row's own operator $(A_p F_p - A_m F_m)/dV$ over the control volume between the two points the upwind difference uses. MEASURED corrections to the briefs: `T_equation` has no caller outside `post_process_adv` (no marching solve to protect; the bitwise property holds anyway); and ADV-STATIC's 569 K prediction had put back $p \operatorname{div}(\mathbf{v})$ alone, not $e \operatorname{div}(\rho \mathbf{v})$; the whole term gives **1.7e7 K at 1.40 R_p and 1.6e9 K at 2.96 R_p** on the static-column snapshot, the closed form $w \sim \rho^{(\gamma-1)} F^{(-\gamma)}$ with $F = \rho v r^2$ falling 8 percent from cell to cell. Tests RED (nine "not a member" compile errors before) / GREEN (closed-form solution $T \sim r^{(-\gamma)}$ on a diverging column to $6.9e-4$ at 400 cells, first-order convergence $5.59e-3 / 2.79e-3 / 1.39e-3 / 6.95e-4$, the adiabat departing by 2.17); advisor `adv_static_limit 17/0`, `physics_probe 1491/0`. **Impact, MEASURED, same golden state post-processed by both binaries:** marching files bitwise in every case; `T_adv` movement `wasp_full_newton` 1.4 percent (the only Newton-converged state, He 2^3S 1.9 percent), `wasp_full` (du stop) 16 percent at 1.088 R_p (He 2^3S 16 percent), `mol_base_handoff` snapshot 301 percent, `lower_profile` snapshot 456 percent (He 2^3S 238x), `hydrostatic_column` 4.2e9 (reaching 2.6e10 K). Restored term against the kept ones: `wasp_full` 67x at 1.033 R_p in 161 of 502 cells, `lower_profile` 25x in 299 cells, `wasp_full_newton` 24x in 2 cells. **The `_adv` temperature of a state that violates continuity is now unbounded above where it was bounded below, and the transit tools read those columns:** on such a state NO steady post-process has a meaning, and what to output there is decision 14 of the target-system document, now put to the user as decision item 9 with a recommendation. Goldens: every `_adv` file moves by design; refresh deferred by the user to this item's gate.
- B5g** (`steady_newton.f90 +467/-35`, `steady_species_rows` suite, design doc sections 18, 18.5): **a residual evaluation is now a function of its arguments.** Three class-(b) caches survived a probe `eval_residual` and were read back by the solve: the elemental rows and scales (`erow_he/escale_he/erow_tr/escale_tr`), the carrier measures (`carrier_relnorm_last...`), and the carrier arrays of diffusive photochemistry (`col_scale_car`, `row_scale_car`, `row_terms`, `headroom_car`, the frozen backgrounds); the Newton ROW SCALE, the acceptance gate, the best-iterate ledger and the choice of the state handed back read them. Fixed on the reader side: `eval_residual` gained `state_is_discarded` and a probe holds/puts back that set (reusing the `save_/restore_carrier_module_state` pair certification.f90 already uses); the loop-top row scale

is taken AFTER the iterate's own evaluation; the gates read a new `carrier_relnorm_state`. The two self-tests also tag their sweeps as candidate sweeps (under the iterate tag they were writing the run's acceptance ledger and `ieq_nonroot_streak`, whose limit stops the run). PROVEN: EXHALE_SPECIES_JAC_TEST=1 and the untested run are the same run, every (JFNK) it line identical over 17 iterations, `Ion_species.txt` byte-identical (RED: 82 differing lines, first at iteration 6). `wasp_full_newton` byte-identical (info 0, 6 iterations, CERTIFIED, 13.30). **What the fix uncovered is bigger than the item:** the `mol_diffusion` element solve that B5e brought to $4.3\text{e-}9$ in 66 iterations now STAGNATES at 1.957 after 17; the runs agree for five iterations and part on a 3 percent difference of `Drow` between two evaluations of the SAME state, because the residual's seed dependence (`resid_sc_probe`) is **0.96 in units of Resid tol** against $||R|| = 1.986$, and tightening the sweep tolerance ($1\text{e-}12$, $1\text{e-}14$ with pass caps 200, 500) never satisfies the increment test from either seed: **the eliminated composition residual of that state has no fixed point it reaches**, and B5e's convergence was a property of one sequence of evaluations. The change is kept (a row scale must be the row's own terms of the same evaluation). Part 2, the carrier arm: all 18 "no Krylov direction" exits and all 514 refused samples were ONE thing, the H2 carrier of cell 500 at $-7.4\text{e-}19$ (a species unknown on its lower bound), not the element budget (the counter's label named one screen of several; corrected, census added); `jv_product` now takes the backward difference $[F(Y) - F(Y - \text{eps } v)]/\text{eps}$ only after every forward halving is refused: 49/0/0 accepted/rejected/ refused where 31/18/18, GMRES truncations $18 \rightarrow 0$, radius $8\text{e-}3 \rightarrow 2.05$; $||R||$ unchanged at 1.873 because the dogleg cuts steps by $2^{-25}..2^{-49}$ against the same bound from the trial side, a bound-constrained unknown space (decision, not repair). `cert_tol_carrier` UNCONFIRMED. Suites (advisor build): `steady_species_rows` 74/0 (88/0 with the self-test pair logs), certification 29/0, `coupled_source_step` 44/0, `carrier_retry` 100/0, `species_face_flux` green; advisor re-ran `wasp_full_newton` in full with the B5g binary: `Hydro_ioniz.txt` and `Ion_species.txt` IDENTICAL to the gate-3 output, CERTIFIED, log10 Mdot 13.30.

- **B5f** (docs/a2_certification_contract_20260906.md section 11.6; no constant moved): the `cert_tol_element` ladder on `mol_diffusion` at `Resid tol` $1\text{e-}8$ / $1\text{e-}9$ / $1\text{e-}10$ gives ONE state: the $1\text{e-}8$ solve meets its target in 66 iterations (element row $4.633\text{e-}11$, reproducing B5e digit for digit, so COST7 leaves the stationary solve untouched); the $1\text{e-}9$ and $1\text{e-}10$ solves abort after 88 iterations with the radius at $9.3\text{e-}23$ (9 of 31 refusals on "the ray difference is below its cancellation floor") and hand back the SAME state, bitwise. The row is floor-limited at $4.63\text{e-}11$ (knee at $||R||$ about $1\text{e-}7$, 1.5 percent movement over the last factor 26); proposed value by the contract's rule $5\text{e-}10$, NOT applied (two rungs are not solves that met their target; a decade above one configuration's floor is not a decade above the row's floor, which the operator's header puts between $1\text{e-}12$ and $1\text{e-}5$ across configurations; no second configuration exists). **Read together with B5g:** B5f ran on the pre-B5g text, and B5g shows the state it anchored on is reached by one sequence of evaluations only (with the probe caches removed the same solve stagnates at 1.957), so this rung stands as a measurement of that state's row, not as evidence that the solve converges. B5f also MEASURED that the coupled route's convergence depends on `eq_sweep_reltol` ($1\text{e-}10$ loses the $1\text{e-}9$ solve at iteration 17), the same fact from the other side. Recorded: the JFNK of a reloaded $1\text{e-}8$ state starts from $||R|| = 1.226$ because the marching loop takes two CFL steps first (the $O(dt)$ kick, eight decades, seen for the first time from a stationary state); the certification prints no cell-resolved record of the species rows, which blocks any floor attribution.

- ADV-REFUSE** (decision item 9, option (a), user 2026-09-08; post_process_adv.f90 +227/-43, write_output.f90 +94/-10, examples/exhale_io.py, EXHALE_transit.py, README.md, README_HOWTO.md, user manual section 4 (PDF rebuilt, 76 pages), docs/postprocess_advection_validity.md, adv_static_limit +13 rows): **the advection post-process refuses the correction in every cell whose own energy-equation term ratio says the flow is not stationary; that cell keeps the run's temperature and the ionization equilibrium at it; a new integer column adv_status at the end of Hydro_ioniz_adv.txt records the reason row by row** (0 corrected; 1 thermal Damkohler; 2 not stationary; 3 ionization conditions (i)-(iii); 4 cell solve failed; five named module parameters in write_output, the # columns line extended, a # adv_status_counts header line counted from the array written). The criterion is enthalpy_flux_term_ratio (new pure function beside thermal_damkohler_number), the restored term over the LARGER kept term, refused on a STRICT > 1 so a stationary flux with round-off is never refused; formed ONCE on the state handed in over every cell, so the refused set cannot move with the iterate and the converged product does not depend on the iteration history; applied to temperature AND composition (a non-stationary cell has no steady advective ionization correction either). write_output gains an optional adv_status dummy (the four 'eq' calls untouched, Hydro_ioniz.txt keeps seven columns); exhale_io.load_hydro now takes names from the header (seven historical keys kept, Run.adv_status exposed, None for an equilibrium file); EXHALE_transit.py prints how many rows inside $1 \leq b \leq \text{Rib}$ are refused, by reason; EXHALE_plots.py and the four other header parsers adapt unchanged. MEASURED (same golden state, "Do only PP"; marching files bitwise everywhere): hydrostatic_column's corrected temperature falls from 2.6×10^4 K to **1182.32 K, exactly the run's own maximum, no row above it** (435 before); status counts 0/1/2/3/4: wasp_full_newton 501/0/2/1/0, mol_base_handoff 470/0/33/1/0, wasp_full 205/0/297/2/0 (160 of the 297 beyond conditions (i)-(iii), matching ADV-STATIC's route-2 estimate of 162; the larger total is the reference point, the state handed in over every cell including the inflow cells), lower_profile 182/0/321/1/0, hydrostatic_column 0/0/481/23/0. Every refused row carries the run's T to the BIT and the equilibrium density of every species exactly (one 1.8×10^{-16} reassociation in lower_profile's He I, pre-existing). Movement against the pre-ADV-ENERGY goldens, T_adv: wasp_full_newton 3.3 percent, wasp_full 8.0, mol_base_handoff 64, lower_profile 1.8x, hydrostatic_column 1525x (0.78 -> 1182 K); species: He III 5.4 percent, He 2^3S 29 percent, He 2^3S 110x (at the base of mol_base_handoff), H II 11x (lower_profile base), He III 16 percent. **Transit effect on wasp_full**: 295 of 500 rows inside the impact-parameter range are refused (printed), and the metrics move at 1×10^{-4} relative or less (He 10830 red 4.102 percent both, blue 2.843 -> 2.844, FWHM 0.899 Å both; largest line movement O I 1302 +0.55 percent relative), because the refused rows sit in the base under the extended wind. Tests RED (seven "not found in module" errors; measured=7 columns on the old binary) / GREEN; advisor adv_static_limit 30/0, physics_probe 1491/0. Worker also fixed a dangling \ref{sec:retiredbase} in the manual (the four retired base keys named instead). Recorded: two documents describe the _adv validity conditions (manual, HOWTO) and had both drifted; four # columns parsers exist in python; dV_cv division now appears twice unguarded (a zero is a broken grid).
- B5h** (ionization_equilibrium.f90 +68/-2 diagnostics, steady_newton.f90 +211/-2 probe only, steady_selfconsistent_residual assertion 4; NO behavior changed, wasp_full_newton and the element

run byte-identical to a control built from the same source with the additions removed): **why one state has two residuals**. MEASURED on mol_diffusion's element configuration with a lockstep two-seed trace of the composition elimination (EXHALE_RESID_SC_TRACE), a nine-seed basin ladder (EXHALE_RESID_SC_BASIN) and a cell acceptance report (EXHALE_IEQ_REPORT_CELL): the amnesty is never used (classes 4 and 6 zero at every pass, every cell class 1 with reaction residual $1e-25$ against $1e-6$), the self-field lag is not it (xuv_self_field_passes 1 -> 20, identical to every digit), the He recombination coupling is not it (off, identical to four digits), the stopping test is not it (a fixed count of one pass per call, chained, the difference frozen from pass 2 to 60). **Two things are**. (1) Two separated ROOTS of the network at the same incident field: from a seed displaced five cells the sweep reaches a composition 8.9x apart in $x(H_2)$ at the H_2 front (cell 214) and 19 percent apart in $x(H\ I)$ at the ionization front (cell 192), bitwise identical from cell 216 outward, both exact roots (class 1, $1e-25$). (2) **A marginal direction: the local-equilibrium elimination does not determine the shielded layer's H_2 CONTENT**. Retention of a seed perturbation of the H_2 partition is 0.77 at the base rising to 0.84 at $r = 1.064$ and falling to 0 outside cell 216, the SAME curve to three digits at perturbations of $1e-6$ and $1e-2$, so the sweep's map has eigenvalue 1 in that direction and the seed dependence is a derivative, linear over four decades with gain $3.2e3$ (the base cell's energy row amplifies the ghost composition by $\sim 4e3$), reaching Resid_tol at a seed change of $3e-12$. READ from the network: the fast $H_2 \rightarrow H_2^+ \rightarrow H_3^+ \rightarrow H_2$ cycle returns every nucleus, so the H_2 row as scaled by its turnover pins the PARTITION and leaves the total to $k15\ n_{HI}^2$ against $k12\ n_{tot}\ n_{H_2}$, a detailed-balance pair that in the shielded layer is far below the turnover; and pinning it would pin the fully molecular limit (0.9997) where the state carries 0.376: the content is upstream and transport data, exactly why it is imposed at the ghosts. **The repair the code already owns, MEASURED**: with Molecular carrier transport: True ($n(H_2)$ a Newton unknown) the same ladder falls from $3.155e-3$ to **9.736e-10** at delta $1e-6$ and $x(H_2)$ of the front cell is bit-identical from all nine seeds; the residual is a function of its arguments below Resid_tol. What remains at large seeds (0.98) is the ionization front's two roots, a physics decision (which root a stationary residual may land on, or a statement that the front is unresolved on this grid), reported. The element solve with H_2 carried then meets B5g's carrier obstruction ($|R|$ flat at 1.986 over 19 iterations, carrier row 0.91, run stopped by PID at 4 min an iteration). Also MEASURED: eq_sweep_reltol below $\sim 1e-10$ is unreachable by construction, the increment measure $\max|df|/\max(|f|, 1e-12)$ reporting round-off of He I and H_2^+ fractions $\sim 1e-20$ in the ionized wind (plateau $6e-11$ for sixty passes), which is what B5f's and B5g's "never met at $1e-12$, $1e-14$ " was. Tests: assertion 4 residual_is_one_number_for_basin RED $3.155e-3$ with H_2 eliminated, GREEN $9.736e-10$ with H_2 carried (the tolerance read from the run's own log); steady_species_rows 86/0 with the self-test pair; B5g's part 1 reproduced as a by-product. A defect of the item's own fixed inside it (a cell-report default of -1 was the lower ghost). **Decision put to the user as item 10**: refuse Coupled carrier solve: True on a molecular configuration without carrier transport, or turn the transport on for it (input_read.f90, not the item's).

- **B5i** (decision item 10, option (a), user 2026-09-08; input_read.f90 +51/-2, grid_and_gates new row coupled_carrier_h2 (+5 assertions), user manual (a Coupled carrier solve entry, which the manual had NOT had; the Molecular carrier transport entry corrected from "inert without oxygen chemistry" to "needs molecular chemistry"), README.md, README_HOWTO.md (new block; the oxygen paragraph's dangling "It" corrected), docs/steady_solver_design.md section 19): **Coupled carrier solve: True with Molecular**

chemistry: True and Molecular carrier transport: False is refused at startup, after every key is resolved (so the oxygen default of the transport key is seen: MEASURED, Oxygen chemistry: True plus the coupled key is accepted, the same with the transport stated False is refused), with a message naming what was asked, why there is no root (B5h's numbers) and the remedy key; error stop 1, exit status 1 asserted (2 is the certification's refusal of a complete run). Scope exactly carrier_in_newton .and. thereis_mol .and. .not. carrier_transport, not gated on the Newton key (the direct steady route EXHALE_PTC=1 runs a coupled solve too, and it is the route B5b to B5h used); the atomic coupled solve with an element row is accepted (asserted). No shipped configuration is refused (MEASURED by grep: no *.inp in the tree sets the key; 62 of 81 B5-series scratch inputs would be, as the decision says). Two molecular cases byte-identical before/after at 200 steps. The parse-site comment that claimed "checked below" for a check that did not exist is corrected. A refused run leaves a zero-byte EXHALE_setup.out and no EXHALE_resolved.out, like every other input_read refusal. Tests RED (the eliminated-H2 coupled run completed and printed a Mdot) / GREEN; advisor grid_and_gates 98/0, physics_probe 1491/0. Advisor corrected docs/input_schema.md K15e ("needs K15d" now qualified to the molecular configuration). Recorded: EXHALE_main.f90 opens EXHALE_setup.out (truncating) before input_read, so any refusal erases the previous run's setup report (fix: open it after the input is accepted); the other "checked below" comments of input_read.f90 deserve a sweep against the checks that exist.

- **HYG-SETUP** (advisor, direct; EXHALE_main.f90): the setup report EXHALE_setup.out is opened AFTER input_read returns instead of before it, so a configuration input_read refuses no longer truncates the report a previous accepted run left in the same directory (B5i's finding). input_read writes nothing to that unit. grid_and_gates green on the advisor build; the marching path is untouched.
- **HYG-CHECKED** (input_read.f90 comments plus one new report block, docs/input_schema.md, README_HOWTO.md, user manual (PDF 77 pages), grid_and_gates new row carrier_transport_inert +4 assertions): every requirement statement of input_read.f90 and every "Needs K.." / "refused with" clause of docs/input_schema.md checked against the code that enforces it. 40 enforced as stated. ONE not enforced (Molecular carrier transport: True with the molecular network off) is now a startup WARNING, not a refusal, by the code's own convention for a present-and-inert key (every consumer is guarded by thereis_mol, MEASURED inert; eight shipped inputs all carry the network). THREE enforced differently from the statement, one with the sense INVERTED: the schema and the HOWTO said Ionization transport is refused with Solver: Newton and with Coupled carrier solve, where the code REQUIRES the second whenever the first is set (K15e does carry a proton row; what is refused outright is the EXHALE_PTC=1 route); "Solver: PTC refuses it" named a solver value that does not exist; the base-grid resolution check lives in write_setup_report (warns below 10 cells per scale height). SIX stale statements corrected: known_keys also feeds the duplicate-key REFUSAL (MEASURED, all 99 entries used, no hole); the sub-Lyman absorbers are not mutually exclusive; Kzz_base/He_Kzz are NOT inert with He_diffusion off (the carrier transport reads kzz_cell), corrected in four places; Molecular reaction heat default True is guarded by with_molecules (a refusal would refuse every atomic run); the coronal cutoff window 0.08-0.13 superseded by coronal_cutoff_width.md 7.2; four K15b

cross-references that pointed at the Stellar LW flux row corrected to K15. MEASURED byte-identical on wasp_full 100 and mol_base_handoff 200 (17 of 17 files). Advisor grid_and_gates 102/0, physics_probe 1491/0. Recorded: all 63 "F" line citations of the schema are stale by 300-1200 lines (marked stale at the convention's definition; recommend deleting the tokens); Coupled carrier solve: True is silently inert in an atomic run without He_diffusion (a report belongs where the row registry is); the user manual has no Ionization transport entry.

- Gate 4 (2026-09-08, 15:39 to 18:45)**, tree after COST7, COST6 (off), B5e, B5g, B5h, HYG-4, ADV-STATIC, ADV-ENERGY, ADV-REFUSE (B5i, HYG-SETUP and HYG-CHECKED landed during the matrix; they touch startup checks and a file-open order only, and were verified on the advisor build with the full grid_and_gates suite, 102/0): shared build remade, **make test all suites green**; make check FAIL against 1e-3 as expected. **The marching files are unchanged from gate 3 to the digit**: Hydro_ioniz.txt and Ion_species.txt carry exactly COST7's movement on every case (identical max-rel values in both logs, the three hp_* cases still identical), so B5g, B5h, the three ADV items, B5i and the hygiene items moved nothing on the marching path, and log10 Mdot is unchanged on all 14 cases. The _adv files move by design: Hydro_ioniz_adv.txt now has eight columns (adv_status, the harness reports SHAPE 8 vs 7), and Ion_species_adv.txt moves by 0.05 (wasp_full_newton, He I at the outer edge) to order one (the molecular cases and the hp_* cases, whose refused base cells now carry the equilibrium densities; the relative measure is element-wise, so a species the old correction had moved off its equilibrium value reads 1). **Goldens to be refreshed by the user, once, for COST7 + ADV-ENERGY + ADV-REFUSE together** (archive first), then make check re-run by the advisor.
- Golden refresh (2026-09-08 19:05, run by the advisor on the user's instruction, the user being away from the machine)**: the previous set archived to backup/regression/golden_pre_cost7_20260908/ (verified identical to golden/ before the refresh), then run_check.sh golden snapshotted the gate-4 outputs of all 16 cases. The refreshed goldens carry COST7 (the stopping test on the estimated error), ADV-ENERGY and ADV-REFUSE (decisions 8b, 9a). Gate 5, a make check on the current tree (B5i, HYG-SETUP, HYG-CHECKED on top, all startup-only), follows.
- HYG-DOCS** (documents only: docs/input_schema.md, user manual (PDF 82 pages), README_HOWTO.md): the 91 stale "F" line tokens (77 lines) and the 37 GUI<n> numbers (MEASURED stale by a uniform 13 lines) are gone; a key is located by its label and the header says so; nine Notes cells that held only a citation got real notes READ from the parse sites. The manual gained the missing Ionization transport entry (in the sense HYG-CHECKED corrected) and the four other keys the schema documented and the manual did not (Reconstruction continuation, Atomic rate set, Caloric EOS, Photoelectron heating); MEASURED afterwards, all 97 schema keys have a manual entry. Three stale manual entries corrected (Resid norm is retired; Resid tol's norm is the row scaling of sections 143/145, not |R|/max|u|; Flux spread tol default 2e-5, not 5e-3); the validity paragraph's "one of three" made four. The _adv validity conditions live in ONE place (manual section 4, \label{sec:advvalidity}), the HOWTO points there; the HOWTO names exhale_io.py as the loader every new script should use (four # columns parsers exist). Advisor: the same three stale statements in parameters.f90's comments corrected (default 1e-5; the row scaling; the retired base valve), and the four

schema rows whose code spans carried a | (a strict renderer split them) rewritten; table widths verified.

- **B5j** (steady_newton.f90 +1061/-41 only; steady_residual.f90 and diffusive_photochemistry.f90 bit-wise unchanged; steady_species_rows +14 rows, steady_selfconsistent_residual assertion 5): **the carrier arm's lower bound, MEASURED as two mechanisms in sequence.** The outermost carrier row is linear in its own unknown: at the hand-off its root is +2.1e-11 with the iterate 4 percent above (positivity-preserving, the Newton right to reduce it); twenty iterations later the row's constant term has changed sign and the root sits at -5.5e-13, below the bound, so the row has no admissible root and the unknown is pinned; and at the cell that binds the solve (215) the root is +2.75e-6 while the Newton has driven the unknown to exactly zero, eleven decades below it, the row reading -1.00000 of its own scale. A species unknown at zero whose row wants it positive cannot be sampled at all (a two-sided difference leaves the box on both sides). Landed: the carrier mass fraction clamped onto [0,1] in write_species_rows_into_composition (the rule the element fractions had); the trial projected onto the faces of the species box with the predicted reduction recomputed for the step taken; Bertsekas (1982) epsilon-active set held for the step and applied to every sampled direction; the probe step cut to the box with the side chosen by the room (subsuming B5g's backward sample); the Cauchy point as the step when the Krylov leg cannot be sampled; a defect of the item's own found and fixed (the accepted trial re-evaluated as $Y + D \cdot s$ rather than as the vector evaluated, whose round trip does not close for a projected component: a carrier at -4.8e-35). MEASURED on mol_carrier, three arms, none converging: baseline 49 iterations ||R|| 1.929, 527 refused samples, cut-backs to 2^{-49} ; (a) bound-aware region 63 it, 1.415, 0 refused, 2^{-4} ; (b) carrier as $\ln n$ (EXHALE_CARRIER_LOG_UNKNOWN=1 (the spelling at B5j; since B5k the log unknown is the default and =0 restores the density)) 65 it, **0.9972**, merit 17.6 -> 0.924, smallest adopted unknown 2.2e-11. (b) is better by every measure and is NOT the default (it changes the unknown space of a route a shipped key selects): decision item 11. **Part 2:** the elimination's stopping test is now the change of the three quantities the residual reads the composition FOR (particle count, H2 caloric share, the radiative pair), EXHALE_EQ_MEASURE=species restoring the old one: at eq_sweep_reltol 1e-12 wasp_full_newton reaches it in 33/49 passes where the species measure never does (plateau 2e-11); the element case's remaining plateau (1.85e-11) is the radiative channel at cells 441-459 wandering, that case's own non-convergence (B5h), not the measure's floor. Seed dependence unchanged to four digits (9.736e-10). **wasp_full_newton moves 1.1e-9 relative**, isolated to the new measure by EXHALE_EQ_MEASURE=species reproducing the control byte for byte: the solve converges FURTHER (||R|| 8.1e-9 -> 1.7e-9 in 8 iterations instead of 6, info 0, CERTIFIED, 13.30); below the 0.1 percent rule, the golden is not touched (the harness will read "within"). Suites (advisor build): steady_species_rows 93/0, steady_selfconsistent_residual 3/0, certification 29/0, coupled_source_step 44/0, physics_probe 1491/0; one "per-iteration" phrase in run.sh reworded. Recorded: a carrier row at an active bound is an error to the certification (row 1.00000 at cell 215) and a constrained optimum to the solver, so a bound-constrained stationary state has a multiplier there, not a zero row (decision item 12); 28 of 64 steps of arm (a) are the Cauchy point alone (Bertsekas' two-metric projection not taken); load_IC refuses a molecular restart written at the input's own He/H by comparing printed digits for equality (7.9307E-02 against 0.0793). Advisor: wasp_full_newton re-run on the matrix path (marching then Newton) with the B5j binary against the refreshed golden: movement 5.9e-9 (Hydro_ioniz.txt) and 9.6e-9 (Ion_species.txt), 13 outer iterations to ||R|| = 4.3e-9, CERTI-

FIED, log10 Mdot 13.30; below the 0.1 percent rule, golden untouched.

- **Gate 5 (2026-09-08, 19:03 to 22:02):** make check on the tree as of B5i, HYG-SETUP, HYG-CHECKED (built 19:03, before B5j landed) against the refreshed goldens: **REGRESSION PASS, all 64 files of the 16 cases byte-identical.** The refreshed goldens are therefore the current tree's own outputs, and the three startup-only items moved nothing. B5j's $1.1\text{e-}9$ movement of `wasp_full_newton` (the Newton converging further under the new increment measure) is not in this gate's binary and will read "within" at the next one; below the 0.1 percent rule it refreshes nothing.
- **B5k** (decision item 11, option (a), user 2026-09-08; `steady_newton.f90` +346/-85, `steady_species_rows` +137/-6 (114/0), design doc section 20, schema K15e, manual entry, PDF 82 pages): **the molecular carrier densities of the coupled steady solve are carried as $\ln n$ by default** (EXHALE_CARRIER_LOG_UNKNOWN=0 restores the density unknown; the element fractions untouched; no certification change, decision 12 (c)). The floor under $\ln n$ is the transport operator's own admissibility floor, $1\text{e-}20$ of the carrier's element budget in that cell (`carrier_headroom`), the fraction `carrier_residual` already puts under its row scale; MEASURED $5.6\text{e-}28$ to $4.6\text{e-}21$ across `mol_carrier`'s column (B5j's arm had `lo = -huge`, unsound: exp underflows to an exact 0). Stated at the site: the column factor n of the Jacobian (both the banded model and the Krylov action probe a RELATIVE change), the region asymmetric in the density ($n \rightarrow n \exp(+\Delta)$), and $\ln 0$. **A real finding at the arming point:** `carrier_headroom` does not exist when the solve reads its controls (allocated inside `carrier_steady_residual`, first reached by the opening `eval_residual`), so the first implementation silently fell back to the density arm on every coupled run; the opening state is now packed as a density and rewritten as logarithms after that evaluation, sound because the residual is a function of the state and not of its coordinates. Floor array made rank 2 (cell AND carrier row; two carriers of one cell have different elements). MEASURED, one binary, the two arms one variable apart: `mol_carrier` density 63 it ||R|| 1.415, 28 Cauchy-only steps, 10 unknowns on a bound, 716 adopted carriers at the floor; **$\ln n$ 66 it ||R|| 0.9957, merit 0.960, 0 Cauchy-only, 0 on a bound, 0 backward products, 0 at the floor, smallest adopted carrier $2.2\text{e-}11 = 2.5\text{e15}$ times its floor;** `mol_diffusion` (H2 carried + element row) density arm 0 accepted steps of 12, log arm 50 accepted, ||R|| 1.986 $\rightarrow$ 1.982. **Neither converges, and the next obstruction is named and is NOT the density arm's:** the log arm's steps are model-accurate (ratio 0.39-1.09) and SHORT (||s|| 0.09-0.17 against a radius of 2.5 sitting at its ceiling for 30 of 68 iterations), cut by 3-5 halvings before the trial is describable, and the refusing screen is the ELEMENT BUDGET (157 samples, against 0 for a negative carrier): the carrier CEILING, which `species_unknowns_outside_their_bounds` deliberately does not treat as a face of the box. Bertsekas' two-metric projection is not what this arm needs (it holds no bound). Gates: `wasp_full_newton` byte-identical twice (full case control vs new, and the reload identical to B5j's binary including every JFNK line); marching bitwise on `mol_carrier` 200 steps. New rows: default is $\ln n$, switch reaches density, the floor identical in both arms, `carrier_is_above_its_own_floor` (RED on the density arm: 0, 716 at the floor). Advisor: `steady_species_rows` 93/0, `steady_selfconsistent_residual` 3/0, certification 29/0, `physics_probe` 1491/0; the two documents that named the old =1 spelling annotated. Recorded: `gm_m = 40` is a literal at the call site in `EXHALE_main.f90` and the log arm sits at it in 58 of 66 iterations (unmeasured whether raising it helps,

the cycle reports iterations used, not the residual reached); carrier_headroom returns huge as an availability sentinel. **Next item B5I:** the element budget as a face of the unknown box.

- **B5I** (steady_newton.f90 +392/-82; diffusive_photochemistry.f90 +51/-6, the headroom accessor only; steady_species_rows suite +161 and +108; docs/steady_solver_design.md section 21; certification.f90, EXHALE_main.f90, steady_residual.f90, ionization_equilibrium.f90, input_read.f90 untouched): **the element budget is a face of the unknown box and the projection treats it exactly.** freeze_species_unknown_box is now the one place the admissible set of a species unknown is written down (one lower and one upper bound per unknown of the flat vector, formed once per outer iteration from the iterate and from the element budget the carrier operator froze at that iterate); the three routines that had each computed their own bounds (species_unknowns_outside_their_bounds, fix_active_species_bounds, largest_step_inside_the_species_box) read it. **A drift between the three was found and fixed on the way:** the projection had read the carrier ceiling from the trial's own density slot while the other two read it from the iterate, so the set a trial was written onto and the set the Krylov cycle was sampled under were not the same set. Faces of a carrier unknown: below, zero (carrier_log_floor in $\ln n$); above, the cell density and the element budget (carrier_headroom, for H2 half the hydrogen the carriers may take), the tighter of the two. The budget is frozen for the step, not evaluated at the trial, which is the carrier operator's own choice (headroom_car assigned under weno_mode == 1, the one evaluation that IS the iterate; refreshing it per trial was tried there and is worse), so box and screen describe one state. The budget face is raised to the iterate where the iterate stands outside it (counted, n_budget_face_at_the_iterate; 0 on both coupled cases with the default step control); it is stored one floating-point step inside the budget (budget_face_inside = 1 - 8 eps, representability in $\ln n$, not a margin). An unknown whose box is a point (zero budget) is held without asking the gradient. EXHALE_SPECIES_BUDGET_FACE=0 restores the entry text's box (measurement switch, default on) [retired by N4b: the budget became a constraint row, EXHALE_ELEMENT_CONSTRAINT_ROWS=0 is the measurement arm now]. pgmres reports the relative residual it REACHED against the tolerance asked; EXHALE_GM_M overrides gm_m (the caller's 40 stays the default and the literal at EXHALE_main.f90 line 5343 is untouched). **MEASURED on mol_carrier (coupled, $\ln n$), face off vs on: 66 vs 52 outer iterations, both ended by the stall detector; ||R|| handed back 0.9957 vs 1.174; trials halved 3 or more times 38 of 68 vs 0 of 53; residual samples refused by the element budget 157 vs 0; trials written onto a face 0 vs 29; adopted carriers above their own budget 0 vs 0 (largest ratio 0.9886 / 0.9878); iterations ending "the model promises no reduction" 0 vs 19; worst Krylov relative residual reached 0.938 vs 0.569 where 0.1 was asked; log10 Mdot 10.63 both.** The halvings and refusals against the budget are gone exactly; ||R|| does not improve; the limit moved to a named place, the Krylov leg of the dogleg. **The Krylov budget is not the limit:** gm_m 40 / 80 / 160 give ||R|| 1.174 / 1.103 / 1.271, worst relative residual reached 0.569 / 0.630 / 0.614, ||s|| of the last accepted steps 0.04-0.24 in all three, log10 Mdot 10.63-10.64; the cycle spends every product and stops at the same 0.6 of its right-hand side, so it stagnates on this operator (preconditioner or operator, not subspace). On mol_diffusion (H2 carried) the face does the same to the step control (halved 6-11 times in 50 of 50 vs 0 of 41; budget refusals 388 vs 0) and the solve stalls where it always has (H2 front, cell 212, element row at 1.000 of its scale; ||R|| 1.982 vs 1.943; log10 Mdot 7.97 both), with the probe

showing instead 419 backward-side products, 26 unknowns held on a bound, 12 Cauchy-only steps and 18 truncated cycles. A defect in the item's own first measurement was found and fixed (the adopted carrier was recorded against the PREVIOUS iterate's budget, which had reported 67 adopted carriers above budget, worst 1.058; the call now follows the iterate's own residual evaluation, arithmetic unchanged iteration for iteration). **Gates:** wasp_full_newton bitwise against the control binary (Ion_species.txt bitwise, Hydro_ioniz.txt identical but for the provenance line, every (JFNK) line identical, $||R|| = 4.299\text{E}-09$, $\log_{10} \text{Mdot}$ 13.30, 13 iterations); marching bitwise on mol_carrier 200 steps; EXHALE_SPECIES_BUDGET_FACE=0 reproduces the entry-text binary exactly; steady_species_rows 118/0 with the new rows RED without the face and GREEN with it; physics_probe PASSED. **Advisor:** the five diffs match the tree against the worker's entry snapshots; forbidden-word scan clean; the current tree builds warning-free (build_lwv); the worker's wasp_full_newton output against the current golden moves $5.87\text{e}-9 / 9.58\text{e}-9$, exactly the B5j movement already on record, so B5l itself moves nothing on the matrix path. No golden refreshed. **Recorded, not repaired (outside this item's files):** carrier_element_headroom states a box side per carrier but no joint bound for carriers sharing an element (the physical constraint $2 n(\text{H}_2) + n(\text{OH}) + 2 n(\text{H}_2\text{O}) + n(\text{H}^+) \leq n\text{H}_{\text{avail}}$ is a hyperplane the code does not state), and OH/H₂O hold hydrogen nuclei their own headroom does not mention, so a state can pass every headroom and still leave the write-back's hydrogen remainder negative (the branch that zeroes H I, H II, H₂⁺, H₃⁺); neither reaches the {H₂} carrier set both measured cases use. Also: the epsilon-active band (one finite-difference step) never engages on mol_carrier with the face (29 trials projected, 0 held; no switch, a wider band unmeasured); largest_step_inside_the_species_box returns zero for the whole direction when one component has no room (18 truncated cycles on mol_diffusion); note_adopted_species_unknowns is not called for a converged state (pre-existing, cannot affect a stall). **Next obstruction on the carrier arm: the Krylov leg (banded preconditioner or operator), not bounds and not subspace.** Per the user's instruction of 2026-09-09 ("leave running things alone, stop code modification for now") no further item is started; B5m (the atomic element-row solve with no Krylov direction) stays planned.

- **Preparation of PLAN_20260909_rev1 (items N-1 and N17; advisor, documentation and file moves only, no source change; user's go-ahead 2026-09-09 with the instruction not to rerun regression checks after simple changes).** docs/PLAN_20260909.md was written from the review of the issues inventory (docs/ISSUES_20260909_review.md), reviewed (docs/PLAN_20260909_review.md) and revised to docs/PLAN_20260909_rev1.md, the plan of record (items N-1 to N17, stages A to E, decisions 13 to 19). Before writing rev 1 the advisor re-read the review's new source claims: the sampled integration-error estimate is printed and dropped and no production caller issues as_reject_int_error (EXHALE_main.f90 2806-2821); after a retry inside the full pass the half passes start from $0.5 \cdot dt_{\text{step}}$ and the clock advances by the saved dt_{step} (1706-1734, 2797-2799, 2820, 2836); the audit harness asserts the OLD failure (if (ieee_is_finite(x(1))) error stop 4); use_tr = (nspec_row > 0) (steady_newton.f90 5763) so wasp_full_newton never enters the trust region, which corrects a statement of the first plan. All confirmed. **N-1:** the B5m reproducer moved from the session scratchpad into the tree as backup/regression/atomic_elem_newton/ (input, metals, setup and resolved reports, run.log, the two handed-back state files, run.sh, README; 6.8 MB, of which run.log 6.2 MB; not in DEFAULT_CASES); the common worker brief became docs/worker_rules.md with the session paths

generalized and three rules added (one owner per source file per increment; affected-path validation only; named outcomes in reports); the review harness docs/audit_20260905/run_issues_review_20260909.sh was run once on the current tree, its output kept as docs/audit_20260905/run_issues_review_20260909.out (zero-radius step norm 0; Cauchy merit 0.5 to 40.5, line minimizer 0; identity GMRES 1; zero-operator GMRES non-finite with reported residual 0; enthalpy ratio 0.5 on a nonzero mass divergence; He/H pair 8.83e-5; hydrogen inventory 2 at the individual ceilings; outer-shell chord 11.83; "defects reproduced, not repaired"), the harness itself untouched. **N17:** ISSUES_20260909.md corrected per rev 1 section 3 (seed retention no longer stated as an eigenvalue of one; B5f marked a withdrawn, uncertified reference; 3.5 scoped to the present fixed-point map; the restart row now states the $\text{heh_dev} > 1\text{e-}6$ branch and the 8.83e-5 discrepancy; carrier_headroom_known noted; hydro_row_max and use_tr bullets; the _adv statement in section 5 made conditional; new section 3.8 on the physical step never being refused for its integration error). code_status_20260905.md: the Huang et al. 2023 shell-opacity note added to the photoionization row; four pre-existing spelling/forbidden-phrase slips fixed on the way (per-cent to percent twice, two per-molecule phrases reworded). Verification: forbidden-word scans of the four documents; the reproducer's run.sh resolves its default binary path; nothing built or rerun.

- **N16a** (PLAN_20260909_rev1 Stage A, the physical step can be refused and the clock is right; EXHALE_main.f90, attempted_step.f90, src/tests/attempted_step/{run.sh,attempted_step_tests.f90}; user 2026-09-09 "N16a 병행"): **the sampled step-doubling comparison is one transaction over one macro-interval H:** the full pass covers H, the two half passes $0.5\ H$ each and end at $t + H$ by construction (pass 3 no longer inherits whatever dt held); a refusal inside any pass, a finite estimate above one, or a nonfinite estimate rejects the WHOLE macrostep (as_reject_int_error now issued by the production caller), restores as_entry, shortens H by the controller and recomputes both trajectories; exhaustion of the transaction's retry budget (n_macro_retry, cap 8) is a named failure that stops with status 2, names the pass and prints the checksum of the restored state (stop_on_exhausted_retry_budget, one report for both exits). The clock, the accepted count and the physical accumulations move only at the adoption boundary below the macrostep loop. **The order of the complete split update is MEASURED as $p = 1$:** at one fixed state the estimate scales as H^2 over five decades ($p + 1 = 1.964, 1.993, 1.9985, 1.9997, 1.9999, 2.0004, 2.0000$ for H from $2\text{e-}2\ \text{s}$ to $2.6\text{e-}7\ \text{s}$; the coarsest $1\text{e-}1\ \text{s}$ is outside that regime and excluded), so attempted_step_error_estimate now returns the retained full step's error $\Delta/(1 - 2^{-p}) = 2\ \Delta$ (err_order_p, err_full_step_factor, convention at the declaration) [superseded by N16d/N16e: $p = 1$ was the base-face leak; the conservative and electron/T rows are $p = 3$, factor 8/7, the species rows $p = 1$] and the reduction is $0.9\ e^{(-1/(p+1))}$. The brief's first procedure (one interval by 1, 2, 4, 8 prescribed steps, ratios of the output rows) was INCONCLUSIVE, p 0.4 to 4.0, because the third difference ($2.2\text{e-}8$ relative) sits at the coupled source step's fixed-point tolerance: the inner error is not below the temporal error there (review F3's condition, not yet enforced). Test hooks, default off: EXHALE_AS_ERR_INJECT[_COUNT], EXHALE_AS_REJECT_IN_PASS, EXHALE_AS_FIXED_DT (seconds; in initialization mode with local stepping it flattens the local steps, stated at the hook). EXHALE_STEP_CLOCK=1 now also prints an err-pass: line per pass and state= (checkpoint checksum) per accepted step. **Tests:** suite RED on the entry binary 117/16 (the sixteen new rows: passes cover one interval, forced error rejects, clock

unchanged, shorter retry accepted, refusal aimed at pass 1/2/3 keeps the endpoint and names the pass, exhaustion in each pass and on the error is named and restores the accepted state by checksum), GREEN on the final binary 133/0; advisor reran the suite with the worker's final binary (md5 5ea530d3251a8b8ce77e72415ddef0ab): 133/0. **Runs:** mol_carrier marching 200 steps byte-identical apart from the provenance line (initialization mode never sets n_err_passes = 3; the guard lines quoted in the report); the suite's case in physical mode, 60 steps: identical through step 19, at step 20 $e = 1.215$ (raw 0.607, the number the old code printed and dropped) rejects, H 1.118 $\rightarrow$ 0.913 ($e = 1.77$) $\rightarrow$ 0.618 s ($e = 0.89$) accepted, t_phys 66.40 $\rightarrow$ 65.92 s; two rejections in 60 steps, all by the integration error. Not run: the matrix, wasp_full_newton, run_fcheck.sh (out of the affected path; the module compiles clean under the suite's -fcheck=all). No manual entry (the manual carries no run-mode or clock text; nothing in it became false). **Advisor:** file scope as briefed; six British per cent (two new, four pre-existing) changed to percent. **Reported, not fixed (outside the item):** (1) the accepted state of a retried macrostep depends on which pass was rejected, up to $9.1e-5$ relative in the base cell's velocity ($5e-6$ elsewhere), with the checkpoint round trip exact, so some state read by the coupled source step is neither saved nor rebuilt (seeds or caches of ionization_equilibrium, constrained_chemical_equilibrium, diffusive_photochemistry): a rejected attempt is not yet without consequence; (2) one interval of the order sequence (H $6.48e-5$ s from the 0.405 controller reduction) sits 2.3x off the H^2 law the other seven follow, unexplained, for N16b; (3) err_rtol = $1e-3$, err_atol = $1e-12$ are flat, unvalidated and cover no species (N16b); the gate now shortens physical-mode steps by about two at the sampled steps on this case.

- **N0, N3, N1, N2** (PLAN_20260909_rev1 Stage A, one worker, four diffs; steady_newton.f90 +738/-143 over the four, new suite src/tests/krylov_and_dogleg/; no other file). **N0:** 14 rows calling the PRODUCTION pgmres, dogleg_step and the factored step-length and predicted-drop code (a test-operator hook in pgmres in the style of carrier_headroom_set_for_test); RED on the entry text 10/7, among them zero_operator_step_is_finite = F, approximate_gradient_length_decreases_the_model = 40.5. **N3:** seven named linear outcomes (gm_tolerance_reached, gm_subspace_exhausted, gm_no_direction_sampled, gm_jacobian_action_unusable, gm_preconditioner_failed, gm_reduced_system_singular, gm_nonfinite); a zero right-hand side returns before any factorization; hypot rotations; rank tested before the triangular division and the leading nonsingular block solved; LAPACK status and finiteness checked after each preconditioner apply and on the returned step; the TRUE scaled residual verified with one product at a breakdown; the cycle exits at a breakdown. Reorthogonalization not added: worst max $|V_i \cdot V_j|$ MEASURED $6.9e-13$ on the carrier fixture. **N1:** absent elements (reservoir abundance zero, the melem_ab ≤ 0 test set_energy_vectors and load_IC use) are no longer registered as rows; initial_trust_region_radius with named statuses, holding the active bounds before its own cycle; a positive floor tr_delta_min = $1e-6 \sqrt{\text{neq}}$ in scaled coordinates (257x and 1157x below the first radius the two fixtures take); tr_dmax follows the radius; "no feasible descent" is info = 3. **N2:** the length along the direction used, $\tau = (r \cdot A \cdot g) / \|A \cdot g\|^2$, the point named an approximate descent point, the paragraph defending the old length by B5e removed, a dropped Cauchy leg with a Krylov residual ≥ 1 exits as tr_no_direction_reduces_model. Suite after diff 4: 31/0, also under -fcheck=bounds,do,mem. **Root cause of B5m (R1's mechanism, its cause):** the 1500 unknowns on a bound of the atomic element

reload were the three elements `metals.inp` does not carry (Si, K, S x 500 cells), registered with an identity row and sitting exactly on 0, so the fraction-to-the-boundary rule zeroed ANY direction with a component there and the first cycle took 0 products; the radius became 0 and stayed so. MEASURED after (worker and advisor agree): unknowns on a bound 1500 $\rightarrow$ 0; initial radius 0 $\rightarrow$ 1.909e-2 from a Krylov step that reached tolerance (3 products, 6.1e-2 against 0.1); first step accepted, ratio 0.995; $||R||$ 1.888 constant $\rightarrow$ 8.3e-2 at iteration 50, then an oscillation between 6.9e-2 and 1.1e-1 with the merit creeping 7.29e-3 $\rightarrow$ 7.23e-3 over 258 iterations (advisor run stopped by its 20-minute cap; 178 accepted, 80 refused; every cycle spends all 40 products). Not converged: the next obstruction is the model quality of the element rows near $||R||$ 1e-1. **Carrier reload** (control reproduces B5l's run exactly): after, 54/1 accepted, $||R||$ handed back 1.174 $\rightarrow$ 1.151, worst achieved Krylov residual 0.569 $\rightarrow$ 0.575, radius range unchanged, $\log_{10}$ Mdot 10.63 both, outcomes now named (8 tolerance reached, 47 subspace exhausted). **The concurrent N16a edits are inert on every fixture** (paired controls bitwise). **wasp_full_newton** (once, final binary): not bitwise, largest relative movement 2.1e-10 (`Ion_species.txt`, Ca I) and 8.8e-11 (`Hydro_ioniz.txt`), `info` = 0, $\log_{10}$ Mdot 13.30; by code path the hypot rotation of N3 (`use_tr` is false there); below the 0.1% rule, identical, no refresh. `lower_profile` (the only default case with `He_metal_diffusion` and a partial reservoir) bitwise to its golden in both arms (it stops at its marching cap). Other suites (worker): `steady_species_rows` 106/0, `certification` 29/0, `steady_selfconsistent_residual` 3/0; `physics_probe` does not include `steady_newton` (`source_closure.py`). **Reported, not fixed:** `largest_step_inside_the_species_box` still annihilates a whole direction when any unheld unknown has no room on the side the direction points (an element physically zero in one cell can trigger it; the outcome is now named and the radius survives, the direction is lost): an active-set rule decision, for N4b. Two dead locals removed from `pgmres`.

- **Stage A gate (advisor, 2026-09-09).** Tree after N16a, N0, N3, N1, N2. `make OBJDIR=build_lwv EXE=EXHALE_lwv.x` warning-free, md5 dd233d6a (identical to the worker's final binary, so the whole tree is what both workers measured). Suites on that build: `krylov_and_dogleg` 31/0, `steady_species_rows` 106/0, `steady_selfconsistent_residual` 3/0, `certification` 29/0, `attempted_step` 133/0 (with the N16a binary, md5 5ea530d3, same source). N7's first two iterations on the atomic reload logged above with named outcomes. Not run, by the affected-path rule: the matrix, `physics_probe`, `run_fcheck.sh`; the shared build/ and EXHALE.x remain behind the sources. No golden refreshed. Gate PASSED; Stage B (N4a, N5, N8a) and the N7 diagnostic list are next.
- **N13** (`PLAN_20260909_rev1` Stage D, transit refusal census; `EXHALE_transit.py`, `exhale_transit_lib.py`, new Python suite `src/tests/transit_census/`; no Fortran, no output format change, so decision 17 is untouched): the census mask $(r \geq 1) \ \& \ (r \leq R_{ib})$ was geometrically wrong (the ray integral uses every shell with $|r| \geq b$); one function `chord_shell_indices` now serves the census, `resonance_depth` and `resonance_spectrum`, and the census reports, line by line at eight impact parameters and at its maximum, the share of the line-center optical depth carried by rows whose correction was refused (`refused_line_center_tau_share`, the trapezoid rule rewritten as a sum over rows so that shares of disjoint subsets add to one), labeled a CONTRIBUTION, not an uncertainty. MEASURED: every

tpm*.txt of wasp_full (10) and mol_base_handoff (5) bitwise before/after; the defect is latent when Rib = r[-1] (both cases as they stand) and real when the domain passes the stellar radius (wasp_full read at EXHALE_TRANSIT_RSTAR_RSUN=0.30: 419 -> 500 sampled rows, 81 above the cap); on wasp_full the rays near b = 1 take 99 percent of their Ly-alpha line-center depth from uncorrected rows. Suite RED on the pre-change copies (11 rows, the mask present and no shared selection), GREEN 11/0 on the tree; advisor rerun 11/0; +0.3 s on the whole script. Advisor: one per-line in the library header reworted. Reported, not fixed: the wavelength-resolved share and the file metadata wait for decision 17; np.loadtxt(..., max_rows=1) at line 896 warns on every run and duplicates the column count of line 325.

- N12** (PLAN_20260909_rev1 Stage D, the upstream caloric energy at the upstream composition; T_equation.f90, post_process_adv.f90, ion_cell_state.f90 (two fields x_h2_up, e_up), src/tests/adv_static_limit/ (new driver upstream_caloric_energy.f90, rows U0-U4)): the upwind term of the _adv energy residual now receives the upstream cell's SPECIFIC internal energy e_up = E(x_H2, up, T_up)/mu_up formed by the caller, and the caloric branch is taken when EITHER side carries H2 (an atomic cell below a molecular neighbor no longer loses the incoming rovibrational energy); the monatomic branch is verbatim when both sides are atomic. **A second, larger defect found and fixed in the same path:** x_h2_cell was assigned only inside the k == 1 stationarity loop (old 650) and read by the temperature loop for EVERY cell (old 1316), so the whole molecular layer was solved at the H2 fraction of the outermost ghost (1.5e-7 on mol_base_handoff, 2.3e-5 on mol_carrier, against 0.84 at the base); advisor confirmed both lines in the pre-change source. The stationarity diagnostic's lower energy e_lo is likewise evaluated at the upstream composition (the ratio's definition, threshold and operator unchanged). Energy convention unchanged: the caloric EOS carries the bound rovibrational ladder only, the 4.478 eV bond lives in the formation reservoir (molecular_reaction_heat.f90), no term is counted twice. **Tests** RED/GREEN (MEASURED): isothermal H2 gradient residual 1.508 -> 0; atomic cell below molecular upstream 1.006 -> 0; dissociation-layer march against the analytic discrete solution, largest departure 0.155 -> 2.4e-11 (RED marched 1000, 892, 782, 615, 560 K against the discrete 1000, 975, 863, 728, 596; GREEN both 1000, 974.76, 943.62, 880.65, 853.52); controls unchanged. Suite 36/0; advisor rerun on build_lwv (md5 8d027f8b, the tree with N12 and N13) 36/0 (a first advisor run reported one no_output FAIL from a relative EXHALE_EXE path, the advisor's mistake, not the suite's). **Impact** (post-process restart Do only PP: True of each case's own state; the before binary is the Stage A binary): wasp_full _adv files bitwise (atomic, both x_h2 zero, monatomic branch verbatim); mol_base_handoff and mol_carrier _adv move in 469 and 467 of 504 rows, T_adv by up to 94.2 percent (218 -> 424 K at r 1.0941) and 103.2 percent (169 -> 343 K), split about 35/45 and 41/46 percent between the stale-composition and the upstream-composition defects; both a warming of the molecular layer (5/2 kT per H2 instead of 3/2); p_adv, cool_adv follow, heat_adv at most 4.8 percent; adv_status changes in one cell of mol_carrier (row 26, 0 -> 2, from the e_lo composition in the ratio); run states (Hydro_ioniz.txt) bitwise in all three. **The molecular _adv goldens of the matrix will move; refresh at the Stage D gate, reported then, not now.** The writer's warning that molecular and oxygen columns are not advection-corrected stays; the product is still not a molecular stationary solution. Advisor: one per-thread phrase in ion_cell_state.f90 reworted; the worker corrected four British spellings in comments. Reported, not fixed: a stale scalar formed in

a k == 1 block and read later is a class worth a sweep of the file; post_process_adv.f90 is named by both N11 and N12 (one owner at a time; N11 waits for decision 16).

- **N9** (PLAN_20260909_rev1 Stage D, the restart He/H discrepancy; load_IC.f90 +84/-12, new grid_and_gates row restart_round_trip (9 assertions); write_output.f90 untouched, no key, no tolerance change): **the reported failure was a message defect**: the refusal printed the He/H of the TOP physical cell while the branch tests the largest departure over all physical cells; on the failing state (B5j/scan500_stag, a coupled-carrier JFNK iterate of the hot-Uranus gate) the top cell agreed with the input to 17 digits and cell 199 was off by 3.96 percent, so the message read "7.9307E-02 against 7.9307E-02 ... cannot be rescaled". Review R5's 8.83e-5 is the distance between the INPUT values of two different runs, not a property of any state. Fixed: the comparison records the cell it attained and report_deviating_heh_cell prints cell, radius, file and input ratios at ES16.9, the departure and the threshold (heh_dev_tol = 1e-6, now named, value unchanged); RED/GREEN on a state with helium raised by half at cell 199. **The round trip is clean** (MEASURED on the 12000-step mol_base_handoff state with EXHALE_DUMP_IC=1): r, v, p bitwise; every species column and every inventory (H, He, O, C nuclei, charge, He/H, H2 ratio) within 4.2e-16; T 5.6e-16; the rho column 4.9e-12 because the loader never reads it and rebuilds mass from the species (perturbing the column by 1e-15 changes nothing, bitwise), so 4.9e-12 is the writer state's own mass-closure defect (4.1e-14 on a 40-step state); a second trip 5.0e-16, no drift; no clock advanced. **The producing path moved the ratio, not the loader** (MEASURED): He/H departs from the input by 1.0e-11 and 8.9e-12 after 12000 marching steps with carrier transport off and on, but by 3.9 to 7.9 percent on every coupled-carrier density-unknown JFNK state and 41 to 43 percent on the ln n states (B5j, B5k): the carrier column is written after the element projection precisely so the Newton's carrier is not rescaled (write_species_rows_into_composition 1439-1469), the sweep reads the nuclei totals from the state (hydrogen_helium_nuclei_density), and with helium diffusion off no row re-imposes He/H. This is the element constraint N4a/N4b are about, seen from the restart side; reported, not touched. Also fixed on the way: parse_labels had no bound on 80-long label arrays (a file with more than 80 columns wrote past them); it now stops with a message (present schema at most 42). Atomic restart path MEASURED unchanged: wasp_full_newton's state reloaded before/after bitwise, no He/H line printed. Advisor: restart_round_trip 9/0 on build_lwv (conda gfortran; the worker's private build used /usr/bin gfortran 13, against the standing toolchain rule, so the advisor's run on the standard toolchain is the one on record). **Reported, not fixed**: the window integrals of the assembled residual are not Lipschitz in the state in the odd-even band above the base (one round trip at 4.9e-16 moves the 1.03-1.10 mass window by 16 percent and flips cell residual signs at cells 183-187; one ulp on every species column reproduces it), so a round-trip residual budget must be stated on the row maxima (N5); restart_grid_guard.sh carries a stale header describing a fixed defect; nothing checks the writer's rho column against its species (decision 15's contract); EXHALE_RESIDUAL=1 runs one sweep unconditionally, so a true "no chemistry on load" evaluation has no entry point yet (N10).
- **N7** (PLAN_20260909_rev1 Stage C, the atomic element-row diagnostic after Stage A; steady_newton.f90 +477/-0, diagnostic prints under EXHALE_ELEM_DIAG=1 (iterations 1, 2, 60; =n moves the third), one extra Jacobian product per selected iteration with the refusal statistics held aside; no arithmetic of the step changed,

MEASURED: every (JFNK) line identical hook off/on over 2027 lines, across four builds). **Verdict, MEASURED on the reload fixture:** the oscillation at $||R||$ $7e-2$ to $1.1e-1$ is NOT a model failure (predicted/actual ratio 0.9946, 1.0001, 1.0003 at iterations 1, 2, 60; the refused iteration's model is right to $3e-4$), NOT a bound effect (0 unknowns held, 0 samples refused in 293 iterations), and NOT limited by the linear solve although the solve fails its tolerance (40 of 40 products, achieved 0.6685 against 0.1, TRUE residual 0.6685 by one extra product, Arnoldi gap $2.3e-4$; the dogleg takes $2.1e-4$ of the Newton leg so that inaccuracy is second order). **Proximate cause: the ray test** (the finite-difference check of the model slope) refuses the step: at the refusing iteration model slope $-9.13e-9$ against ray slope $-1.48e-9$ (factor 6.2 against the allowed 4) with $|f2p - f2m| = 3.0e-12$ only 11x above the floor $eq_sweep_reltol * |f2| = 2.7e-13$ (at iteration 1 the same test has a factor 1371 of room and the slopes agree to 0.5 percent); 82 of the 91 refusals in 293 iterations are this test, 3 its sign form, 6 a real over-prediction. Limit cycle read off the radius: $9.78e-4$ accepted $\rightarrow 1.96e-3$ accepted $\rightarrow 3.91e-3$ refused by the ray test $\rightarrow$ cut to $9.78e-4$, period three; the merit falls $1e-4$ of itself per accepted step, so $Resid\ tol\ 1e-8$ is seven decades away. **Underlying cause: the merit and the gate are different functionals.** At iteration 60 the eight element rows hold 0.437 of the squared merit (He 0.305) and NOTHING of $||R||$ (energy row holds 1.000 of $||R||$, 0.560 of the merit); their own certification measures sit at $1.2e-3$ to $7.4e-3$ against $cert_tol_element = 1e-8$; accepted steps swing $||R||$ by 40 percent ($7.8e-2$ to $1.09e-1$) while the merit stands at $7.291e-3$ to four digits. At iteration 20 the brief's option (c) is met outright: $||R||$ 1.400 set entirely by the mass row, the energy row 0.810 of the merit, seven of eight elements worst at cells 12 or 17. **The Cauchy leg is worth nothing here:** $||sU||$ $6.6e-9$ against a radius $3.9e-3$ ($g = B^T r_0$ from the band, $||A\ g||$ $1e4$ times $r_0 \cdot A\ g$), predicted decrease $1.3e-9$ against the dogleg's $7.6e-9$; one product per iteration for $1.7e-6$ of the step. **The band is not the defect** (0 zero, nonfinite or round-off columns, $dbgtrf$ info 0), though it conditions worse along the solve ($dbgcon$ $7.3e-8 \rightarrow 9.4e-10$, smallest pivot always a trace element of cell 500). **gm_m probe:** unlike the carrier arm, the larger subspace changes the achieved residual on this operator (first divergence at iteration 17: 40 of 40 at 0.1035 against 48 of 80 at 0.09999; deep in the stall 0.67 against 0.29) and the solve stalls WORSE ($m = 80$ plateau $||R||$ 0.27 to 0.34, merit $1.57e-2$, against $7.7e-2$ to $1.1e-1$ and $7.3e-3$ at $m = 40$): longer steps lift the ray probe off its floor and the refusal becomes an honest over-prediction, and the solve still goes nowhere. **Proposals, not implemented:** P1 measure the ray test's floor from two evaluations of one state instead of assuming $eq_sweep_reltol * |f2|$; P2 do not let the slope comparison overrule a reduction ratio within [0.998, 1.001] (every ray-size refusal inspected had one; the over-prediction guard stays live); P3 drop or rescale the banded-gradient leg where $||sU||$ is below a small fraction of the radius (one product saved); P4 a GMRES restart buys $m = 80$'s reduction at $m = 40$'s memory but does not fix the stall; **P5 the acceptance functional and the merit must be reconciled** (a merit that reads the element rows on their certification scales, or a separate control for them): a design decision, decision item 20 (section below). Also fixed in the same file: the `tr_exit_text/gm_outcome_text` buffers (48 and 64 characters) truncated two 67-character messages; all three are 80 now. **Reported, not fixed:** the worst `j, k` on the (JFNK) `it` line is the merit's largest scaled row while $||R||$ on the same line is a certification maximum, and the two need not be the same row (iteration 20: $||R||$ is the mass row's, worst is an energy cell); the brief's belief that `Solver: Newton 100.0` caps the iterations is wrong (the third word is `newton_switch`; the JFNK cap is the literal 500 at `EXHALE_main.f90` 3509 and 3000

on the direct route, no key or hook; runs were capped by wall clock); `atomic_elem_newton/README.md` is right but incomplete (the refused trials have a correct model). Advisor: rebuild warning-free, `krylov_and_dogleg` 31/0 on the tree with N7 and N9.

- **N4a** (`PLAN_20260909_rev1` Stage B, one elemental constraint map with three contexts; new `src/modules/lower_atmosphere/element_inventory.f90` (745 lines, one SRC line in the Makefile), `diffusive_photochemistry.f90` (its three element densities, hydrogen budget and five carrier box sides now derived from the map with unchanged arithmetic), `steady_species_rows` suite, `docs/element_inventory_contexts_20260909.md` corrected; `steady_newton.f90` untouched): **the map reports, nothing enforces it yet**, so the Stage B gate may not claim from this item that trial construction, chemistry and certification enforce one constraint (N4b). `nuclei_per_particle` is the one place a stoichiometric coefficient is read (`bsp_nH/nHe/nO/nC`, `mion_elem`); `element_inventory_of_cell` gives the full inventory grouped carried/locked/reserved/closure with the budgets and the shared-constraint breach as a MAGNITUDE (against the budget the sides spend, and against the budget the chemistry closes with, separately); `element_inventory_of_candidate` evaluates a trial's carriers against the frozen element densities; `EXHALE_INVENTORY_REPORT=1` prints it at the operator's entry. **Physical judgment:** the separate box sides are a wrong feasible set by stoichiometry (two carriers of one element spend one budget); a cell at both ceilings breaches by exactly 1.0 of the available hydrogen (MEASURED). **Seven corrections to the advisor's draft**, each checked in the source: `nO_free/nC_free` include the neutral stage while `cbg_nOion` excludes it (the F7 gap as a number: the oxygen box side spends the element total while `carrier_source` closes against the total less the ion stages, so the box admits states whose source rows clamp free O at zero; N4b's feasible set); `nH_avail` is floored; `H I` and `He I` are the closure, not rows; `CO` is fixed data of the local solve (its destruction row is the carrier operator's); **the shared sums ARE enforced on the marching path by `limit_to_element_budget`** (what has no shared statement is the coupled stationary solve, whose `eval_residual` compares a COUNT of cells); **`max(rest, 0)` in the write-back CREATES nuclei** (with `rest < 0` the carriers keep what they hold and the closure cannot go negative, so the written element total exceeds the handed one by `|rest|`; only `element_census_verify` at `EXHALE_ELEMENT_ASSERT 1` or `2` sees it, off by default); absent elements are no longer registered (N1). **Tests** 149/0 (advisor on `build_lwv`: 145/0, the four coupled-log rows unset), RED shown by `EXHALE_INVENTORY_SHARED=0` (the three infeasibility rows fail in that arm); all nine molecular goldens classified: worst carrier breach exactly 0.0, tightest hydrogen occupancy 0.96 to 0.99 (oxygen_chemistry occupancy 1.0 is carbon's, 175 of 500 cells with all C in CO). **The N9 finding caught live:** inside a coupled carrier JFNK the He/H departure from the reservoir grows $1.4\text{e-}11 \rightarrow 4.9\text{e-}5$ (third iterate, the first with a Newton $n(\text{H}_2) \rightarrow 1.1\text{e-}1$ (thirteenth) while every carrier budget stays at breach 0.0, so the fix belongs at the write-back of each iterate (the element row or a projection there), not at the end of the solve; the two saved coupled states have carrier breach 0.0 and He/H departures $4.0\text{e-}2$ and $6.7\text{e-}1$. **Marching:** `mol_carrier` 200 steps byte-identical (two independent before/after pairs); `mol_diffusion` 12000 steps byte-identical over the 7511 steps both arms had covered at report time (step trace character for character, the step-7001 snapshot files byte-identical apart from provenance); the pair finished later in the session scratch and the advisor compared its end: `Hydro_ioniz.txt` and `Ion_species.txt` at 12000 steps IDENTICAL apart from the provenance

line, $\log_{10}$ Mdot 10.57 in both arms. Reported, not fixed: element_ratio_gate = $1e-9$ now exists in two modules because element_census_tolerance() is private. Rule breach reported by the worker: one pkill -f on its own shell pattern (it matched only its own shell; the other workers' runs were verified alive). Advisor: five British percent in the two files fixed; the advisor's earlier pgrep for the worker's binary name missed its renamed scratch copies, a false "finished" the worker corrected.

- N7b** (N7 proposals P1 and P2, and the JFNK iteration hook; steady_newton.f90 +249/-40, krylov_and_dogleg +12 rows): **P2 opens the cycle; P1 is implemented and MEASURED inactive, and the measurement refutes N7's reading of the refusals.** The ray test's decision is extracted into ray_slope_verdict (public, tested), its floor is $\max(\text{eq_sweep_reftol} * |f2|, \text{ray_floor_factor} * \text{measured_reproducibility})$ with the reproducibility measured once per outer iteration by two evaluations of the merit at one state (cost one residual evaluation per outer iteration, +1.5 percent on the carrier reload): MEASURED $1.8e-15$ at a merit of 14.95 and 0 to $4e-19$ at $2.7e-5$, four to seven decades BELOW the assumed part ($2.6e-13$), which therefore governs everywhere; the refused trials' $|f2p - f2m|$ ($1e-10$ to $3e-10$) stand two to three decades above the floor, so they are real differences and NOT the probe near its cancellation floor as N7 and ISSUES 3.1 (d) said (corrected there). What they are: at the refusing iterate the ratio is 0.999 while the model slope ($-5.5e-7$) and the ray slope ($+1.7e-7$) have OPPOSITE signs; the model slope is set by the banded-gradient leg, which is not in the step ($|s|$ $6.6e-9$ against a radius $3.9e-3$) but carries $|A_g|$ $3.1e3$ into r_0 . A s, so the slope test is right that the slope is wrong and the leg is what is wrong (N7 P3). ratio_sound_band = $1e-2$: within it the ray verdict is formed and printed but not acted on; tr_ratio_below_acceptance untouched. EXHALE_JFNK_MAXIT read once per solve, resolved by jfnk_outer_iteration_cap, printed with its origin. **Atomic reload, 150 iterations, before -> after:** accepted 106 -> 129, ray-size refusals 35 -> 1, ratio-below-eta refusals 6 -> 15 (the guard takes over), radius $9.8e-4$.. $9.8e-2$.. $2.7e8$ (no longer binds; the accepted step is the interior Krylov point at $|s|$ $1.0e-5$, ratio 0.72), $|R|$ $7.27e-2$ -> $6.92e-2$, merit $7.26e-3$ -> $7.08e-3$, achieved linear residual still 0.69 against 0.1 with 40 of 40 products: **the limiter is now the linear solve and the model behind it** (N7 items 4 and 7, P3/P4), not the step length. No convergence claimed. **Carrier reload:** every numeric (JFNK) line identical; 20 of 55 iterations had carried a STALE refusal label ("the model promises no reduction" printed on ACCEPTED iterations because refuse was reset only inside .not. model_ok; pre-existing, fixed): earlier reports quoting "iterations that ended on N: the model promises no reduction" counted that defect wherever N exceeds the refused count (B5I's 19 and 20 included). wasp_full_newton BITWISE on all ten output files (three log lines differ: two wall clocks, one new cap line). Tests RED on the extraction stage (4 of 43 fail: the measured floor rows and the sound-ratio rows), GREEN 43/0; advisor rebuild warning-free, krylov_and_dogleg 43/0, binary md5 6d4bf63e identical to the worker's final. Two dead locals removed (md5 unchanged). Advisor: four British neighbor spellings in comments fixed. Advisor decision: P1's measured floor stays (correct in form, one evaluation per iteration). **Reported, not fixed:** the ray test runs before the acceptance test, so a catastrophic over-prediction (ratio -533, -1493 at iterations 47, 50) is named a slope disagreement; ray_h = $1e-3$ is not a measured interval; tr_dmax doubles without an absolute cap and with P2 runs to $2.7e8$ (harmless, the interior point is taken, but delta stops being a diagnostic); n_tr_exit has no index guard.

- **N5** (PLAN_20260909_rev1 Stage B, the residual contract; steady_newton.f90 +877, ionization_equilibrium.f90 +274/-6, EXHALE_main.f90 replay driver +134/-60, src/tests/residual_determinism/ rewritten with a suite run.sh (cases: the two in the tree plus EXHALE_RESID_RELOADS=<dir>[:<dir>], binary EXHALE_RESID_EXE)): **the old determinism suite had a hole**: it replayed the THREE-unknown residual in every configuration (0 of 1500 entries, the species-row registry is empty before a solve); the replay now registers the rows and returns 0 of 1500 / 2000 / 2500 / 5500 entries differing (3, 4, 5, 11 unknowns per cell) after three trials, a discarded trust-region trial and a jv_product probe with the active bounds held (**contract 1 GREEN on four configurations**; a rejected macrostep is not reachable from the driver, the one part unmeasured). **Contract 2, conditional closure convergence, MEASURED and it exceeds the certification tolerances**: the judged row maximum moves, per unit relative change of the eliminated partition's seed, by $4.2e2$ cert_tol_carrier on the carrier reload, $3.1e1$ cert_tol_element and $1.7e1$ Resid_tol on the atomic element reload, $6e-8$ on the element + carrier reload; a seed above $2.4e-3$ (carrier) or $3.3e-2$ (atomic) moves the judged row past its tolerance; the retention of a scaled seed at the fronts is ABOVE one (1.39, 3.69, 3.46) where B5h had 0.77 at the base. Two suite rows stay FAIL by design as the standing measurement (closure_spread_within_the_row_tolerance_{atomic_elem_newton, det_car}, 61.5 and 8.5 against < 1). **Contract 3, branch discovery** (EXHALE_RESID_BRANCH_REPORT=1, roots grouped by a branch-datum distance $1e-1$ anchored on B5h's two roots); with H2 transported B5h's two-root H2 front does NOT reproduce (x(H2) at the front bitwise identical over 10 seeds, the transported value replaces the seed in the H2 row); the H I partition shows one root on the carrier reload and TWO on the element + carrier reload (9 and 1 seeds, $1.05e-1$ apart); whether the second is a distinct branch or a slow iteration is N6's question. **Chemical decay against transport** on the resolved gradient length: H2 $t_{adv} * \lambda$ 3.68 (base), $4.0e-3$ (front), 0.82 (wind); H2+, H3+, HeH+ 3 to 12 decades faster than transport everywhere, so the transported-H2 / eliminated-ion split stands and no promotion is proposed (conservation modes are removed by the parameterization itself: the neutral stage closes each element). **Rounding floor of the species rows in the judged measure is exactly epsilon** ($2.2e-16$) against tolerances $1e-8$, 7.65 decades apart, condition numbers 2 to $1e3$: compensated summation (the Al-Refaie et al. 2024 reading) is not a lever. Validation: krylov_and_dogleg 43/0, steady_species_rows 145/0, mol_carrier 200 steps bitwise, wasp_full_newton bitwise on all ten files (one new log line, N7b's cap). Advisor: rebuild warning-free, md5 913e6579 identical to the worker's final; krylov_and_dogleg 43/0; residual_determinism 5 PASS / 1 FAIL, the FAIL being the atomic closure-spread row by design (a first advisor run with the wrong variable name EXHALE_EXE tested the stale shared EXHALE.x, the advisor's mistake); two British percent in ionization_equilibrium.f90 fixed, the worker fixed five neighbor. The worker's session was cut once by the Opus limit (09:50 KST) and resumed; nothing was lost. **Reported, not fixed**: a discarded evaluation restores n_eq_sweeps_last, so a caller reads the previous state's pass count; report_accepted_cell_state writes inside the parallel region without a lock; carrier_diffusion_coefficient reads the marching operator's pct_Dco, which the stationary residual never fills (a pure stationary path would read zero; diffusive_photochemistry.f90); the global count shadows the intrinsic.
- **N16b** (PLAN_20260909_rev1 Stage E brought forward, species-aware integration error; attempted_step.f90

+546/-28, EXHALE_main.f90 macrostep code +125/-7, attempted_step suite +289): the step-doubling estimate reads four row classes on their own scales (conservative rows; transported species fractions, the f_sp columns carrier_state builds fc from: H+, H2, OH, H2O, CO; element mass fractions under He_diffusion; electron fraction and T through the caloric EOS), the species scale being the species' own value with an absolute floor of element_ratio_gate (1e-9) of the largest density the species can reach in the cell, read from the N4a map's stoichiometry; the flat 1e-12 no longer judges any species row (a carrier at 1e-20 changed by 1 percent: e = 20 on its own scale, 2e-10 on the flat floor). **The inner nonlinear error is collected cell by cell** (the coupled pair's safety-factored geometric factor on the cell's own increment, and the carrier substep's row residual) and the estimate carries the interval e_lower .. e_upper the three inner errors leave it in: reject when above one at the bottom, accept when below one at the top, UNRESOLVED in between (named outcome, counted in the end-of-run block); the brief's fraction rule for each class (1/10) is kept as the report, not as the verdict, because applied literally it called EVERY estimate of a healthy run unresolved (species difference 6e-17 five decades below the pair's own error) and would have made F1's rejection inert. **p = 1 confirmed on every class** (atomic case, eight levels over five decades: species p + 1 1.9987, 1.9799; electron/T 1.9943 to 1.9997; conservative 1.9809 to 2.0004; molecular case coupled ladder H = 5, 2.5, 1.25 s: species 1.949, 2.299, electron/T 1.912, 1.983), so the retained-step factor 2 holds for all rows. On the molecular case at H <= 0.4 s the species class reports itself unresolved (inner over difference 2.7e5); review F3's condition, MEASURED and detected. **The 60-step physical run is unchanged to 17 digits** (t_phys, checksum, both files): the conservative rows still decide it. mol_carrier marching bitwise. **N16a's 9.1e-5 state leak IDENTIFIED** (not fixed, files not owned): Apply_BC.f90 base_face_W and base_face_lower_W (lines 48, 50; written only by Apply_BC, read by Rec_BC in the first Reconstruct of a retaken pass BEFORE the pass's first Apply_BC, so the base face's left state is the refused attempt's) and caloric_eos.f90 caloric_mixture_active, nk_per_mass, x_h2, molecular_cell (125, 145-147; written only inside get_species_densities, which rebuild_state_from_checkpoint calls AFTER U_to_W, so the restored p and T use the refused attempt's caloric composition); the two reinforce each other at the base cell, which is the measured signature (v 9.1e-5, T 6.8e-8, cooling 1.7e-5). Row rejected_pass_leaves_no_state_behind stays RED at 9.1162e-5, labeled with the files and lines. N16a's 2.3x outlier reproduced and bounded (only at controller-formula levels, conservative and charge classes, always the base cell; 0.665 and 0.53 of the H² prediction), not explained; two candidates named (the unbounded inner error of the mass/momentum rows through the second half's coupling; the base-face leak). err_rtol, err_atol unchanged (the CFL step 20 would need 1.21e-3; the estimate RISES at the first shortened interval, a second reason not to anchor on this case). Tests 155/1 (the one FAIL by design), RED on the entry text 150/6 (an estimate that was the solvers' own error rejected ten macrosteps there; now 0, named). Advisor: suite rerun with the worker's final binary (md5 7bd041dd) 155/1; four modelled in EXHALE_main.f90 comments changed to modeled. **Reported, not fixed:** the mass and momentum rows carry no inner error; the carrier substep's number is a row residual, not a solution error (conditioning unmeasured); NCO_col, k_co_diss, theta_co_shield are outside the checkpoint by an unstated exception (read only after their write within a sweep).

- **N16d** (the checkpoint's restore is complete; EXHALE_main.f90 rebuild_state_from_checkpoint 4 code lines, comments in Apply_BC.f90, caloric_eos.f90, attempted_step.f90, the suite row extended to both

files): **closed by rebuilding, not saving.** The five leaking quantities (base_face_W, base_face_lower_W; caloric_mixture_active, nk_per_mass, x_h2, molecular_cell) are functions of (u, f_sp), so the rebuild now runs `rho = u(1,:)`, `get_species_densities`, `Apply_BC`, then `U_to_W` and `comp_T_from_p` (composition first because the pressure map and the boundary read it; boundary second so the face states are the restored state's). Every one of the six call sites follows a restore, so no restore-free path can move. **MEASURED:** rejected_pass_leaves_no_ state_behind 9.1162e-5 -> 0.0 bitwise on both files; suite 156/0 (advisor rerun with the worker's final binary md5 abdf17f7: 156/0). **The fix changes the measured order of the conservative rows:** the full-versus-two-half difference was 40x the real splitting difference (the leaked base face carried an $O(H)$ error into one face of one RK stage of an $H/2$ step, an $O(H^2)$ contribution) and now falls as H^4 (p + 1 3.9966 to 4.0104 over the ladder), the local-error order of the three-stage Runge-Kutta; the electron/T rows follow it; the transported species stay H^2 (unchanged to all printed digits). N16a's 2.3x outlier is **EXPLAINED:** the two controller-formula levels now sit on the H^4 law to 0.1 percent; the leaked base face carried the difference of two intervals whose ratio was not the ladder's. **Consequence not yet fixed (N16e, launched):** `err_order_p = 1` was calibrated on the leak; the conservative and electron/T estimates carry a factor 2 where 8/7 is right (inflated 1.75x) and the reduction exponent is -1/2 where -1/4 is; the species class is genuinely $p = 1$, so the order is a property of each class. `mol_carrier` marching 200 steps bitwise (0 refused attempts, no restore); with one injected refusal the initialization-mode state moves $2.1e-5$ (v at r 1.0185) and $1.2e-5$ (He 2^3S), the fix of the same defect with both channels live (atomic hydrostatic_column had the base-face channel alone). `wasp_full_newton` not run (no restore unless an attempt is refused; whether it has zero refusals was not measured). The reload route does not share the rebuild (checked). Advisor: two Per-cell phrases in `caloric_eos.f90` reworded. **Reported, not fixed:** the same composition-before-pressure ordering stands at three other sites of EXHALE_main.f90 (1075-1078, 5162-5166, 5312-5326) and `upmap_bc_roundtrip` measures an identity its comment says is not one (N16e); the added `Apply_BC` adds one boundary evaluation per restore to the run-total diagnostic counters.

- **N20** (decision 20 a with N7's P3; `steady_newton.f90` +314/-84, `krylov_and_dogleg` +13 rows, `steady_species_rows` +1 row; `certification.f90` unchanged, `cert_scale_floor` was already public): **one functional, implemented exactly, with a measured consequence that is a decision, not hidden.** `merit_row_scale_from_certification` is the one denominator a species-row-route row is read through; `cell_row_scales` builds the hydrodynamic rows through `residual_row_scale(k,j,u)`, the element rows through their certification scale, the carrier rows through the term sum the certification divides by; base cells of element rows keep the unknown's own scale (Dirichlet reservoir); the three-unknown route is untouched by construction ($Dr = D$ when `nspec_row <= 0`). **MEASURED:** on both reloads every element row's largest scaled merit row EQUALS its certification measure at the same cell and the merit's largest scaled row IS $||R||$'s row (before: factor 2.5 apart at iteration 1, 112 at iteration 60). **P3:** the banded-gradient leg is dropped from the step AND its image when shorter than `cauchy_leg_min_fraction = 1e-3` of the radius (measured $1.4e-5$ to $8.7e-7$ on both fixtures, so everywhere), so the slope handed to `ray_slope_verdict` is the step's own; products saved: zero, because `tau_c` needs `A g` (a lazy leg would save one per iteration, P-N20-1, reported). A quotient verdict at `ratio <= eta_accept` is named `tr_ratio_below_acceptance`. `tr_dmax` capped at $1e3 \sqrt{\text{neq}}$ ($7.4e4$

on the atomic fixture; binds on neither fixture now, would have bound the 2.7e8 of N7b). **The consequence:** the carrier reload improves 2.6x (best $||R||$ 1.151 $\rightarrow$ 0.446, both arms on the stagnation detector, +16 percent evaluations, `dgbcon` 5.4e-10 $\rightarrow$ 1.9e-11) while **the atomic element reload no longer solves:** the certification row scales span 11.4 decades (3.7e-10 .. 92.7) against 8.3 before, `dgbcon` 7.3e-8 $\rightarrow$ 8.9e-10 at iteration 1, the Krylov point collapses to $||s||$ 2.5e-15 inside a radius 19.6 at iteration 11 and the solve aborts on the no-descent watchdog at $||R||$ 1.777 (before: 7.38e-2 at 150 iterations). A third arm (element/carrier rows on certification scales, hydrodynamic rows on the state scales) does not collapse but is worse than the entry text on both fixtures (0.741, 1.026): the hydrodynamic reconciliation is what helps the carrier arm and kills the atomic one. The merit's row scaling IS the operator's ($r_0 = F/D_{row}$, $A = D_{row}^{-1} J D$), so the remaining freedom is the column scaling D and the preconditioner: N8a's, launched with the atomic collapse as its RED. Tests RED on the entry objects (the element-row scale row 4.0 against 1.0; the ratio-naming row), GREEN `krylov_and_dogleg` 56/0, `steady_species_rows` 145/0, certification 29/0; `wasp_full_newton` BITWISE on all ten files (two wall-clock lines). Advisor: rebuild warning-free, `steady_newton.o` md5 6f3aef50 identical to the worker's, the three suites 56/0, 145/0, 29/0. Two pre-existing defects fixed (a dead local; [diag 7] printed at `cell 0` for the carrier). **Advisor decision:** (a) stays in the tree; the atomic arm's collapse is a conditioning problem of the scaled operator and is the first deliverable of N8a, not a reason to give the step control a functional other than the gate's; the state of the atomic arm is reported as WORSE than before N20 until N8a lands. Reported, not fixed: EXHALE_main.f90 1182 should pass `u` to `cell_row_scales`; docs/p54_base_layer_mass_flux.md 10.4 now has a second measurement behind it.

- **N16e** (the order of each row class is its own integrator's; `attempted_step.f90` +106/-50, EXHALE_main.f90 +60/-21, `attempted_step` suite +75/-15): `err_order_p(n_err_class) = [3, 1, 1, 3]` (hydro, species, element, charge/T) with the measured ladders at the declaration (conservative 3.9966 to 4.0104; electron/T 3.9962 to 3.9867; species 1.9987, 1.9799; the element class is assigned the species' order by construction, said so); `err_full_step_factor` is a class's own (8/7 for the RK3-carried rows, 2 for the first-order split ones), the estimate records the deciding class (the one whose lower bound reached the gate) and `attempted_step_reduced_dt` uses that class's $-1/(p+1)$. MEASURED: the conservative and electron/T estimates fall by exactly 4/7 (inflation 1.7500004 removed), the 60-step physical run's trajectory does not move (N16d had already removed the step-20 rejection, so the brief's expected "1.21 rejected $\rightarrow$ 0.69 accepted" belonged to N16b's entry text, not to this item's; stated); on the forced-error ladder the controller with exponent -1/4 asks for 0.846 and 0.226 of the interval where -1/2 asked 0.601 and hit the 0.2 floor, so the floor no longer does the sizing. Stale sites: EXHALE_RESIDUAL=1 and `physical_handoff_check` NOT reordered, the composition cannot have moved there (`init` and `equilibrate_loaded_composition` both end with `get_species_densities`), the invariant is now stated at both; `update_map_begin_step`'s install branch reordered (composition before `U_to_W`; EXHALE_UPDATE_MAP on `mol_carrier` moves 0.0); `upmap_bc_roundtrip` renamed `upmap_bc_interior`, required exactly zero (`Apply_BC` writes only the ghosts), MEASURED 0. Tests 163/0 (156 before; the unit rows cannot be run RED against a scalar `err_order_p`, the RED is the entry suite's own contradiction, $p = 1$ asserted and $p + 1 = 4$ measured in one run; the whole-binary row 0 of 2 $\rightarrow$ 2 of 2). `mol_carrier` marching bitwise. Advisor: suite rerun with the worker's final binary (md5 0ae8d8d9) 163/0. **Reported, not fixed:**

update_map_begin_step's install branch owes the next Reconstruct an Apply_BC (same family as N16d, a diagnostic route); err_atol = 1e-12 still floors the conservative rows on a code-unit density; the mass/momentum inner error is still unbounded (the deciding row's interval is a point); no fixture has the species class deciding a step, so the exponent switch is covered by unit rows only.

- N8a** (PLAN_20260909_rev1 Stage B, scales, boundary equations, derivatives and the inner budget, with N20's atomic collapse as its RED; steady_newton.f90, krylov_and_dogleg +17 rows, steady_species_rows +2 rows; certification.f90 and steady_residual.f90 unchanged, no tolerance moved): **the collapse was one exactly empty Jacobian column, not the spread of the row scales.** At iteration 11 of the atomic reload the carbon unknown of cell 500 had been driven to zero, its column scale sat on the absolute floor 1e-20, the probe step $\sqrt{\text{eps}} * 1e-20$ underflowed the residual difference, the column was exactly 0 and the banded preconditioner singular (dgbcon 3.98e-27, smallest pivot 3.7e-19); the 2.5e-15 Krylov point is what a triangular solve with that factor returns. Iteration 1's 8.9e-10 was only the 7.3-decade spread of the certification row scaling (no empty column). **Two repairs, the model untouched to the last bit** ($r_0 = F/D_{\text{row}}$ and A unchanged, MEASURED by the scaled column norms): (a) element_unknown_column_scale = $\max(|x|, 1e-4 | \text{reservoir} |, 1e-20)$, the arithmetic at the function (the column is resolvable while the probe step stands far above $\text{eps} * X_{\text{reservoir}}$); (b) equilibrate_the_preconditioner: two-sided equilibration of the band that is FACTORIZED only, powers of two, banded_preconditioner_solve applying C dgbtrs(R v). Attribution (first solve, accepted/refused; empty columns at it 11; dgbcon at it 11): entry 14/16, 1, 4e-27; floor alone 69/16, 0, 4.9e-11; both 114/36, 0, 1.1e-9. **Atomic reload (EXHALE_JFNK_MAXIT=150):** entry abort at $||R|| 1.777$ (no descent, 30 iterations); floor alone 1.189 (stagnation, 86); both **3.800e-2 at the 150 cap, still descending** (2.5e-1 to 3.8e-2 over the last thirteen iterations), dgbcon at iteration 1 8.9e-10 $\rightarrow$ 1.65e-6, the brief's target (the pre-N20 7.4e-2) passed by 1.94x with one functional kept; refusals by name shift from ray-floor to ratio-below-eta (35) and one slope-size. **Carrier reload** 0.446 $\rightarrow$ **0.1513** (2.95x), dgbcon 1.9e-11 $\rightarrow$ 8.7e-6, +30 percent evaluations. The model's own column equilibration (EXHALE_COLUMN_EQUIL=1, default off) was built, MEASURED and NOT adopted: it moves the trust-region coordinates, the radius runs to the cap and trials ascend by 4e4 times the predicted decrease. **First cell rows declared once** (species_row_base_equation: element row = Dirichlet reservoir, READ from element_transport_residual; carrier row = control-volume balance, READ from carrier_residual's $j == 1$ branch; no equation changed). **Base derivatives:** the directional-derivative sweep at the base is a clean V with its minimum 4.8e-8 at $\sqrt{\text{eps}}$, the step build_banded_jac_full takes; whole-system rows 8.2e-8. **Closure map at the base** (measure_the_closure_map_at_the_base, four evaluations once per solve): the sweep exits after ONE pass at 9e-12, four decades inside the 1e-8 asked, so the reference is a pass count; quotients 2.1e-4 (atomic, attenuation) and 5.34 (carrier); outer_residual_closure_target (what the outer residual may keep, what the ray floor reads) is now separate from eq_sweep_reltol (what the sweep is asked = target / measured amplification; 1.87e-9 on the carrier reload). **B5h's ~4e3 base amplification is not reproduced on either fixture** and is not claimed. Manufactured states in krylov_and_dogleg (columns spanning eight decades, pivot ratio 5e-18 $\rightarrow$ 0.54 after equilibration; exactness rows: the map and the unknown-space step unchanged to the last bit). Tests krylov_and_dogleg 73/0, steady_species_rows 147/0, certification 29/0; the new binary with both changes off reproduces the entry arm to every digit; wasp_full_newton BITWISE (ten files; nspec_row

= 0 short-circuits every new path, the hydrodynamic column scales untouched). **Advisor:** rebuild warning-free, steady_newton.o md5 c7621824 identical to the worker's, the three suites 73/0, 147/0, 29/0; the atomic reload rerun on the advisor binary reproduces the first solve to the digit ($||R||$ 3.800e-2 at 150, info = 1), and the run's SECOND stationary solve (the code re-enters the Newton from the handed-back state, another 150 iterations) reaches $||R||$ 5.273e-4, flux spread 2.3e-6, still not certified (Log10 Mdot 10.10; exit 2), 32 min at 16 threads: the arm now descends four decades in 300 iterations where it stood still since B5m. Not converged. **Reported, not fixed:** n_eq_sweeps_model = 1 so the pass count, not eq_sweep_reltol, controls the closure on both fixtures; EXHALE_JFNK_MAXIT caps each solve, not the run; fix_active_species_bounds lost its D argument (the epsilon-active width now on state_column_scale); the [diag 4] true-residual row is not valid for long steps (its finite-difference product is out of range). The fixture README updated by the advisor.

- **Advisor measurement after N8a (atomic reload, EXHALE_JFNK_MAXIT=600, 16 threads, 18 min).** One solve with the cap at 600 stops on the stagnation detector at iteration 224 with $||R||$ 2.671e-2 (info = 2; 170 accepted, 55 refused, all but three by "the reduction ratio is below eta"; log10 Mdot 10.06), WORSE than the same binary's 150 + 150 route, where the run's re-entry into a second solve from the handed-back state reached 5.273e-4. [The advisor's reading here was WRONG and N21 corrected it: the run does not re-enter the Newton from the handed-back state; info = 1 sends it back to MARCHING for 2000 steps (the log's Newton finish at step 2002), and the second solve starts from the marched state at $||R||$ 6.06e-2. The two decades are gained by the marching, not by a fresh trust region; N21 measured every reset and none helps.] Not converged either way.
- **N4b** (decision 14 route (i), a step that moves along the element-budget surface; steady_newton.f90, element_inventory.f90 +94 (the derivatives of the map, element_constraint_derivatives), steady_species_rows +35 rows; diffusive_photochemistry.f90 untouched, the frozen budgets recovered exactly through the public carrier_headroom): **the acceptance row is met by ten decades.** On the carrier reload the He/H nucleus ratio against its reservoir stays at 1.39e-11 at every one of 68 iterates where the entry text drifted from 1.39e-11 to 2.016e-1 (20 percent of the ratio the run was given) over 79 iterates: the write-back of each iterate now returns the element totals it was handed (the N9 / N4a finding closed where those reports said it belonged). Best $||R||$ 0.1513 -> **0.0899** at 14 percent fewer residual evaluations (5690 -> 4891). The shared constraint $2 n(\text{H}_2) + n(\text{H}^+) + n(\text{OH})$

- $2 n(\text{H}_2\text{O}) \leq n\text{H}_{\text{avail}}$ (and the O and C rows) replaces the five separate corners as the step limit (each corner is implied by the row, so nothing is lost and the direction along the surface is gained); a nonlinear feasibility screen by magnitude at every candidate; a restoration phase for an infeasible start (on the element-and-carrier reload 52 trials restored over 1165 cell-steps, worst violation 1.630 -> 2e-16; 6 candidates refused by magnitude; 167 write-back cells whose closure could not return an element total, counted). ****Stated, not claimed**:** on both reload fixtures the shared hydrogen row is never within a probe step of being tight (occupancy 0.985 to 0.989, active band 1e-8), so the active-set PROJECTION onto the constraint hyperplane, the part route (i) is named for, fired on ZERO

iterations of either run; its correctness is stated by the suite rows, and the measured improvement comes from the write-back, the budget leaving the coordinate box and the shared step limit. The atomic element reload (element rows, no carrier row) is identical line for line over its 150-iteration first solve ($\|R\| 3.800e-2$ both). Tests steady_species_rows 182/0 (RED arm EXHALE_ELEMENT_CONSTRAINT_ROWS=0 fails 20 rows), krylov_and_dogleg 73/0, certification 29/0; mol_carrier marching 200 steps bitwise; wasp_full_newton bitwise ($\|R\| 4.213e-9$, info = 0, log10 Mdot 13.30). **Advisor**: rebuild warning-free, steady_newton. omd542fcf223 and binary md59c9ba732 identical to the worker's; the three suites 73/0, 182/0, 29/0; wasp_full_newton's two matrix files compared by the advisor against the Stage A reference: IDENTICAL. Two earlier builds of the item were discarded and their arms rerun (report section 8). Certification unchanged (decision 12 c). **Stage B gate**: N4a map, N4b constrained step and element-conserving write-back, N5's three contracts measured (contract 2 fails by design on two reloads), N8a scales and closure map: the gate's rows are in; what is NOT met is convergence of any species-row solve, which is Stage C's criterion, not Stage B's. N4b's user-visible switches: EXHALE_SPECIES_BUDGET_FACE is retired and replaced by EXHALE_ELEMENT_CONSTRAINT_ROWS (=0 restores the five corners), EXHALE_ELEMENT_WRITE_BACK added (the element-conserving write-back); no input key, output column or unknown layout changed. Also found by N4b: $dc/d(nd) = c/nd$ exactly (the constraint is homogeneous of degree one in the density, so an active row cannot push the mass unknown); two earlier builds of the item were discarded on the atomic fixture's line-for-line check (a blocked probe component zeroed unconditionally made the sampled operator nonlinear, factor 33 regression; the probe sides compared before capping). The advisor annotated the retired switch in the B5l entry and indocs/steady_solver_design.md'.

- **N21** (what limits the species-row solves now; steady_newton.f90 +417, krylov_and_dogleg +31 rows): **no reset of the trust-region state helps, and the advisor's premise was wrong.** The run's second stationary solve does not start from the handed-back state: info = 1 returns the run to MARCHING for about 2000 steps and the second solve enters at $\|R\| 6.06e-2$ (advisor's own log: Newton finish at step 2002, then step 4002, the third solve reaching $4.26e-4$ on the stagnation detector), so the two decades the 150 + 150 route gains are gained by the marching between the solves. MEASURED on the atomic reload from the SAME 150-iteration prefix, EXHALE_JFNK_MAXIT=300: continue (control) 227 iterations, stagnation, $\|R\| 2.671e-2$ (170/55, reproducing the 600-cap run); resetting the pseudo-transient shift, the acceptance memory or the closure map: bitwise identical logs (three exact no-ops: dtau already sits on its floor dtau0, the Grippo window is reported and never acted on on the coupled route, the closure map re-measures to amplification 1.000); resetting the radius and ceiling $2.851e-2$; returning to the best iterate $2.921e-2$; all five together (what a re-entry does) $3.161e-2$; a second Krylov cycle of 40 $2.927e-2$; a restart every 10 iterations without improvement runs to the 300 cap at the same $2.671e-$

2. Carrier reload: control reproduces N4b (8.993e-2), periodic restart 1.153e-1, second Krylov cycle 1.728e-1, both worse. **The periodic restart trigger is delivered default OFF** (`tr_restart_stall_default = 0`) by the brief's own condition. **The linear solve is not the limiter**: after N8a the atomic cycle reaches a median 0.157 of its right-hand side over iterations 1 to 150 (N7: 0.57 to 0.69) but 0.74 to 0.82 over 151 to 225; the carrier cycle 0.963; doubling the products halves the linear residual and makes the outer solve WORSE on both fixtures; in the second atomic solve a cycle reaching 0.98 coexists with a two-decade fall of $||R||$. The radius reset is worse for a stated reason: the carried radius 9.85e-9 accepts steps at ratio 1.000 while a re-initialized 1.15e2 over-predicts by 39x then 3392x and costs four refused iterations. READ: the band is refactored every outer iteration (a re-entry changes the preconditioner only through the shift) and the Krylov cycle starts from zero every iteration. Tests RED (13 symbols absent from the entry module), GREEN krylov_and_dogleg 104/0, steady_species_rows 182/0; with everything disarmed the delivered binary reproduces the entry text's 20-iteration solve line for line; `wasp_full_newton BITWISE` (ten files). Advisor: rebuild warning-free, `steady_newton.o md5 cdbdf535` and binary `e140284a` identical to the worker's, the two suites 104/0, 182/0; the advisor's own `n8a_reload` log confirms the marching between the solves. **The finding that names the next item (N22)**: the predicted decrease stops scaling with the step below about 1e-7 (from iteration 140: the step falls 4x and pred stays 7.099e-4; every longer step of the resulting period-two cycle is an ascent of 3e-3 against a promise of 3e-4, ratios -4, -8, -13, refused, quartered, then accepted at ratio exactly 1.000); the production dogleg geometry is cleared by two new rows (quartered radius quarters the step and the predicted decrease to 2.5e-1 and 2.50009e-1), so what remains is `trust_region_step`'s bookkeeping at those lengths or a precision floor in $r_0 \cdot A_s$ and $||A_s||^2$; the separating measurement is one diagnostic line ($||A_s||$, $r_0 \cdot A_s$ beside $||s||$ at two radii of one iterate). README of the fixture updated by the advisor.
- **N13b** (decision 17 option a, transit spectra carry their validity; `EXHALE_transit.py`, `exhale_transit_lib.py`, `transit_census` +8 rows; Python only): one reader `read_adv_validity` locates the status columns by name in the `# columns` line and distinguishes three cases: schema 2 (N11's two fields: a row is refused for the census when `adv_T_status` is not 0, the composition field reported beside it), schema 1 (today's single column), legacy (no schema: validity UNKNOWN, and both the console and the saved metadata say UNKNOWN rather than 0 refused). Every `tpm_*.txt` gains a comment metadata block (`transit_schema 1`, the input profile and its schema, its `adv_input_certified` / stationarity operator / model restrictions lines copied, its provenance lines, the tool's own identity by git hash and file mtime, the census for that line with the eight-point refused-share ladder and its maximum, the CONTRIBUTION-not-uncertainty sentence, the `EXHALE_TRANSIT_*` overrides in effect); the two hand-written products get the same block. MEASURED: numerical columns bitwise on all 10 `wasp_full` and 5 `mol_base_handoff` products; the schema-2 read tested on synthetic files and on a relabeled `wasp_full` profile (the relabeling is the worker's, not N11's mapping; the production file comes with N11); every consumer in the tree (`python/make_transit_figures.py`, the WASP-52b and LHS 1140 b notebooks, two LHS scripts) reads through `np.loadtxt` or skips `#`, none edited; the planet folders' `tpm_*.txt` untouched (mtime 2026-08-19). Suite RED 8 rows (no reader existed), GREEN 19/0; advisor rerun 19/0. Fixed on the way: the second `np.loadtxt(..., max_rows=1)` column count (NumPy warning on every run) replaced by one rule. Reported: the refused share is taken over the temperature field only (a companion share over the composition

field would be the honest partner); still line center only; README_HOWTO.md should mention the metadata block (advisor added the sentence).

- **N11** (decision 16 option a; post_process_adv.f90, the _adv writer of write_output.f90, examples/exhale_io.py, EXHALE_plots.py, adv_static_limit +driver M1-M5 and rows R8-R9): **stationarity of the input state is judged by the certification's own mass operator** (assemble_residual, residual_row_scale, the mass row $|R_1|/s_1$ against $\text{adv_mass_row_tol} = \text{cert_tol_mass}$), the enthalpy term ratio kept as a printed sensitivity diagnostic that refuses nothing; **two status fields** adv_T_status , adv_comp_status (0 corrected, 1 retained, 2 failed, 3 unsupported, 4 not_evaluated) in BOTH _adv files, columns 8 and 9, with write_adv_validity_header writing # adv_schema 2, the two legends, # adv_status_counts T <5> comp <5>, # adv_input_certified <T|F> <reason> (the certification's verdict on this very state), # adv_stationarity_operator face_flux_mass, # adv_product (a ONE-WAY correction on the input's density and velocity field; a converged scalar temperature solve does not make a row self-consistent), # adv_model_ restrictions; adv_unsupported where the omitted molecular and oxygen species carry more of the cell's particle count than the carried ones (structural, no threshold); exhale_io exposes both fields and the header (load_adv_header, load_adv_status), a legacy file reads schema 1 with both fields ADV_UNKNOWN, never a translation of the old column; EXHALE_plots.py shades the refused rows. **MEASURED, the physics of the item:** the ratio screen and the mass row are not interchangeable and the ratio is the weaker: on the analytic outflow fixture the ratio is $1.1\text{e-}13$ everywhere while the mass row stands at $2.1\text{e-}7$ (RED of the old criterion); on wasp_full the ratio refused 297 of 503 rows, the mass row refuses 503 of 503 (largest 1.04 at $r\ 1.032$); mol_base_handoff 33 vs 503 (0.245 at $r\ 4.21$); wasp_full_newton (the one certified state) 2 vs 3. So on the two relaxation snapshots every row is retained and the _adv product IS the run's own state (T_{adv} equal to T in 504 of 504 rows; the old product differed by up to 6.4 and 207 percent); on the certified state the change is confined to the base (p , $T\ 7\text{e-}5$; species $7.7\text{e-}2$ at $r\ 1.0002$). Run states bitwise in all three; ρ , v in the _adv files moved by exactly zero. Tests 48/0 (advisor rerun on build_lwv, md5 4320f5b7 identical to the worker's: 48/0). No golden refreshed; every _adv golden will move. **Advisor decision (N11b, launched):** the criterion is the right OPERATOR but the wrong TOLERANCE for a conditional correction. $\text{cert_tol_mass} = 3\text{e-}12$ is the tolerance of a certified stationary state; the advective correction is a first-order local correction whose own error is of the order of the mass row's measure (the fractional change of the face mass flux across the cell), so a du-stopped or snapshot state with a mass row of $1\text{e-}3$ is exactly what the tool was built for and is what review R6 calls the "approximate local use" with "a tolerance tied to the required error". N11b writes the mass row measure itself as a column of Hydro_ioniz_adv.txt, sets $\text{adv_conditional_tol} = 1\text{e-}2$ (the correction is described as accurate to that fraction, stated in the header beside adv_input_certified), and keeps the certification verdict as the separate statement it is; the product is then a CONDITIONAL correction with its condition written in every row. Advisor: two per-field and two percent (one in a printed message) fixed. **Reported, not fixed:** assemble_residual leaves $R(:, 1-\text{Ng})$ undefined (RK_rhs assigns $2-\text{Ng}..N+\text{Ng}$ only; no caller reads it today; $\text{RK_rhs.f90}/\text{steady_residual.f90}$); README.md 154, README_HOWTO.md 959 and the manual section 4 still describe the single adv_status column (N11b updates them with the final schema); a suite row for corrected needs a converged fixture.

- **N11b** (the `_adv` product is a `CONDITIONAL` correction with its condition in every row; `post_process_adv.f90`, the `_adv` writer of `write_output.f90`, `exhale_io.py`, `EXHALE_plots.py`, `EXHALE_transit.py`, `exhale_transit_lib.py`, `adv_static_limit` (53 rows), `transit_census` (21 rows), `README.md`, `README_HOWTO.md`, manual section 4): the row decision is taken against `adv_conditional_tol = 1e-2` (the fraction a corrected row is accurate to in the mass flux, the first-order argument at its declaration), the measure $m_j = |R_1|/s_1$ is written as column 10 `adv_mass_row` of `Hydro_ioniz_adv.txt` and named in # columns, the header carries # `adv_conditional_tol` and # `adv_mass_row`, and # `adv_input_certified` stays the certification's verdict with `cert_tol_mass`; schema stays 2 (readers are header-driven; the one place that counted columns from the end of the row was a test row, fixed to locate by name). **MEASURED** against the N11 binary: corrected rows (T) `wasp_full` 0 -> 215 of 504, `mol_base_handoff` 0 -> 341, `wasp_full_newton` 500 -> 502 (the two others are ghost rows below the energy loop's first cell); the column equals the certification's own printed measure on the same state to every digit (1.942392 at cell 432); run states and ρ , v of the `_adv` files bitwise. `exhale_io` returns `adv_mass_row` and the tolerance; the transit metadata block copies the tolerance line and the largest measure; the census still refuses on `adv_T_status`. Tests 53/0 and 21/0 (advisor rerun on `build_lwv`, md5 3131cab0 identical to the worker's: 53/0, 21/0). **Every `_adv` golden moves, far on the two snapshot cases: refreshed at the Stage D gate.** For the record: the first-order argument bounds the dropped term; in some corrected cells the KEPT enthalpy-flux term of the mass divergence still dominates the energy equation (452 of 503 cells of `hydrostatic_column` against 390 retained), and whether one percent of the mass flux bounds the temperature error of such a cell is not measured; a corrected row is not close to the run's own row and is not expected to be (the certified state moves T by $6.7e-2$ at m_j $3e-16$). Advisor: one percent in `README.md` fixed; `docs/postprocess_advection_validity.md` marked stale with a pointer.
- **N10** (decision 15 option a, the restart contract; `input_read.f90` +137, `load_IC.f90` +551, `write_output.f90` +13, `EXHALE_main.f90` +166, `grid_and_gates` +1 row script of 20 assertions, `docs/input_schema.md` K43 and appendix D.2, manual entry; PDF not rebuilt): the key Restart intent: trajectory | relaxation | stationary [evaluate|equilibrate] with the three consistency refusals (default trajectory in phys, relaxation otherwise, so no existing input changes behavior); both state files carry the eight-line block (`restart_schema` 1, `reservoir`, `species_columns`, `grid`, `constants`, `options` with 20 switches, `t_phys[s]`, `source`), built and parsed in `load_IC.f90` (one definition), every number ES23.16; no block = legacy, loaded as before and marked provenance unknown in the log and in every file the run writes (the mark is inherited); a disagreeing reservoir (within `heh_dev_tol`), `grid`, `constants` or `options` is a refusal naming the field, `species_columns` and `source` are reported; a pair whose halves carry different blocks is refused. `stationary_state_of_the_loaded_restart`: WENO3 fixed, the N16d rebuild order, one equilibrium sweep (part of the residual's definition), the loaded composition's departure from that sweep's root printed, the residual and the certification held to the FILE's own certified=, then `evaluate` writes back and stops or (default) enters `steady_wind_with_element_diffusion` at once; no CFL step. **MEASURED**: the atomic element fixture AS LOADED stands at $|R|$ 1.8924 (the marched hand-off read 1.888: the two CFL steps); the certified `wasp_full_newton` state is NOT the fixed point of its own sweep (particle count moves $2.1e-11$, species $5.0e-10$, cooling $3.6e-10$: the cell solve's $xtol = \sqrt{\epsilon}$ and the file's digits), reported at every evaluation rather than

equilibrated away; stationary evaluate takes 0 steps and returns the certification unchanged; stationary on that state enters the JFNK at $7.133\text{e-}9$ and exits `info = 0` with no step. Tests 20 rows GREEN (RED on the entry binary: the key warned and ignored, one step marched, no block, no refusal); `grid_and_gates` 130/0 (advisor rerun 131/0 on `build_lwv`). `mol_carrier` 200 steps numeric-line bitwise (eight new # lines); `run_check.sh` line 297 excludes # lines, so the block cannot change a matrix verdict; `wasp_full_newton` cold numeric-line bitwise against a control binary of the same tree without N10 (its $5.8\text{e-}9$ / $9.4\text{e-}9$ movement against the golden is B5j's recorded movement, not N22's as the worker supposed). **Fixed on the way:** the header loop matched columns as a SUBSTRING (`index(line, 'columns') > 0`), so the `_adv` files' NOTE prose or the new `species_columns` field would have replaced the species labels silently; now the first token (`comment_field_is`); the same hazard in `restart_round_trip.sh`'s Python readers fixed. Deviations from the design, stated: no EXHALE_STATIONARY_EVALUATE_ONLY env hook (the second word is the one answer); the provenance unknown mark is not yet in EXHALE_setup.out (`write_setup_report.f90` not owned; one line); `species_columns` reported not refused (the by-label species map is a documented path); source is the git revision, not an executable md5. **Consequence for the user (decision item 21):** the exact options comparison forbids the arm-ladder workflow (converge without an option, restart with it on); legacy pairs are unaffected, every new rung would be refused; the disciplined fix is a key naming the tokens allowed to differ. Also corrected: `docs/input_schema.md` appendix D listed the retired `valve/fluxconst` fields and the wrong `base_bc` values. Reported: Run mode is absent from `input_schema.md`; the EXHALE_PTC=1 route writes no setup report; a re-evaluated state writes `sec_ion_step=0`.

- **N22** (the predicted decrease stops scaling below $||s|| \ 1\text{e-}7$; `steady_newton.f90` +195/-10, `krylov_and_dogleg` +16 rows): **all three candidates of the brief refuted by measurement, and the cause named:** the trials that break the scaling are exactly the trials whose species unknown was written onto a face of its box (`nout = 1`), and that write is a move of the state no model of the step contains (11 of 11 such trials refused with a fixed ascent of $3\text{e-}3$ over five decades of step; 10 of 10 others accepted at ratio 1.000). When the radius is cut the SCALED step quarters but the step in the unknowns the residual reads, $||D \ s||$, moves 0.4 percent, so the model image and `pred` are unchanged to seven digits; $r_0 \cdot A \ s = -||A \ s||^2$ to every digit (`pred = 0.5 (r_0 \cdot A \ g)^2 / ||A \ g||^2, radius-free, the interior minimizer along the leg: not a defect); a fresh Jacobian action at the same step reproduces pred to every digit (no stale cache); the probe displacement is $\sqrt{\text{eps}}(1 + ||Y||) = 1.8\text{e-}7$ whatever the direction, nine decades above the residual's reproducibility (no floor; the reverse: the stalling step is $2.7\text{e-}3$ of the probe arc). The fix: hold_the_step_where_it_leaves_the_species_box zeroes the step at an unknown it would carry out of its box (the iterate is feasible, so the trial is, and the step is one the operator can be sampled along); probe_length_of_the_jacobian_action is the one definition of the probe arc; a diagnostic line for each trial under EXHALE_ELEM_DIAG. Atomic reload, cap 300, before -> after: 227 -> 113 iterations (both stagnation), accepted/refused 170/55 -> 91/19, negative ratios 49 -> 14, best $||R|| \ 2.353\text{e-}2$ -> $1.333\text{e-}2$, handed back $2.671\text{e-}2$ -> $1.356\text{e-}2$; the period-two cycle is gone (iterations 95-98: pred doubles with the step, all accepted at 0.998 to 1.001, steps $1\text{e-}2$ to $1\text{e-}1$ instead of $1\text{e-}9$); the closing refusals are ray/floor verdicts at POSITIVE ratios. Carrier reload BITWISE (the branch never fires); wasp_full_newton BITWISE (ten files; the probe-arc refactor is shared through jv_product and this settles`

it). Tests: the twelve-rung ladder rows are GREEN at entry (the geometry was never the cause), the new rows RED by four absent symbols, GREEN 120/0; steady_species_rows 182/0. Advisor: rebuild warning-free, steady_newton.o md5 d70a078b identical to the worker's delivered object, the two suites 120/0, 182/0. Not converged. **Reported, the next item (N23)**: the approximate-gradient leg is 0.7 percent of the step and supplies ALL of its model image ($||sU||/\delta 6.9e-3$ passes `cauchy_leg_min_fraction = 1e-3` while $||A s|| = \tau_c ||A g||$ to four digits): the criterion must be the share of the IMAGE, not the length ratio (N7b's finding at a length the threshold admits).

- **N10b** (decision 21 option a; load_IC.f90 +319/-20, input_read.f90 +145/-1, write_setup_report.f90 +62/-6, grid_and_gates +1 script of 11 rows, docs/input_schema.md K44 and D.2, manual entry): the key Restart option change: `<token>[, ...]`; the options field is compared TOKEN BY TOKEN from one table (`opt_name(20)`, `opt_value`, the emitted text bitwise unchanged), only the named tokens may differ, every other difference refuses the load naming the token with its from `->` to; six layout tokens (metals, mol, oxychem, carrier, carrier_newton, iontrans) may never be named (they decide the rows the state carries: a cold start, not a restart); an unknown token, a grid/reservoir/constant word, the key with no token or without Load IC? True are input errors; the change is written into both halves of the new state as `# option_change` from `->` to at restart of `<source>`, inherited (history of 32 lines); EXHALE_setup.out gains Restart provenance: UNKNOWN | UNKNOWN by descent | states its configuration and the permitted changes (N10's deviation 2 closed; the "by descent" wording was added because the first wording was measured false on a ladder's second rung). **Correction of the advisor's brief**: LW is a FLUX (Stellar LW flux), not an option token, so the `armA_noLW -> armA_LW` rung never needed this key; and every state written before the Jupiter-radius unification (those two included) is refused by the grid guard at $2.25e-2$ in the outermost cell, so the arm directories are configurations, not restart sources. The ladder step was measured on the `armA_noLW` configuration with `base_ir`, `mol_ir` flipped: refused by token without the key, accepted with it, the provenance line in both halves. Tests RED 11/11 at entry (the key was an unrecognized line), GREEN 11/11; grid_and_gates 141/1 with the one FAIL attributed (below); `mol_carrier` 200 steps numeric-line bitwise (the rebuilt `# options` line bitwise). Also fixed: N10's test stage A read the LIVE `wasp_full_newton/output/`, which the matrix rewrites (a false failure of the legacy-mark row as soon as the pair carried a block); it now strips the block from its own copy. **Advisor**: rebuild warning-free, md5 b10641e5 identical to the worker's; grid_and_gates 141/1, the FAIL being `stationary_evaluate_rebuilt_within_sweep_budget` reading the live output/ of `wasp_full_newton`, which the running Stage D matrix had left as a mid-run relaxation snapshot ($3.2e-3$ against $1e-8$; the same pair evaluated by the entry and the N10b binary is bitwise, so not N10b's); the advisor pinned the certified state as `backup/regression/wasp_full_newton/IC/` (copied from the golden, `certified=T`, README there) and pointed the row at it. Reported: `restart_grid_guard.sh`'s stale header (third time); `sec_ion_step=0` on a re-evaluated state; whether `molbase`, `he_diff`, `he_metal_diff` belong with the layout tokens is a reviewer's question (the worker's reading: they add operators, not rows).
- **N23** (the approximate-gradient leg admitted by its share of the model image; steady_newton.f90 +220/-5, krylov_and_dogleg +16 rows): two constants `cauchy_leg_image_share_max = 0.5`,

cauchy_leg_step_share_max = 0.1, two pure routines (cauchy_leg_shares_of_step_and_image, cauchy_leg_is_image_without_step: image share above the first AND step share below the second, so a leg that IS the step keeps its image), and a branch in the dogleg's trial loop that DROPS such a leg from the step and its image and rebuilds the dogleg at no extra product (drop, not rescale: a rescale would either shorten A SU without SU, which is not a step of the operator, or move the step off the dogleg path); the model slope handed to ray_slope_verdict is the slope of the step taken in both branches. **MEASURED, the reading ([diag 9])**: at the reload's second solve quartering the radius moves the promise by 0.9959 with the leg admitted and by 0.2501 with it dropped (six iterates: 0.9959 to 0.8938 against 0.2501 to 0.2517), so the promise follows the radius again where it did not. **Honest impact: the new rule fires ZERO times on both fixtures and all three runs are BITWISE** (atomic 113 iterations, 1.356e-2; carrier 8.993e-2; wasp_full_newton 13.30): on the post-N22 trajectories the length test reaches every image-dominant leg first (117 length drops, 0 image drops); the shares printed at every trial say the image test is the right criterion nonetheless (0.9986 of the image at 3.4e-5 of the step; eleven of seventeen read trials satisfy the predicate on their own), and the leg N22 saw at iteration 145 sat on a trajectory the N22 hold removed. Tests RED (four symbols absent), GREEN 136/0; steady_species_rows 182/0; -Wall warnings 6 before and after. Advisor: rebuild warning-free, the two suites 136/0 and 182/0 on build_lvw (the object md5 differs from the worker's delivered one by a post-measurement comment edit that moved line numbers, which the I/O statements carry; the worker's line-for-line reproduction of the solve covers it). Not converged; the rule is not claimed to improve either fixture. Reported: cauchy_leg_min_fraction is now a proxy for a test the code can make directly (letting the image test alone decide is the next increment, not taken because no fixture exercises the difference); the trial diagnostic's evaluations enter the cost counters.

- **N24** (what refuses the remaining trials of the atomic element arm; steady_newton.f90 +51/-2, krylov_and_dogleg +13 rows): **the obstruction is the linear model, and the row is named**. Atomic reload, cap 300: 110 trust-region iterations (the "113" of earlier reports counted three summary lines), 91 accepted / 19 refused. The 19 are two populations, neither a step-control defect: 14 ascents at LONG steps (||s|| 1.6 to 51, five with zero species unknowns held, so N22's mechanism is not what is left) and five closing refusals at positive ratios (||s|| 2.3e-4 to 8.9e-7 with the promise quartering exactly while the true decrease keeps a step-independent floor of 3e-11 to 1.6e-10). One accepted step (outer 59, ratio 0.948, ||R|| 7.37e-2 -> 2.12e-2) moves the binding row from the energy equation to **the sodium element row of the outermost cell (cell 500, r 4.06 Rp)** and the solve never leaves it: from outer 60 to 110 that row holds 0.435 of the squared merit and the whole judged distance (5.3e-2 of its own scale, 5.36e6 at cert_tol_element 1e-8) while ||R|| stands at 1.333e-2. At outer 105 everything else is sound (0 held, 0 refused samples, 0 empty columns, dbgcon 1.25e-6, step ratio 0.9998); what fails is the linear model: 40 of 40 products, true relative residual 0.58 against 0.1, **the Arnoldi image 21 percent away from the operator** (1.6e-6 at iteration 1, 2.3e-4 at N7's iteration 60), ||sN|| 8.6e2 against a radius 5.9e-2, so the region uses 7e-5 of an inaccurate direction; the stuck row is at once the smallest scaled column and the smallest scaled row of the band. The stagnation detector read the judged distance, which IS that row; ||R|| was not falling (1.337e-2 -> 1.333e-2 over 32 iterations). **The image-only leg test** (EXHALE_CAUCHY_LEG_BY_IMAGE=1, default off) measured and NOT adopted: worse on both fixtures (atomic 2.30e-2 against 1.36e-2; carrier 0.324 against 0.090, to the cap), the arms parting at outer 7, so N23's

"the rules agree" was that trajectory's property. Deliverable 3 STOPPED as briefed (linear solve / model / one row's physics, not the step control). Tests RED (two symbols absent), GREEN 149/0; steady_species_rows 182/0; the delivered binary reproduces the entry solve line for line over 925 lines; wasp_full_newton BITWISE. Advisor: rebuild warning-free, 149/0, 182/0 on build_lwv (the advisor's binary of N23 had been built from an earlier revision of steady_newton.f90, as the worker showed; the suites and fixtures were the check). **Proposal (N25):** (1) explain the 21 percent Arnoldi gap (EXHALE_GM_ORTH0; Jv reproducibility by two products); (2) ask the cycle for the direction ON THE BALL (Steihaug-Toint truncation) instead of solving to 0.1 a leg four decades beyond the radius; (3) the sodium row's own terms at cells 490 to 500: the element operator gives cell 500 a zero-gradient outer ghost, so nothing outward constrains it (a boundary question of the element operator, binary_element_diffusion.f90); (4) cert_tol_element 1e-8 is what makes this row the whole judged distance (UNCONFIRMED, ISSUES 3.1 e; not moved). Reported: the (JFNK) it line prints k=* for a species row without naming the element (on this arm the identity was the whole diagnosis); the stagnation messages quote ||R|| where the detector read the judged distance.

- **Stage D gate and golden refresh (advisor, 2026-09-09 21:14 to 2026-09-10 00:27, make check on the shared build/; refresh 00:40).** Tree after every item to N24 (N25 was running on a private build). Verdict against the goldens of 2026-09-08 19:05: on all 16 default cases the two marching files are **data identical** (15 cases bitwise; wasp_full_newton WITHIN at 5.78e-9 / 9.41e-9, B5j's recorded movement, which the 0.1 percent rule counts as identical) and all 32 _adv files FAIL by SHAPE (10 against 8 and 40/36/43 against 38/34/41 columns: the two status fields and the measure column of N11/N11b), so make check exits 2 as expected. The _adv movement is the physics of N12 (stale composition, upstream energy), N11 (mass-operator stationarity) and N11b (conditional tolerance): T_adv changes in 477 of 504 rows on mol_base_handoff (max 61.6 percent, 364 -> 588 K at r 1.093), 479 of 504 on mol_carrier (max 138 percent, 379 -> 901 K at r 1.098), 215 of 504 on wasp_full (max 2.3 percent, 5637 -> 5505 K at r 1.131). **Goldens refreshed:** the previous set archived as backup/regression/golden_pre_stageD_20260910/, run_check.sh golden snapshotted the outputs the gate had just produced (17 cases, including the three hp_* and oxygen_chemistry extras), and the advisor compared the 68 snapshot files against the case outputs with the comparator's own #-filter: 68 identical, 0 differ (this comparison, not a three-hour rerun, is the "check again" of the standing rule, since a snapshot is a copy; the user asked that regression checks not be repeated). The shared build/ and EXHALE.x are now current with the tree (md5 f6ccb498 at the gate's build; N23/N24 built after it touched steady_newton.f90 only, so the marching path of the matrix is the gate's). State files now carry the N10 metadata block and the _adv files the N11/N11b header; both are # lines the comparator ignores.
- **N25** (the Arnoldi gap, a direction on the ball, the sodium row at the outer boundary; steady_newton.f90, krylov_and_dogleg +25 rows, binary_element_diffusion.f90 a default-off print only, 33 lines, no arithmetic): **(1) the 21 percent Arnoldi gap is the NONLINEARITY of the finite-difference action across directions**, shown positively: every one of the 40 Arnoldi columns matches its own product to 1e-16 to 3e-16, two products of one direction agree exactly, the basis's orthogonality loss is 4.4e-13, yet the action of $V_1 + V_2$ differs from the sum of the two actions by 1.1e-4 and the assembled gap tracks the step length (8.5e-5 and 2.5e-4

per unit of $||sN||$ at outer 1 and 105; $||sN||$ 12 $\rightarrow$ 865), the signature of a curvature term over the fixed probe arc $\sqrt{\epsilon}(1 + ||Y||) = 1.8e-7$. Reorthogonalization NOT adopted (loss $1e-13$ to $3e-12$ against the $1e-8$ level; as a measured arm EXHALE_GM_REORTHO=1 it is worse on both fixtures: 1.487 and 0.43). **(2) Steihaug-Toint truncation** is in as EXHALE_KRYLOV_ON_THE_BALL=1, default off: products per iteration 33 $\rightarrow$ 7 (atomic), 40 $\rightarrow$ 1.5 (carrier); atomic better on the judged distance ($1.69e6$ against $5.31e6$, lowest $||R||$ $9.4e-3$ against $1.33e-2$) but worse on $||R||$ handed back ($1.69e-2$ against $1.36e-2$); carrier worse (0.486 against 0.090); the brief's condition fails, default stays off. **(3) The sodium row of cell 500 is not a well-posed outflow condition, and the reason is a DEFECT of the stationary residual evaluation, not of the element operator:** the outer ghost of the element mixing ratio is FROZEN at the pre-solve composition ($1.73111e-6$, the metals.inp reservoir the handed-in state carried) throughout the solve while cell 500 falls to $1.0e-11$, so at the stagnating iterate the ghost is $1.7e5$ times the cell it "continues"; the marching path refreshes that ghost (the second solve, entered at step 2002, has ghost = cell 500 again); the accepted step of outer 59 did not move $fx(500)$ at all (it moved the outer face mass flux); the scaled Jacobian row is dominated by the momentum of cells 499/500 and the sodium of 499 while its own diagonal is $1.3e-4$ of the largest entry (the row is nearly independent of its own unknown). The physically stated condition (zero gradient at the top, no diffusive flux) requires the ghost to follow the ITERATE; the repair is in the composition write-back / ghost policy of eval_residual, which is N26 (launched). **(4)** the (JFNK) it line names the element (worst row: element Na of cell 500) and the stagnation messages quote the judged distance and dj_best. Tests RED (eight symbols absent), GREEN 174/0; steady_species_rows 182/0; default path unchanged (atomic 109 iteration lines identical to the digit; carrier identical); wasp_full_newton BITWISE. Advisor: rebuild warning-free, 174/0, 182/0 on build_lwv; three British neighbor in the element operator's comments fixed. Not converged; the ball arm's best iterate is chosen by the judged distance while the control is $||R||$ (the two rank iterates differently, which limits arm comparisons).

- **N26** (the upper ghost of a transported element or carrier follows the iterate; steady_newton.f90 +135/-20, steady_species_rows +7 rows): in write_species_rows_into_composition (the ONE place a trial composition is assembled for the residual: eval_residual and the three best-iterate restores) the transported columns of the upper ghosts take the outermost cell's value by the rule the marching path uses (READ: $Y_{etr}(N+1:N+Ng, q) = Y_{etr}(N, q)$ in species_advection_stage; $f_{try}/f_c(N+1:N+Ng, ic) = \dots(N, ic)$ in the carrier operator; Apply_BC never touches f_{sp}), the element totals of a ghost restored after the carrier copy, a ghost deficit clamped as the operator clamps its own; lower ghosts untouched (Dirichlet reservoir); the three-unknown route returns before reading anything. The mixing ratio follows the mass fraction exactly (equal He mass fraction at N and its ghost gives equal n_{He}/n_H , hence equal n_X/n_H ; MEASURED $1.4e-16$). **MEASURED, atomic element reload (EXHALE_JFNK_MAXIT=300):** $||R||$ handed back $1.356e-2 \rightarrow 3.719e-4$ (36x), judged distance $5.36e6 \rightarrow 6.21e4$ (86x), flux spread $6.6e-4 \rightarrow 5.5e-7$, 91/19 $\rightarrow$ 235/18 accepted/refused, the sodium row of cell 500 binds on **0 of 252** iterations (was 51 of 109) and no row of cell 500 is ever the worst; the solve now stagnates on the INTERIOR helium row of cell 246 (r 1.155), info = 2, uncertified. **Carrier reload:** binding cell (H2 of cell 205, interior) and judged distance ($4.807e7$) unchanged to four digits, $||R||$ handed back $8.993e-2 \rightarrow 2.519e-1$ (worse by 2.8x, a path

- difference of a stagnating solve, reported not judged), flux spread down 12 percent. `mol_carrier` marching bitwise (no caller outside `solve_steady_jfnk`); `wasp_full_newton` BITWISE. Tests RED on the entry text with only the export added (4 rows: ghost mass fraction 6.7e4 off, mixing ratio 1.0e5 off, the outermost element row of a flat column 1.19e-2 instead of 0, the carrier ghost 0.80 off), GREEN 189/0 (a non-vacuous row keeps the stale ghost and reads 1.19e-2 in both arms); `krylov_and_dogleg` 174/0. Advisor: rebuild warning-free, `steady_newton.o` md5 4bd3138a and binary e56609ff identical to the worker's, 174/0 and 189/0 on `build_lwv`; one neighbors fixed. **Reported, the next small item (N26b)**: the relaxation solvers `solve_trace_element_in_hydrogen` and `element_diffusion_step` refill their working arrays' upper ghosts before every pass while `element_transport_residual`, documented as the same operator, reads the ghost from `f_sp` with no refill, so the operator's boundary depends on its caller (`binary_element_diffusion.f90`, two lines). Not claimed: convergence; an explanation of the collapse of the outermost cells to 1e-11 of the reservoir.
- **N26b** (the element operator refills its own upper ghosts; `binary_element_diffusion.f90` +26/-0, new suite `src/tests/element_operator/` with 5 rows): `element_transport_residual` now writes $X_{he}(N+1:N+Ng) = X_{he}(N)$, $f_X(N+1:N+Ng) = f_X(N)$ and $wY_{tr}(N+1:N+Ng) = wY_{tr}(N)$ (the third line the relaxation twin gets for free through `project_elements`) so the operator poses ONE outer boundary (zero gradient, no diffusive flux at faces 0 and N) whatever the caller left in the ghosts; the lower ghosts stay the Dirichlet reservoir; the header states the boundary. Both outermost rows (N-1 and N, WENO3 reaches cell N+1 at face N-1) are covered. MEASURED: the relaxation's own fixed point handed to the residual with stale ghosts read 1.19e-2 of its scale in the outermost row before and 1.5e-13 after (the interior level is 1.3e-13); the same column with stale and with zero-gradient ghosts now gives the two outermost rows bitwise. Suite RED 3 of 5 on the entry text, GREEN 5/0; certification 29/0; the atomic reload's first two JFNK iterations identical but for the merit's reproducibility going 2.8e-14 -> 0 exactly (the residual no longer reads a ghost the sweep leaves at a different last bit), the outputs moving at 7.9e-12 (T) and 1.6e-10 (an `_adv` status/measure column) because N26's write-back hands over the ghost to 2e-16 and the refill to the bit; `mol_diffusion` marching 200 steps BITWISE (the marching path reaches the relaxation twin, never the residual). Built and measured in an isolated source tree with `steady_newton.f90` pinned, because N27 was editing it. Advisor: rebuild warning-free, `element_operator` 5/0, certification 29/0 on `build_lwv`; one percent fixed. Reported: the write-back's ghost is the cell's to round-off, not exactly (any other consumer of those ghosts reads a 2e-16 gradient).
 - **N27** (the helium row of cell 246 and the H2 row of cell 205 read side by side; `steady_newton.f90` +191/-3 diagnostic only, `steady_species_rows` one row restated): **the helium row of cell 246 is held by the LINEAR SOLVE, not by its physics, its boundary, a bound, the closure or the tolerance.** At outer 245 the row is +2.43e-22 = diffusive (settling) +2.40e-22 plus advective +3.3e-24, same sign (they add); the Newton removed the advective term (a factor 7200 since outer 1) and barely touched the diffusive one (x1.28); the measure 6.17e-4 is the maximum of a flat plateau (5.9e-4 at 240 to 5.4e-4 at 253), not a bad cell; 0 held, 0 refused samples in 246 iterations, band healthy (`dgbcon` 7.5e-7), merit reproducibility exactly 0, the trust region inert (ratio 0.933, 0 cuts, aN 1.0) and the Krylov cycle spends 40 of 40 products on a direction whose true relative residual is **0.9917** against 0.1 with the Arnoldi image 14.7 percent off the operator; the step buys 4.7e-5 of the merit per iteration.

The helium Jacobian triple is a proper tridiagonal stencil with the diagonal largest (the unknown controls its row; the wind dominates the row's entries). The element rows hold 0.98 of the merit (He 0.63, Fe 0.36); $||R||$ is the energy row at $1.37e-4$; the stagnation detector is honest (merit and $||R||$ fall while the judged distance rises $6.12e4 \rightarrow 6.21e4$). **A number that does not fit N25's reading:** the Arnoldi gap is 0.147 at $||sN||$ $1.8e-2$, 8 per unit against N25's $2.5e-4$ per unit, so the gap is not a function of the assembled length alone. **The H2 row of cell 205** is the same shape on a coarser front: measure 0.480 (not a cancellation, half its own terms), $f(H2)$ falling $1.4e-1 \rightarrow 1.1e-3 \rightarrow 4.9e-4$ over cells 195-207 (logarithmic slope -757), $f(HI)$ flat (1.7 percent), so cell 205 is NOT a two-root HI front (consistent with N5); linear 40/40 at 0.97, gap $3.1e-2$. Deliverable 3 STOPPED (nothing in the step control to change at the binding iterate). Deliverable 4: `cert_tol_element` would have to be $6.2e-4$ (4.7 decades; far above the $1e-5$ the operator's header gives as its loosest floor); the closure's reproducibility ($3.1e-7$ in the row's measure per unit seed, N5) is not what holds it. Proposals: re-measure the Arnoldi gap at three iterates (it no longer tracks $||sN||$); the untried freedom is the COLUMN preconditioner (row scaling is fixed by decision 20 a); whether the detector should stop a solve whose merit still descends while the judged row does not is an acceptance decision; the carrier front is a grid statement. Tests 189/0, 174/0; advisor rerun 189/0, 174/0. **Incident, found by N27 and CONFIRMED by the advisor:** with N26b's ghost refill in `element_transport_residual` the atomic element arm parts from N26's arm at outer 10 and stands at $||R||$ 1.199 at iteration 40 where N26's arm was at $3.21e-2$ (advisor run, cap 40, current tree: 1.888, 1.891, 1.677, 1.484, 1.518, 1.582, 1.199 at iterations 1, 5, 10, 15, 20, 30, 40 against N26's 1.888, 1.891, 1.678, 1.955, 1.758, 0.912, 0.0321); N27's own diagnostic is inert (hook on = hook off). N26b's validation (the first two iterations, the certification suite, `mol_diffusion` marching) did not reach it, and the advisor's verification of N26b did not rerun the arm: the advisor's mistake. N26b's refill is therefore NOT accepted as it stands; N26c (launched) finds why an operator that poses its own boundary breaks a solve whose caller already hands it that boundary to $2e-16$, or reverts the refill. N27 restated the row `a_stale_upper_ghost_is_read_by_the_outermost_row` to its negation because N26b made the original false; N26c reconciles the two rows with whichever text stands.

- **N26c** (why N26b's ghost refill broke the atomic solve; `binary_element_diffusion.f90` refill REVERTED with the header restated, `element_operator` rows restated (6), N26's `steady_species_rows` row restored): **the refill changes the residual by one element's outermost row in its twelfth digit and nothing else, and the atomic element arm does not survive a last-bit change of the upper ghosts, whatever makes it.** MEASURED: the caller's ghosts already equal the cell to 1 to 4 units in the last place (x_{he} $9e-16$, f_X $1.9e-15$, w_{Ytr} $6e-16$ over the whole solve); the two texts' outermost rows agree to 17 digits on every element but nitrogen's, which differs by $9.8e-13$ relative ($6.8e-17$ in its own measure); no other ghost of the routine is read (face coefficients stop at N-1); the band's stencil already carries the dependence of rows N-1, N on cell N through the write-back. **A control that multiplies the same three ghost quantities by $1 + 2e-16$ and poses no boundary at all breaks the arm as badly** ($||R||$ at iteration 40: caller's ghosts $3.212e-2 = N26$ to the digit; refill 1.199; one-ulp control 1.125; the three part at outer 10-11 and from each other). Where it amplifies: the Krylov leg at its one-digit tolerance (relative residual reached $8.86e-2 / 8.89e-2 / 8.93e-2$ at outer 10, pred 0.4 percent apart, then 2.7x apart at outer 11, then different branches of the acceptance at outer 12: accepted 0.874 / REFUSED -1.98 / accepted 85); the solve's own merit reproducibility prints $1.4e-14$, a hundred times the ghost movement.

The refill is physically right and is reverted anyway: no version of it can be bit-inert on the production caller, and the arm's acceptance is a trajectory no last-bit change preserves. Delivered text reproduces N26's atomic arm line for line over 252 records ($|R|$ 3.719e-4, judged 6.22e4, He of cell 246, 27543 samples, exit 2) and N26's carrier arm over 63 (0.2519, 4.806e7, H2 of cell 205); mol_diffusion marching bitwise; suites certification 29/0, element_operator 6/0 (restated to ASSERT the measured caller dependence, 1.194e-2, so the suite exits 0; the advisor accepts that choice), steady_species_rows 189/0, krylov_and_dogleg 174/0; advisor rerun of the four on build_lvw (md5 3201310a, the worker's delivered binary): 6/0, 29/0, 189/0, 174/0. **The finding that matters beyond this item:** the atomic element arm is NOT a reproducible quantity; its $|R|$ at a fixed iteration count is set by which Arnoldi vector crosses a one-digit tolerance at the last bits of the residual, so "reproduces the previous arm" rejects correct changes as readily as wrong ones, and what the arm can carry is named outcomes (binding rows, refusal counts, flux spread), not $|R|$ at iteration N. Recorded in ISSUES 3.1 (d) and in the fixture's README. The asymmetry of the element operator (the relaxation twins write their ghosts, the residual takes the caller's) stays in the tree, documented and measured (1.19e-2 on a stale caller). Advisor: a cp -irp of the N4b worker hung on its interactive prompt since 2026-09-09 (PID 61163, cwd the repository root, copying from the session scratch) was killed by PID. CORRECTION within the hour: killing the cp let the worker's shell run its NEXT statement, which re-applied N4b's first (discarded) patch onto the delivered element_inventory.f90 (832 -> 926 lines, a duplicate routine, would not compile); the N4b worker, woken by the stale task, rebuilt the file from its saved entry copy plus the delivered patch and it reproduces diff_element_inventory.diff line for line; the advisor confirmed 832 lines, one routine, a warning-free build (md5 3201310a, N26c's delivered binary) and element_operator 6/0, steady_species_rows 189/0. Lesson written into docs/worker_rules.md: \cp -f / \mv -f always, and a stale command is cancelled as a task, not killed as a process.

- **Direction after N26c (user, 2026-09-10: "권고안대로 진행").** The advisor's recommendation adopted: (b) N8b, anchoring the species-row certification tolerances on the two candidate states by the review's five measurements (reproducibility floor, derivative convergence, a manufactured steady solution of the element operator, grid refinement of the candidate state, conserved-flux error), reported per row class and regime with a PROPOSAL for the user (decision item 22) and no value moved; and in parallel (c) N28, the code items of ISSUES 3.7/3.8 (run-wide JFNK cap, one element_ratio_gate, the update map's install branch owing an Apply_BC, the undefined innermost ghost column of assemble_residual, the PTC route's setup report, sec_ion_step on a re-evaluated state, the stale test header, Run mode in the input schema). (a), the column preconditioner, waits for N8b's numbers. The user manual PDF was rebuilt by the advisor (85 pages; it was two keys behind its source: Restart intent, Restart option change).
- **N8b** (anchoring the species-row certification tolerances; certification.f90 +173 additive, no value moved; steady_newton.f90 one diagnostic; certification suite 29 -> 42 rows; memo docs/certification_tolerance_anchoring_20260910.md): **1e-8 is refused by no floor of the arithmetic, the closure or the representation (five to eleven decades below it); what refuses it is the DISCRETIZATION and the ELEMENT FLUX CONSERVATION.** MEASURED on the two candidates:

reproducibility exactly 0; closure-seed floor $5.3\text{e-}15$ (the state's own sweep seed $1.7\text{e-}10$); state-perturbation amplification 0.23 to 0.30 in the wind but $1.1\text{e}3$ to $2.0\text{e}3$ at $r < 1.01$ (floors $5\text{e-}17$ and $4.4\text{e-}13$); derivative: a clean V at the base ($4.5\text{e-}7$), a PLATEAU at $6.4\text{e-}5$ over four decades of probe length at the binding cell 245 (the part of the derivative outside the band) so the smallest row a Newton step resolves there is $3.8\text{e-}8$; manufactured relaxed column at 801/401/201 cells $5.4\text{e-}13$ / $2.47\text{e-}6$ / $7.42\text{e-}6$, order 1.59 (the flat column $2.7\text{e-}14$); grid refinement of the candidate at r 1.1515: the He/H row $5.97\text{e-}4$ ($N = 500$) $\rightarrow$ $2.48\text{e-}4$ ($N = 1000$), remap control $1.6\text{e-}5$, so **about 78 percent of the $6.0\text{e-}4$ is discretization** (a C2 interpolant is needed; below r 1.03 the remap dominates); element flux spread 39.3 in the layer, $2.6\text{e-}2$ in the wind, $2.2\text{e-}3$ over the escape window against the $5.5\text{e-}7$ mass-flux spread the arm is quoted with. Memo section 7 is the proposal (decision item 22, added by the advisor to To_be_determined): (a) $1\text{e-}5$ in the wind with the layer reported, (b) keep $1\text{e-}8$ and state the fixtures cannot certify, (c) two regimes gating; no proposal certifies either candidate. **Two defects found outside the item's files, reported:** (1) **the candidate's species do not carry the run's own density:** mass closure $6.68\text{e-}3$ at the top against $1.9\text{e-}9$ on the three-unknown control; reproduced in the suite on a column closed to $6\text{e-}16$ that reads $3.37\text{e-}3$ after ONE relax_element_composition, so project_elements does not return to hydrogen the mass it takes from helium (N29, launched); (2) the writer-to-loader round trip does not return the state: the atomic candidate reloads at $||R||$ 0.486 instead of $3.7\text{e-}4$ because load_IC rebuilds rho from the species (a consequence of 1). Tests 42/0 (RED at the build level: the suite does not compile against the entry objects); the default certification report byte-identical (anchoring under EXHALE_CERT_ANCHOR=1). Advisor: rebuild warning-free, certification 42/0; one neighborhood and two percent fixed.

- **N28** (the code items of ISSUES 3.7; EXHALE_main.f90, RK_rhs.f90, steady_residual.f90, element_census.f90, element_inventory.f90 and attempted_step.f90 (the duplicate constant), grid_and_gates +2 rows and headers, docs/input_schema.md K45): (1) EXHALE_JFNK_MAXIT caps the RUN: the three call sites of steady_wind_with_element_diffusion pass one named default jfnk_outer_iterations_default = 3000 (the marching hand-off's literal was 500, the one default that moved; wasp_full_newton converges in 13 and cannot reach either), and under the variable the run takes its FIRST stationary solve and refuses later entries by name (MEASURED: three solves of 2 iterations $\rightarrow$ one); (2) element_ratio_gate has one owner, element_census.f90 (public, element_census_tolerance() public with its override), read by element_inventory.f90 and attempted_step.f90; (3) update_map_begin_step's install branch calls Apply_BC in the N16d order (the EXHALE_UPDATE_MAP diagnostic's second block on mol_carrier moves 5 percent in the mass and energy rows: the fix of a defect, a boundary of one state under an interior of another); (4) assemble_residual defines the innermost ghost column (dF, S, R zero at 1-Ng; two callers read it in whole-array expressions); -fcheck clean on the certification driver and on a 20-step mol_carrier (the run that reaches it); (5) the EXHALE_PTC=1 route writes its setup report (was 0 lines; new row RED/GREEN); (6) a Restart intent: stationary evaluate keeps the file's sec_ion_step (2402 $\rightarrow$ 2402, was 0; new row RED/GREEN); (7) restart_grid_guard.sh's stale header rewritten and the script made to honor EXHALE_TEST_OUT; (8) Run mode documented as K45. Item 9 (n_eq_sweeps_last on a discarded evaluation) is steady_newton.f90's, reported. Validation: mol_carrier 200 steps and wasp_full_newton cold BITWISE on every output file (the one log difference: cap 500 $\rightarrow$ 3000, from the caller);

grid_and_gates 146/0; attempted_step 162/1 with the FAIL proven pre-existing (rebuilt from entry sources: same row); advisor rerun on build_lvw: grid_and_gates 146/0, attempted_step **163/0** (the N28b fix below had reached the tree by then). **Found, reported, N28b launched:** bg_cell (type(ion_rates)) is allocated without initialization and the type has no default values, so before the first sweep the checkpoint checksum reads 8.7e244 and the row a_perturbed_state_is_not_the_checkpoint could never fire (a perturbation of 1e2 lost in the round-off of 1e244). Advisor: three forbidden phrases fixed (per-element in a printed line, neighboring, per-region). One deviation stated by the worker: an extra private -fcheck build (the shared run_fcheck.sh runs make distclean).

- **N28b** (bg_cell allocated and never initialized; ion_cell_state.f90 the type only, ionization_equilibrium.f90 the allocation site, attempted_step suite +1 row and a deterministic dirty-heap setup): **the defect is real, wider than bg_cell, and includes a LOGICAL.** Three grid-sized objects of ionization_equilibrium came out of allocate holding the recycled block: bg_cell and ieq_rate_cell (both type(ion_rates), no default values) and the real array heat_chem (the one array left out of the routine's zero list). MEASURED with a probe that frees a 16 kB block before the allocation: T_K, P_HI 8.7e244 and **x_h2_fixed TRUE** (an imposed H2 partition the caller never asked for, read from garbage, standing in the array the carrier transport reads), on a fresh heap all zero (which is why N28 saw the dead row fail and the advisor's rerun saw it pass: the defect appears only when a freed block is recycled). Fix: ion_rates carries its zero in every one of its 37 fields (34 reals, 3 logicals) as default initializers, stated as the invariant "a cell not yet swept has no gas, no rates and no imposed partition", which reaches bg_cell, ieq_rate_cell, bg_cell_adopted, bg_cell_best, the closure scratch and the four checkpoint copies at once; heat_chem = 0 at the allocation. Two comments that claimed a default initializer would not reach threadprivate copies were corrected (MEASURED with a four-thread program: it does, as OpenMP specifies). Tests: RED on the entry text with the new driver 68/2 (the_unswept_background_carries_no_rates -3.4e250; a_perturbed_state_is_not_the_checkpoint h1 == h0), GREEN 164/0 (the intermediate arm located heat_chem at 8.7e244 by a field-by-field peek). mol_carrier 200 steps bitwise (every reader of the three objects is gated on bg_ready/ieq_rates_ready, set after the loops that write them; heat_chem has no physics reader, only the checkpoint); wasp_full_newton not run (atomic: the arrays never enter a physics path). Advisor: rebuild warning-free, attempted_step 164/0. Reported: three test fixtures enumerate all 37 fields of bg_cell by hand (now redundant) and element_census_tests.f90 set two fields and left the rest to the allocator (now covered by the type).
- **N29** (the element projection does not conserve mass; binary_element_diffusion.f90 +258/-113, element_operator suite 6 -> 14 rows): **the leak is the mass the write-back is told to hand back, not the write-back.** project_elements took the mixture mass as msum = m₁ n_H + m_{He} n_{He} with m₁ = mass_per_H_nucleus_without_He(), the RESERVOIR mass per hydrogen nucleus (the trace metals at melem_ab), while the cell carries the metals at whatever the trace transport left; the projection then wrote the difference into hydrogen at every call. MEASURED at one cell of the certification column (metal/H 0.024401 against the reservoir's 0.028227): sum m_i f_i 1.0000000 -> 1.0000636 after one step, helium's

gain exceeding the hydrogen and metal loss by $6.4\text{e-}5$: mass created. A second leak of the same kind: the trace-metal arm wrote metal densities with no counter-flux in the hydrogen it diffuses through ($3.0\text{e-}3$ on the suite column with the metals frozen). **Fix:** `mixture_mass_split` reads the mass the cell's species actually carry (`bsp_mass` and `melem_A`, no species weight restated), `project_elements(f_sp, Xhe, msum, scale_metals)` scales component 1 as a whole, every helium-free species included (CO too), the metals written first and the hydrogen group closing the mixture; one projection after the metal loop puts the metals' mass change back on hydrogen; `settling_coefficient` takes its component-1 mass from the same split. Metals off is bitwise by construction (`m_1 = m_H = 1` exactly). **MEASURED:** certification column closure after the relaxation $3.37\text{e-}3 \rightarrow 2.3\text{e-}15$; the handed-back atomic candidate $6.44\text{e-}3 \rightarrow 1.94\text{e-}9$ (the file's own precision, the three-unknown control's $1.93\text{e-}9$); the writer-to-loader round trip of that state now returns it (writer ||R|| 1.112 and reload 1.1123 on the SAME row, against $3.2\text{e-}2$ vs 0.4385 on different rows before: memo section 6 closed); `mol_diffusion` and `wasp_full_newton` BITWISE (metals off / no element row); **lower_profile MOVES and should:** its stored golden is $3.6\text{e-}4$ out of closure (O/H $4.61\text{e-}4$ against the reservoir $4.90\text{e-}4$), a 1500-step cold start goes $6.77\text{e-}3 \rightarrow 3.14\text{e-}9$ in closure with T at 1.6 Rp moving 1.3 percent and H I 6.3 percent: **a golden that moves for a physics fix, refreshed at the next gate.** Atomic reload cap 40 by named outcomes: closure $6.44\text{e-}3 \rightarrow 1.94\text{e-}9$, 32/8 $\rightarrow$ 31/9, refusals 0 in both, binding row energy of cell 2 $\rightarrow$ cell 4, flux spread $2.3\text{e-}3 \rightarrow 1.4\text{e-}2$, same stop. Tests RED 4 rows on the entry text ($3.44\text{e-}3$, $4.78\text{e-}4$, $3.44\text{e-}3$, $3.82\text{e-}4$), GREEN 14/0; certification 42/0, diffusion_tests 35/0, krylov_and_dogleg 174/0. **The worker's control discipline:** the advisor's EXHALE_lwv.x is rebuilt at every verification and is a build of the live tree, so the worker built its own control (the delivered objects with only the entry-text `binary_element_diffusion.o`) and discarded three comparisons that had used the advisor's binary as "before"; workers are to build their own controls (added to the rules below). **Advisor:** `steady_species_rows_tests.f90` referenced `element_ratio_gate` through `element_inventory`, which since N28 imports it with use, only and does not re-export it, so the suite would not build (N28's validation did not include this suite; the advisor's did not either): one use `element_census`, only: `element_ratio_gate` line added by the advisor; rebuild warning-free (md5 a9627a14), `element_operator` 14/0, certification 42/0, `steady_species_rows` 189/0, `krylov_and_dogleg` 174/0; one percent fixed. Reported: the metal arm's counter-flux is closed on hydrogen but not on momentum or internal energy (a trace approximation, named); `element_census_verify` gates the element ratios and only reports the mass closure (the $6.4\text{e-}3$ state passed every gate it has): whether it should gate is a decision.

- **N8c** (the five anchors re-measured on the states after N29; the memo docs/certification_tolerance_anchoring_20260910.md only, 344 $\rightarrow$ 524 lines, no source file changed): **the reading is unchanged and is now taken on states worth measuring.** Post-N29 candidates (advisor-verified: no src file newer than the brief): atomic reload 275 iterations, 254/22, ||R|| $2.271\text{e-}4$, binding row He/H of cell 245 (r 1.153), flux spread $9.8\text{e-}7$, **mass closure $1.95\text{e-}9$** (was $6.68\text{e-}3$), 11 of 13 entries refuse; carrier reload byte-identical to N26's (no metals, no element row; the corrected projection is never reached), closure $1.93\text{e-}9$. The anchors: two evaluations of one state 0 exactly; the state's own sweep seed $1.7\text{e-}10 \rightarrow 1.6\text{e-}14$ (the loader no longer repairs a composition whose mass missed the density), closure floor $5\text{e-}19$;

state-perturbation amplification $1.2e3$ to $2.0e3$ at $r < 1.01$ and 0.4 to 0.7 elsewhere; derivative: a clean V at the base ($7.6e-8$), the PLATEAU at $6.6e-5$ at the binding cell stays (smallest resolvable row $3.9e-8$ there, $6.5e-9$ in the wind; both taken at the first JFNK iterate of a relaxation restart, two CFL steps from the loaded state, stated); manufactured column order 1.589 , closure $2.3e-15$ (was $3.4e-3$), extrapolated $1.52e-6$ at the production spacing; grid refinement of the candidate $5.93e-4$ ($N = 500$) $\rightarrow$ $2.47e-4$ ($N = 1000$), ratio 2.40 , remap control $1.5e-5$, Richardson limit $1.31e-4$: **78 percent of the row is discretization**; element flux spread in the layer **39.3 $\rightarrow$ 9.79** (He), $37.3 \rightarrow 9.09$ (H), the wind $2.6e-2 \rightarrow 2.7e-2$, the escape window $2.0e-3$. Memo: sections 3 (mass closure) and 6 (the reload) marked CLOSED by N29 with the readings; section 5 rewritten; section 7's proposal re-stated with the new floors and **option (a) supported as written** ($1e-5$ gating at $r \geq \text{cert_regime_wind_r}$, the layer reported and not gating); two of its three prerequisites discharged, the layer's flux conservation (9.8) standing. No proposal certifies either candidate (wind element row $3.05e-4$ and $7.2e-2$ against a proposed $1e-5$). certification $42/0$. Reported: the iron wind row of the atomic candidate rose $18x$ between the two chaotic states (a size, not a trend); the derivative anchor is not measured at the loaded state itself (needs the probe under stationary evaluate, steady_newton.f90).

- **Decision 22 (user, 2026-09-10): option (a).** cert_tol_element and cert_tol_carrier become $1e-5$, gating, for cells at $r \geq \text{cert_regime_wind_r}$ (1.20); the layer's element and carrier rows ($r < \text{cert_regime_layer_r}$, 1.10) are reported and do not gate until the layer's element flux conservation (9.8 after N29) is repaired; the hydrodynamic tolerances do not change; a wind-certified state says so in one line. Anchor: the operator's stated floor and a decade above the manufactured column's discretization at the production spacing ($1.52e-6$); no existing state is admitted by it ($3.0e-4$, $7.2e-2$). Implemented as N30 (launched).
- **Gate after N29 and the lower_profile refresh (advisor, make check 2026-09-10 07:58 to 11:03; refresh 11:10).** Tree after N28, N28b, N29 (N30 running on a private build). Against the 2026-09-10 00:40 goldens: **15 of 16 default cases data identical on all four files** (wasp_full_ newton now identical too, its golden being the 00:40 snapshot); **lower_profile FAILS on all four, by design** (N29's mass fix: its golden was $3.6e-4$ out of closure; v moves up to 4.8 percent at r 3.18 in 85 of 504 rows, ρ 1.0 percent in 4 rows, T 0.5 percent in 2 rows; the _adv measure column 57 percent at one row; $\log_{10} \text{Mdot}$ 7.29). The failing case was identified by its own section of the log (the advisor's first parse attributed the summary line to hp_front, which PASSES). Refreshed: golden/lower_profile archived as golden_pre_n29_20260910/lower_profile, run_check.sh golden lower_profile snapshotted the gate's output, the four files compared identical to the case output with the comparator's filter. No other golden touched.
- **N30** (decision 22 a; certification.f90 +296/-54, steady_newton.f90 +50/-30, certification suite $42 \rightarrow 57$ rows, steady_species_rows $189 \rightarrow 194$, docs/input_schema.md the cert_reason row): cert_tol_element and cert_tol_carrier are gone as single numbers; cert_tol_element_wind = cert_tol_carrier_wind = $1e-5$ with one accessor each (cert_tol_*_at(r), a step at cert_regime_wind_r = 1.20 ; cert_tol_reported_only = huge below), the anchoring written at the declaration with the memo cited; **one reduction** certification_species_row_gate decides a species row over the GATED cells (no gated cell = refused, never a satisfied row); the entries carry gated and reported maxima with their cells; the report prints

gated at $r \geq 1.2$ against $1e-5$: `<measure>` at cell `<j>` and, for a certified state with species rows, the one line certified IN THE WIND ... the rows below it are reported and do not gate: worst `<measure>` at cell `<j>`; the state file carries `cert_reason=certified_in_wind` (the field already existed). **The band $1.10 \leq r < 1.20$ is reported and does not gate** (the decision was silent; the band holds the candidates' binding cell at $r = 1.153$ where the discretization is $4.6e-4$, so a $1e-5$ gate would ask the residual to fall 1.5 decades below the error of its own discrete equation; no anchor for the band exists). `distance_from_certification` and `certified_row_measures` read the gated measures and the accessors, so **the trust region's ranking changes and both arms move** (that is the decision's intended reach): atomic reload 275 $\rightarrow$ 166 iterations at the same $\log_{10} \text{Mdot}$ 10.10, binding row He/H $2.896e-4$ at cell 288 (a WIND cell) against $1e-5$, NOT certified; carrier reload 63 $\rightarrow$ 76 iterations, H2 $7.284e-2$ at cell 285 against $1e-5$, NOT certified; the hydrodynamic tolerances, the H(n=2) level balance ($4.8e-10$ against $1e-6$) and `steady_gates_met`'s own carrier gate (`carrier_resid_th`, a volume ratio, not this tolerance) untouched. `wasp_full_newton` bitwise, still certified with no reason. Tests RED (the superseded rule refused a layer- $1e-2$ / wind- $5e-6$ row and a wind- $5e-5$ row alike, on a layer cell; four judged-distance rows moved onto the new divisor), GREEN 57/0, 194/0, `krylov_and_dogleg` 174/0. No key added (the radii are properties of the flow, constants of certification.f90). Advisor: rebuild warning-free, 57/0, 194/0, 174/0 on `build_lwv`; the stale comment in `EXHALE_main.f90` (580) rewritten; `write_output.f90`'s `# adv_input_certified` T branch now carries the reason (a wind-certified state keeps its qualification in the `_adv` header); superseded notes appended to `a2_certification_contract`, `steady_solver_design` and `element_inventory_contexts`; one neighborhood fixed. Reported: whether `steady_gates_met` should read the anchored tolerance is a separate decision on a shared interface.

- **DOC1 to DOC3** (the documentation brought to the stage-2 state, 2026-09-10, three concurrent documentation workers, no source touched, no binary run): **DOC1** `EXHALE_BC_and_IC.tex` (29 pages): the whole lower-boundary chapter described the retired component-wise ghost closure (Dirichlet density, one-way valve, isothermal ghost pressure, four keys that `input_read.f90` now refuses with error stop); rewritten as the characteristic (p , s) face condition of `base_boundary.f90` (branches by face Mach number, the smoothstep of width `base_face_mach_blend`, the Gauss-Legendre ghost averages, the three surviving controls), the upper boundary as `free_outflow_ghost`'s isothermal continuation, a new section on the transported columns' boundaries (N26, N26c, N8a, N29), the checkpoint rebuild order (N16d), and the old test matrix retitled as the retired closure's, not reproducible by today's binary. `composition_restart_and_base_handoff.tex` (18): thirteen base species in the nuclei count (OH, H2O, CO added), every `load_IC.f90` line citation re-derived, a new section for the restart contract (intent, the eight-line metadata block field by field with refuse/report on each, provenance unknown, the grid guard, Restart option change), the round-trip and projection evidence of N9 and N29 labeled as read from this log. `transmission_spectrum.tex` (11): a new section for `read_adv_validity` (schema 2, the five statuses, `adv_conditional_tol`), the `chord_shell_indices` census with the CONTRIBUTION statement, and the `transit_schema 1` block. **DOC2** `steady_solver_memo.tex` (16): a new section 7 for the species-row solve as it stands (merit on certification scales, the trust region with named outcomes and the leg rules, the constraint rows, the empty-column repair and the equilibrated preconditioner, the three arms measured and disarmed, the linear solve as the limiter with the additivity finding, the ulp-level chaos, the residual

contract, the certification by regime), eleven hooks added to its table. `molecular_hydrogen_treatment.tex` (23): the `_adv` row corrected (the energy residual reads the upstream caloric energy since N12), the layer table RE-MEASURED on the current `mol_lyman_werner` output (144/184/476 cells, H2 peak $5.15e12$, 1 percent edge $4.48e9$ against the stale 227/251, $4.63e13$, $4.42e9$), thirteen of sixteen cases pinned, a new section on the element budget, the H3+ departure factors marked STALE in place, the `mol_ir_bands` peak re-measured ($2.148e-8$ at $r\ 1.0059$ against the stale $2.934e-7$). `lower_atmosphere_coupling.tex` (28): a new section 7.4 on the element operator (N29's conservation, N26b/N26c's boundary, N8c's flux spread table as the open item); two pre-existing overfull boxes fixed. `EXHALE_user_manual.tex` (88): certification by regime in section 2.7, the Run mode key entry, eleven hooks, the coupling header example replaced by the line the code writes (`cert_reason`), `Cooling_breakdown.txt` columns 12 to 14 (H2, H2O, CO infrared) restored to the column list. **DOC3** `README.md` (sixteen-case matrix, 25 suites, the restart contract, the transit metadata, certification stated, nineteen examples), `README_HOWTO.md` (`Resid tol` default $1e-5$ not $1e-3$, `Flux spread tol` default $2e-5$ on the face flux not $5e-3$, a certification table, a restart section, an eleven-row hook table, the sixteen cases and 25 suites), the NEW docs/`code_status_20260910.md` (418 lines, replaces the 2026-09-05 file as the read-first document; section 3.3 states what is and is not certified and what limits the rest), the project `CLAUDE.md` pointer line, `TO_BE_DONE.md` (an index of what is open, three headings corrected, 35 em-dashes removed), `steady_solver_design.md` section 22. All eleven PDFs rebuilt, `grep -c '^!' 0` on every log. Advisor: every default and constant quoted checked against the source (`flux_spread_th_default` $2.0e-5$, `resid_max` $1e-5$, `cert_tol*_wind` $1e-5$, `cert_regime_wind_r` 1.20, `ratio_sound_band` $1e-2$, `tr_delta_min_dimensionless` $1e-6$, the `Cooling_breakdown` header); one constant name corrected in the memo; pre-existing British spellings fixed in the workers' files, in `ISSUES_20260909.md` and in the handoff; the 51 em-dashes of the workspace `CLAUDE.md` replaced. **Reported by the workers and CONFIRMED by the advisor** (ISSUES 3.7): the base reservoir is stated once at startup and the refreshes of `dp_bc` and `ntot_bc` after every sweep are dead writes (the boundary itself is right: the reservoir equals the user's base pressure by construction); `golden/mol_ir_bands/` carries no `Cooling_breakdown.txt`; `examples/README.md` covers part of nineteen folders under the wrong planet; the .md twins of two .tex documents have parted from them; the `max(rest, 0)` write-back clamp (N4b) restated. ISSUES section 5 rewritten to the state after N30; the handoff opens with a dated 2026-09-10 state block. The carrier reload is still a scratch recipe (`B5k/mkrun.sh carrier`), not a pinned fixture: to pin with N31.

- **N31** (why the finite-difference Jacobian action is not additive across directions, and two probe arms; `steady_newton.f90` +563/-3, of which the arithmetic is two probe rules and the rest is a default-off hook, `krylov_and_dogleg` 174 -> 190 rows, new fixture backup/regression/`carrier_elem_newton`): **the action is not additive because the residual carries a NON-SMOOTHNESS FLOOR five to nine decades above the double-precision rounding of its own rows, and the difference quotient along the PRECONDITIONED Krylov directions divides that floor by an increment five decades smaller than a generic direction produces.** N25's reading (curvature over the fixed arc) is refuted at the iterates that matter and confirmed at one that does not. MEASURED on the atomic element reload with the new hook `EXHALE_JV_ADDITIVITY=1` ([diag 13], at outer 1, 20 and the last of a capped run, on the first two Arnoldi directions and on two directions off the basis,

at a tenth, one and ten times the probe arc): the defect of the Arnoldi pair scales as the arc to the power **-1.026** at outer 1 and **-0.999** at outer 40 (rounding), the off-basis pair as **+1.125** and **+2.109** (curvature, and $3.2\text{e-}8$ and $2.1\text{e-}8$, negligible), and at outer 20 both are curvature at cells 1 and 2. The second difference of the residual on the defect-holding rows confirms it directly: ratios 0.774 and 1.676 at outer 1 where a smooth residual gives 4 and 4, and the floor stands **9.5e5** (outer 1) to **2.0e9** (outer 40) times epsilon times the residual on those same rows, so it is not the arithmetic of the assembly. READ, the only tolerance in the atomic path that can set a floor of that size: `ionization_equilibrium.f90` line 969 hands `hybrd1` an `xtol` of `sqrt(dpmpar(1)) = 1.49e-8` (lines 1432, 2110, 2114), a statement about the step and not the residual, so the composition is a piecewise map of the state; `eval_residual` fixes the number of composition passes but not `hybrd1`'s own iteration count. **A decision of the feasible set is ruled out:** on all six probes of the three iterates, components blocked 0, forward side, and the step taken equal to the nominal step. **The localization gives no unknown class to fix:** at outer 1 the squared defect is 0.31 mass, 0.64 energy, 0.03 momentum, 0.02 helium, under $1\text{e-}4$ in the eight trace elements, and nine tenths of it sits in 35 of the 500 cells between 1 and 75. The scale reading is a uniform shortfall: the largest unknown of the state is displaced by $2.2\text{e-}8$ of itself (about `sqrt(epsilon)`) and almost every other by $1\text{e-}10$ to $6\text{e-}9$, one to two decades short, with no unknown near a bound (nearest face $2.9\text{e-}5$ against a displacement of $4.5\text{e-}14$). The model that follows, `defect ~ floor / (eps ||Jv||)`, predicts both pairs at outer 1 within a factor three. **Two arms, both default off.** `EXHALE_JV_COLUMN_SCALE=1` (`probe_step_on_the_column_scales`, `eps0 = sqrt(eps)(1 + ||Dc-1 Y||) / ||Dc-1 v||`) is the brief's remedy and it is WORSE, for the reason the diagnosis gives: the Krylov direction's weight sits on unknowns of small column scale, so the rule SHORTENS the arc by 42 and the defect rises $8.430\text{e-}6 \rightarrow 1.466\text{e-}4$ at exponent -0.965, the 1/arc law reproduced by a change of the arc made for another reason. `EXHALE_JV_PROBE_ARC=<x>` (`jv_probe_arc_scale`, default 1) is the lever the measurement points at, and it wins the linear question and loses the Newton by the same change: at $x = 100$ the defect of the Arnoldi pair falls $8.430\text{e-}6 \rightarrow 2.240\text{e-}6$, the exponent turns +1.139 and the second differences become 4.065 and 4.022, so the rounding regime is gone and the quotient is truncation-limited, while the atomic reload then ABORTS at outer 33 (`info = 2`, `||R|| 1.504`, flux spread $5.902\text{e-}2$, 12 steps refused by the model with the actual reduction 26.4 times the predicted, radius $2.028\text{e-}7$) where the control runs 144 iterations to $3.641\text{e-}4$ and $5.557\text{e-}6$ with no model refusal. At $x = 20$ it aborts at outer 43 at 1.524 with 21 model refusals and a radius of $2.076\text{e-}13$; the carrier reload runs its 40 iterations at both values and hands back 1.006 at $x = 100$ and $9.076\text{e-}1$ at $x = 20$, each with 2 model refusals, against the control's $9.572\text{e-}2$ with none. **At $x = 5$, which does NOT stall**, the binding regime is reached (outer 58, `||R|| 3.087e-3`, element Fe of cell 146, against the control's $3.386\text{e-}3$ and element Fe of cell 147 at outer 60) and THERE the Krylov cycle still spends 40 of 40 products at a relative residual of 0.898 to 0.980 against the 0.1 asked, indistinguishable from the control's 0.935 to 0.959. **The substance of the item:** the additivity of the linear operator wants a LONGER arc (defect $\sim$ floor / arc) and the trust region's model wants a SHORTER one (truncation $\sim$ arc, and the ray test refuses a step whose model and true slope differ in size), the arc already sits at the Newton's balance point, and every direction was measured to lose, so NO probe rule fixes this. What can is lowering the residual's own floor, which is not double precision. Column arm named outcomes at cap 40, arm against control: atomic `||R||` at 40 1.674 against $7.607\text{e-}2$, handed back 1.633 against $5.377\text{e-}2$, flux spread $4.985\text{e-}1$ against $3.555\text{e-}3$, 30/10 against 32/8; carrier `||R||` at 40 $1.357\text{e-}1$ against $1.070\text{e-}1$, handed

back 1.263e-1 against 9.572e-2, flux spread 7.718e-2 against 4.635e-2, 31/9 against 33/7, SAME binding row (carrier H2 of cell 205), no new gm_no_direction_sampled and no refusal of any class in either. The one thing the column arm improves is the count of Krylov cycles reaching the tolerance asked (40 of 40 against 26 of 40), which is the REDUCED problem's residual on an operator that is not the Jacobian and is precisely why that count must not be read as progress. NEITHER ARM IS ADOPTED: both fail the first clause of the acceptance rule outright. **Default path unchanged:** with every hook off the atomic reload's 40 iteration lines are identical to the control's to the digit, the carrier reload's binding row and ||R|| at 40 are the control's, and wasp_full_newton is BITWISE but for the provenance timestamp; with EXHALE_JV_ADDITIVITY=1 on, the 40 iteration lines are still identical, so the hook is inert armed as well as disarmed. Tests: krylov_and_dogleg RED on the control objects (seven symbols absent), GREEN 190/0; steady_species_rows 192/0, certification 57/0, element_operator 14/0, residual_determinism 5/1 with the same row failing by design in the control build. **Fixture pinned (deliverable 0):** backup/regression/carrier_elem_newton now holds the carrier reload that existed only as a scratch recipe, with run.sh, a README and the control of the unmodified tree at cap 40 (||R|| 1.969, 1.984, 1.921, 6.800e-1, 1.070e-1 at 1, 5, 10, 20, 40; handed back 9.572e-2; flux spread 4.635e-2; info = 1; 33 accepted, 7 rejected; 40 of 40 products every cycle; binding row carrier H2 of cell 205; 2754 samples admitted, none refused). A capped run of that fixture never reaches a mass-loss rate. **Defect fixed on the way:** the new probe rule was first written with a local integer n , which is the grid size N of global_parameters under Fortran's case-insensitive rules, so its own size guard fell through on every call; the new suite rows caught it at once and the local is now nunk. Not claimed: that the floor is hybrd1's xtol -- that is READ from the source and not measured, and the experiment that closes it (tighten that xtol, re-measure the floor) belongs to ionization_equilibrium.f90, which this item does not own. The comment of the suite row the_probe_arc_stands_above_the_residual_noise was restated: the arc IS nine decades above the residual's reproducibility as a length of the whole vector, but the conclusion drawn from it about the quotient does not follow. Advisor: rebuild on build_lwv (one linker warning, pre-existing), krylov_and_dogleg 190/0, steady_species_rows 194/0, residual_determinism one row FAIL by design; atomic reload at cap 40 with every hook off, 8 threads: 1.888, 1.891, 1.819, 1.583, 1.861, 8.964e-1, 7.607e-2 at 1, 5, 10, 15, 20, 30, 40, handed back 5.377e-2, flux spread 3.555e-3, identical to the worker's control; the carrier fixture's README control checked against its own run.log; the fixture README is the reference for every later carrier-arm number (the 0.25 quoted after N26 was a scratch-recipe run, not this fixture). Accepted. **Next item N32:** the residual's floor (the inner solves of the composition and of the temperature stop on step or bracket tolerances, newton_dense, hybrd1 xtol sqrt(eps), brent_root 1e-10); measure whether the additivity floor follows them.

- **N32** (which inner tolerance sets the residual's non-smoothness floor; two default-off tolerance hooks and a widened [diag 13], ionization_equilibrium.f90 +47/-2, T_equation.f90 +42/-3, steady_newton.f90 about +130 inside the additivity hook, krylov_and_dogleg 190 -> 203 rows): **NEITHER. The floor follows neither the composition solve's stopping tolerance nor the temperature bracket's, and it is not in the chemistry at all: it sits in the HYDRODYNAMIC rows of the residual.** New hooks EXHALE_IEQ_TOL=<x> (replaces tol = sqrt(dpmpar(1)) for the run; new function composition_solve_tolerance, cached once by ioniz_eq, announced only when armed) and EXHALE_TEQ_TOL=<x> (the bracket of brent_root, now a

module variable at its old $1e-10$ through `temperature_bracket_tolerance`). MEASURED on the atomic element reload at outer 1 (cap 40, 8 threads, one entry state for every arm): the additivity defect of the Arnoldi pair is $8.174e-6$ at exponent -1.063 (default $1.49e-8$), $7.452e-6$ at -0.995 ($1e-10$), $9.348e-6$ at -1.059 ($1e-12$) and $9.546e-6$ at -1.064 ($1e-14$), and the second difference on the defect rows stays on its floor at $5.6e-12$ to $1.6e-11$ with no trend in the tolerance: **the law is FLAT**, over six decades. The measurement is exactly reproducible (three control runs, and 1 thread against 8, give $8.174e-6$ and $1.362e-11$ to every printed digit), so the ± 17 percent between arms is a real movement of the state and not a fall. `EXHALE_TEQ_TOL=1e-13` is inert to the bit through all 40 iterations, which confirms as MEASURED what was READ, that `brent_root` is reached only from `post_process_adv` and never from `eval_residual`. **Why the composition cannot be the floor**: quadratic convergence pins its root at the SQUARE of the step tolerance. On the reported cell the accepted $x(H^+)$ is $2.037266074305163e-12$ at the default and $2.037266074305300e-12$ at $1e-14$, agreeing to 13 significant digits ($7e-14$ relative), and the pinned fixture's own `run.log` records an accepted reaction residual of $2.91e-16$ at maximum for the steady sweeps; the floor implies a jitter of about $1e-9$ relative, five decades above that. The tightest tolerance costs 30 percent more wall time per residual evaluation (0.090 s against 0.066 s at 4 threads, 17 evaluations each) and buys nothing. **Where the floor IS**, from the widened hook: the energy SOURCE, which is where excited_hydrogen, `lya_rt`, the cooling table interpolation, the write-back clamps and `comp_T_from_p` all end up, is smooth on the arc and five decades below the floor (second difference of heat - cool on the defect cells $3.039e-17$, $2.474e-16$, $1.046e-15$ at half the arc, the arc and twice it, ratios 8.1 and 4.2, textbook h squared, against the residual's own $1.362e-11$ on the same rows); by row class over the whole column the floor is mass $1.801e-11$, energy $3.090e-11$, momentum $5.946e-12$ against element He $4.269e-12$ and every trace element at $1e-16$ to $2e-13$; nine tenths of it sits in 29 of 500 cells between cell 1 and cell 58. The mass row carries no source at all, so no part of the chemistry can be its cause. The competing explanation, that the row is a small difference of large terms and its floor is their rounding, is REFUTED: the largest of heat and cool on those cells is $3.424e-3$ against $||F||$ $5.507e-3$, a ratio of 0.62, and epsilon times it is $7.6e-19$. **Deliverable 2 (the stationary-solve-only polish arm) was NOT BUILT**: the brief conditions it on deliverable 1 naming a tolerance, and none is named. **A correction to the brief and to N31**: on this fixture `solve_ieq` and `newton_dense` never run. The helium triplet is on (the fixture's `EXHALE_setup.out`) and `metals.inp` is present, so the sweep calls `hybrd1(ion_system_HeH_TR_metals, ...)` directly; MEASURED `nt_calls` 0 over the whole hook and no "ioniz-eq solver" line in the fixture's reference `run.log`. `hybrd1` returns `info = 1` at the default, at $1e-10$ and at $1e-14$ alike, so there is no `info = 2/3` and no fallback to count, and `newton_dense`'s hardcoded step test $1e-11 \times xscale$ is not in this path. **Default path unchanged**: `wasp_full_newton BITWISE` in `Ion_species.txt` and identical but for the provenance timestamp in `Hydro_ioniz.txt`, every (JFNK) line identical; the atomic reload's 40 iteration lines identical to the control's with every hook off. Tests: `krylov_and_dogleg` RED on the control objects (both new functions absent), GREEN 203/0 with 13 rows added; `steady_species_rows` 194/0, certification 57/0, `element_operator` 14/0, `adv_static_limit` 53/0, `coupled_source_step` 30/0, `physics_probe` 1491/0, `residual_determinism` 5/1 with the same row failing by design as in the control. Noticed, not fixed: the fixture README does not say the helium triplet is on, which is what selects the direct `hybrd1` branch and is what sent this item's brief to the

wrong routine; and newton_dense applies the caller's tolerance to fnorm while its step test uses a hardcoded $1e-11 \times \text{xscale}$ no caller can set, which would be the real floor on a fixture that does reach it. **Next item:** the floor is in the flux assembly, not in ionization_equilibrium.f90 -- the candidates left are the reconstruction where the limiter is not the frozen WENO3 weight, the Riemann wave-speed estimate, Apply_BC, and the transport coefficients of the diffusion operator, which are the one remaining way the composition reaches a hydrodynamic row. Advisor: rebuild on build_lwv; krylov_and_dogleg 203/0, steady_species_rows 194/0, adv_static_limit 53/0, residual_determinism the same single row FAIL by design; atomic reload at cap 40 with every hook off, 8 threads: the N31 ladder to the digit (1.888, 1.891, 1.819, 1.583, 1.861, 8.964e-1, 7.607e-2; handed back 5.377e-2, spread 3.555e-3), so the worker's 1.112 at iteration 40 is the thread-count sensitivity of N26c, not a change of the path. The atomic fixture README now states the triplet is on and which branch that selects; newton_dense's hardcoded step test recorded in ISSUES 3.7. Accepted; deliverable 2 correctly not built. **Next item N33:** the jump itself, scanned along the Arnoldi direction across four arcs and split by term and operator (base boundary, reconstruction, Riemann flux, caloric inversion, composition read, transport sources).

- **N33** (where the residual's non-smoothness floor is: the term, the operator and the branch; a default-off multi-scale jump scan [diag 14], steady_newton.f90 +710, krylov_and_dogleg 203 -> 220 rows): **THERE IS NO BRANCH. The floor is the ROUNDING OF THE FLUX ASSEMBLY, raised above the last bit of the row by two cancellations that compound.** The Roe flux of a face is built from the JUMPS of the two reconstructed states, and in the nearly hydrostatic base layer those jumps are $3.2e5$ to $5.9e6$ times smaller than the states themselves, so the interface flux carries the last bit of $O(1)$ quantities and not of its own value; the row is then the flux difference divided by the cell volume, $r^2/dV = 5.12e3$, one over the $1.955e-4$ width of a base cell. The bound $\epsilon \times (\text{face state}) \times r^2/dV$ is what a correctly rounded assembly cannot go below, and MEASURED it IS the floor: $1.675e-12$ against $1.240e-11$ at cell 1, $2.520e-13$ against $1.644e-12$ at cell 84, $4.326e-15$ against $2.366e-14$ at cell 167, $4.241e-17$ against $4.409e-16$ at cell 251 -- the bound falling 4.6 decades with the measured floor following it, ratio 5.5 to 12 throughout, and the ten cells holding the largest second difference are cells 1 to 23, every one a base cell. Beyond cell about 250 the bound falls below the residual's own curvature and stops binding, which is what a floor does. New hook EXHALE_RESID_JUMP_SCAN=1 samples F at 65 points of $Y + t \cdot v$ on ten windows, each a quarter of the one before, from two probe arcs down to two arcs over 4^9 ; one window is not enough, because at the standard arc every interval already holds thousands of steps and none stands above its neighbors. **The measurement:** on the atomic element reload at outer 1 the tracked row is the energy row of cell 1, its median first difference falls by exactly four whenever the spacing does (the smooth part is a straight line), and its second difference does not fall at all -- $1.0e-11$ to $1.6e-11$ over 5.4 decades of spacing, $1.8e-5$ down to $6.9e-11$ -- which is a step of F and not a kink or curvature. Isolated steps appear once the smooth increment per interval falls below the floor: 28 of 64 intervals at $3.05e-5$ arcs and 61 of 64 at $7.63e-6$ arcs, at least $4.0e6$ stepping intervals per probe arc, the largest moving its row by $2.625e-12$, $2.749e-9$ of the row. **The term and the operator,** at the largest step (cell 9, mass row): S of the mass row is identically zero and the whole step is in dF ; every quantity upstream of the flux -- the ghosts, T , n_{tot} + n_e and the four reconstructed face states -- moves by one to eight units in the last place of an $O(1)$ number with no outlier, in exact multiples of $2.220e-16$; the first quantity that steps is the interface flux, by $4.803e-16$

where its neighbors move by $3.2\text{e-}17$, and $4.803\text{e-}16 \times 5116.3 = 2.457\text{e-}12$ against the row's measured $2.625\text{e-}12$. **No branch fires anywhere near it:** faces falling back from Roe to HLLE 0 and unchanged, face states the positivity limiter scaled 0 and unchanged, so the base boundary's Newton, characteristic_branch_weight, the Mach cap, the ghost quadrature, the stencil selection, the wave-speed branch and comp_T_from_p are all continuous here. The off-basis direction carries the SAME floor, $9.1\text{e-}12$ to $1.4\text{e-}11$ flat over 5.4 decades, and isolates no step at any window, because its smooth increment never falls below it: the floor is a property of the residual and not of the direction, and what the preconditioned direction does is shrink the numerator of the difference quotient by five decades. The carrier reload reproduces the law (floor $1.532\text{e-}11$, bound $1.382\text{e-}12$, ratio 11.09), so it is the discretization at the base and not one fixture. **N31 corrected:** the floor's size stands, but "five to nine decades above the rounding of its own rows" compared it against $\epsilon \quad ||F||$ on the row, which is the wrong reference for a row built as a flux difference over a cell width; against the right one it is a factor of ten. N32's localization is confirmed and made quantitative. **Deliverable 2 NOT BUILT:** the brief conditions the arm on a branch being named and there is none. The treatment the measurement does name, for the advisor, is a WELL-BALANCED flux difference at the base -- assemble the rows from the departure of the face states from the local hydrostatic isentrope, which base_boundary.f90 already integrates, so the balance cancels analytically instead of in floating point -- in Num_Fluxes.f90, RK_rhs.f90 and Source.f90, none of them this item's files; the cheaper partial lever is the base cell width, on which the floor depends as one over it. **Default path unchanged:** the atomic reload at cap 40 gives all 338 (JFNK) lines identical to the control's through done info=1 $||R|| = 1.112\text{E+}00$; with the scan armed the outer-1 line is the control's to every digit; wasp_full_newton BITWISE in Ion_species.txt, Ion_species_adv.txt and the three breakdown files, differing only in the provenance timestamp in the two hydro files, log10 Mdot 13.30 on both. Tests: krylov_and_dogleg RED on the control objects (resid_jump_scan_on absent), GREEN 220/0 with 17 rows added; steady_species_rows 194/0, element_operator 14/0, certification 57/0, residual_determinism 5/1 with the same row failing by design and the same measured $3.354\text{e+}01$ on the control binary. Noticed, not fixed: global_parameters declares integer :: count, which shadows the intrinsic COUNT in every scope that uses the module, so count(mask) written anywhere fails to compile -- the same class as the t0 and g2s incidents and worse in that the shadowed name is an intrinsic; renaming a public global was out of scope. Advisor: rebuild on build_lwv; krylov_and_dogleg 220/0, steady_species_rows 194/0; atomic reload at cap 40 with every hook off, 8 threads: the N31 ladder to the digit ($7.607\text{e-}2$ at 40, handed back $5.377\text{e-}2$, spread $3.555\text{e-}3$), no [diag 14] line; two British spellings in the new comments fixed. Accepted; deliverable 2 correctly not built (there is no branch to smooth). The finding is structural and reaches the marching discretization, so it is put to the user as DECISION 23 (well-balanced flux differencing recommended; quad accumulation of the flux difference as the control experiment; coarser base cells; accept the floor). The count global shadowing the intrinsic recorded in ISSUES 3.7 (HYG). **Next item N34** (runs without a decision, changes no default): the control experiment, quad accumulation of the face fluxes and their difference inside the stationary residual only, as a measured arm.

- **N34** (the control experiment of decision 23: the hydrodynamic rows of the stationary residual in quadruple precision). EXHALE_RESID_QUAD=1 sends the STATIONARY evaluations of assemble_residual -- the tag ieq_sweep_state_kind puts on an iterate or a probe; the marching stages reach RK_rhs directly and are un-

touched -- through a quadruple-precision instantiation of the whole flux assembly, from the conserved state through the caloric EOS, the PLM limiter and the WENO3 weights, the Roe flux with its entropy fix and HLLE fallback, the geometric and gravitational sources and the division by the cell volume, rounded to double once, at the row. One source text (hydrodynamic_rows_body.inc, 1111 lines, kind-generic) instantiated twice by hydrodynamic_rows.f90; the composition, the heating and the cooling stay double and enter converted exactly. **The double instantiation is the production operator BIT FOR BIT**, asserted three ways: a suite row on a stated grid (largest difference exactly zero in dF, S and the four face states, tolerance 0), and two whole 40-iteration reloads under EXHALE_RESID_QUAD=2 -- the atomic (all 355 (JFNK) lines identical) and the carrier, which is the molecular one and so exercises the generic caloric EOS (all 364 identical). The quadruple instantiation reaches the analytic mass row of a uniform state to 1e-30 before its rounding, where the double one stands at its own. **Verdict: the floor is confirmed and is NOT removed.** On the atomic reload at outer 1 the tracked row's non-smoothness floor falls only 1.240e-11 to 5.917e-12, and cell by cell it lands on $\epsilon_{\text{DOUBLE}} \times \text{face state} \times r^2/dV$ -- ratio 0.93 to 1.4 over the base cells against 5.7 to 8.9 in the control -- and not on the quadruple bound, which is 1.224e-30, 4.8e18 below what is measured. So N33's amplifier (the face cancellation 3.2e5 times the r^2/dV 5.1e3) is confirmed a second way and its attribution completed: four fifths of the floor is the arithmetic INSIDE the assembly, the rest is the double representation of what the assembly is handed and hands back -- the ghosts Apply_BC writes, the sweep's temperature and particle count, and the double row itself -- which precision inside one operator cannot reach. The detector isolates no step at any of the ten windows (control: 28 of 64 and 61 of 64 at the two narrowest), but the second difference is still flat over 5.4 decades and the row still changes sign 39 times in 64 intervals: a staircase with a smaller rise. **What the arm does buy is the fidelity of the Krylov model:** the additivity defect falls 8.174e-6 to 1.009e-6 with the exponent in the arc still negative (-1.063 to -0.901, so still rounding and not curvature), and at the binding iterate of the atomic reload at cap 40 the relative gap of the Arnoldi image falls 1.188e-1 to 3.813e-3, a factor 31, the returned relative residual then being the true one to four digits (0.2114 against 0.2114; the control returned 0.1122 for a true 0.1617). The element rows, which do not pass through the assembly, are unmoved to four digits (C 1.367e-13 to 1.366e-13), which is the control inside the control. **The Newton does not converge and the Krylov cycle still stalls.** Atomic cap 40, arm on: 40 of 40 products at 0.21 against the requested 0.10, $|R|$ 8.413e-1 handed back against the control's 1.112 with trajectories parting at iteration 9 (the arm is chaotic at the ulp level, so that is not by itself a result). Atomic cap 250: both stop on the stagnation detector, control at iteration 167 at 3.047e-4, arm at 136 at 4.096e-4, both with 40 of 40 at 0.983 to 0.989, both binding on the element Fe rows of cells 132 to 137, both NOT CERTIFIED with 11 entries of the inventory refusing. Carrier cap 40: unchanged in every named outcome -- binding row the H2 carrier row of cell 205 throughout, 40 of 40 at 0.9886 against 0.9729, $|R|$ 9.336e-2 against 9.427e-2 -- as it must be, since that row is assembled by carrier_steady_residual and never sees the flux. Cost MEASURED: the flux assembly alone 46x at 8 threads and 74x at one (500 cells, 3.628e-5 s to 1.681e-3 s), the whole atomic reload at cap 40 1.11x (271 s to 302 s), because a stationary residual is mostly the ionization sweep. **Nothing adopted, no default changed:** wasp_full_newton bitwise but the provenance timestamp, log10 Mdot 13.30 on both; the atomic and carrier reloads identical line for line with the arm off. Tests: krylov_and_dogleg RED on the control objects, GREEN 232/0 with 12

rows added; steady_species_rows 194/0, element_operator 14/0, certification 57/0, energy_update 53/0, acceptance_classes 22/0, adv_static_limit 53/0, species_face_flux 1/0, spectrum_type 154/0, grid_and_gates 146/0, attempted_step 70/0, coupled_source_step 30/0, residual_determinism 5/1 with the same row failing by design and the same $3.354e+01$ on the control binary. One file outside the brief's list: caloric_eos.f90 gained three read-only accessors (the mixture ratios and the H2 rovibrational table nodes) with no arithmetic and no behavior change, because the generic pipeline cannot evaluate the molecular caloric EOS without them; the molecular whole-run bitwise comparison is the evidence that they changed nothing. Noticed, not fixed: backup/regression/atomic_elem_newton/README.md still names the helium row of cell 246 as the binding row at cap 250 from N26/N27, where the measurement on this tree is the element Fe rows of cells 132 to 137 at iteration 167 and $|R|$ 3.047e-4. **For decision 23:** extra precision confirms the diagnosis and does not cure it; the amplifier has to go (well-balanced flux differencing, or a coarser base cell, which buys one factor of the width), and even with the hydrodynamic rows made model-faithful the two rows that actually bind -- the element rows of the atomic reload and the H2 carrier row of the carrier reload -- are not hydrodynamic rows and are untouched by any of it. Advisor: rebuild on build_lwv (the new hydrodynamic_rows.f90 in SRC); krylov_and_dogleg 232/0, steady_species_rows 194/0, energy_update 53/0; atomic reload at cap 40 with every hook off, 8 threads: the N31 ladder to the digit $(7.607e-2)$ at 40, handed back $5.377e-2$). The caloric_eos.f90 accessors (read-only, outside the brief's list) accepted on the molecular reload's bitwise evidence. Accepted; nothing adopted. Decision 23 restated with this result: option (b) retired as a remedy, (a) still the right discretization but not by itself the cure of the species-row solve; the atomic fixture README gained a 2026-09-10 state section (binding rows Fe of cells 132 to 137 at cap 250 on this tree). **Next item N35** (runs without a decision): the conditioning of the preconditioned operator on the element and carrier rows at the binding iterate, and what the banded preconditioner misses there.

- **N35** (what holds the Krylov cycle on the species rows once the rounding floor is removed: the conditioning of the preconditioned operator and what the banded preconditioner misses). Three default-off measurement hooks on the linear system of one outer iteration, adopting nothing: EXHALE_KRYLOV_SIZE_SCAN=1 ([diag 16]) reruns the SAME cycle at 40, 80, 160 and 320 products and the last two again with the basis orthogonalized twice, with EXHALE_GM_HISTORY=1 printing the reduced residual after every product and the true one every twenty; EXHALE_PRECOND_SPECTRUM=1 ([diag 15]) takes 200 Arnoldi products of $A_z M^{-1}$ from a deterministic start, diagonalizes the Hessenberg (dgeev) and reports the Ritz values of the whole operator and of its compressions onto the species and the hydrodynamic rows, with the localization of the three smallest Ritz vectors; EXHALE_BAND_DIFFERENCE=1 ([diag 17]) measures $(A - A_{\text{band}}) v$ on those Ritz vectors and on the first two Arnoldi directions, splits the columns they live on into the part inside the band, the band's disagreement there and the part outside, with the radiation live and frozen, and reads the binding row's band row by class as a fraction of its diagonal. **Verdict: the band misses nothing, and the operator itself is near-singular on the species rows.** Its action agrees with the full finite-difference one to $4.0e-5$ to $1.5e-3$ relative on every direction tested on both fixtures; the columns hold 99.4 to 100.0 percent of their norm inside the band (the largest outside part is 0.6 percent, element He of cell 308); and what remains is not on the rows that bind at all, but 0.999 in the momentum row of the LAST cell (carrier) and 0.83 to 0.89 momentum plus 0.11 to 0.17 energy in five

to seven cells (atomic), the element and carrier rows taking 0.0000 to 0.003 of it. So the two remedies the brief conditioned on -- a rank-few radiation correction by the Woodbury identity, or a wider `kl_jac` -- would each repair something already right, and **nothing was built. What IS wrong, MEASURED at the binding iterate of both fixtures.** (1) The preconditioned operator is well conditioned on the hydrodynamic rows and near-singular on the species rows: the Ritz spectrum of the compression onto the hydrodynamic rows has a magnitude ratio 43.4 (carrier, carrier H2 of cell 205 binding) and 158 (atomic, energy of cell 52 binding), none and three of 200 below $1e-2$; onto the species rows, $6.56e5$ (carrier, smallest $1.493e-6$, nineteen below $1e-2$, one below $1e-4$) and $1.06e4$ (atomic, smallest $9.366e-5$). The smallest Ritz vector is 0.71 to 0.91 carrier rows on the front cells 199 to 205 and 458 to 466 (carrier) and 0.9955 element rows over 145 cells (atomic), and the carrier fixture's near-zero part is a complex pair, not a real null direction. The attribution is there from outer 1 ($2.17e5$ against 39.6). And the discretization statement behind it: the band row of carrier H2 of cell 205 has diagonal $1.2396e-1$ against off-diagonal magnitudes $4.1026e2$, **3.304e3 times the diagonal to the hydrodynamic unknowns of its own stencil and 5.65 times to the neighboring carriers**, so the binding row's own unknown is three decades below its coupling to another class and there is no diagonal for a preconditioner to stand on. (2) **A larger subspace makes the step worse, not better**, which is neither of the brief's two alternatives. The reduced least-squares residual falls monotonically with the size while the TRUE residual of the returned step stops falling at 60 to 80 products and rises: atomic with the quad arm on, reduced $2.658e-1$, $1.781e-1$, $1.144e-1$, $9.958e-2$ at 40, 80, 160 and 182 products against TRUE $2.851e-1$, $2.575e-1$, $3.006e-1$, $3.203e-1$ -- the cycle announces its tolerance reached at 182 and hands back a step leaving 0.32, three times what was asked; carrier, reduced $9.677e-1$, $8.001e-1$, $4.639e-1$, $1.434e-1$ against TRUE $9.731e-1$, $8.638e-1$, $9.668e-1$, **1.472e+00**. With the quad arm off the true residuals are a factor 3 to 6 worse at every size (atomic $7.856e-1$, $3.905e-1$, $7.580e-1$, $2.004e+00$) and the shape is the same. Reorthogonalizing the basis changes nothing at any size, the measured loss of orthogonality being $0.000e+00$, and the Arnoldi image of each column at the same iterates is exact to $3.30e-16$ and $3.77e-16$: the recursion is not the defect. What fails is additivity -- the action of the sum of the first two directions departs from the sum of their actions by $1.357e-3$ (atomic) and $7.601e-5$ (carrier), the assembled 40-product step's image by $1.068e-1$ and $2.256e-1$ -- because the matrix-free action is the secant of a nonlinear residual over a fixed arc and the step it is extrapolated to grows from 0.31 to 11.4 while the arc stays $5.2e-6$. Tests: `krylov_and_dogleg` 251/0 with 19 rows added (RED on the control objects, which do not carry the routines), among them the Ritz values of a stated upper triangular 6 by 6 spanning six decades reproduced to $1.171e-12$ with the recursion carried to the dimension of the space, the column split summing to the whole difference with an empty band and with an exact one, and the four hooks off by default; `steady_species_rows` 194/0, `element_operator` 14/0 (the three suites whose driver links `steady_newton`). Default path: the atomic reload at cap 40 with every hook off, new binary against my control, all 354 (JFNK) lines IDENTICAL through `||R||` $5.377E-02$, `flux spread` $3.555E-03$, the N31 ladder to the digit; `wasp_full_newton` bitwise but the provenance timestamp, `log10 Mdot` 13.30 on both. Cost MEASURED: the size ladder is 1082 products per diagnostic iteration (the carrier reload at cap 40 from about 3.5 to about 9 minutes at 8 threads), the spectrum 600 and the band difference about 15. Noticed, not fixed: the frozen-radiation column of element He of cell 410 carries $7.267e-4$ of its $2.816e-1$ norm outside the band where the WENO3 stencil says it should carry none, 0.26 percent

and two decades below anything in this item's conclusions; the likely route is the base boundary state recomputed from the iterate, not confirmed. Also corrected in the reading: the preconditioner is `build_banded_jac_full`, a coloring of the FULL residual, not of the frozen one -- `build_banded_jac` on `frozen_residual` is called only by the standalone check in `EXHALE_main.f90` -- so "inside the band" carries the coloring's contamination by the non-local response, and the frozen column is a separate control rather than the band's own approximation. **For the advisor:** the next question is not the preconditioner. The species rows' row scaling (the binding row's diagonal $3.3e3$ below its hydrodynamic coupling is set by the certification row scale of decision 20 a divided by the column scale, and using one scale for the merit and another for the model is a decision); the cycle has an optimal length around 60 to 80 products on both fixtures and the step control asks for a tolerance the operator does not honor; and `gm_resid_rel` is verified against the operator only at a breakdown or a rank loss, where the measurement says it needs verifying whenever the cycle ran more than a few tens of products. Advisor: rebuild on `build_lvw`; `krylov_and_dogleg 251/0`, `steady_species_rows 194/0`; atomic reload at cap 40 with every hook off, 8 threads: the N31 ladder to the digit ($7.607e-2$ at 40, handed back $5.377e-2$). Accepted; deliverable 4 correctly not built. The two levers named are the LINEAR model's row scaling (a freedom decision 20 a did not fix: the merit and gate keep the certification scales, the linear inner product may differ) and a Krylov cycle that returns the step its TRUE residual chooses (the optimum lies at 60 to 80 products; the reduced residual lies past it). **Next item N36:** both as measured arms, default off.

- **N36** (the two levers N35 named: the row scaling of the LINEAR model, separate from the merit's, and a Krylov cycle that returns the step its TRUE residual chooses). Two default-off arms on the linear solve, adopting nothing. `EXHALE_MODEL_ROW_EQUIL=1` (`unit_infinity_norm_row_scaling_of_the_band`) forms the diagonal E that brings every row of the banded model to unit infinity norm as a power of two, applies it to the band that is factored and, inside `pgmres`, to the right-hand side and to every action of the operator, and maps the image and the relative residual BACK to the certification scales at every exit, so that the merit, the gate, the trust region's predicted decrease and the certification keep the scales decision 20 a fixed and only the linear model's inner product changes; `ab` itself is untouched, so the merit's gradient is where it was. `EXHALE_GM_TRUE_RESIDUAL=1` makes the cycle measure its candidate step against the operator from 20 products on, every 10 and at every point where it would otherwise end, keep the best, and stop on two consecutive rises or on the tolerance IN THE TRUE NORM; what comes back is that step, its own image from the product that measured it, and the true residual as `gm_resid_rel`, under a new eighth outcome `gm_true_residual_turned`. **Verdict: both arms work as specified, both fail the acceptance, both stay off.** (1) **The row scaling is a real freedom and taking it is worse.** The cycle then minimizes a functional the merit does not read, and the two disagree by a factor 1.2 to 673 MEASURED (`[row equil]`, equilibrated against certification: atomic $9.956e-2 / 4.111e-1$ at outer 2 and $9.937e-2 / 1.218e-1$ at outer 40; carrier $2.353e-1 / 5.730e+1$ and $1.286e-1 / 8.657e+1$), so the cycle announces the $1e-1$ it was asked for while handing back a step that leaves 86.6. It cannot touch what binds: the preconditioned operator $A_{eq} M_{eq}^{-1} = E (A B^{-1}) E^{-1}$ is SIMILAR to the unequilibrated one, and the measurement at the same iterate agrees (hydrodynamic Ritz ratio 39.62 against 39.60 carrier, 57.02 against 56.36 atomic; the count of species-row Ritz values below $1e-2$ unchanged at 18 and 2), while the banded factorization is bit for bit the same matrix because `equilibrate_the_preconditioner` already equilibrates both sides; and a row scaling leaves

every ratio WITHIN a row where it was, so the binding carrier row's diagonal $1.2396e-1$ against $4.1026e2$ of coupling, $3.304e3$ times the diagonal to the hydrodynamic unknowns of its own stencil, is exactly N35's number still. What it costs: the carrier reload at cap 40 goes from $||R||$ $9.572e-2$, info=1, binding carrier H2 of cell 205 to 1.955, **info=2 with four entries of the inventory refusing**, and the true residual of the returned step at 40 products from 0.973 to $1.050e2$. (2) **The true-residual cycle is the right instrument and it does not turn the solve.** At the binding iterate it caps the cycle where the step stops improving: on the ladder 40/80/160/320 the control's returned step gets WORSE with size (atomic TRUE $2.851e-1$, $2.575e-1$, $3.006e-1$, $3.203e-1$; carrier $9.731e-1$, $8.638e-1$, $9.668e-1$, **$1.472e+00$**) while the arm stops at 80 and at 120 products and returns $2.660e-1$ and $8.244e-1$, never worse than the control's best at any size and at fewer products. Two cycles of the atomic reload at cap 40 name the new outcome. But the best true residual any subspace reaches is 0.27 and 0.82, the Newton does not descend past its plateau, and $||R||$ handed back at cap 40 is 1.060 against the control's $5.377e-2$ (atomic; the trajectory parts at iteration 15, the first cycle to pass 20 products) and IDENTICAL on the carrier, where a subspace of 40 always makes the last candidate the best and the arm changes only the number reported (true $9.870e-1$ against reduced $9.839e-1$). **Acceptance, by the named clauses:** the true residual below 0.1 within 80 products on both fixtures fails for both arms (arm 1 atomic $8.79e-2$ but carrier $7.51e+1$; arm 2 atomic $2.66e-1$, carrier $8.64e-1$), and the refusal count at cap 40 does not fall by half (atomic 9 to 10 for both arms; carrier 8 to 7 and 8 to 8), with no new refusal class and no new gm_no_direction_sampled anywhere. Corrected mid-item and re-measured: the first true-residual arm left the reduced tolerance in place as a stop and so discarded a step it never measured (carrier outer 1, the 80-product cycle ended at product 49 and handed back product 40's step at TRUE $1.259e-1$ where product 49 reaches $9.724e-2$); with the reduced residual no longer ending a cycle, the numbers above are the corrected arm's. Tests: krylov_and_dogleg RED on the control objects (nine symbols absent), GREEN 264/0 with 13 rows added, among them the equilibrated solve returning the same step as the unequilibrated one to $9.90e-16$ on a stated dense system solved over the whole space, the equilibrated cycle reporting the CERTIFICATION residual of a truncated solve to eight digits, and a stated operator whose action stops being one linear map after 40 products, on which the cycle names gm_true_residual_turned and stops before its subspace is spent; steady_species_rows 194/0, element_operator 14/0, residual_determinism 5/1 with the same row failing by design and the same measured $3.354e+01$. **Default path:** the atomic reload at cap 40 with every hook off, all 354 (JFNK) lines IDENTICAL to my control through $||R||$ $5.377E-02$, flux spread $3.555E-03$; the carrier reload all 364 identical; wasp_full_newton bitwise but the provenance timestamp, log10 Mdot 13.30. With the arms ON wasp_full_newton stays CERTIFIED: arm 1 $||R||$ $3.043e-9$ against $1.667e-9$ with a largest relative movement $5.2e-10$ in the output, arm 2 bitwise in Ion_species.txt with 8 extra residual evaluations, its cycles never reaching the 20 products the arm needs. **For the advisor:** what the two arms rule out is the linear algebra. The norm the cycle minimizes is free and wrong to change; the length of the cycle is free and already at its optimum; the operator's species block is near-singular at every scaling and the step it can produce leaves 0.27 to 0.82 of the residual whatever the subspace. The remaining lever is the one N35 named and N34 confirmed the arithmetic cannot reach: the DISCRETIZATION of the rows that bind. Advisor: rebuild on build_lwv; krylov_and_dogleg 264/0, steady_species_rows 194/0; atomic reload at cap 40 with every hook off, 8 threads: the N31 ladder to the digit ($7.607e-2$ at 40, handed back $5.377e-2$). Accepted; both

arms stay off. **The linear algebra is exhausted as a lever** (N31 to N36: probe rule, inner tolerances, extended precision, band coverage, row scaling of the linear model, cycle length). STOPPED here for the user: decision 23, and the next question is the discretization of the species rows that bind at the front (the carrier row of cell 205 whose own diagonal is $3.3e3$ below its hydrodynamic coupling). code_status_20260910.md section 5 task 1, ISSUES 3.1 and 5, the handoff and decision 23 restated accordingly.

- **N37** (decision 23 (a): a well-balanced flux difference for the near-hydrostatic layer, as the default-off option Well balanced:). With the key on the reconstruction, the Riemann jumps and the pressure force carry the DEPARTURE from the cell's own local hydrostatic equilibrium instead of the state: within cell j that equilibrium is the constant-density one through its own (ρ_j, p_j) , $p_{eq,j}(r) = p_j - \rho_j (\phi(r) - \phi(r_j))$, which assumes no thermal, entropy or compositional stratification and so preserves ours (Kaeppeli and Mishra 2016, A&A 587, A94, sections 2.1.1 and 2.1.3; the momentum source as the face-pressure difference of the cell's own equilibrium is their 2014 paper, J. Comput. Phys. 259, 199, eq. 2.26). The pressure stencil acts on the neighbor's pressure measured against that equilibrium continued THROUGH the face (this cell's density up to the face, the neighbor's beyond it), which is what vanishes on the discrete equilibrium; the two cell pressures are differenced first, while that is exact, and the hydrostatic terms added to the difference. The momentum row is then assembled from the face pressure measured against the same equilibrium and source returns $S(2) = 0$: with $A^+ P_{up} - A^- P_{dn} - (A^+ - A^-) p_j = -\rho_j [A^+ (\phi_i(j) - \phi_c(j)) + A^- (\phi_c(j) - \phi_i(j-1))]$ identically, the equilibrium's flux difference and its gravitational source cancel in the ALGEBRA and the cell's own $O(1)$ pressure never enters the row. Mirrored in the kind-generic copy, so EXHALE_RESID_QUAD=2 is the production operator bit for bit under both key values (suite row, largest difference exactly zero, tolerance 0). **Exact preservation, MEASURED**: on the scheme's own discrete equilibrium (the isothermal solution of the face-matching condition, the spherical form of their eq. 42) the momentum row falls from $7.0e-5$ (PLM) and $2.5e-3$ (WENO3) of the local weight ρg at $N = 250$ to $4.1e-14$ and $3.9e-14$, and stays at $4.5e-14$ to $5.5e-14$ at $N = 500, 1000, 2000$ while the base scheme falls at its design order; the mass and energy rows reach $1e-16$ against $1e-8$. Twelve new assertions in hydrostatic_residual, plus eleven rows in krylov_and_dogleg (with no gravity the arm is the base scheme to $8e-16$ on a uniform state and to $1.4e-16$ on a Sod tube, for both reconstructions; and the same exactness on a second grid and geometry). **What it does NOT reach, and this was written into the design before the code**: N33's non-smoothness floor does not move (energy row of cell 1, $1.009e-11$ with the arm against $1.046e-11$ without; cells 84, 167, 251 the same to a factor 1.3; six to ten times $\epsilon \times \text{face state} \times r^2/dV$ either way), because a cell pressure is itself a rounded $O(1)$ double and every jump built from two of them steps by its last bit whatever the grouping. What does move is the additivity defect at ten times the probe arc, $2.410e-7$ with the arm against $3.151e-5$ without. **The stationary solves are worse with the key on, and the reason is not the arm**: store_row_terms scales the momentum row by $\max(|dF(2)|, |S(2)|, |Smom|)$, the arm returns $S(2) = 0$, so the row IS its own scale and every normalized momentum residual is exactly 1 (MEASURED on the carrier reload as loaded: $k=2$ is $1.000000E+00$ with the arm against 1.954 without, while the mass row is the same number to seven digits and the energy row to four). The merit then carries no momentum information: $||Fs||_2$ rises from 16.5 to 28.1 at the same state, and $28.1^2 - 16.5^2 = 517$ is the 500 momentum rows pinned at unity. The atomic reload at cap 40

reaches 1.415 against the control's 1.112 and the carrier reload 1.732 against 9.34e-2; at cap 250 the arm's solve aborts at iteration 61 on "no descent step in 12 consecutive iterations" with $||R||$ and the worst scaled row both exactly 1.000 and that row the MOMENTUM of cell 1, which is what a merit whose momentum rows are all pinned at one looks like from inside the line search. `wasp_full_newton` with the key on says the same from the other side: it is NOT CERTIFIED on exactly one entry, the momentum row at 1.000E+00, while its flux spread is 1.539e-15 against the control's 6.233e-14, its energy row 9.911e-12 against 1.667e-9, its mass row 1.228e-13 against 1.453e-13 and $\log_{10} \text{Mdot}$ 13.30 either way, the profiles moving by at most 1.2e-4 (velocity). **The mechanical column drains by the same amount with the arm as without** (hydrostatic_column, max lv/c_s 0.796 against 0.792 at 300 steps and 2.776 against 2.777 at 3000): that case's state is not the scheme's equilibrium and its drain is set by the outer boundary, as B4 recorded. **Default path:** with the key absent or False every path is bit for bit my control (`wasp_full_newton` reload, the atomic reload's 196 (JFNK) lines, hydrostatic_column at 300 steps, `mol_carrier` and `mol_diffusion` at their maxsteps); the one line that moves in every output file is the # options field, which now carries the 21st token `wellbal`, nameable in `Restart option change`. EXHALE_resolved.out and the parse dump each carry one new line. Key documented as K46. **For the advisor:** the option is kept in the tree and stays off; before it can ever be default on, the momentum row's reference scale has to become the pressure force the arm cancels analytically, $|\rho_j (A+ (\phi_i(j) - \phi_c(j)) + A- (\phi_c(j) - \phi_i(j-1)))|/dV$, which is two multiplications from what `RK_rhs` already holds but sits in an acceptance interface this item does not own. Advisor: rebuild on `build_lwv`; krylov_and_dogleg 279/0, grid_and_gates 162/0 (four rows FAIL when the shell rows are pointed at the stale shared EXHALE.x: pass EXHALE_EXE with EXHALE_OBJDIR), steady_species_rows 194/0, certification 57/0, element_operator 14/0, residual_determinism the same single row FAIL by design; atomic reload at cap 40 with every hook off and the key absent, 8 threads: the N31 ladder to the digit (7.607e-2 at 40, handed back 5.377e-2). The regression comparison strips comment lines, so the new `wellbal` token in the # options header is invisible to make check. Accepted as a default-off option; the momentum-row reference scale in `store_row_terms` (the arm's $S(2) = 0$ makes the row its own scale) is recorded in ISSUES 3.7 as the prerequisite of any default-on decision; the publisher PDFs of the two Kappeli and Mishra papers remain wanted (the institutional versions were read in full).

- **N38** (the discretized element and carrier rows at the front: why the row that binds is nearly independent of its own unknown). One default-off measurement hook, adopting nothing. `EXHALE_FRONT_ROW=1` ([diag 18]) takes the species row that carries the largest scaled residual and its two neighbors and prints, at the selected outer iteration: every TERM of the row on its certification scale, with the advective term split per FACE by masking the face mass flux (the divergence of a face species flux is exactly zero where the mass flux it multiplies is, so the masked call leaves one face standing alone and the SAME operator is used, never a copy); the reconstructed face fraction read back out of that masked divergence beside the donor cell's own; the chemical source of a carrier from the sweep's own rates; the band's entries for the row, to its own unknown, to the same unknown in the neighbors and to every hydrodynamic unknown of the three cells, each as a number and as a multiple of the diagonal; those same entries SPLIT BY TERM, by one-sided differences of the row's own terms along the same column with the same step `build_banded_jac_full` uses; the cell Peclet number, the logarithmic gradient

and the chemical against the advective time scale; and the column scale of every unknown of the binding cell beside the unknown itself. **Verdict: the binding row's diagonal is NOT small -- it is the size of the row -- and the 3.3e3 of N35 is a CHOICE OF COLUMN SCALE that the Krylov cycle cannot see.** (1) Every hydrodynamic entry of the row is its ADVECTIVE term, on all three fronts measured: carrier H₂ of cell 205 (carrier reload, outer 40) momentum of cell 205 -7.40478e+1 of which the diffusive part is 2.0e-5 and the chemical -2.3e-5; element He of cell 191 (atomic, outer 1) momentum of cell 190 -1.86758e+4 with a diffusive part 1.9e-6; element Fe of cell 137 (atomic cap 250, outer 140, one of the rows N34 named) momentum of cell 136 -1.25980e+4 with 2.0e-6. Summed over the reported stencil the hydrodynamic columns carry **3389, 4176 and 2424 diagonals** and the neighboring species unknowns 5.855, 1.052 and 1.005, which is N35's 3.304e3 and 5.646 reproduced. Each species sub-block is meanwhile a clean diffusive tridiagonal (15.385/-7.772/-7.612 helium, 17.432/-8.861/-8.564 iron): N27's reading of an interior element row holds at the front too, and it is the wind columns that bury it. (2) **The size of that coupling is one over the Mach number**, because the momentum column scale is $|\rho v| + \rho c_s$, the SOUND speed in a deeply subsonic layer: MEASURED scale over unknown **3.839e+2** (carrier front, momentum 6.83274e-5 against scale 2.62332e-2), **1.361e+5** (helium front) and **5.149e+4** (iron front), against 1.000 for the mass, the energy and every species unknown. The arithmetic closes to a factor 1.4 to 1.6: one face carries -0.5322 of the carrier row and $383.9 \times 0.5322 = 204$ against the measured 129; one face carries -0.1969 of the helium row and $1.361e5 \times 0.1969 = 2.68e4$ against the measured 1.87e4. (3) **The three other candidates are measured and are not the cause.** The species column scale's reservoir floor is INACTIVE everywhere it was looked at (the carrier unknown is $\ln n$ with column scale exactly 1.0, the default; the helium and iron unknowns' scales are their own magnitudes to five digits), so raising or lowering element_column_scale_floor by a decade changes the diagonal by nothing. A stiff advective/chemical cancellation in the diagonal is a factor 1.8 and not decades (carrier diagonal $1.19228e-1 = \text{diffusive } +2.17367e-1 \text{ and advective } -9.89614e-2$, with the chemical derivative $+9.44223e-4$, 0.8 percent, although the chemical time scale is 220 times the advective one), and there is no cancellation at all in the helium diagonal. The limiter is a live switch (the reconstructed face fractions stand at 0.587 and 0.657 of the donor at the carrier front, and frozen at first order the advective part of the diagonal changes SIGN, $-9.89614e-2$ to $+2.50409e-1$) but it moves the ratio by less than 2, because freezing it raises the momentum coupling by the same kind of factor. N5's two-root front was not re-run: cell 205 is not one. (4) **The test that settles it:** the existing default-off column equilibration (EXHALE_COLUMN_EQUIL=1) is exactly the remedy branch (i) names, and MEASURED at outer 1, where both arms stand at the same iterate, it brings the binding row's hydrodynamic coupling from 3389 to **0.993** (carrier) and 4176 to **2.926** (atomic) while the Krylov cycle returns the SAME step at the SAME residual to every printed digit (carrier 40 of 40, reduced 1.257e-1, TRUE 1.259e-1; atomic 8 of 40, 7.821e-2, TRUE 7.824e-2) and the species-row Ritz count below 1e-2 is unchanged at 18 and 2. It cannot be otherwise: a column scaling E is applied to the operator and to the band built FROM it, and $(A - E)(\text{band}(A) - E)^{-1} = A - \text{band}(A)^{-1}$ IDENTICALLY. N36 showed a row scaling is a similarity; this shows a column scaling is an identity, so **both sides of the scaling are now exhausted**. What the arm costs, since it also reshapes the trust-region ball and the probe steps: cap 40 ends at $||R||$ 1.822 (carrier) and 1.592 (atomic) against 9.572e-2 and 5.377e-2. Acceptance by the named clauses: the Ritz ratio below 1e3 FAILS

(2.07e5, 1.45e3), the cycle reaching 0.1 within 40 products FAILS on the carrier, named outcomes are lost on both; it stays off. **Deliverable 3 built nothing**, by the branch N35's deliverable 4 took: the remedy branch (i) names is either already the default (ln n) or already measured to be invisible. Tests: krylov_and_dogleg RED on the control objects (attributed_jacobian_entry absent), GREEN **279/0** with 6 rows added (the hook off by default; a stated term set composing its own row and its two faces composing its advective term through species_row_from_its_terms, the expression the measurement itself uses; the attributed entry of a term linear in the unknown equal to its coefficient times the scaling, with a zero probe step and a row with no scale attributing nothing) -- 279 rather than 270 because N37 added nine rows to the same file, a brief overlap reported and not resolved; steady_species_rows 194/0, element_operator 14/0, steady_selfconsistent_residual 3/0 on the wasp_full_newton log, residual_determinism 5/1 with the same row failing by design and the same 3.354e+01. Default path: the atomic reload at cap 40 with every hook off, all 354 (JFNK) lines IDENTICAL to my control through ||R|| 5.377E-02, flux spread 3.555E-03; the carrier reload all 364 identical at 9.572e-2; wasp_full_newton done info=0 ||R|| 1.667E-09 on both binaries with all 28 lines the same; and with the hook ON both reloads still end at the control's values, so the measurement is not an operation on the state. Cost MEASURED: one residual evaluation for each of twenty columns at each of three outer iterations, about four minutes on the carrier reload at cap 40 and 8 threads. Also corrected in the reading: N35's "there is no diagonal for a preconditioner to stand on" does not follow from the band row -- the diagonal is the size of the row, the disparity beside it is a column scale, and the near-singularity of the species-row compression of A^{-1} is a separate fact that the band-row ratio is not evidence for. Advisor: verified on the same final build (the suites above); atomic reload identical. Accepted; deliverable 3 correctly not built. N35's sentence "no diagonal for a preconditioner to stand on" is withdrawn: the diagonal is the size of the row, the 3.3e3 is the momentum column scale (the sound speed) against a layer Mach number of 1e-3 to 1e-5, and column scalings are invisible to the cycle, so the near-singularity of the species-row compression is a fact of its own.

- **Close of the species-row solver program** (2026-09-10 evening, user's instruction: "어떤 것을 해도, 끝이 없어 보이니, 엄청난 차이가 나오지 않는 한 이번 건만 마무리 하고 끝내기 바래", then "있는 그대로 정리하고, md tex 파일들 update 후 commit/push"): N37 and N38 were finished as briefed and no further item is started. State of the tree at the close, verified on the advisor's rebuilt build_lvw: krylov_and_dogleg 279/0 (174 at the morning commit), grid_and_gates 162/0, steady_species_rows 194/0, certification 57/0, element_operator 14/0, residual_determinism one row FAIL by design; the atomic reload at cap 40 with every hook off and every new key absent gives the N31 ladder to the digit (1.888, 1.891, 1.819, 1.583, 1.861, 8.964e-1, 7.607e-2; handed back 5.377e-2, flux spread 3.555e-3); the wasp_full_newton reload ends info = 0, ||R|| 1.667e-9, the number every worker's paired control gave (its output is not compared with the cold-start golden, which is a different path: sec_ion_step 2402 against 0). Everything N31 to N38 added is a default-off hook, arm or key: EXHALE_JV_ADDITIVITY, EXHALE_JV_COLUMN_SCALE, EXHALE_JV_PROBE_ARC, EXHALE_IEQ_TOL, EXHALE_TEQ_TOL, EXHALE_RESID_JUMP_SCAN, EXHALE_RESID_QUAD, EXHALE_KRYLOV_SIZE_SCAN, EXHALE_GM_HISTORY, EXHALE_PRECOND_SPECTRUM, EXHALE_BAND_DIFFERENCE, EXHALE_MODEL_ROW_EQUIL, EXHALE_GM_TRUE_RESIDUAL, EXHALE_FRONT_ROW, and the input key Well balanced: (K46); the marching path and the goldens are untouched (the # options header gains the wellbal token, which the regression

comparison strips with every comment line). New fixture backup/regression/carrier_elem_newton. Documents closed to this state: ISSUES_20260909.md 3.1, 3.7 and 5; code_status_20260910.md 3.3 and task 1; session_handoff_20260910.md (new, the handoff of record); To_be_determined_by_user_20260906.md decision 23 with its outcome; steady_solver_design.md 22.4; input_schema.md K46 and the manual's key table (PDF rebuilt, 90 pages); README.md (the key). The analysis the user asked for, of why the coupled solve fails and what a different approach would be (a segregated solve, the CETIMB way read from Koskinen et al. 2013 section 2.1.3 and 2022 Appendix B, a different treatment of the layer), is docs/solver_approach_analysis_20260910.md. Reported, not fixed (ISSUES 3.7): the count global shadowing the intrinsic; newton_dense's hardcoded step test; the dead dp_bc/ntot_bc refreshes; store_row_terms's momentum scale under the new key; fortdep.py blind to include; golden/mol_ir_bands/ without Cooling_breakdown.txt; examples/README.md; the two .md twins.

8. PLAN_20260911: the partitioned stationary route (2026-09-11)

Plan of record: docs/PLAN_20260911_partitioned_solver.md. User instruction of 2026-09-11: read docs/solver_approach_analysis_20260910.md, its review docs/solver_approach_analysis_20260910_review.md and docs/solver_partition_experiment_20260911.md, correct the errors they establish, run the solver experiments, and adopt the partitioned route in the production solver if it measures better than the coupled species-row solve. This reopens the stage-2 solver program that the 2026-09-10 instruction had closed after N37 and N38. Items P1 to P9, then P10 to P12 on the same day's further instruction to fix the open items the plan had left to the user, then P14 and P15 on the two findings of that afternoon's Codex adversarial review and P16 to P18 on the user's two decisions of the same afternoon (P13 and P19 are the record itself); the rules every worker on this tree follows are docs/worker_rules.md.

The items

- **P1 (defect D1): a failed element composition solve is no longer handed back as an update** (binary_element_diffusion.f90 +311/-23, src/tests/element_operator/element_operator_tests.f90 +190/-13; every line count of this section is MEASURED from `git diff --numstat` on the tree at the close). `solve_mass_fraction` returns an explicit success result and no longer accepts the last halved trial of a line search that never descended; `element_diffusion_step` takes an optional status and, when asked, restores the composition it was handed unless the candidate is finite, inside $[0,1]$ to $1e-12$ before the range clip, and no farther from the density the step held fixed than the entry composition was, to $1e-10$; `relax_element_composition` steps on a copy, discards a refused step, retries at half the length within a bounded budget and reports three outcomes by name. MEASURED (P1) on the archived D1 reproduction (atomic_elem_newton, species mode, omega 1): a returned composition missing rho by $1.768571764708571E-01$, a nonfinite chemical refresh and exit status 1 become the entry composition restored at $1.046888214610824E-15$, an admissible refresh at

1.10297235E-15 and exit status 0. The first inadmissible operation of that failure, which the experiment report left unlocated, is now named: the composition solve drives X out of $[0,1]$ and the range clip used to absorb it (element_step_out_of_bounds), and the mass discrepancy is downstream of the clip. Tests: element_operator 14/0 at entry, **24/0** after, **22/2** on a RED build with the four acceptance actions disabled (an_unsolved_step_is_refused, an_unsolved_step_leaves_the_entry_composition measured 1.31725E-02); diffusion_tests 35/0, steady_species_rows 194/0, certification 57/0 unchanged. Impact (MEASURED, P1, control built from the entry text of the one file): mol_diffusion and lower_profile, 300 steps, one thread, Ion_species.txt byte-identical and Hydro_ioniz.txt identical but the provenance run= timestamp, on both cases; the marching caller passes no status, so none of the new tests runs there. One fixture of the same suite was corrected while there (the_relaxation_fixed_point_under_a_callers_ghost built a column whose species missed rho by 1.321E-01): on the suite's own mass-closed column the worst interior element row of the relaxation fixed point moves from 1.62384E-02 to 1.35062E-13, so the fixed point on a mass-closed column really is a zero of the stationary element row.

- **P2 (defect D2): the carrier movement bound now constrains the returned state** (diffusive_photochemistry.f90 +225/-81, src/tests/carrier_retry/carrier_retry.f90 +261/-1). The displacement is defined once (the largest change of any solved carrier column over the physical cells, divided by the largest H2 mixing ratio of the outer entry state, a quantity that cannot move while the pass runs), it is enforced before acceptance on every trial, a refused trial is undone and halved down to $2^{**}(-10)$ of the cell transport time, and the pass ends on one of five named outcomes. MEASURED (P2) on carrier_elem_newton through the archived driver: the entry text returns drift = 2.8126757844770485e-02 at trust = 0.01, reproducing the 0.02812676 of docs/solver_partition_experiment_20260911.md section 7.2 exactly; the repaired routine returns 9.9956436739254070e-03 at the same setting and 2.9845242387380568e-03 at trust = 0.003, so the bound holds and it scales (a factor 3.33 smaller setting gives a factor 3.35 smaller movement). The halving ladder was measured on an instrumented copy: 0.4412 at the full cell crossing time, then 0.2523, 0.1445, 0.0779, 0.0404, 0.0205, 0.0103, 5.155e-3, 2.581e-3, 1.291e-3 and 6.458e-4 at the tenth halving, so the movement is proportional to the trial length and a bound of 1e-4 keeps nothing on this column. Tests: carrier_retry **111 PASS / 10 FAIL** on a RED control carrying the entry logic behind the new interface, **122/0** after; carrier_returned_state_acceptance 36/0, attempted_step 70/0, carrier_constraint_attribution 10/0, carrier_reference_scales 14/0, species_masses 9/0, certification 57/0, adv_static_limit 53/0. Impact: mol_carrier, 300 steps, one thread, Ion_species.txt byte-identical and Hydro_ioniz.txt identical but the provenance timestamp, which is the expected result because the marching loop calls photochemical_transport_step directly and never this routine. The cost is stated: a bounded pass advances less, so the frozen H2 residual after the composition update is larger (4.4708e-01 at trust 0.01 against the entry text's 3.1897e-01, from 4.9275e-01 before), and the state handed on is correspondingly closer to the entry state.
- **P3 (defect D3): the stop test of the stationary solve is now the acceptance test of the state** (steady_newton.f90 +203/-29, src/tests/steady_completion_flag/run.sh +61/-15, src/tests/krylov_and_dogleg/krylov_and_dogleg_tests.f90 +127/-1). The loop top of solve_steady_jfnk and of solve_steady_ptc no longer stops for success on steady_gates_met alone:

it stops only when the gate AND every row the certification judges the carried system by are within their own tolerances, measured on the iterate by the same evaluation and the same formula the acceptance at return uses. A gate met while a certified row refuses prints one line naming the row, its measure, its own tolerance and its cell, and the iteration continues within its budget. MEASURED (P3) on the D3 reproduction (carrier_elem_newton, first hydrodynamic solve): the entry text stopped at the loop top of iteration 14 with `info = 0` and was refused at return on a mass row of $5.954\text{e-}12$ against $3\text{e-}12$; after the repair the same solve continues, 27 of its 40 iterations meet the gate with a row refusing, and it ends `info = 1` on its pass budget with mass $5.945\text{e-}12$, momentum $5.991\text{e-}13$, energy $2.481\text{e-}09$. So the second of the two admitted outcomes: the mass row of this arm does not come below $3\text{e-}12$ in 27 further iterations (it stands at $5.9\text{e-}12$ to $7.9\text{e-}12$ throughout, consistent with the N33 rounding floor), and the flag is now honest about it. Cost of the added test, MEASURED with a temporary clock pair in a private build: 27 calls, mean 3.19 ms against a mean outer iteration of 0.73 s, 0.44 percent of one outer iteration, and only behind the gate, so an iteration below the gate pays nothing. Tests: krylov_and_dogleg 279/0 on the control objects and **290/0** after (11 new rows asserting the predicate the stop test reads); steady_completion_flag RED on both real logs at entry (an `info = 0` log with no statement that the stop test read the certified rows; a refusal at return with no stop statement) and GREEN after; certification 57/0, carrier_returned_state_acceptance 37/0, attempted_step 71/0 unchanged. Impact: the wasp_full_newton reload ends `info = 0`, `||R||` $1.667\text{e-}09$, flux spread $6.233\text{e-}14$, 8 outer iterations, `log10 Mdot` 13.30 on both binaries, the certification block line for line identical, every output byte-identical but the provenance line of the two Hydro_ioniz* files and one extra loop-top stop: line in the log; the carrier_elem_newton reload at cap 40 gives every (JFNK) line identical because the gate is never met on that arm. No golden can move: READ from the ten DEFAULT_CASES, none of their golden logs carries a (JFNK) start or (PTC) start line at all.

- **P4: the momentum row's reference scale under Well balanced: True** (RK_rhs.f90 +90/-2, steady_residual.f90 +59/-2, src/tests/grid_and_gates/hydrostatic_residual.f90 +318/-11, that suite's README.md +44/-1). Under the arm store_row_terms scaled the momentum row by $\max(|dF(2)|, |S(2)|, |S_{\text{mom}}|)$ with $S(2) = 0$ by construction, so every scaled momentum residual was exactly 1 (N37, ISSUES 3.7). The reference is now $\max(|dF_2|, |S_2|, |S_{\text{visc}}|, \text{equilibrium_pressure_force}(j))$, the last term the magnitude of the pressure force of the cell's own hydrostatic equilibrium, which is the right-hand side of whichever cancellation identity the assembled row belongs to (PLM $|\rho_j [A+ (\phi_i(j) - \phi_c(j)) + A- (\phi_c(j) - \phi_i(j-1))]|/dV$, WENO3 $|\rho_j (\phi_i(j) - \phi_i(j-1))|/dr$; Kaeppeli and Mishra 2016, A&A 587, A94, eq. 16 in spherical geometry). It has one definition, equilibrium_pressure_force_of_state, called by RK_rhs, by positivity_limited_fluxes and by assemble_residual (the last so that the kind-generic EXHALE_RESID_QUAD arm, which returns the well-balanced row but not the term it is read against, does not scale the row by a stale weight). The numerator is untouched; with no gravity the term is zero and the scale is the row's own terms exactly (MEASURED 0 for all four scheme/solver pairs). MEASURED (P4): wasp_full_newton reloaded with the key ON is **NOT CERTIFIED** on the entry text on exactly one entry, the momentum row at **1.000E+00** at cell 1 (which reproduces the N37 record to the digit, spread $1.539\text{E-}15$, energy $9.911\text{E-}12$, mass $1.228\text{E-}13$, `Mdot` 13.30), and **CERTIFIED**

after, `info = 0`, `||R||` 9.719E-10, momentum row 9.719E-10 against 1.0E-08 at cell 500, `log10 Mdot` 13.30. Tests: `grid_and_gates hydrostatic_residual` **28 PASS / 12 FAIL** on the entry-text objects (the twelve failures at exactly 1.000 for all four pairs) and **40/0** after, of which 16 are new rows measuring the row divided by `residual_row_scale(2, ...)`, the one expression the certification reads; the whole suite 162/0 at entry (READ, N37) and **178 PASS / 0 FAIL** after; `krylov_and_dogleg` 279/0 and certification 57/0 unchanged. Default path (key absent): `wasp_full_newton reload` and `mol_carrier` at 300 steps byte-identical on both binaries apart from the provenance line, every (JFNK) line identical, key-off `info = 0`, `||R||` 1.667E-09, `Mdot` 13.30. The arm's effect on the two solver fixtures is large and in one case negative, and both are recorded: the carrier reload's handed-back `||R||` falls from 1.721 to **1.026E-01**, the key-off level (READ: N37 9.34E-02), with a measured momentum row of 1.404E-02 at cell 499 instead of a pinned 1; the atomic reload stops earlier (`info = 2` at iteration 24) and hands back 1.702 against the control's 1.415 at cap 40, because the merit no longer carries 500 momentum rows pinned at one (`||Fs||` 2 17.1 against 28.2). A defect found while there and fixed: the allocation guard in `RK_rhs` tested `face_flux` for the well-balanced arrays `face_q_up/face_q_dn` as well, and `face_flux/face_p` are also allocated by `store_the_interface_fluxes`, so a stationary evaluation reached before any marching stage could write an unallocated array; the guards are now separate and each tests its own array, with no arithmetic on any path.

- P5: the analysis document corrected** (`docs/solver_approach_analysis_20260910.md`, +463/-115, 114 to 592 lines). All nine corrections the review established are in place, each with an inline `[corrected 2026-09-11: ...]` mark citing the review section or the experiment measurement behind it: the spatial truncation error is not a lower bound on the algebraic residual (the frozen-background H2 wind row measures 2.449021e-13 on the same 500-cell grid where the withdrawn argument put a 5e-4 floor); Ritz-magnitude ratios of a masked compression are neither condition numbers nor the species Schur complement; partitioning moves the coupled iteration outside the Krylov solve with outer matrix $(1 - \omega) I + \omega D^{-1} C A^{-1} B$ and the outer loop is not new code (`steady_wind_with_element_diffusion`); the CETIMB route is renamed as informed by CETIMB and gains the two conditions the review adds; the well-balanced arm removes the equilibrium's algebraic cancellation and not the floor; Koskinen et al. 2013 lower boundary is section 2.1.1; the three-unknown path uses its existing globalization and not the trust region (`use_tr = (nspec_row .gt. 0)`); section 7 recast with the 2026-09-11 numbers; a new section 9 indexing the twelve changes and a new section 5.5 on partitioned components inside a coupled method (Knoll and Keyes 2004, J. Comput. Phys. 193, 357). Three numbers of the original disagreed with their sources and are corrected, one of them wrong: the N4b row read "carrier arm 1.4 -> 0.09" where the measurement is 0.151 -> 0.090. Citations were verified independently in the published text (Koskinen 2013 section headers by `pdftotext -layout`; the Knoll and Keyes bibcode by the ADS API).
- P6: the outer iteration as a stated contract** (`src/EXHALE_main.f90`, +448/-131). `steady_wind_with_element_diffusion` now says in its own header which equations are alternated with the wind and why (a species the run moves by an operator-split transport step, the diffusing elements always and the molecular carriers where `Coupled carrier solve` is off, has a stationary balance that no hydrodynamic solve touches), and it accepts a state only by the certification of the FULL set of active equations evaluated on the refreshed state it would hand back. A hydrodynamic `info = 0` while an alternated species row refuses is

not an accepted state, and a nonzero hydrodynamic flag no longer ends the loop by itself, because both `info = 1` and `info = 2` hand back a state the composition update still makes progress from. The endings are named (`outer_certified`, `outer_state_not_finite`, `outer_element_update_refused`, `outer_no_progress`, `outer_pass_budget`), the P1 status is consumed, and the quantity the passes are controlled by is the worst gated species row of the certification: where it fails to fall over a pass both updates are shortened (the element under-relaxation omega halved, floor 0.125; the carrier movement bound halved, floor `carrier_trust_floor = 1.0d-3`) and `outer_no_fall_max = 3` consecutive passes without a fall end the loop. The last pass takes no composition update, because an update exists to be consumed by the next hydrodynamic solve and taking it would replace a stationary wind by a state the update itself spoiled. The composition drift and the volume-weighted carrier residual are reported and are not tests. With no alternated species the body is one steady solve and the state refresh, and the flag is the solve's own, which is what an atomic run and a coupled carrier solve both do. Default path MEASURED on the P6 run pair (`wasp_full_newton.ctrl` against `.new`): `info = 0`, `||R|| 1.667E-09`, flux spread `6.233E-14`, CERTIFIED, `log10 Mdot 13.30`, and `output/Ion_species.txt` and `output/Hydro_ioniz.txt` byte-identical between the two.

- **P7: the measurement.** Below.
- **P8: the corrections propagated to the documents and tests that carried them** (`docs/code_status_20260910.md`, `docs/ISSUES_20260909.md`, `docs/certification_tolerance_anchoring_20260910.md`, `src/tests/steady_selfconsistent_residual/run.sh`, `src/tests/steady_species_rows/steady_species_rows_tests.f90`). The withdrawn inference (a spatial truncation error of $5e-4$ at $N = 500$ floors the algebraic residual, so $1e-5$ cannot certify) is corrected in place in the three documents that still carried it, marked with the review section and the measured $2.449021e-13$ beside it, and decision 22 (a) is kept as the standing decision in every one of them with only its justification changed; the grid-refinement anchor and its remap control are relabelled as consistency diagnostics; "permanently uncertified until the operator's order or the grid changes" is withdrawn and quoted where it stood; ISSUES 3.7's `store_row_terms` item is struck through and marked RESOLVED by P4, and the resolved table gains D1, D2 and D3 with their measured before and after. Two test contract errors, both RED before and GREEN after: `steady_selfconsistent_residual` assertion 2 read "a hydrodynamic row above its tolerance implies `info = 2`", which conflates not certified with one flag and already failed on the entry text for any budget-limited log (`FAIL flag_of_carrier_elem_newton measured=info=1 reference=info=2 -> PASS`, with `wasp_full_newton` still PASS and a synthetic `info = 0` copy still FAILing); and `relaxation_drift_covers_every_element` built its column by swinging each metal mixing ratio about the reservoir on a fixed hydrogen fraction, so its species missed the density the column hands the operator by **1.66042E-02**, which is now **6.58484E-16** on a column of mass fractions that sum to one, the suite going 194/0 to **195/0** with a new row measuring the closure by the code's own mass policy.
- **P9: this section**, `docs/session_handoff_20260911.md`, the status section of the plan, the dated sections of the two fixture READMEs, the reopening paragraphs of `docs/code_status_20260910.md` section 3 and `docs/ISSUES_20260909.md` section 5, and the new entries of `TO_BE_DONE.md`.
- **P10: the best-iterate ledger reads the hydrodynamic rows against their own certification tolerances**

(steady_newton.f90 +131/-55, src/tests/krylov_and_dogleg/krylov_and_dogleg_tests.f90 +135/-1, READ from the P10 report; the whole uncommitted diff of the two files now stands at +335/-84 and +268/-1, MEASURED by `git diff --numstat`, P3's share included). Items P10, P11 and P12 were added after the plan had run, on the user's instruction of 2026-09-11 to fix the open items the session had left to the user ("fix all of them", and "the pressure term too"); P10 is the third entry of the plan's "what remains". distance_from_certification divided the hydrodynamic slot by the run's `Resid tol`, so the ledger that chooses the state handed back could not see a mass row above $3e-12$ while $||R||$ was below $1e-8$. The slot now arrives as a distance formed by ONE new public routine, `hydrodynamic_distance_from_certification`, `max(mass/3e-12, momentum/1e-8, energy/1e-6)` on the windows `resid_relnorm` uses, so $d < 1$ is exactly the row condition the loop-top stop of P3 and the acceptance at return impose, and `keep_this_iterate` ranks on that distance on every route including the three-unknown one (the `nspec_row <= 0` branch that ranked on $||R||$ is gone). `certified_row_measures` hands the plain $||R||$ of the same evaluation back through a new optional argument rather than leaving a caller to re-derive it, and the iteration line prints the two side by side. `resid_tol_of_solve` is KEPT, with one reader, the EXHALE_RESID_BRANCH_REPORT block, where `rmax(1)` is deliberately the plain mixed-row measure and dividing it by one row's tolerance would be a ratio of two different rows. Tests: `krylov_and_dogleg` 290/0 at entry, **300 PASS / 9 FAIL** on a RED build in which both distances read `maxval(rc)/1.0d-8` (the superseded reading, at the `Resid tol` of these fixtures), **309/0** after, the 19 new rows being the block `judged_distance_reads_each_row_against_its_own_tolerance`; `certification` 57/0, `steady_completion_flag` 3/0, `steady_selfconsistent_residual` 3/0 unchanged. Impact MEASURED (P10) against a private build of the entry text of the one file: `wasp_full_newton` reloaded ends `info = 0`, $||R||$ $1.667E-09$, `log10 Mdot` 13.30 with **all eleven output files byte-identical**, the ledger handing back the same iterate and the only number that moves being the reported best-iterate distance, $1.749E-01$ (which is $||R||_{best}/1e-8$) to $8.469E-02$ (the same iterate's worst row over its own tolerance, so 8.5 percent of it). The carrier reload shows the re-ranking doing what it is for: pass 2 hands back a mass row of **4.85E-12 at $||R||$ 2.459E-09** where the control took $6.98E-12$ at $1.593E-09$, and pass 3 **4.96E-12 at 2.005E-09** against $8.17E-12$ at $1.723E-09$, with the H2 carrier row and the worst gated species row identical pass by pass because the carrier relaxation sits outside the hydrodynamic solve. The HD 209458 b element reload runs its twelve passes in 1064 s against the control's 1043 s (2 percent on a shared machine, not a claim) with every one of the twelve worst gated species rows and its cell identical to the control, and ends on the same single refusing entry, the hydrodynamic mass row at **8.893E-10 at cell 3** against the control's $1.054E-09$ at cell 16; the solves reporting a stalled best iterate fall 5 to 3 and those ending on "no descent direction exists" 11 to 10. What stays open, and what this measurement settles: the mass row does NOT reach $3e-12$ and that state still does not certify, because at $1e-9$ to $1e-12$ the row IS the N33 rounding of the base-layer flux difference. MEASURED (P10, finding 3): at pass 1 the ledger's best distance is $4.407E+02$, a mass row of $1.32E-09$, while the same state re-evaluated at its own composition after the best-iterate restore carries **4.11E-09**, so ranking iterates by that row is ranking by the luckiest rounding of a quantity that moves when the state is evaluated again, which is why the pass-by-pass mass rows scatter rather than descend in the control and in the new run alike. The entry is an acceptance question, not a solver item, and no further solver item follows from it.

- P11: the momentum row's reference scale is the largest PHYSICAL term of the momentum equation** (RK_rhs.f90, steady_residual.f90, src/tests/grid_and_gates/hydrostatic_residual.f90 and that suite's README.md; the whole uncommitted diff of the four files stands at +285/-2, +151/-33, +736/-12 and +94/-1, MEASURED by `git diff --numstat`, with P4's +90/-2, +59/-2, +318/-11 and +44/-1 inside it). The fourth entry of the plan's "what remains", and the "pressure term" of the user's instruction. The scale was built from the pieces of the discretization, $\max(|dF(2)|, |S(2)|, |S_{\text{mom}}|)$ plus P4's equilibrium pressure force under the well-balanced key; it is now built from the three left-hand terms of $d(\rho v)/dt + \text{div}(\rho v v) + dp/dr + \rho d\phi/dr = S_{\text{visc}}$, kept beside `face_flux/face_p` as module arrays of `RK_integration`, filled on both reconstructions, under both key values and on the positivity-repair path, and gathered by one routine, `momentum_row_terms_of_cell`, with `momentum_row_terms_of_state` filling a whole state and `reconstruction_continuation_rhs` blending the three terms with the same λ as `dF` and `S`. The residual $R(2) = dF(2) - S(2) - S_{\text{visc}}$ is untouched on every path, and the split is a rearrangement of the assembled row and not a model of it: $F(2) - p_{\text{out}}$ is the momentum flux without the pressure exactly for every flux function, and the geometric term is recovered from the source as $p_{\text{geom}} = S(2) + \text{weight}$ so that no second definition of the cell pressure exists. MEASURED (P11), on a state with structure in density, pressure and velocity, $\max_j |r_{\text{arm}} + dp/dr + \rho g - (dF_2 - S_2)|$ over the scale is **0 under WENO3 and 1.7e-16 (default) / 5.3e-15 (key on) under PLM**, for both Riemann solvers, so the three terms add up to the row they scale. The defect is closed on the states where the terms are known in closed form: with no gravity and a uniform pressure at rest, where every term of the equation vanishes, the PLM scale read **1.000 of the geometric pressure term $(A^+ - A^-) p_j/dV$** , that is $2 p/r$, with both Riemann solvers at every N , and now reads the tiny floor 2.22507E-308; on supersonic uniform flow at $M = 3$, where the ram divergence is the only term, it stood **6.66667E-02 = $1/(\gamma M^2)$ above it** and now agrees with it to 1.9e-14 (PLM) and 4.2e-13 (WENO3); on the discrete hydrostatic equilibrium the PLM scale was **2.79144E-01, 28 percent BELOW the weight it balances**, because $S(2)$ is the weight minus the geometric term and no term of the max was the weight itself, and the departure $(w - s)/w$ is now 0. P4's own zero-gravity diagnostic closes with it: the departure of the well-balanced scale from the base scheme's on the same state falls from **9.43654E-01 to 3.74E-13** under PLM with ROE (2.76E-13 with HLLC), the two having agreed under WENO3 all along (1.7e-13 to 2.7e-13). Tests: `hydrostatic_residual` **50 PASS / 10 FAIL** on the entry-text objects and **60/0** after, the ten being the four scheme/solver pairs of P4's corrected `zero_gravity_momentum_scale_is_the_rows_own_terms` (which had compared the arm's scale with $\max(|dF_2|, |S_2|)$, the SUM of the two terms the row holds, 7.50E-01 FAIL to 0) and the PLM pairs of the three new identity statements; the whole `grid_and_gates` suite **188/10 to 198/0** (198 = P4's 178 + 20 new rows); `krylov_and_dogleg` 290/0, `certification` 57/0 and `adv_static_limit` 53/0 before and after. Two quantities are REPORTED and not asserted, with the reason: the two-sided departure of the scale from the weight on the arm's own equilibrium is not round-off for the base scheme (MEASURED 8.5e-4 at $N = 250$ falling to 2.3e-6 at $N = 2000$, PLM with ROE), because that state is the discrete equilibrium of the ARM's reconstruction and HLLC's face pressure carries a dissipative part of size dr/H times the weight; and the halving statement subtracts the unperturbed row of the same cell, since the undifferenced ratio is not 2 (MEASURED 1.0012 at $N = 250$, 1.586 at $N = 2000$) while the imbalance the perturbation ADDS halves

to 2.000000 within 5e-6. Impact: nothing that was certified stops being certified. `wasp_full_newton` reloaded with the key OFF is **CERTIFIED** on both binaries, `info = 0`, `log10 Mdot` 13.30, with the momentum row moving **8.469E-10 to 1.165E-09** of its 1.0E-08 tolerance, the mass row 1.453E-13 to 2.017E-12 of its 3.0E-12 (so this reload now certifies with less margin on the mass row) and `||R||` 1.667E-09 to 1.495E-09; the outputs are not byte-identical, because the merit and the loop-top gate both divide by the momentum scale and the two solves accept different iterates, and the states differ by at most **2.6E-09 relative** over every column (9.3e-10 in `Hydro_ioniz.txt`, 2.6e-09 in `Ion_species.txt`). With `Well balanced: True` the same reload is **CERTIFIED** on both, the momentum row moving **9.719E-10 to 9.621E-11** with all eleven output files byte-identical and the `||Fs||2` ladder identical iteration by iteration: at cell 500, `r = 4.62`, the ram divergence exceeds the cell's weight by about a factor ten, which is why the measure of the certified state falls by that factor on a supersonic outer cell. The marching path of `mol_carrier` at 300 steps and one thread is **byte-identical in all seven output files**, `log10 Mdot` 7.95, the `run.log` differing in exactly eight lines, every one of them a report of the momentum measure (the worst-row entry 1.8164E+00 at cell 438 to 1.9788E+00 at cell 449) and none of them a state. On the carrier reload every pass line is identical but the momentum entry, which falls **1.32E-12 to 3.22E-13** after pass 1 (4.47E-13 and 3.54E-13 at passes 2 and 3), the state as loaded reading 3.622E-01 to **3.311E-01** in its scaled momentum row, and all four output files byte-identical. What stays open: the kind-generic EXHALE_RESID_QUAD arm still hands back neither the momentum terms nor the face departures, so on that arm (default off, a rounding control experiment nothing is adopted from) both are read from the last `RK_rhs` evaluation rather than from the state just assembled; the fix belongs in `hydrodynamic_rows.f90`, which this item does not own.

- **P12 (with P12b): two test-hygiene defects, and the same shape in the sibling scripts** (`src/tests/test_columns.f90` new, 125 lines; `attempted_step/run.sh` +18/-5, `carrier_returned_state_acceptance/run.sh` +18/-5, `carrier_retry/run.sh` +18/-5, `carrier_reference_scales/run.sh` +22/-6, `carrier_constraint_attribution/run.sh` +22/-6, `constrained_network_layout/run.sh` +22/-6, `species_masses/run.sh` +22/-8, `element_operator/run.sh` +5/-0, `steady_species_rows/run.sh` +5/-0, and the two drivers, `MEASURED` by `git diff --numstat`; of the drivers' totals the item's own share is READ from the P12 report, the use line, the renamed calls and 66 lines out of `element_operator_tests.f90` and 43 out of `steady_species_rows_tests.f90`). The fifth entry of the plan's "what remains", both halves of it. The **build_stamp fallback**: the trigger is narrower than "a tree that has never been built", it needs such a tree AND a work directory outside `$root/build`, which is the concurrency-safe usage the worker rules mandate, because `source_closure.py` walks `$root/build` recursively and a default work directory inside it swept the fallback into the closure by accident. Seven `run.sh` now select `${EXHALE_OBJDIR:-$root/build}/build_stamp.f90`, write the fallback into their own work directory only when that is absent, and compile whichever of the two is in force as the FIRST file of the closure with the closure filtered of any second copy, which is the shape `src/tests/physics_probe/run.sh` already had; `grep -ln 'if [ ! -f "$root/build/build_stamp.f90" ]' src/tests/*/run.sh` now returns nothing. Four of them gained a private object directory, `EXHALE_TEST_OBJDIR`, without which the no-build verification cannot be run on them at all (`species_masses`, which had no override of any kind, `carrier_reference_scales`,

carrier_constraint_attribution, constrained_network_layout). RED and GREEN, MEASURED in four arms: at HEAD with an external work directory the compile dies on Cannot open module file 'build_stamp.mod' and each script prints its one failure line, **0 PASS / 1 FAIL, exit 1**, in attempted_step, carrier_returned_state_acceptance, carrier_reference_scales, carrier_constraint_attribution, carrier_retry and species_masses; after the fix every suite gives the count it gives at HEAD with a default work directory, **70/0** (attempted_step), **36/0** (carrier_returned_state_acceptance), **14/0**, **10/0**, **122/0**, **9/0** and **8/0** (constrained_network_layout, which has no utilities.f90 in its closure, so it has no RED to show and its change is the shape alone), in a tree with no build/ and in the repository against a private build, and the diff of the PASS/FAIL lines before and after is empty for all of them. Which stamp each arm used is MEASURED too: in the no-build arm every work directory holds a build_stamp.o compiled from the fallback, and in the arm with a build tree no fallback file was written at all, so the generated stamp was taken, which the superseded scripts could not do because they looked only in \$root/build. The **column written twice**: the new module test_columns holds column_carrying_its_own_density(f_c, q_h2, he_taper, metal_swing), named for what it builds, with its header stating the invariant $\sum_i m_i n_i = \rho$ and the round-off it holds to, and column_mass_closure(rho_c, f_c), the measurement moved out of element_operator_tests.f90 unchanged; the two drivers call them at 15 and 2 sites and use utils, only: calc_rho is gone from both, it having been there only for the moved code. The shared construction reduces to each former copy exactly and not approximately (metal_swing = 0 gives swing = 1.0d0, q_h2 = 0 leaves isp_H2 at zero, he_taper = .false. selects the constant the steady driver used, and zsum accumulates in the same element order), and the statement is the measurement: element_operator **24/0** and steady_species_rows **195/0** before and after, with the diff of the two full logs, diagnostic lines included, empty in both, so every printed digit is unchanged (the_column_is_mass_closed_as_built 4.10591E-16, drift_column_carries_its_own_density 6.58484E-16, the relaxation's 27 steps and returned drift 1.7790E+00 among them). The file is in no source list of the Makefile (grep -c test_columns Makefile = 0) and no dependency generator sees it; the two run.sh that need it compile it immediately before their driver. Nothing stays open from this item: its own report found nothing further, and the four sibling scripts and the hardcoded work directory it reported are what P12b then closed.

Two changes outside the item list, made by the advisor while reading these three:

- src/tests/steady_species_rows/steady_species_rows_tests.f90, the two rows P10 reported as stating the superseded semantics. In judged_distance_is_the_worst_row_over_its_tolerance the hydrodynamic slot now arrives as a distance (rows(1) 2.0d0, 0.5d0 in place of 2.0d-8, 5.0d-9), with a comment saying so, and judged_distance_follows_the_residual_target, which asserted the property the item removes, is replaced by hydrodynamic_distance_reads_each_row_against_its_own_tolerance: a mass row of 1e-9 beside a momentum row of 1e-13 and an energy row of 1e-8 gives 1e-9/3e-12, the mass row being what the distance reports. The two resid_tol_of_solve assignments and that import are gone.
- the Coupled carrier solve key comment in src/modules/files_I0/input_read.f90 (+17/-13, MEASURED). It stated that the alternation "was measured not to converge and not to be able to" (section 139), which

is narrower than it reads after P7. It now names the default (False), says what the alternation is judged by (the certification of the refreshed state in `steady_wind_with_element_diffusion`), cites the two reloads of this section for what it reaches, and keeps True as the coupled measurement arm.

Four more items ran in the afternoon of the same day: **P14** and **P15**, the two findings of the Codex adversarial review of the working tree (the review and the verification of each finding are the subsection below), and **P16** and **P17**, the user's first decision of the afternoon (the mass tolerance re-anchored on the fixture's measured rounding floor) and its consequence for the best-iterate ledger. **P18** followed and closed the same day.

- P14: every ending of the stationary outer iteration hands back one consistent state**
 (src/EXHALE_main.f90, `steady_wind_with_element_diffusion` only, +84/-40 against the entry text; src/tests/grid_and_gates/output_state_consistency.sh +181/-4; MEASURED by P14 with `diff -u` and `git diff --numstat`). Codex finding 1, and it is real. On a stagnation ending the composition had been updated while the particle densities and the temperature written beside it had not, so the state the run wrote was one composition update ahead of its own particle count and temperature. MEASURED (P14) on the forced stagnation of the hot-Uranus carrier reload: re-evaluating the written state moves its temperature by **1.317089E-08** relative at cell 295, where the same fixture ending on its pass budget, which takes no update, moves by **5.921E-16**; the written temperature of the worst cell is $1.768302856793E+03$ where the same conserved state and composition give $1.768302833502E+03$, and the worst species column is $2.543E-03$ (He 2^3S , the stage answering to the temperature with the steepest exponential). Four changes. The progress control (the `sp_worst` test, `n_no_fall`, the omega and trust halving, the setting of the stagnation ending) now stands between the certification of the pass and the update the next solve would consume, guarded so that a certified or non-finite pass reaches neither, and the update's own gate therefore excludes the stagnation ending by itself: the pass that ends the iteration takes no update. The ending is announced with the other endings, below the summary line and the diagnostics, so the pass that ends on it prints its summary, its carrier diagnostics and its H2 front like every other pass. A pass that takes no update says which of the two reasons it was. And the element path's update now refreshes the particle densities and the temperature between `relax_element_composition` and the equilibrium sweep (`get_species_densities`, `comp_T_from_p`), which is the shape the carrier path already had; before it, the sweep equilibrated the new composition at the old temperature. The header states the invariant. One behaviour change to record because it is not a reordering: the shortened step of a pass whose measure did not fall is now consumed by THAT pass's update, where before it reached the next pass's. RED and GREEN on the new second arm of `output_state_consistency.sh`, which runs the stagnation (`EXHALE_CARRIER_TRUST=1e-4` `EXHALE_OUTER_PASSES=8` `EXHALE_JFNK_MAXIT=5`, the setting that stagnates; the brief's cap of 40 does not, the worst gated H2 row falling through the hydrodynamic solve alone), asserts by name that the ending reached is the stagnation one, and then measures the Restart intent: `stationary evaluate round trip over the 500 physical cells against` **1e-12**: `measured=1.317089e-08` FAIL on the entry text, `measured=6.499247e-14` PASS after. The tolerance is justified by measurement and not by a snapshot: the states handed back by an ending that takes no update measure $6.5E-14$ and $5.9E-16$, and the defect measures $1.3E-08$, four decades above the allowance. The suite goes **199 PASS / 1 FAIL** on the entry text to **200 / 0**. Impact: `wasp_full_newton`

reloaded is unmoved (info = 0, CERTIFIED, ||R|| 1.495E-09, flux spread 1.9113E-12, log10 Mdot 13.30 on both binaries, all eleven outputs identical apart from the provenance run= timestamp); the carrier reload at three passes is identical pass line by pass line to the control and to the P7 record; the element reload's pass 1 is identical in every digit and the ladder parts from pass 2, because the sweep of pass 1's update now runs at the relaxed composition's own temperature (momentum 1.60E-12 against 4.24E-14 at pass 2), both arms ending on the pass budget with the same 9 refusing equations and the same base-layer mass row as the worst refusing entry.

- P15: the kind-generic rows hand back the face departures their momentum row was built from** (src/modules/time_step/hydrodynamic_rows.f90 +75/-15, src/modules/time_step/hydrodynamic_rows_body.inc +51/-17, src/tests/krylov_and_dogleg/krylov_and_dogleg_tests.f90 +255/-1 measured against the file as it stood when the item started; MEASURED by P15). Codex finding 2, and the third entry of the plan's "what remains". Under Well balanced: True the momentum row's pressure-gradient term, the momentum row scale and every measure read against that scale are formed from the module arrays face_q_up/face_q_dn of RK_integration, which only RK_rhs and the positivity repair ever filled; the kind-generic rows kept their own departures as locals and store_the_interface_fluxes wrote back face_flux and face_p alone, so on the arm those three quantities belonged to the last RK_rhs evaluation or to the zeros of the first allocation. The departures now leave the kind-generic text as two OPTIONAL intent(out) arguments of hydrodynamic_flux_difference_and_source, converted to double at the one rounding on the way out exactly as the flux and the face pressure are, both public wrappers ask for them, and store_the_interface_fluxes stores them under the key. The reconstruction continuation CAN reach the generic arm (assemble_residual chooses the arm from generic_precision_rows_arm() alone and the generic text carries its own blending branch), and on it the terms would be wrong for two reasons rather than one, the single-scheme weight w_face_p and the fact that the two schemes reconstruct different faces so no single pair of departures satisfies the blended row identity; the_generic_text_is_usable therefore now refuses recon_lambda_on .and. 0 < recon_lambda < 1, ahead of every arm evaluation. That branch ends in error stop and has no executed row, which is stated rather than worked around. Twelve new rows, each on a state the arm assembles AFTER a residual was assembled on a different state, both reconstructions and both arm values: MEASURED RED **2.098877E+01** (PLM) and **2.032204E+01** (WENO3) on the pressure gradient against the production term, **3.262392E+00** and **3.428893E+00** on the row scale, **1.190363E+01** and **1.181676E+01** on the row rebuilt from the stored face data, and GREEN at **0 bitwise** for the generic-double instantiation, which IS the production operator bit for bit, and 2.6E-16 to 2.7E-15 for the quadruple one at its 1e-13 bound. krylov_and_dogleg **309/0 to 321/0**; certification 57/0 and steady_species_rows 195/0 unchanged. Impact: wasp_full_newton reloaded is unmoved with the key off, and with Well balanced: True and EXHALE_RESID_QUAD=2 it stays CERTIFIED at the certified momentum row **9.621E-11**, P11's key-on production value, on both binaries, because the JFNK and PTC loops reset the sweep kind to marching before the certification of the handed-back state and the arm is consulted only for a non-marching sweep kind (READ, steady_newton.f90); the brief's expectation that the control's certification reads another state's departures is corrected by that measurement. What the defect

did move is the measure INSIDE the loop, where the arm is armed: the convergence measure of iteration 1 is **1.756E-03** on the control against **1.753E-03** after, the momentum row divided by the pressure-gradient term of another state and overstated by 0.17 percent, and that measure is what the loop-top stop of P3 and the ledger of P10 read on every iterate. $||F_s||_2$ is identical iteration by iteration, iterations 2 to 7 are identical line for line, the accepted state is the same state, and mol_carrier marching is byte-identical.

- P16: the hydrodynamic mass-row tolerance is anchored on the row's measured rounding floor**
 (src/modules/time_step/certification.f90 +212/-2, src/modules/time_step/steady_residual.f90 +177/-1, src/tests/certification/certification_contexts.f90 +105/-0, docs/certification_tolerance_anchoring_20260910.md +302/-0 appended; MEASURED by P16 with `git diff --numstat`, the share of the other uncommitted item in steady_residual.f90 separated by attributing each hunk). The user's first decision of the afternoon, and it CLOSES the first entry of the plan's "what remains": the tolerance is now $\max(\text{cert_tol_mass}, \min(\text{cert_tol_mass_ceiling}, \text{cert_mass_round_margin} * \text{floor}(j)))$ with $\text{cert_mass_round_margin} = 10$, the ceiling 1 (a row at 1 is a mass flux changing by the whole of itself across the cell, not an operative branch on any state measured), $\text{cert_tol_mass} = 3\text{e-}12$ unchanged and now the FLOOR of the tolerance, and the verdict on that row taken cell by cell through the pure mass_row_cell_verdict. **The floor expression the item's brief specified could not serve, and the report of that is part of the result:** $\text{eps} (|A^+ F_{\text{up}}| + |A^- F_{\text{dn}}|)/dV / \text{residual_row_scale}(1, j)$ is algebraically between one and two ulps at every cell, because the row's scale IS the larger of the two addends and the dV cancels, and MEASURED it is **4.4409E-16 on all 500 cells of all four states** looked at, so ten times it never rises above $3\text{e-}12$ and the item as written would have been inert. The cause is the one N33 established: the interface flux carries the last bit not of its own value ρv but of the $O(\rho c_s)$ quantities the Riemann solve differences, so the floor is about eps/Mach and not eps . The estimate adopted is $\text{eps} \rho (|v| + c_s) A / dV$ over the row's own scale, and it bounds the row's measured non-smoothness within a factor **1.5931** (wasp_full_newton as loaded, base Mach 3.937E-03), **1.8857** (atomic_elem_newton as loaded, 5.062E-06), **2.6806** (carrier_elem_newton as loaded, 2.400E-04) and **1.7733** (the HD 209458 b state this item certifies, 2.767E-06), where the specified expression is short by two to six decades. The margin is measured, not chosen: $c_{\text{round}} = 10$ is a round number 3.7 times above the largest of those four ratios and inside N33's independent reading of the same floor against its own bound, 5.5 to 12 (READ). The measurement itself is in the code: EXHALE_MASS_FLOOR_SCAN=1 (new, default off) adds one ulp to the density of every physical cell, runs the whole flux assembly again and reports the STEP each mass row takes beside both estimates, and the state given is then assembled once more and reproduces its own mass rows to 0.00E+00 on every fixture. Tests: certification **57/0** on the entry driver against the entry objects, **0 PASS / 1 FAIL** with exit 1 on the new driver against the entry objects (the compile naming the three new symbols), **70/0** after, the eleven new rows asserting that a floor below the fixed value leaves exactly $3\text{e-}12$ and names the fixed tolerance, that a floor above it gives exactly $c_{\text{round}} * \text{floor}$ and names the rounding anchor of that cell, that a row at half the anchor is WITHIN where the superseded rule transcribed inside the test refuses it, that a row above the anchor still refuses, that the ceiling binds, and that the margin stands above every ratio measured. Seven further suites are identical row for row in both arms, and a -fcheck=bounds,do,mem build runs both reloads

with no runtime trap (the floor estimate reads the ghost cell at $j = 1$, which is why it was checked). **Impact.** `wasp_full_newton` reloaded is unmoved: CERTIFIED in both arms, mass row $1.212\text{E-}12$ at cell 500 with the verdict taken there against the FIXED $3.0\text{E-}12$ at distance 0.404, $\log_{10}$ Mdot 13.30, outputs byte-identical apart from the provenance timestamp; its floor is below $3\text{E-}13$ at every cell, so the anchor cannot reach that case, and the cold-start `wasp_full_newton` of `make check` cannot be affected either, its stop having no row binding (READ from `backup/regression/wasp_full_newton/run.log`) and a tolerance that only ever rises cannot move it. **The HD 209458 b element reload CERTIFIES, at pass 12,** outer pass 12: ACCEPTED -- every active equation of this state is within its own tolerance, where the control (a build of the entry text of the two files in the same tree) ends the same twelve passes NOT CERTIFIED on the one entry it had before, the hydrodynamic mass row at $7.159\text{E-}10$ against $3.0\text{E-}12$ at cell 13. On the certified state the largest mass row is $7.703\text{E-}10$ at cell 2 and the verdict is taken at cell 34 ($r = 1.00665$), $6.542\text{E-}10$ against **4.6E-09, the rounding anchor of that cell**, distance 0.141; the eight elemental wind rows and the worst gated species row of all twelve passes are identical to the control to three digits, cell included, and the two arms part at pass 6, the first pass whose hydrodynamic solve stops on the loop-top test because its mass row is inside its own tolerance (`info = 0` where the control's is `info = 2`). The anchored tolerance still refuses states: the mass row is admitted at 11 of the 12 passes and refused at pass 3 at distance 69.9. On the hot-Uranus carrier reload the base cells are admitted by their own anchor at every pass and the refusal moves OUT to cell 180 ($r = 1.0707$), where the estimated floor is below $3\text{E-}12$ and the row carries signal, refusing $5.067\text{E-}12$ to $9.549\text{E-}12$ against the fixed $3.0\text{E-}12$: the refusal has moved from rounding to signal. The H2 carrier row is unchanged pass by pass in both arms ($7.38\text{E-}02$ at cell 290, $7.22\text{E-}02$ at 289, $7.06\text{E-}02$ at 288), and the three passes cost 15.47 s against the control's 69.39 s. The derivation, both candidate estimates, the measured ratios and all three impact measurements are the appended section "**Anchor (6), 2026-09-11: the mass row's rounding floor**" of `docs/certification_tolerance_anchoring_20260910.md`; its nine earlier sections are byte-identical.

- **P17: the best-iterate ledger reads the mass row against the cell's own tolerance** (`src/modules/time_step/steady_newton.f90` +146/-32, `src/tests/krylov_and_dogleg/krylov_and_dogleg_tests.f90` +98/-0; MEASURED by P17). P16's own outside-scope finding, and the consequence of the anchor for the ledger: the hydrodynamic slot was formed as `rc(1)/cert_tol_mass` from a triple of row maxima, so the ledger ranked the mass row against $3\text{E-}12$ while the stop test and the acceptance read the cell's own tolerance. The slot is now $\max_j |R_{kj}| / \text{residual_row_scale}(k, j) / \text{tol}_k(j)$ over the whole physical column, with $\text{tol}_1(j)$ taken from the certification's own rule at that cell through P16's public `mass_row_cell_verdict` (so the anchor formula is not restated in the solver module) and the fixed momentum and energy tolerances on the other two rows; that is the same functional the acceptance applies cell by cell, so $d < 1$ in the ledger is now exactly the hydrodynamic part of the verdict. `rnorm` is unchanged and is now formed only when the caller asks for it, which the two loop-top calls do and the three trial calls do not. The superseded `hydrodynamic_distance_from_certification` keeps its name, signature and body and is documented as what it is, a BOUND from above on the distance (`tol_mass_at(j) >= cert_tol_mass` at every cell), because its removal would break the compile of a test file P17 does not own; P18 removes it. MEASURED RED twice: the new driver does not build against the

entry-text objects (exit 1, Symbol 'hydrodynamic_distance_from_certification_by_cell' ... not found in module 'steady_newton'), and a private build in which the new function's body is replaced by the superseded reading and nothing else is altered gives **327 PASS / 7 FAIL** with the seven rows naming the column (the_wind_cell_holds_the_mass_rows_distance 3.333333E+02 against 3.333333E+00, the_base_cell_is_a_decade_inside_its_own_tolerance 3.333333E+02 against 1.000000E-01, is_a_certified_hydrodynamic_state F against T among them). GREEN krylov_and_dogleg **321/0 to 334/0**, and seven further suites identical in both arms. **Impact.** The single number that says the item worked, on the HD 209458 b element reload: the ledger's best judged iterate: distance over the twelve passes runs **2.2E+02 to 6.3E+05 in the control and 1.321E-01 to 1.868E-01 after**, while the acceptance CERTIFIES the pass-12 state in both arms at a mass-row distance of 0.141 (control) and 0.132 (new, verdict at cell 50, $r = 1.00977, 4.245E-10$ against the $3.2E-09$ anchor). The control's ledger therefore ranked every iterate of every solve as hundreds to hundreds of thousands of times outside certification, on a row its own acceptance admitted; the ledger and the acceptance now speak about one quantity. The trajectories part at pass 2 and the elemental relaxation is not damaged (the worst gated species row descends $2.61E-03$ to $7.73E-06$ against the control's $2.61E-03$ to $6.84E-06$, both inside $1e-5$ at pass 12, and the certified state's elemental rows are SMALLER in the new arm on every element, Fe $5.086E-05$ against $8.966E-05$, C $3.70E-06$ against $1.00E-05$, O $3.34E-06$ against $1.24E-05$). The cost is not measurable: 909 outer iterations at 1.206 s against 861 at 1.216 s, so the 500 rounding-floor evaluations added per judged residual cost nothing this measurement can separate from noise. More restarts and fewer stagnations, which is the reading of a ledger that now stops improving where $|R|/3e-12$ used to keep moving on rounding noise: "best iterate has not improved" fires in 7 solves against 4, "no descent direction exists" ends 11 against 8, and the control's one STAGNATED solve is gone, while no solve fails and every pass still returns a state. On the hot-Uranus carrier reload the consequence is a verdict: the hydrodynamic solve returns **info = 0 at all three passes** against the control's 0, 2, 2; the mass row's verdict is WITHIN at passes 2 and 3 (cell 1, $1.826E-11$ against the $2.5E-11$ anchor, and cell 2, $6.437E-12$ against the same) where the control refuses at cell 180; the state of the last pass has **1 refusing entry, the H2 carrier row alone**, against the control's 2; and the best judged distance runs 0.5282, 0.8819, 0.2578 against 1.630, 3.501, 1.840. The H2 carrier row is identical pass by pass in both arms and in the P7 record ($7.38E-02, 7.22E-02, 7.06E-02$ at cells 290, 289, 288), which is what must hold with the carrier relaxation outside the hydrodynamic solve. wasp_full_newton reloaded is unmoved, and for a stated reason: its estimated floor is below $3e-13$ at every cell, so the cell's tolerance IS the fixed value everywhere and the maximum of the ratios is the ratio of the maxima; the two run logs are identical line for line except the Execution Time, every written file is byte-identical except the provenance timestamp, and the best judged distance is $6.722E-01$ in both arms. A separate fix in the same file, reported on its own: write_gate_met_while_a_row_refuses printed the largest measure of a row and its cell beside the tolerance of the BINDING cell, a pair of numbers belonging to two different cells, and now names the binding cell throughout where the row records one, which is what the certification's own refusing entry list already did; no suite parses that line and the carrier rerun on the final binary reproduces every reported digit.

- **P18: one entry point for the hydrodynamic distance** (steady_newton.f90 +23/-50,

krylov_and_dogleg_tests.f90 +49/-25, steady_species_rows_tests.f90 +22/-16; P17's own outside-scope finding). With the ledger reading the mass row cell by cell (P17), the fixed-tolerance form hydrodynamic_distance_from_certification(rc) had no production reader left and survived only through test rows. It is DELETED, with its public export and the then-unused cert_tol_mass import of the module; every test row that read it restates its fact through hydrodynamic_distance_from_certification_by_cell on a one-cell column whose rounding floor (1e-30) puts it under the fixed tolerance, so the restatement is the same double division and every literal reference is unchanged. MEASURED (P18): grep -rn hydrodynamic_distance_from_certification src/ | grep -v _by_cell returns nothing; krylov_and_dogleg 334/0 before and after with no row lost (two renamed: the_fixed_tolerance_alone_reports_the_base_cell, and_the_fixed_reading_is_never_below_the_distance), steady_species_rows 195/0, certification 70/0, steady_completion_flag 3/0; a compile RED (the entry-text drivers against the delivered objects fail on the missing symbol), a semantic RED of the cell-by-cell fact (327/7, P17's seven rows) and a semantic RED of the restatement itself (329/5, the fixture's floor raised into the anchored regime) all discriminate. Impact: the wasp_full_newton reload gives info = 0, CERTIFIED, ||R|| 1.495e-9, log10 Mdot 13.30 in both arms, all eleven output files byte-identical, the two run logs differing in the execution-time line alone. Documents that named the deleted function as current fact (certification_tolerance_anchoring_20260910.md, session_handoff_20260911.md) were corrected by the advisor.

- P20: the code-size appendices moved to this log and were re-measured** (docs/Update_EXHALE_appendix.tex new, docs/Update_EXHALE_stage1.tex appendix removed and its \date fixed at 2026-09-04, src/utils/update_log_to_tex.py postamble, pointer lines in README.md and README_HOWTO.md). The two appendices, the code-size summary against the original ATES and the list of source inherited unchanged from it, were stranded in the stage-1 log and stale at 2026-08-30. They are now a hand-maintained fragment the generator \inputs after the converted body, so appending to this log keeps them. Re-measured 2026-09-11 with python3 src/utils/codesize.py (MEASURED, all of it): the Fortran tree is **205 files / 76,259 SLOC**, 21.7x the original ATES (47 files / 3,514 SLOC), against 111 files / 21,018 SLOC and 6.0x at the 2026-08-30 edition; the core excluding the Wind-AE port is 169 files / 71,503 SLOC, of which the production build is 95 files / 50,762 SLOC (14.4x) and the src/tests/ drivers are 74 files / 20,741 SLOC, the rule the script applies being every .f90 under src/ outside src/modules/wind_ae/. 123 new core modules (49 production, 74 test drivers), 38 of the 47 original files modified, and the files **byte-identical with upstream are down from 13 to 8**: hybrd, hybrd1 and fdjac1 took the unit_step_floor difference-step argument, Source.f90 the well-balanced arm (N37), J_inc.f90 the single-spectrum-type decision, and speed_estimate_HLLC.f90 was deleted as dead code, so one upstream file is now removed rather than none. Update_EXHALE_stage1.pdf 464 -> 460 pages, Update_EXHALE.pdf 139 -> 158 pages (MEASURED, pdfinfo).
- The measurement of record after P14 to P17** (advisor, private snapshot build of the tree with P14 to P17, 8 threads, MEASURED): the HD 209458 b element reload, partitioned, 12 passes, ACCEPTED -- every active

equation of this state is within its own tolerance at pass 12 (worst gated species row 7.73e-6, He/H partition at cell 289; mass 9.26e-10, momentum 2.44e-14, energy 2.17e-8); the hot-Uranus carrier reload, 3 passes, hydro info = 0 at every pass (mass 7.33e-12 / 1.29e-11 / 6.44e-12 admitted by the cell's anchor), the one refusing entry the H2 carrier row (7.06e-2 at cell 288).

Three changes outside the item list, made by the advisor while reading P14 to P17:

- `assert_written_state_is_the_accepted_one` (`src/EXHALE_main.f90`) carries an ABSOLUTE floor of 1e-12 beside its 1e-10 relative test. P14 reported the false alarm: the spread is a difference of O(1) numbers over their mean, so two evaluations of one state agree only to the rounding of that difference, and on the partitioned carrier reload a state flat to fourteen digits read **3.0982E-14 as written against 2.7487E-14 accepted** while the relative test alone called that a change of u. 1e-12 stands a decade above that rounding and four decades below the smallest spread a gate has ever accepted (READ, the comment at the test).
- the three rows of the `grid_and_gates` suite that wrote to a fixed directory under `$ROOT/build/tests/grid_and_gates` and ignored `EXHALE_TEST_OUT` now honor it (`momentum_row_from_fluxes_only.sh`, `base_level_single_statement.sh`, `sed_coverage_stop.sh`; READ, each now takes `${EXHALE_TEST_OUT:-$ROOT/build/tests/grid_and_gates}`). Two concurrent runs of the suite, which is what a before-and-after measurement is, collided there. P14 fixed `output_state_consistency.sh`, the fourth, because its item owned it.
- the paragraph at `assemble_residual` (`src/modules/time_step/steady_residual.f90`) that stated what the arm does not hand back is rewritten to what P15 made true: the kind-generic rows return the momentum row and store the interface fluxes, the face pressures and, under the well-balanced arm, the face departures they were built from, but neither the three terms of the momentum equation nor the equilibrium pressure force the row is read against, so the one expression of each is evaluated there for the state just assembled; and a reconstruction continuation never reaches the arm, the generic text refusing it.

The pseudo-time start of the stationary restart route

Found while running P6 and settled by the advisor. Restart intent: stationary entered the stationary solve at the CFL interval of the reloaded state. A state written by a run is already relaxed, and a continuation started at the CFL step from it does not reach the root: MEASURED, 8 threads, the hot-Uranus carrier reload's hydrodynamic solve is handed back at ||R|| 2.685E-01 after 40 iterations and the next four passes enter at 1.133, 1.018, 1.026 and 1.025 and hand back 2.976E-01 to 3.579E-01 (<scratchpad>/P6/runs/carrier_p6/run.log), while the HD 209458 b element reload stops on its stagnation detector at ||R|| 1.490 and the pass is then refused because the element relaxation finds no admissible advance (<scratchpad>/P6/runs/atomic_partition.new/run.log). With the start at 1.0, the value the marching hand-off uses, the same two solves reach ||R|| 2.507E-09 at iteration 13 and hand back **1.498E-09** (carrier) and hand back **1.167E-08** (atomic). The advisor therefore made 1.0 the default of the Restart intent: stationary route (`src/EXHALE_main.f90` line 5234), with `EXHALE_PTC_DTAU0` as the override for measurement; the direct steady route keeps its CFL start, which is the pseudo-transient continuation it was designed as. Verified with no environment variable set: the run prints stationary restart: `dtau0 = 1.00E+00` and reproduces

the first pass to every printed digit (<scratchpad>/P7/carrier_default_dtau0/run.log). No regression case sets Restart intent: at all (MEASURED: the grep over backup/regression/*/input.inp returns nothing), so this default moves no golden; it moves any user run that sets the key.

P7: the measurement of record

One binary (EXHALE_P6.x, built from the merged tree), both fixtures, Restart intent: stationary, EXHALE_PTC_DTAU0=1.0, OMP_NUM_THREADS=8, 500 cells. The partitioned arm is Coupled carrier solve: False with EXHALE_OUTER_PASSES setting the pass budget and EXHALE_JFNK_MAXIT the hydrodynamic cap; the coupled arm is the same fixture with Coupled carrier solve: True. Wall times of the partitioned runs are the sum of the seconds the outer pass lines report, which agrees with the run directory's file timestamps to under a second; the coupled carrier control prints no total, so its time is the timestamp span of its run directory. Every number MEASURED from the log named.

route	fixture	passes (hydro cap)	wall time	hydro mass / momentum / energy	worst gated species row	certified?
state as loaded	carrier, hot Uranus	--	--	2.284e-1 / 1.954 / 1.871	H2 1.166e-1 at cell 492	no, 4 entries
partitioned	carrier	8 (40)	173 s	8.75e-12 / 1.66e-12 / 2.41e-9	H2 6.217e-2 at cell 282	no, 2 entries
partitioned	carrier	40 (40)	831 s	8.121e-12 / 3.377e-12 / 2.730e-9	H2 2.234e-2 at cell 443	no, 2 entries
coupled	carrier	one solve (250), stopped by the stagnation detector at iteration 105	321 s	1.231e-2 / 1.009e-1 / 5.801e-2	H2 7.815e-2 at cell 283	no, 4 entries
state as loaded	atomic, HD 209458 b	--	--	1.509 / 3.348e-1 / 1.892	He/H 5.840e-1 at cell 291	no, 11 entries
partitioned	atomic	5 (80)	528 s	1.05e-9 / 4.40e-13 / 1.13e-8	O 4.64e-4 at cell 268	no, 9 entries
partitioned	atomic	12 (80)	1043 s	1.054e-9 / 1.665e-13 / 1.501e-8	He/H 5.468e-6 at cell 288	no, 1 entry

route	fixture	passes (hydro cap)	wall time	hydro mass / momentum / energy	worst gated species row	certified?
coupled	atomic	3 (250), ended in the fourth	2392 s	1.16e-6 / 2.89e-6 / 2.83e-6	He/H 8.29e-5 at cell 295	no, run not completed

Sources: <scratchpad>/P7/carrier_elem_newton_part1/run.log, carrier_part40/run.log, carrier_coupled250/run.log, atomic_elem_newton_part1/run.log, atomic_part12/run.log, atomic_coupled250/run.log. The atomic coupled control alternates the element relaxation too, because He_diffusion is on, and its three completed passes cost 1380.59 s, 482.04 s and 528.98 s; it was ended in the fourth. The (EXHALE_main) outer pass N: line of every pass carries the hydrodynamic flag, the worst gated species wind row with its cell, the three row maxima, omega, the carrier trust and the seconds, and the closing (certification) block names what refuses. Those <scratchpad>/ paths are the session scratch root and are TEMPORARY: they will be gone, and everything quoted from them is recorded here, in the two fixture READMEs and in docs/session_handoff_20260911.md. The runs reproduce from the pinned fixtures by the recipes in those READMEs.

What the ladders show. On the atomic fixture the worst gated species row falls 2.61e-3, 3.81e-3, 1.77e-3, 9.45e-4, 4.64e-4, 2.36e-4, 1.20e-4, 5.97e-5, 3.07e-5, 1.50e-5, 7.94e-6, 5.47e-6 over the twelve passes (the one rise, pass 1 to pass 2, is the alternation's own feedback across a hydrodynamic solve and is why a single rise does not end the loop), the refusing entries fall 9 to 1, and in the last pass every elemental wind row is within 1e-5: He/H 5.468e-6 at cell 288, and the metals 1.582e-6 (Mg) to 3.720e-6 (O). On the carrier fixture the H2 wind row falls 7.38e-2 to 2.23e-2 over the 40 passes with the binding cell moving 290 to 443, which is the H2 front advancing about one cell in a pass and not a residual settling.

The decision. The plan's adoption rule is that the route is adopted only if it certifies where the coupled solve does not, or reaches a strictly lower joint residual in no more wall time. Neither fixture certifies, so the first branch is not met on either. The second branch is met on both: on the atomic fixture the partitioned route stands strictly below the coupled control on every judged row (mass 1.054e-9 against 1.16e-6, momentum 1.665e-13 against 2.89e-6, energy 1.501e-8 against 2.83e-6, worst gated species row 5.468e-6 against 8.29e-5) in 1043 s against 2392 s for three passes; on the carrier fixture the partitioned route at 8 passes stands strictly below the coupled control on every judged row (8.75e-12 against 1.231e-2, 1.66e-12 against 1.009e-1, 2.41e-9 against 5.801e-2, H2 6.217e-2 against 7.815e-2) in 173 s against 321 s, and the 40-pass continuation goes on to 2.23e-2 for 831 s. The partitioned route is therefore what the production outer loop runs. Coupled carrier solve already defaults to False (carrier_in_newton = .false., src/modules/init/parameters.f90 line 419), so no default changes; True remains the coupled measurement arm, and what this session added on the partitioned side is the joint acceptance contract of P6 and the three defect repairs P1 to P3 that made a pass mean something.

What did not certify, and why. Entry 1 was CLOSED the same afternoon by the user's first decision (P16, anchor (6)); entries 2 and 3 are stated as they stand:

1. **CLOSED 2026-09-11 by P16.** The atomic element reload refused on the hydrodynamic mass row alone, 1.054E-09 against 3.0E-12 at cell 16 ($r = 1.0031$), on the three-unknown solve of this fixture. Every one of its twelve hydrodynamic solves ends with $||R||$ between 8.980E-09 and 1.501E-08 against the run's Resid tol of 1e-8, and the last one ends on no descent direction exists for the banded model at this state at $||R||$ 1.501E-08. So the alternation has done its work (the element rows are inside their tolerance) and what is left is the base-layer mass row of the hydrodynamic solve, which is the N33 rounding floor territory and not a species-coupling question. **P10 settled which of the two it is.** With the best-iterate ledger reading that row against its own 3e-12 (P10), the same twelve passes end at 8.893E-10 at cell 3, the same order, and the row is MEASURED to carry no signal at that level: the ledger's best distance of pass 1 is 4.407E+02, a mass row of 1.32E-09, while the same state re-evaluated at its own composition after the best-iterate restore carries 4.11E-09. At 1e-9 to 1e-12 the row is the rounding of the base-layer flux difference, so ranking iterates by it is ranking by rounding noise and no ledger can bring it down. The entry was therefore an ACCEPTANCE question, the 3e-12 tolerance against the base-layer rounding floor of this fixture, and not a solver item; the user decided it the same afternoon, and with the mass row judged at $\max(3e-12, 10 * \text{floor}(j))$ the same twelve-pass recipe **CERTIFIES at pass 12** (MEASURED by P16 and again by P17, which certifies at pass 12 in both of its arms). The verdict is taken at a wind cell, not at the base: cell 34, $r = 1.00665$, 6.542E-10 against the 4.6E-09 anchor of that cell, distance 0.141 (P16), and cell 50, $r = 1.00977$, 4.245E-10 against 3.2E-09, distance 0.132 (P17). The base-layer floor of this fixture is what the anchor is measured on: see the decisions subsection below and docs/certification_tolerance_anchoring_20260910.md, "Anchor (6)".
2. The carrier reload refuses on the H2 wind row, and the user's second decision of the afternoon ran the continuation that says what that row IS. MEASURED over 110 passes (the table and the recipe are in the decisions subsection below): the H2 front does NOT stop, $x_2 = 0.5$ moving 1.0726 to 1.1230 R_p over 108 passes at about 5E-04 R_p in a pass with no sign of slowing, so the continuation is unfinished after 110 passes and 0.05 R_p . And the worst gated row stops being the front after pass 40: it stands at **2.2E-02, nearly uniform in r**, and its cell walks outward through the wind, 443 to 500, reaching the outer boundary at pass 108 while the value does not fall. That floor is a property of the operator and not of the front: the bounded carrier pass (trust 1e-2 on the H2 mixing-ratio maximum, met by one transport step) advances the front but leaves the wind cells, whose H2 mixing ratio is orders of magnitude below the maximum, at one implicit step in a pass, so their own H2 balance never relaxes, and the drift measure, absolute in the grid maximum, does not see them. The consequence for the method, recorded and not adopted: a bounded front advance and a relaxation of the wind cells' H2 balance are two different operations, and the second needs its own step (a cell-relative measure, or a wind-window relaxation after the bounded pass) before the H2 wind row can be judged at 1e-5. Whether the front stops at all inside 2 R_p is a physical question and stays open (the P23 comparison against Koskinen et al. 2022 Model A).
3. The hot-Uranus mass row sits on the same floor throughout the carrier run, 4.48E-12 to 1.57E-11 against 3.0E-12 over the 40 passes, with the momentum row at 4.14E-13 to 5.13E-12 and the energy row at 1.75E-09 to 4.56E-09. The mass row of that fixture is the rounding of the base-layer flux difference recorded in N33, which is the reading P16 then made operative: under anchor (6) the base cells of this fixture are ADMITTED by their own anchor at every pass and the refusal moves out to cell 180 ($r = 1.0707$), where the floor is below 3e-12 and the

row carries signal.

The Codex adversarial review (2026-09-11)

Run on the working tree after P1 to P13, that is on the merged text of the morning's items and not on the committed tree. Two findings, both verified in the source by the advisor before any item was written, and both closed the same afternoon. The findings are quoted as they were written.

1. **[high]** "정체 종료 시 조성과 출력용 상태가 서로 달라집니다 (src/EXHALE_main.f90:6503-6514)": on a stagnation exit the composition had been updated while the particle densities and the temperature were not refreshed on the element path, so the certified state and the written state differed; the recommendation was to move the progress control before the composition update. Verified in the source, reproduced as a measured inconsistency of 1.317089E-08 in the temperature of the written state, and **closed by P14** with exactly that ordering: the progress control now stands between the certification of the pass and the update, so a pass that ends the iteration takes no update, and the element path's update refreshes the particle densities and the temperature before its equilibrium sweep.
2. **[medium]** "고정밀 경로가 다른 상태의 압력 정보를 인증에 사용합니다 (src/modules/time_step/steady_residual.f90:256-258)": with EXHALE_RESID_QUAD the generic rows stored only face_flux and face_p, so momentum_row_terms_of_state read the previous RK_rhs call's face_q_up/face_q_dn (zeros on the first call); the recommendation was to return the terms from the generic computation and to refuse the continuation combination. Verified in the source, and **closed by P15**, both halves: the arm now hands its own face departures back along the route the fluxes take, and the continuation combination is refused, because the generic arm CAN be reached on a continuation (assemble_residual chooses the arm from generic_precision_rows_arm() alone and the generic text carries its own blending branch) and no single pair of departures satisfies the blended row identity.

Neither finding moves a certified state of the default path: P14's wasp_full_newton reload and P15's, with the key off and with Well balanced: True and EXHALE_RESID_QUAD=2, are CERTIFIED with every output byte-identical apart from the provenance timestamp. What each moves is stated in its own bullet above.

User decisions of 2026-09-11 (afternoon)

Two decisions were taken after the open items of the plan were put to the user.

Decision 1: the mass tolerance is re-anchored above the fixture's measured rounding floor (item P16, and the ledger consequence P17). The tolerance of the continuity row is now

$$\text{tol}(j) = \max(\text{cert_tol_mass}, \min(\text{cert_tol_mass_ceiling}, \text{c_round} * \text{floor}(j)))$$

with cert_tol_mass = 3e-12 unchanged as the FLOOR, the ceiling 1, and c_round = cert_mass_round_margin = 10. The floor estimate is $\epsilon \rho (|v| + c_s) A / dV$ over the row's own scale, which is one ulp in the wind and about $\epsilon \rho / \text{Mach}$ in a layer whose flux is a tiny residue of the momentum the cell carries at its own fastest signal speed.

The brief's first expression for the floor, $\text{eps} (|A+ F_{\text{up}}| + |A- F_{\text{dn}}|)/dV / \text{residual_row_scale}(1, j)$, is inert, and that is a measured result and not a judgement: the row's scale IS the larger of the two addends and the dV cancels, so the expression is algebraically between one and two ulps at every cell, MEASURED at **4.4409E-16 on all 500 cells of all four states** examined, and ten times it never rises above $3e-12$. The adopted estimate bounds the step each mass row actually takes when one ulp is added to the density of every physical cell (EXHALE_MASS_FLOOR_SCAN=1, new and default off) within a factor

state	base Mach	step over the adopted estimate	step over the brief's expression
wasp_full_newton as loaded	3.937E-03	1.5931	1.109E+02
atomic_elem_newton as loaded	5.062E-06	1.8857	2.4365E+05
carrier_elem_newton as loaded	2.400E-04	2.6806	5.0915E+03
the HD 209458 b state P16 certifies	2.767E-06	1.7733	1.4201E+06

All MEASURED by P16, largest ratio over the 500 cells of each state. $c_{\text{round}} = 10$ is a round number 3.7 times above the largest of those ratios and inside N33's independent reading of the same floor against its own bound, 5.5 to 12 (READ). The outcome: the **HD 209458 b element reload CERTIFIED at pass 12**, where the control built from the entry text of the two files ends the same twelve passes NOT CERTIFIED on the hydrodynamic mass row ($7.159E-10$ against $3.0E-12$ at cell 13), and **wasp_full_newton unmoved** in both arms (CERTIFIED, mass row $1.212E-12$ at cell 500 judged against the FIXED $3.0E-12$ at distance 0.404, $\log_{10} \dot{M}$ 13.30, outputs byte-identical apart from the provenance timestamp), its floor being below $3E-13$ at every cell. The anchor still refuses states: the HD 209458 b mass row is refused at pass 3 at distance 69.9, and on the hot-Uranus carrier reload the refusal moves from the base cells, admitted by their own anchor, out to cell 180 where the row carries signal. The derivation and all three impact measurements are the appended section "Anchor (6), 2026-09-11: the mass row's rounding floor" of docs/certification_tolerance_anchoring_20260910.md.

Decision 2: the long continuation of the hot-Uranus carrier reload. The pass budget was set to 400 and the run went to pass **110**, where the progress control ended it ("the worst gated species row has not fallen in 3 consecutive passes"), the carrier trust having been halved $1.0E-02$ to $5.0E-03$ to $2.5E-03$ at passes 109 and 110. Recipe: backup/regression/carrier_elem_newton, Load IC? True, Coupled carrier solve: False, Restart intent: stationary, EXHALE_OUTER_PASSES=400 EXHALE_JFNK_MAXIT=40 OMP_NUM_THREADS=8. Binary: the advisor's snapshot build of the merged tree after P1 to P12 and BEFORE P14 to P17, so every hydrodynamic solve of this run returns $\text{info} = 1$ on the base-layer mass row ($5E-12$ to $1E-11$ against $3E-12$) and the run predates anchor (6). All numbers MEASURED from its run.log.

pass	worst gated H2 row	cell	H2 front $x_2 = 0.5$ [R_p]	$x_2 = 1e-2$ [R_p]
1	7.38E-02	290	1.0726	1.0966
10	5.86E-02	281	1.0755	1.1017
20	4.07E-02	274	1.0796	1.1070
30	2.62E-02	267	1.0827	1.1140
40	2.23E-02	443	1.0871	1.1199
50	2.20E-02	473	1.0917	1.1262
70	2.19E-02	482	1.1017	1.1413
90	2.18E-02	490	1.1126	1.1563
100	2.17E-02	494	1.1184	1.1643
108	2.17E-02	500	1.1230	1.1727
110	2.17E-02	500	1.1230	1.1727

Two regimes. Passes 1 to 30: the worst gated row is at the front's outer edge, its cell moving 290 to 267 (r about 1.28 to 1.25), and it falls $7.4E-02$ to $2.6E-02$ as the front relaxes. From pass 40 the worst row is no longer at the front: it sits at $2.2E-02$ and its cell walks outward through the wind, 443 to 500, reaching the outer boundary at pass 108 while the value does not fall. The front itself keeps moving outward at about $5E-04$ R_p in a pass, $x_2 = 0.5$ from 1.0726 to 1.1230 over 108 passes, with no sign of stopping.

So three statements, the first two measured and the third a reading of them. (a) The front has NOT reached a stationary position in 110 passes: 0.05 R_p travelled and the continuation unfinished. (b) The remaining wind residual of $2.2E-02$ is nearly uniform in r and is NOT the front. (c) The reading: the bounded carrier pass, trust $1E-02$ on the H2 mixing-ratio maximum and met by one transport step, advances the front but leaves the wind cells, whose H2 mixing ratio is orders of magnitude below the maximum, at one implicit step in a pass, so their own H2 balance never relaxes, and the drift measure, absolute in the grid maximum, does not see them. The progress control ended the run because after pass 40 the worst gated row measures that wind floor and not the front.

Consequence for the method, recorded and NOT adopted: a bounded front advance and a relaxation of the wind cells' H2 balance are two different operations, and the second needs its own step, a cell-relative measure or a wind-window relaxation after the bounded pass, before the H2 wind row can be judged at $1e-5$. The physical question whether the front stops at all inside 2 R_p is separate and stays open (the P23 comparison against Koskinen et al. 2022 Model A).

Gates

Advisor verification of the merged tree (READ from the closing brief): snapshot build with the default toolchain; element_operator 24/0, carrier_retry 122/0, krylov_and_dogleg 290/0, certification 57/0, steady_species_rows 194/0 and 195/0 after P8; grid_and_gates 158 PASS with 3 FAIL, all three missing files of the snapshot copy (inputdata, examples) rather than assertions (P4 measured the same suite 178/0 in the live tree); the wasp_full_newton reload info = 0, ||R|| $1.667e-9$, CERTIFIED, log10 Mdot 13.30, unchanged. make check (advisor, 2026-09-11 09:40 KST, the shared build rebuilt from this tree): REGRESSION PASS, every DE-

FAULT_CASES output and wasp_full_newton byte-identical to the goldens; no golden moved, because no default case enters the stationary solve or the fixed-wind relaxations.

After P10 to P12 (advisor, merged tree, private snapshot build, 8 threads, MEASURED 2026-09-11): the thirteen suites that link the changed modules all PASS (krylov_and_dogleg 309/0, grid_and_gates 187/1 with the one FAIL a WASP-52b SED file absent from the snapshot copy, steady_species_rows 195/0, element_operator 24/0, carrier_retry 122/0, certification 57/0, attempted_step 70/0, carrier_returned_state_acceptance 36/0, species_masses 9/0, carrier_reference_scales 14/0, carrier_constraint_attribution 10/0, constrained_network_layout 8/0, steady_completion_flag 3/0). The two fixtures re-run on that binary reproduce P10's and P11's records: the HD 209458 b element reload, 12 passes, worst gated species row $2.61\text{e-}3 \rightarrow 5.47\text{e-}6$ with only the hydrodynamic mass row refusing ($8.893\text{e-}10$ against $3\text{e-}12$ at cell 3, $r = 1.0006$); the carrier reload, 3 passes, H2 $7.38\text{e-}2 \rightarrow 7.06\text{e-}2$ at cells 290 $\rightarrow$ 288, mass $8.12\text{e-}12 / 4.85\text{e-}12 / 4.96\text{e-}12$, momentum $3.22\text{e-}13 / 4.47\text{e-}13 / 3.54\text{e-}13$.

Second make check after P10 to P12 (advisor, 2026-09-11 15:15 KST, the shared build rebuilt from the merged tree): REGRESSION PASS, every DEFAULT_CASES output and the cold-start wasp_full_newton byte-identical to the goldens; no golden moved.

After P14 to P17, each count MEASURED by the item that owns the file and READ from its report: grid_and_gates **200/0** (P14, the second arm of output_state_consistency.sh; 199/1 on the entry text), krylov_and_dogleg **321/0** after P15 and **334/0** after P17, certification **70/0** after P16 (57/0 on the entry driver), with steady_species_rows 195/0, adv_static_limit 53/0 and the other suites each item ran unchanged in both of its arms. The two rows that FAIL by design or by fixture are unchanged: residual_determinism's closure_spread_within_the_row_tolerance_atomic_elem_newton at $3.354\text{E+}01$ (N33, by design) and steady_selfconsistent_residual's handback_matches_the_accepted_iterate on a reload log that prints no iterate norm.

Third make check after P14 to P18 (advisor, 2026-09-11 21:10 KST, the shared build rebuilt from the final tree): REGRESSION PASS, every DEFAULT_CASES output and the cold-start wasp_full_newton byte-identical to the goldens; no golden moved.

Reported by the workers and not fixed

- P1: relaxation_drift_covers_every_element in steady_species_rows builds a column whose species miss its own density by $1.660\text{E-}02$ (closed the same day by P8); relax_element_composition damps helium alone, which belongs to the outer composition control; element_diffusion_step's trace-metal arm re-seeds an exhausted cell and project_elements clamps grp_new at zero, the only place in the step that can create or destroy mixture mass, now covered by the closure test and not investigated further.
- P2: the carrier relaxation reaches into the steady residual for its advecting flux (carrier_rows_advect = .true.) while the element operator is given its flux by the caller, so the routine cannot be called at all unless someone else assembled a mass row; residual_row_scale in steady_residual.f90 carries a comment block that is the history of a change and not the physics; the fixed-point exit of the carrier relaxation is unreachable in production while the H2 front propagates, and loosening relax_tol to make it reachable is not a fix.

- P3: ~~distance_from_certification divides the hydrodynamic slot by the run's Resid tol and not by each row's own certification tolerance, so the best-iterate ledger cannot see a mass row above $3e-12$ while $||R||$ is below $1e-8$, which is exactly the state D3 is about (it did no harm on the D3 run, where the returned iterate's mass row is $5.945e-12$ against the last iterate's $7.914e-12$, but the ranking does not guarantee it) (fixed 2026-09-11, P10); steady_selfconsistent_residual assertion 2 (closed the same day by P8); attempted_step/run.sh and carrier_returned_state_acceptance/run.sh write a fallback build_stamp.f90 into their own work directory, where the compile cannot find it, so a tree that has never been built fails with Cannot open module file 'build_stamp.mod' (fixed 2026-09-11, P12, in those two and in five sibling scripts).~~
- P4: the kind-generic rows do not form the equilibrium pressure term themselves (closed from assemble_residual, but the two places must stay one expression if the arm is ever made to write its own face data back); ~~the default path's PLM momentum scale carries a coordinate artifact, the geometric pressure term $(A^+ - A^-) p_j / dV$ entering the max separately although it is one physical term with the flux divergence, so at zero gravity with a uniform pressure the scale is $2 p / r$ where the physical force is zero (MEASURED: the arm's scale, which is the row's true content, sits 0.944 below the base scheme's for both Riemann solvers at every N; under WENO3 the two agree to $2E-13$). Fixing it moves the default momentum measure at every cell and therefore the acceptance of every recorded state, so it is a user decision. (fixed 2026-09-11, P11, on the user's instruction: the scale is now the largest of the three physical terms, the zero-gravity artifact reads 2.22507E-308 in place of 1.000, and the acceptance movement is measured, wasp_full_newton still CERTIFIED.)~~
- P5: the same truncation-as-floor reasoning stood in three other documents (closed the same day by P8); the analysis document's own title is still stronger than its corrected conclusion; docs/session_handoff_20260910.md and docs/steady_solver_design.md carry statements of the same family, not read line by line; the experiment's private build is retained at /tmp/exhale_partition_20260911.ZxXJEB per its own section 9.
- P8: docs/certification_tolerance_anchoring_20260910.md still opens with cert_tol_element and cert_tol_carrier at $1e-8$, which decision 22 (a) superseded, and whether that memo is a dated record or a live reference is a decision; docs/code_status_20260910.md section 3 described the stage-2 solver program as closed (this session's item P9 marks it superseded); ~~mass_closed_column exists only in src/tests/element_operator/element_operator_tests.f90 and two suites now build the same column by hand, so the recipe belongs in one place a test can call (fixed 2026-09-11, P12: src/tests/test_columns.f90).~~
- P9: the Coupled carrier solve key comment in src/modules/files_I0/input_read.f90 states that the alternation "was measured not to converge and not to be able to" (section 139). The first half is what P7 measured against: the alternation with the joint acceptance contract of P6 brings every elemental wind row of the atomic fixture inside $1e-5$ and moves the carrier front monotonically, so the comment's reading is at least narrower than it states. Not my file this increment; it needs the sentence rewritten to what section 139 actually measured (the old alternation, without the acceptance contract, and its sign error on the front's motion). (fixed 2026-09-11, advisor.)
- P10: two rows of src/tests/steady_species_rows/steady_species_rows_tests.f90 state the superseded semantics, feeding a raw $||R||$ into the slot that now takes a distance and asserting that the hydrodynamic slot

follows the run's Resid tol (MEASURED 193 PASS / 2 FAIL on the new objects) (fixed 2026-09-11, advisor); dj that the ledger ranks with is the accepted trial's own rows while the judged rows line prints the loop-top evaluation, so the reported best distance can sit below the minimum of the printed lines (wasp_full_newton: 8.469E-02 against a printed minimum of 1.178E+00), which is entry-text behavior and not a defect; in the EXHALE_RESID_BRANCH_REPORT block one divisor for the whole class stays, because the block aggregates spreads across seeds whose maximum can sit in different rows.

- P11: ~~the kind-generic rows still hand back neither the momentum terms nor the face departures (hydrodynamic_rows.f90's store_the_interface_fluxes writes face_flux/face_p and nothing else), so on the EXHALE_RESID_QUAD arm the well-balanced face departure and, on a reconstruction continuation with $0 < \lambda < 1$, which of the two pressure conventions the blended flux carries are read from the last RK_rhs evaluation instead of from the state just assembled; the shape of the fix is two weights in place of the use_plm branch, and the arm is default off and a rounding-control experiment nothing is adopted from.~~ (fixed 2026-09-11, P15, which was also the Codex review's second finding: the arm stores its own face departures, and the continuation combination is refused rather than weighted, because the two schemes reconstruct different faces and no single pair of departures satisfies the blended row identity. The momentum terms are still not handed back by the arm, and they are not read from a previous evaluation either: the one expression of each is evaluated at the call site for the state just assembled, which is the P4 entry above.) The marching stop of mol_carrier reports a momentum measure of 1.979E+00 against a scale that is now the largest physical term in the layer, which is commensurable with the mass and energy entries in a way it was not; whether the marching stop should quote it at all is an acceptance question.
- P12: nothing further. The four sibling scripts carrying the same dead fallback and the work directories hardcoded with no override, which the P12 report listed as outside its own two files, are what P12b closed the same day.
- P14: ~~assert_written_state_is_the_accepted_one raises a false alarm whenever the flux spread reaches the rounding level (MEASURED on the partitioned carrier reload, 3.0982E-14 as written against 2.7487E-14 accepted on a state whose mass flux is flat to fourteen digits, while the same run at a spread of 1.7736E-03 passes the test); it needs an absolute floor beside the relative one, or the warning will keep accusing the best states the solver produces and three rows of the grid_and_gates suite write to a fixed directory under \$ROOT/build/tests/grid_and_gates and ignore EXHALE_TEST_OUT, so two concurrent runs of the suite collide there (both fixed 2026-09-11, advisor, the floor at 1e-12 and the three rows named in the paragraph above). What is left is one line of documentation: src/tests/grid_and_gates/README.md gives one line for each test name, and the output_state row now asserts two things where the README says one.~~
- P15: quad_rows_calls and quad_rows_seconds are incremented in hydrodynamic_rows_in_quadruple_precision and public, and no file reads them (grep over src/), so the arm's measured cost, which the module header calls a measured number and not an estimate, reaches no run summary; relatedly, nothing in run.log says whether EXHALE_RESID_QUAD took effect, so P15 had to establish that its arm run was an arm run indirectly, from the iteration-1 measure moving between the two binaries and from the sweep-kind reset READ in steady_newton.f90. A line in the setup report, or those two counters reaching it, would make an arm run self-documenting. And the continuation refusal the item added has no executed row: it ends in error stop,

which cannot be asserted from inside the driver process, and no production path reaches it, the key being default off and the continuation controller disarming before the JFNK finish (READ).

- P16: the control arm of the HD 209458 b reload ends with a state its own certification refuses and **exits 0**, because the outer loop's ending is the pass budget and no stationary claim is made. That is the run-mode contract working as written, but a reader of the shell status cannot tell that case from a certified one.
- P17: ~~hydrodynamic_distance_from_certification has no production reader left and cannot be deleted without breaking the compile of a test file the item does not own~~ (item P18, above). Two fixture observations, no action taken: running `grid_and_gates` twice at once with distinct EXHALE_TEST_OUT makes the `flux_spread` sub-test report `mass_flux_spread_recomputed_from_output measured=no_dump_line`, so the two invocations share something outside EXHALE_TEST_OUT (each arm on its own gives 200/0), which is worth a line in that suite's README for whoever next runs two builds side by side; and the log-driven block of `steady_species_rows` requires a (`species_jac_test`) line in the log, so the best judged `iterate <= min judged rows` seen invariant it states is checked only on logs made with the species Jacobian probe armed, and P17 had to apply that comparison by hand (0 of the 12 solves of its new arm report a best above their own printed minimum, against 1 of 12 in the control). A fixture log carrying that probe would be worth pinning.

Code-size summary: EXHALE vs. the original ATES (2026-06-07; updated 2026-09-11)

A line-by-line comparison of the Fortran source against the pristine upstream code (ATES/ATES-Code-main) quantifies the extent of the in-house extension. *Pure code lines (SLOC)* exclude blank lines and comments (the comment stripper is string-aware, so a `!` inside a quoted string is not treated as a comment); the line-by-line diff additionally normalizes whitespace and case, so reformatting and comment-only edits are *not* counted as changes. The measurement script is `src/utls/codesize.py`, which reprints every number of this appendix and the byte-identical list of Table 9. Measured 2026-09-11, i.e. after the whole of stage 1 and the stage-2 series through item P19 of section 8: the acceptance and certification contracts, the attempted-step controller, the partitioned stationary solver, the oxygen and carbon chemistry, the molecular infrared bands and the test suites those series brought with them.

What the script counts as *core*. Every `.f90` file under `src/` except those in `src/modules/wind_ae/`, which is the ported Wind-AE IC generator and is listed separately. That rule puts the standalone drivers under `src/tests/` in the core figure, and they are now a large part of it, so the table splits the core row into the production build and those drivers. Only the production files are compiled into EXHALE.x; each test suite builds its own driver from `src/tests/<suite>/run.sh`.

	Fortran files	code lines (SLOC)
Original ATES (ATES-Code-main)	47	3 514
EXHALE core (excl. Wind-AE port)	169	71 503
of which the production build	95	50 762 ($\approx 14.4\times$)
of which the src/tests/ drivers	74	20 741
net change (core)	+122	+67 989 ($\approx 20.3\times$)
Wind-AE IC-generator port (wind_ae/)	+36	+4 756
EXHALE total	205	76 259 ($\approx 21.7\times$)

Line-by-line diff, core (comments + blanks excluded; whitespace/case normalized): **69 179** code lines added in EXHALE (new modules + the new side of modified lines) and **1 190** removed from the original (deleted + the old side of modified lines), for **70 369** total changed lines. At the file level, of the 47 original files **38 were modified**, **8 are unchanged** (all eight byte-for-byte; the list is Table 9) and **one was removed**: flux/speed_estimate_HLLC.f90, Toro’s adaptive p_* estimate, was used by Num_Fluxes.f90 and never called, because the HLLC branch builds its own Davis/Einfeldt speeds inline. **123 new core modules** were added. Of those, 49 are part of the simulation build:

New module	Purpose	SLOC
time_step/steady_newton.f90	stationary solve of the hydrodynamic rows: residual $F(Y) = 0$ on the physical cells, Newton/Krylov with pseudo-transient continuation, line search, best-iterate ledger	9435
lower_atmosphere/diffusive_photochemistry.f90	vertical transport of the molecular carriers (H_2 , OH, H_2O , CO) solved together with the chemistry that makes and destroys them	2485
lower_atmosphere/molecular_infrared_data.f90	band and line data of the molecular infrared coolant (generated by cooling_data/molecular_infrared_bands.py)	2182
functions/binary_element_diffusion.f90	conservative binary H/He element transport (stage-resolved pair coefficients, donor-cell drift, metals slaved to H)	1898
time_step/attempted_step.f90	the attempted-step controller: one checkpoint, one trial, one adoption boundary, and the rollback contract	1608
time_step/certification.f90	one evaluator of the equations a run actually solves, and the contexts that read its verdict	1273
nonlinear_system_solver/constrained_chemical_equilibrium.f90	chemical equilibrium of the molecular H/He network in logarithmic number densities, element conservation as explicit residual rows	1173
lower_atmosphere/h2_self_shielding_table.f90	H_2 Lyman–Werner dissociation cross section with overlapping lines, and its two branching ratios	952
time_step/energy_semi_implicit.f90	implicit source step for the thermal energy of a cell, at fixed composition and heating	545
time_step/steady_residual.f90	finite-volume stationary residual $R = du/dt$ and its norms, shared by the diagnostic, the convergence monitor and the stationary solver	509
time_step/viscous_conduction.f90	Navier–Stokes viscous momentum force, its dissipation, and thermal conduction (Crank–Nicolson)	504
radiation/excited_hydrogen.f90	non-LTE $H(n=2)$ 2s/2p equilibrium (Christie, Arras & Li 2013) and the Balmer feedback	474
lower_atmosphere/element_inventory.f90	one stoichiometric map of the H, He, O and C nuclei of a cell, and the three constraints a composition state is judged by	448

New module	Purpose	SLOC
radiation/charge_exchange.f90	charge-exchange network (Huang et al. 2023, Table 4)	446
files_IO/lower_atmosphere_profile.f90	lower-atmosphere profile reader: log- p interpolation, elemental reservoirs, $K_{zz}(p)$	367
states/caloric_eos.f90	caloric equation of state: internal energy density, pressure and temperature, with the H ₂ level ladder	340
lower_atmosphere/molecular_reaction_heat.f90	net chemical heat of the collisional reactions of the H ₂ /He network	339
lower_atmosphere/oxygen_rates.f90	rate coefficients and thermodynamic data of the oxygen-hydrogen chemistry, with their validity ranges	316
functions/element_census.f90	one definition of the elemental and charge content of a composition state, and the invariant checks built on it	307
nonlinear_system_solver/System_HeH_mol.f90	coupled H/He + molecular (H ₂ /H ₂ ⁺ /H ₃ ⁺ /HeH ⁺) system, turnover-scaled rows	303
radiation/electron_energy_degradation.f90	degradation of a fast photoelectron in a partly ionized H/H ₂ /He gas: the heat fraction and the ionizations of each target	287
radiation/lya_rt.f90	Ly α transfer by the escape-probability closure, evaluated in line each step	257
lower_atmosphere/molecular_infrared_cooling.f90	infrared exchange of the molecular layer: H ₂ O and CO vibration-rotation bands, H ₂ quadrupole and magnetic-dipole lines	231
states/base_boundary.f90	the lower boundary as a characteristic condition at the face, the number of conditions set by the Mach number there	199
functions/h2_photo_channels.f90	H ₂ photoabsorption resolved into mutually exclusive final-state channels, with one stoichiometric table	199
lower_atmosphere/lyman_werner.f90	H ₂ photodissociation in the Lyman and Werner bands (the Solomon process)	195
radiation/opacity_models.f90	opacity-model dispatcher for the photoionization cross sections (A/C/P/T)	180
files_IO/metals_input_read.f90	parse metals.inp: trace-metal abundances at runtime	177
functions/composition.f90	single point turning (ρ, f_{sp}) into every species density, n_e and n_{tot} , and converting pressure and temperature	162
flux/species_face_flux.f90	advection of the species mass fractions on the same faces, with the same mass flux, as the density	162
time_step/hydrodynamic_rows.f90	the hydrodynamic rows of the stationary residual in double precision, instantiated from the kind-generic text	161
nonlinear_system_solver/System_HeH_metals.f90	coupled H/He+metals ionization (+ analytic Jacobian)	150
lower_atmosphere/mol_rates.f90	molecular reaction rates (Koskinen et al. 2022, Table 1)	150
lower_atmosphere/h3p_cooling.f90	optically thin H ₃ ⁺ infrared cooling (Miller et al. 2013)	148
nonlinear_system_solver/System_HeH_mol_metals.f90	molecular + metals merged ionization system	141
nonlinear_system_solver/newton_solver.f90	damped Newton with an analytic Jacobian for the ionization systems, with the MINPACK hybrd1 retry	128
init/species_table.f90	metadata table of the metal stages and of the base, molecular and oxygen species	128
lower_atmosphere/co_photodissociation.f90	CO photodissociation on the 912–1201 Å band	127
nonlinear_system_solver/ion_residual_core.f90	shared ionization-residual rows and analytic-Jacobian pieces of the System_* family	127

New module	Purpose	SLOC
lower_atmosphere/water_photolysis.f90	H ₂ O and OH photodissociation in the four far-ultraviolet bands of the oxygen option	111
lower_atmosphere/co_self_shielding_table.f90	CO self-shielding function $\Theta[N(\text{CO}), N(\text{H}_2)]$ (Visser, van Dishoeck & Black 2009)	110
nonlinear_system_solver/ion_cell_state.f90	named-field cell state of the ionization residuals, in place of the position-packed params	93
lower_atmosphere/lower_column.f90	analytic lower-atmosphere column from the 1-bar level to the base pressure (Koskinen et al. 2022)	90
flux/low_mach_dissipation.f90	gated fourth-difference stress damping the two-cell velocity mode a contact-resolving flux leaves undamped	87
files_IO/opacity_input_read.f90	parse opacity.inp	85
radiation/hydrogen_n2_rates.f90	atomic data and rate coefficients of the H $n=2$ manifold and of the Ly α line, in one place	76
nonlinear_system_solver/System_HeH_TR_metals.f90	merged He 2 ³ S+metals system	68
lower_atmosphere/h2_vibrational_relaxation.f90	the fate and the size of the vibrational energy left in an H ₂ molecule, for both of its drivers	55
nonlinear_system_solver/constrained_equilibrium_probe.f90	standalone re-evaluation of one saved constrained-equilibrium state, without the hydrodynamics	15
production new-module total		30 003

The other 74 are the standalone test drivers under `src/tests/`, which did not exist in ATES at all. They state what a routine is allowed to return and are run by suite, not by the production binary:

Test driver	What it asserts	SLOC
tests/krylov_and_dogleg/krylov_and_dogleg_tests.f90	what the linear solve and the step geometry of the stationary solver are allowed to return	2675
tests/steady_species_rows/steady_species_rows_tests.f90	the unknown space of the stationary system, one row per transported balance	1806
tests/diffusion_tests.f90	the binary H/He element-diffusion operator: analytic limits and conservation	1051
tests/grid_and_gates/hydrostatic_residual.f90	the momentum row returns zero on a hydrostatic state	967
tests/carrier_retry/carrier_retry.f90	the carrier-local checkpoint and the rejection-and-retry controller, on a real molecular column	842
tests/certification/certification_contexts.f90	the certification evaluator and its contexts, against the predicates they replace	790
tests/species_face_flux/species_face_flux_tests.f90	species advection on the face mass fluxes, and the element rows of the diffusion operator	557
tests/physics_probe/weno3_reconstruction_order.f90	convergence order of the face reconstruction against the exact face value	482
tests/element_operator/element_operator_tests.f90	the boundary the element transport operator is measured under	471
tests/spectrum_type/loaded_sed_table_semantics.f90	what a row of a loaded SED table means, and what the production consumers make of it	417

Test driver	What it asserts	SLOC
tests/attempted_step/attempted_step_ tests.f90	the checkpoint and the step-size policy of the attempted-step controller	415
tests/a2_m1/a2_m1_rate_check.f90	the oxygen rate coefficients against their published sources	406
tests/spectrum_type/balmer_band_ quadrature.f90	quadrature of the Balmer-continuum band integrals of excited_hydrogen	367
tests/energy_update/conduction_floor_ tests.f90	what the Crank–Nicolson transport stage returns, refuses and conserves	360
tests/physics_probe/metal_photoionization_ fits.f90	every metal photoionization cross section against the published fit, and only where it is valid	343
tests/element_census_tests.f90	the elemental and charge invariants of a composition state	340
tests/energy_update/energy_update_ tests.f90	the residual-controlled temperature solve of the implicit energy source step	334
tests/spectrum_type/loaded_sed_threshold_ edges.f90	the photon grid built from a loaded SED, and the rates formed on it	304
tests/physics_probe/species_formation_ energy_table.f90	the formation-energy table of the energy ledger, and the reaction energies that are its differences	275
tests/physics_probe/lya_band_ transmission.f90	the stellar Ly α band reaching a cell through the atomic hydrogen column above it	272
tests/spectrum_type/wasp121b_sed_ sensitivity.f90	the production band quantities of the three WASP-121 b spectrum candidates	262
tests/coupled_source_step/coupled_source_ tests.f90	the coupled temperature-composition source step	260
tests/physics_probe/lyman_werner_cell_ mean.f90	the Lyman–Werner rate a cell carries is the mean of the local rate over that cell	234
tests/physics_probe/lya_beam_cell_mean.f90	the stellar Ly α field of a cell is the beam mean over the cell’s own optical depth	229
tests/spectrum_type/balmer_continuum_ field.f90	the two scalar Balmer-continuum rates of H($n=2$)	227
tests/physics_probe/positivity_limiter_ scaling.f90	the positivity scaling of the reconstructed face states (Zhang & Shu 2010)	223
tests/carrier_returned_state_acceptance/ carrier_returned_state_acceptance.f90	what makes a returned carrier state a solution, and what does not	222
tests/grid_and_gates/photon_grid_ threshold_edges.f90	every active absorber’s threshold against the bin partition of the photon grid	213
tests/physics_probe/photoionization_field_ substitution.f90	the order in which the field and the composition that attenuates it are brought into agreement	209
tests/adv_static_limit/stationarity_by_ the_mass_row.f90	which test decides that the state the post-process was handed is stationary in a cell	199
tests/physics_probe/co_destruction.f90	the CO destruction rates, the shielding table, the deposited energies and the cell mean	195
tests/physics_probe/recombination_ coefficient_fits.f90	every recombination coefficient against the published fit	189
tests/spectrum_type/photon_grid_n2_ floor.f90	the photon grid built when the excited-hydrogen coupling is armed	184
tests/carrier_reference_scales/carrier_ reference_scales.f90	the element a transported carrier is made of, and whether its chemistry rows can be differentiated	181

Test driver	What it asserts	SLOC
tests/carrier_constraint_attribution/carrier_constraint_attribution.f90	which residuals belong to the solver and which to a constraint	179
tests/adv_static_limit/enthalpy_flux_of_mass_divergence.f90	the third term of the advected energy equation	178
tests/grid_and_gates/free_outflow_boundary.f90	the outer free-outflow boundary on a supersonic wind, and its ghost state	174
tests/spectrum_type/planck_photon_field.f90	the incident flux put on the photon grid for the photospheric blackbody type	173
tests/a2_m2/a2_m2_kinetics_check.f90	the kinetics quantities the oxygen option adds on top of the rate audit	163
tests/physics_probe/lya_escape_probability.f90	the escape probability of the two-level closure against the published slab solution	163
tests/e1_h2/e1_h2_channel_check.f90	the mutually exclusive H ₂ photoabsorption channels	161
tests/adv_static_limit/upstream_caloric_energy.f90	the upstream energy of the advected energy equation	154
tests/physics_probe/photoionization_field_self_consistency.f90	the field one cell is solved in, against the composition that cell returns	148
tests/grid_and_gates/photon_grid_quadrature.f90	the photoionization rates of one unattenuated cell against the band quadrature	146
tests/physics_probe/co_helium_ion_sink.f90	He ⁺ + CO as a species sink, not only as a heat source	145
tests/physics_probe/riemann_wave_speeds.f90	signal speeds at a face, and the state one conservative update with that flux leaves behind	143
tests/physics_probe/helium_level_cooling_ledger.f90	which helium level each cooling channel of eval_cool is charged to	139
tests/physics_probe/heating_channel_closure.f90	one assembly of the volumetric heating rate, and the closure of its channels	136
tests/acceptance_classes/acceptance_class_contract.f90	one meaning of the ionization acceptance classes for every consumer	132
tests/physics_probe/opacity_model_p_ledger.f90	the radiative ledger of a cell whose cross sections are broadened by opacity model P	124
tests/physics_probe/photoevent_energy_ledger.f90	the energy an isolated photoabsorption event has to account for	117
tests/physics_probe/xuv_cell_mean_attenuation.f90	the beam attenuation the photoionization rate of a cell is evaluated with	115
tests/grid_and_gates/grid_window_indices.f90	the two index statements that decide which cells a quantity is read from	101
tests/physics_probe/h2_rovibrational_identity.f90	the identities tying the H ₂ partition function of the chemistry to the internal energy of the caloric EOS	96
tests/adv_static_limit/stationary_flux_refusal.f90	the sensitivity of the advected energy correction to a non-stationary mass flux	91
tests/physics_probe/ionization_threshold_turn_on.f90	one definition of each ionization threshold, and every cross section turning on exactly there	90
tests/grid_and_gates/grid_width_identity.f90	the cell-width and total-width identities of the radial grid	83
tests/physics_probe/h2_channel_detector_ratio.f90	the measured proton-to-H ₂ ⁺ ratio of Chung et al. (1993) against the four channels	83

Test driver	What it asserts	SLOC
tests/physics_probe/fine_structure_escape_probability.f90	the escape probability of the ground-term fine-structure lines, and the depth it is fed	80
tests/physics_probe/atomic_mass_and_radius_constants.f90	the helium mass and the astronomical constants of the input readers, one definition each	79
tests/physics_probe/h3p_cooling_limits.f90	the limits, the continuities and the published values of the H_3^+ cooling rate	79
tests/physics_probe/assertion_report.f90	one printed verdict line per assertion, and the failure count the driver exits on	74
tests/adv_static_limit/thermal_damkohler_limit.f90	the validity condition of the advection-corrected energy solve, as a function of the flow speed	72
tests/a2_m2/a2_m2_g3_run_check.f90	with every photolysis rate at zero, the solved O/OH/H ₂ O partition is the chemical equilibrium	71
tests/physics_probe/absorbed_fraction_switch.f90	$(1 - e^{-d})/d$, the fraction of the photons entering a cell that the cell absorbs	71
tests/constrained_network_layout/transported_proton_constraint_layout.f90	which species are unknowns of the continuation, and whether the system built is square	71
tests/test_columns.f90	synthetic columns whose species carry the mass density they are measured against	67
tests/physics_probe/h2_infrared_line_populations.f90	the LTE populations weighting the H ₂ quadrupole lines are those of the caloric EOS ladder	65
tests/physics_probe/wind_ae_bridge_composition.f90	the composition an EXHALE input.inp hands the Wind-AE solver	54
tests/physics_probe/voronov_carbon_ionization.f90	the electron-impact ionization of C ⁰ and C ⁺ against the published Voronov (1997) rows	51
tests/physics_probe/water_photolysis_lyman_alpha_yields.f90	the H ₂ O branching on the band that contains Ly α , against the published yields	49
tests/physics_probe/h3p_recombination_branching.f90	the two product channels of H_3^+ dissociative recombination	44
tests/energy_update/cooling_models.f90	analytic cooling laws for the tests of the implicit energy update	43
tests/species_masses/carrier_background_masses.f90	the mass the carrier transport gives each of its background collision partners	36
test-driver total		20 741
new-module total		50 744

So the 123 new core modules contribute 50 744 of the 69 179 added code lines; the remaining 18 435 added (and 1 121 of the 1 190 removed, the other 69 being the deleted HLLC estimator) are edits spread across the 38 modified files. The ten heaviest, by added code lines, are the main driver (EXHALE_main.f90, 3 498 added, 246 $\rightarrow$ 3 625 SLOC: the marching loop, the attempted-step adoption boundary, the run-mode contracts and the written-state assertions), the ionization equilibrium and its diagnostics (ionization_equilibrium.f90 2 274 and util_ion_eq.f90 1 803: the metal, molecular and oxygen networks, the acceptance classes, the photoionization field and the heating channels), the cooling module (Cool_coeff.f90, 2 250 added, 442 $\rightarrow$ 3 966 raw lines: CHIANTI metal-line fits, Fe tables, fine-structure statistical equilibrium, Penning and molecular rate coefficients), input parsing (input_read.f90, 1 753), the state reader and writer (load_IC.f90 1 063, write_output.f90 928, write_setup_report.f90 751), the advection-corrected post-processor (post_process_adv.f90, 647) and the cross sections (cross_sec.f90, 438). Those ten carry

15 405 of the 18 435. The rest of the edits are in the reconstruction, the fluxes, the boundary conditions, the grid, the time step, the energy equation and the System_* ionization systems, which wire the metals, the molecules, the opacity dispatcher, the Roche potential, the transonic/auto/Wind-AE IC, the excited-H and Ly α physics, the element diffusion and the Brent/Newton solver upgrades into the original solver.

In total the EXHALE core is now $\approx 20.3\times$ the original code base (3 514 $\rightarrow$ 71 503 SLOC), or $\approx 14.4\times$ counting only the production build (50 762 SLOC), touching almost all of the original files (38 of 47 modified, one deleted, plus 123 new); with the ported Wind-AE IC generator the full Fortran tree is 205 files / 76 259 SLOC ($\approx 21.7\times$). Beyond Fortran, the tree also carries the Python tooling (GUI, the EXHALE_transit.py transmission post-processor, plotting/analysis, the lower-atmosphere adapters and the elemental-flux closure driver) and 181 816 lines of documentation (187 md/tex files: the memos, the manual and this log) under docs/.

Fortran source inherited unchanged from ATES

Of the 47 upstream Fortran files (ATES/ATES-Code-main), **8 are byte-for-byte identical** in EXHALE (re-measured 2026-09-11): the numerics EXHALE reuses without touching, which are now the linear-algebra part of the included MINPACK nonlinear solver plus the gravity pre-evaluation. Table 9 lists them with their role. No file is identical after whitespace and case normalization but not byte-identical, so the normalized-unchanged set and the byte-identical set are the same eight.

Five files left the byte-identical set since the 2026-08-30 edition of this table, each for a documented reason rather than drift.

- nonlinear_system_solver/hybrd.f90, hybrd1.f90 and fdjac1.f90 carry a new unit_step_floor argument threaded from hybrd to fdjac1, which selects the forward-difference step of the Jacobian: `.false.` is MINPACK's own $h = \varepsilon|x_j|$ with the exact-zero fallback, `.true.` floors it at unit magnitude, $h = \varepsilon \max(1, |x_j|)$, which is the rule for unknowns already scaled so that unity is their natural size (the logarithmic unknowns of the constrained chemical equilibrium). The params dummy also became assumed-size, so one transport array serves systems of different length. hybrd1 passes `.false.` and is therefore unchanged in behaviour.
- states/Source.f90 gained the well-balanced arm of 2026-09-10 (item N37 of the stage-2 solver program, docs/ISSUES_20260909.md): under `Well balanced: True` the gravitational source and the geometric pressure term are not evaluated here at all, because they cancel the equilibrium part of the momentum flux difference identically and RK_rhs assembles the row from what is left (Käppeli & Mishra 2014, J. Comput. Phys. 259, 199, eq. 2.26). The file also reads its pressure through `caloric_eos` instead of a constant γ .
- radiation/J_inc.f90 carries the single-spectrum-type decision: `Spectrum type:` now states the spectrum of the whole photon grid, the XUV and the part below 13.6 eV alike, and no band is filled from a type the input did not select. The file also became the one definition of the eV-to-Hz conversion and of the Balmer-continuum head $e_{\text{th,H I}}/4$.

The two wave-speed estimators are no longer in the “effectively untouched” category either. `flux/speed_estimate_HLLC.f90` was deleted as dead code, as recorded above. `flux/speed_estimate_R0E.f90` was rewritten (99 code lines

added, 43 removed) into a star-state estimate that returns a status: the primitive-variable, two-rarefaction and two-shock branches of Toro (2009) chapter 9 are selected by the pressure ratio, the vacuum condition of his eq. 4.76 is tested first, and a face with no admissible star state is reported rather than given a square root of a negative ratio.

Table 9: The 8 Fortran files byte-identical between the original ATES (ATES-Code-main) and EXHALE (measured 2026-09-11), grouped by role.

File	Purpose
<i>MINPACK linear algebra and constants</i> (Argonne; dir nonlinear_system_solver/)	
qrfac.f90	QR factorization via Householder transformations
qform.f90	Accumulate the orthogonal factor Q from the QR factorization
r1updt.f90	Rank-one update of the QR factorization
r1mpyq.f90	Apply the Givens-rotation sequence of the rank-one update
dogleg.f90	Dogleg (trust-region) step combining Gauss–Newton and steepest descent
enorm.f90	Euclidean norm of a vector (overflow/underflow safe)
dpmpar.f90	Machine floating-point precision / range constants
<i>Grid setup</i>	
init/set_gravity_grid.f90	Pre-evaluate gravity at cell centers and interfaces